# Gene Essentiality from the Cell Graph

The backbone against flux balance analysis and Flux Cone Learning,

and why the solve helps the prediction where the metabolic module does not

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**Abstract**

Gene essentiality is the one endpoint on which a flux model has a published, non-trivial score: on the released Merzbacher 2025 yeast split, flux balance analysis reads an AUROC of 0.69 over all 1,121 labeled genes and Flux Cone Learning 0.73 on the 223 held-out genes. This document scores the cell graph transformer on the same held-out genes, puts FBA on every medium the ontology defines and a classical ladder of models on the same split, and asks two questions. The first is how far the held-out AUROC goes: a width-12 transformer reading the chromosome-pathway indicator and ProtT5, with FBA's four growth ratios on its logit, scores 0.893 [0.84, 0.94] under Flux Cone Learning's own fold protocol, 0.15 above it on the same genes, and every part of the gain is an input the ontology computes for every gene at no training cost: a two-layer MLP on the same three inputs reads the transformer's per-cell number in six seconds and 0.886 as the same fold ensemble. The second is why the solve helps and the metabolic module, which is built on the same genome-scale model, does not: routed through the module's growth column the prediction never beats the backbone alone, and the rounds locate the reason in the layer's constraint set, whose enzyme-capacity ceilings as parameterized contain no growing flux, so its alternating projections never reach the polytope FBA optimizes over and a linear program under its bounds is infeasible. On the genome-scale model's own bounds the module carries the solve, whether handed it as a warm start or solving for it in the forward pass with the gradient through the dual, and the gated module cell reaches the level of the solve appended as an input. Every number here is at a validation-selected epoch on a pinned test set, with a gene bootstrap and the across-seed range, and every table is generated from the result files.

*Section status:* ✓ ready ■ author-reviewed, keep stable ✗ not done, agents may edit freely

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# 1 The endpoint, its labels and its splits ✗

## 1.1 Vocabulary ✗

**AUROC** Area under the receiver operating characteristic, computed as the Mann-Whitney rank statistic with mid-ranks on ties: the probability that a randomly drawn essential gene scores above a randomly drawn nonessential one.

**backbone** The cell graph transformer over the nine-graph perturbed gene graph with `prot_T5_all` node features unless another encoding is named, read at the pooled perturbed tokens. The *pooled* arm is this backbone with a scalar head and no metabolic module.

**chromosome-pathway indicator** The 197-dimensional per-gene encoding `normalized_chrom_pathways` (`sgd_gene_graph_hot.py`): five SGD numeric gene features scaled to $[0, 1]$ (protein length, molecular weight, isoelectric point, the median and the median absolute deviation of the gene's expression), a one-hot of the gene's chromosome, and a multi-hot over the SGD pathways it is annotated to. *chrom-pathway* in figure labels; the name understates it, since five of its columns are continuous gene properties and two of those are expression.

**FBA affine** The readout that adds $w^\top r + b$ to the backbone's logit, where $r$ holds the deleted gene's knockout-to-wild-type growth ratio on each of four media from one FBA solve per gene; the *fba* arm.

**fold ensemble** Five models trained on the five cross-validation files (each a different fifth of the 787 labeled non-test genes held for validation), their logits averaged on the held-out genes; Flux Cone Learning's own test protocol.

**gene bootstrap** Resampling the scored genes with replacement (2,000 draws) and reporting the 2.5 and 97.5 percentiles of the statistic.

**GO multi-hot** The per-gene encoding over Gene Ontology terms (`go_term_hot.py`): one column per term, a gene's row marking the terms SGD annotates it to, propagated to every ancestor through *is_a* and *part_of* (the true-path rule), the three namespace roots dropped, and a term kept only if ten or more genes carry it: 2,362 columns over 5,501 annotated genes. *GO direct* keeps the annotated terms alone (2,887 columns, two or more genes); *GO BP*, *MF* and *CC* keep one namespace (1,580, 430, 352). The SGD file carries no evidence codes, so annotations inferred from mutant phenotypes are in the vector.

**held-out genes** The 223 genes Merzbacher 2025 held out of its own training (`test = 1` in the released split), of which 195 resolve as single-deletion genotypes in the build. Every model number in this document is on those 195 unless the gene set is named.

**layer that solves** The metabolic module with its flux replaced by the growth optimum of the linear program under the layer's own bounds, one solve per genotype in the forward pass, the bound duals as its gradient. Its *elastic* form lets the model's one forced flux (ATP maintenance) go unmet at a price, so the program is feasible for every genotype by construction.

**message passing** The update of a node's representation from an aggregate of its neighbors' representations, $h_i' = \gamma(h_i, \bigoplus_{j \in \mathcal{N}(i)} \phi(h_i, h_j))$; graph convolution (GCN), graph attention (GAT, GATv2), GraphSAGE and the graph isomorphism network (GIN) are choices of $\phi$, $\bigoplus$ and $\gamma$. *Relational* message passing (RGCN, RGAT) keeps one weight or one attention per edge type over a multigraph.

**metabolic module** The enzyme-constrained thermodynamic flux layer over yeast-GEM 9.0.2 that maps the perturbed representation to a flux vector by $K$ alternating projections onto the mass balance $Sv = 0$ and the box of bounds; here it is read at the objective column `r_2111` (growth). The *anchored* layer tightens the GEM's box by the enzyme-capacity ceilings and the second-law directions; the *stoichiometry-only* layer keeps the GEM's box.

**parsimonious FBA** Two linear programs: the growth optimum over the polytope $\{Sv = 0,\ \ell \le v \le u\}$ with the deleted gene's reactions closed through the gene-protein-reaction rules, then the flux of least total absolute value at that optimum. *pFBA*; *the solve* when the object is the optimum rather than the method.

**released labels** The 1,121 essentiality calls in the released split. The column named `essential` is a viability flag (1.0 marks a nonessential gene); it is inverted once and the four class counts are asserted against the paper's Methods.

**selected epoch** The argmax of a centered five-epoch moving average of the validation AUROC. The held-out genes are scored once, at that epoch.

**solve-derived encodings** Per-gene vectors computed from pFBA rather than from sequence: the deletion's own flux vector, its difference from the wild-type vector, and eight statistics of the gene's reactions under the wild-type solve and its flux variability (`ess_reference_flux_table.py`).

**warm start** The flux layer's proposal written as a reference flux plus a null-space residual of relative size $\alpha$, $v = v_{\text{ref}} + \alpha N z$, before the alternations; with a per-genotype table the reference is the deletion's own pFBA flux.

**Weisfeiler-Leman test** Color refinement: every node starts with a color, and each iteration replaces it by a hash of its own color and the multiset of its neighbors' colors; the multiset of colors after $k$ iterations is the graph's 1-WL signature. Different signatures prove two graphs non-isomorphic; equal signatures prove nothing. A message-passing network is at most as expressive as the test.

### 1.2 Where the labels come from ✗

The build is a union of the SGD essential-gene list (1,140 genotypes, all essential) and the single-mutant fitness screens (Costanzo 2016, Kuzmin 2018; 4,540 genotypes), 5,656 single-deletion genotypes in all. No fitness value enters any objective: the head is one logit scored by binary cross entropy against a 0/1 label, and every genotype whose gene carries no label arrives masked. Two label sources are used and named throughout:

**Table 1:** The split files, one per label source: genes per split with the essential count in parentheses. The test block is the released held-out genes under either source; validation is a class-stratified fifth of the remaining labeled genes at seed 0. Under the genome source the released label wins on every disagreement (12 genes).

| split file | labels | train (ess) | val (ess) | test (ess) |
|---|---|---|---|---|
| `ess_fcl_resplit0_val0.2_s0` | released (1,121) | 627 (99) | 157 (25) | 198 (32) |
| `ess_fcl_resplit1_val0.2_s0` | released (1,121) | 627 (99) | 157 (25) | 198 (32) |
| `ess_fcl_resplit2_val0.2_s0` | released (1,121) | 627 (99) | 157 (25) | 198 (32) |
| `ess_fcl_resplit3_val0.2_s0` | released (1,121) | 627 (99) | 157 (25) | 198 (32) |
| `ess_fcl_resplit4_val0.2_s0` | released (1,121) | 627 (99) | 157 (25) | 198 (32) |
| `ess_fcl_resplit5_val0.2_s0` | released (1,121) | 627 (99) | 157 (25) | 198 (32) |
| `ess_fcl_resplit6_val0.2_s0` | released (1,121) | 627 (99) | 157 (25) | 198 (32) |
| `ess_fcl_resplit7_val0.2_s0` | released (1,121) | 627 (99) | 157 (25) | 198 (32) |
| `ess_fcl_resplit8_val0.2_s0` | released (1,121) | 627 (99) | 157 (25) | 198 (32) |
| `ess_fcl_resplit9_val0.2_s0` | released (1,121) | 627 (99) | 157 (25) | 198 (32) |
| `ess_fcl_val0.2_s0` | released (1,121) | 629 (100) | 158 (25) | 195 (31) |
| `ess_genome_heldout_val0.2_s0` | released + SGD essential + viable fitness (5,572) | 629 (100) | 158 (25) | 4674 (974) |
| `ess_genome_val0.2_s0` | released + SGD essential + viable fitness (5,795) | 4,368 (879) | 1,093 (220) | 195 (31) |

### 1.3 The protocol inherited from 027, and what it got wrong ✗

The wave-9 protocol pinned the held-out genes and let the data module draw train and validation at 80/10/10 over the remaining 5,461 records. Of those, 926 carry a label, so the validation set held 75 labeled genes with about 12 essential, on which a single AUROC read has a Hanley-McNeil standard error near 0.07; 10 percent of the labeled genes sat in a random test block nothing reports; and 4,535 unlabeled records were forwarded every epoch under a False mask. A split file replaces the draw: it names every record's split, is class-stratified, carries the labels it was stratified on, and the trainer drops every record it does not name.

## 2 The comparators, computed rather than quoted ✗

### 2.1 Flux balance analysis on four media ✗

The published FBA comparator is cobrapy `single_gene_deletion` with glucose as the carbon source, scored on all 1,121 labeled genes. The phenotype it is compared against was called on rich medium, and the model here is scored on the 195 held-out genes, so the comparison as published is neither on the same medium nor on the same genes. Table 2 recomputes it on yeast-GEM 9.0.2 under the exchange bounds the model ships with and under the three media the ontology defines, with the essentiality score being the negative knockout-to-wild-type growth ratio, rounded to $10^{-3}$ so that the roughly 900 deletions that grow at 0.99 to 1.00 of the wild type rank as the tie they are (without the rounding two identical runs differed by 0.03 AUROC on the held-out genes).

**Table 2:** FBA single-gene deletion on yeast-GEM 9.0.2 (cobrapy 0.30.0), scored as the negative knockout-to-wild-type growth ratio rounded to $10^{-3}$. AUROC on all 1,121 labeled genes, on the 223 held-out genes, and on the 195 held-out genes the model scores with a 2,000-draw gene bootstrap interval. 1121 of the labeled genes are in the GEM; the rest keep wild-type growth. Bold marks the highest value in a column.

| medium | WT growth (1/h) | lethal | all 1,121 | held-out 223 | our 195 [95% CI] |
|---|---|---|---|---|---|
| shipped bounds (glucose minimal, aerobic) | 0.086 | 186 | 0.707 | 0.633 | 0.654 [0.56, 0.75] |
| SM, synthetic minimal | 0.314 | 160 | 0.707 | 0.642 | 0.666 [0.57, 0.76] |
| SC, synthetic complete | 0.543 | 91 | **0.719** | **0.653** | **0.666** [0.57, 0.76] |
| YPD approximation (no peptone) | 0.535 | 104 | 0.715 | **0.653** | 0.664 [0.57, 0.76] |

The medium moves the number of lethal deletions by a factor of two and the AUROC by about 0.01. The published 0.69 on all labeled genes is reproduced to within 0.02.

## 2.2 yeast-GEM's own benchmark, and accuracy against balanced accuracy ✗

yeast-GEM ships its own essentiality benchmark and prints it in its README: on 1,107 genes, an accuracy of 0.902 with 933 true non-essential, 65 true essential, 94 false non-essential and 15 false essential calls. Its release file (`results/external/yeast-GEM-9.1.1/`, version 9.1.1 at commit `2d594ae`, 2026-08-30) says how: the Kennedy synthetic complete medium is written onto the exchanges (every exchange closed, 28 uptakes at $-0.5$, glucose at $-20$, fifteen inorganic uptakes open), each gene is deleted through the gene-protein-reaction rules, and a deletion is inviable when its growth ratio is below $10^{-6}$; the genes are the SGD verified ORFs in the model, and the labels are the Stanford deletion project's inviable list of 2011, scored against the model at that one rule, a protocol the repository attributes to Heavner and Price (2015). The released split's labels are the same truth: on the 1,104 genes both name, the two sources agree on 1,102.

Table 3 puts every scorer in this document at that rule. Two things are visible at once. The first is that accuracy is the wrong statistic for this endpoint: 86 percent of the genes are nonessential, so a scorer that calls nothing essential reads 0.856 on yeast-GEM's set and 0.841 on the 195 held-out genes, and yeast-GEM's 0.902 is that floor plus four and a half points; its recall on the essential class is 0.41 and its balanced accuracy 0.70. FBA on the previous release, 9.0.2, under the ontology's own synthetic complete recipe reads 0.907 on the released labels with a recall of 0.46, the same rule on the same truth with a different medium composition, so the 9.1.1 number is reproduced to within half a point. The second is where the transformer sits at a threshold: at its loss's decision boundary it reads 0.856, the majority floor, with a recall of 0.68 and a balanced accuracy of 0.78, and at the threshold that calls 31 genes essential it reads 0.867 with precision and recall both at 0.58. The released Flux Cone Learning predictions at $P(\text{viable}) < 0.5$ call eight genes essential on the 195, all eight correctly, for a recall of 0.26. A ranking statistic is what separates the scorers, and the balanced accuracies, 0.63 to 0.72 for the solvers and 0.78 for the transformer, order them the way the AUROCs do.

**Table 3:** Every scorer at the rule yeast-GEM reports its benchmark with: a deletion is called essential when the criterion in the rule column holds, and the call is scored against the label as a confusion matrix with essential as the positive class (TP/FN/FP/TN). Accuracy is followed by the majority class's share of the gene set in parentheses; precision and recall are for the essential class; balanced accuracy is the mean of recall and specificity. yeast-GEM 9.1.1's row is its own release file on its own labels (the Stanford deletion project's inviable list over the model's verified ORFs); those labels and the released split agree on 1102 of the 1,104 genes both name. Every other row is on the released labels. The transformer is the round-12 fold ensemble at the loss's decision boundary and at the threshold that calls as many genes essential as the labels hold. Bold marks the highest value in a column.

| scorer | rule | gene set | ess/n | TP/FN/FP/TN | acc (maj) | prec | rec | spec | bal |
|---|---|---|---|---|---|---|---|---|---|
| yeast-GEM 9.1.1 (its own benchmark) | growth ratio < 1e-6, Kennedy SC medium | 1,107 verified ORFs, Stanford inviable list | 159/1,107 | 65/94/15/933 | 0.902 (0.856) | 0.812 | 0.409 | 0.984 | 0.696 |
| FBA, yeast-GEM 9.0.2, shipped | growth ratio < 1e-6 | 1,121 labeled (released labels) | 157/1,121 | 82/75/104/860 | 0.840 (0.860) | 0.441 | 0.522 | 0.892 | 0.707 |
| FBA, yeast-GEM 9.0.2, shipped | growth ratio < 1e-6 | 195 held-out (released labels) | 31/195 | 11/20/11/153 | 0.841 (0.841) | 0.500 | 0.355 | 0.933 | 0.644 |
| FBA, yeast-GEM 9.0.2, SM | growth ratio < 1e-6 | 1,121 labeled (released labels) | 157/1,121 | 78/79/82/882 | 0.856 (0.860) | 0.487 | 0.497 | 0.915 | 0.706 |
| FBA, yeast-GEM 9.0.2, SM | growth ratio < 1e-6 | 195 held-out (released labels) | 31/195 | 11/20/7/157 | 0.862 (0.841) | 0.611 | 0.355 | 0.957 | 0.656 |
| FBA, yeast-GEM 9.0.2, SC | growth ratio < 1e-6 | 1,121 labeled (released labels) | 157/1,121 | 72/85/19/945 | **0.907** (0.860) | 0.791 | 0.459 | 0.980 | 0.719 |
| FBA, yeast-GEM 9.0.2, SC | growth ratio < 1e-6 | 195 held-out (released labels) | 31/195 | 11/20/2/162 | 0.887 (0.841) | 0.846 | 0.355 | 0.988 | 0.671 |
| FBA, yeast-GEM 9.0.2, YPD | growth ratio < 1e-6 | 1,121 labeled (released labels) | 157/1,121 | 73/84/31/933 | 0.897 (0.860) | 0.702 | 0.465 | 0.968 | 0.716 |
| FBA, yeast-GEM 9.0.2, YPD | growth ratio < 1e-6 | 195 held-out (released labels) | 31/195 | 11/20/2/162 | 0.887 (0.841) | 0.846 | 0.355 | 0.988 | 0.671 |
| Flux Cone Learning (released predictions) | P(viable) < 0.5 | 223 released held-out (released labels) | 31/223 | 8/23/0/192 | 0.897 (0.861) | **1.000** | 0.258 | **1.000** | 0.629 |
| Flux Cone Learning (released predictions) | P(viable) < 0.5 | 195 held-out (released labels) | 31/195 | 8/23/0/164 | 0.882 (0.841) | **1.000** | 0.258 | **1.000** | 0.629 |

| scorer | rule | gene set | ess/n | TP/FN/FP/TN | acc (maj) | prec | rec | spec | bal |
|---|---|---|---|---|---|---|---|---|---|
| FBA (released predictions) | growth < 1e-6 | 195 held-out (released labels) | 31/195 | 11/20/11/153 | 0.841 (0.841) | 0.500 | 0.355 | 0.933 | 0.644 |
| transformer, round-12 fold ensemble | logit > 0 | 195 held-out (released labels) | 31/195 | 21/10/18/146 | 0.856 (0.841) | 0.538 | **0.677** | 0.890 | **0.784** |
| transformer, round-12 fold ensemble | top 31 logits (label prevalence) | 195 held-out (released labels) | 31/195 | 18/13/13/151 | 0.867 (0.841) | 0.581 | 0.581 | 0.921 | 0.751 |

## 2.3 Flux Cone Learning on the same genes ✗

Recomputed from the authors' released per-gene predictions: 0.7308 on the 223 held-out genes (published 0.73) and 0.742 on the 195 genes the model scores. On those same 195 genes the released FBA predictions score 0.640, so the published pair overstates FBA by about 0.06 relative to a head-to-head on one gene set. The paper prints a third number, 0.78 ± 0.065 in the legend of its Figure 3a, which is the mean over five cross-validation folds drawn inside its 898 training genes; a model scored on the held-out genes is compared with the 0.73 line, never with that one.

## 2.4 The classical ladder on the same split ✗

The fitness and expression lines each carry a ladder of models that need no graph and no training beyond a solve, and essentiality gets the classification form of it: FBA's growth ratio with no fitting, then four model classes ($L_2$ logistic regression, a random forest, a radial-basis support vector classifier, cosine nearest neighbors) on each of twenty-four gene encodings, every pair fit twice, with and without the four FBA growth ratios appended. The encodings are the seventeen the fitness ladder carries (codon frequency, the chromosome and pathway indicator, six nucleotide windows, the two FUDT embeddings, the species language model, CaLM, ProtT5 and ESM2 with and without dubious genes, and their concatenation, `mixed`) plus seven controls: a one-hot gene identity, and standard normal vectors of dimension 1, 10, 100, 1,000, 6,607 (the one-hot's width) and 8,192, drawn once at seed 0. A one-hot rung must score chance on held-out genes, since every held-out column is zero in training, and a random vector must score chance at every width; either scoring above chance would mean the split leaks. Every fitted rung is fit on the split file's 629 training genes, its one hyperparameter chosen by five-fold stratified cross-validation inside those genes, and scored once on the 158 validation and 195 held-out genes, on which nothing was chosen. The transformer cells in Sections 3 and 4 train on the same 629 genes and score on the same 195. Three further encodings come from the solve itself rather than from the sequence, added after round 8 (`ess_reference_flux_table.py`): the deletion's own parsimonious FBA flux vector on YPD (4,131 reactions), that vector's difference from the wild-type vector, and eight numbers about the gene's own reactions under the wild-type solve and its flux variability at 90 percent of the optimum (reaction count, reactions carrying flux, the largest and the summed absolute flux, the largest forced flux, the smallest and largest variability range, and the deletion's growth ratio). Figure 1 shows all 216 fitted rungs; Table 4 keeps, per encoding, the model class whose cross-validation AUROC is highest.

**Table 4:** The classical ladder on the released-label split (`ess_fcl_val0.2_s0.json.gz`). Per encoding, the model class whose five-fold cross-validation AUROC inside the 629 training genes is highest, with and without FBA's four growth ratios appended; the validation genes and the 195 held-out genes (31 essential) are scored once, on genes no fit touched, the held-out number with a 2,000-draw gene bootstrap. FBA alone is not fitted. Every model and encoding is in Figure 1. Bold marks the highest value in a column.

| encoding | best model (chosen) | dim | CV | val | held-out [95% CI] |
|---|---|---|---|---|---|
| FBA growth | medium=shipped | – | – | 0.683 | 0.654 [0.56, 0.75] |
| FBA growth | medium=SM | – | – | 0.710 | 0.666 [0.57, 0.76] |
| FBA growth | medium=SC | – | – | 0.726 | 0.666 [0.57, 0.76] |
| FBA growth | medium=YPD | – | – | 0.714 | 0.664 [0.57, 0.76] |
| one_hot_gene | logistic (C=10) | 6,579 | 0.504 | 0.508 | 0.506 [0.50, 0.52] |
| one_hot_gene + FBA | random forest (max_depth=8) | 6,583 | 0.775 | 0.730 | 0.679 [0.59, 0.77] |
| random_1 | random forest (max_depth=4) | 1 | 0.598 | 0.560 | 0.577 [0.46, 0.69] |
| random_1 + FBA | cosine kNN (k=10) | 5 | 0.806 | 0.699 | 0.689 [0.56, 0.80] |
| random_10 | cosine kNN (k=50) | 10 | 0.549 | 0.481 | 0.455 [0.35, 0.56] |
| random_10 + FBA | logistic (C=0.001) | 14 | 0.786 | 0.684 | 0.596 [0.46, 0.72] |
| random_100 | random forest (max_depth=None) | 100 | 0.522 | 0.376 | 0.454 [0.34, 0.57] |
| random_100 + FBA | random forest (max_depth=4) | 104 | 0.773 | 0.662 | 0.560 [0.42, 0.70] |
| random_1000 | random forest (max_depth=4) | 1,000 | 0.475 | 0.490 | 0.509 [0.40, 0.62] |
| random_1000 + FBA | random forest (max_depth=None) | 1,004 | 0.778 | 0.749 | 0.655 [0.53, 0.78] |
| random_6607 | cosine kNN (k=1) | 6,607 | 0.549 | 0.477 | 0.575 [0.49, 0.66] |
| random_6607 + FBA | random forest (max_depth=None) | 6,611 | 0.734 | 0.713 | 0.628 [0.50, 0.75] |
| random_8192 | cosine kNN (k=50) | 8,192 | 0.523 | 0.515 | 0.588 [0.49, 0.69] |
| random_8192 + FBA | random forest (max_depth=8) | 8,196 | 0.768 | 0.748 | 0.680 [0.57, 0.79] |
| codon_frequency | SVC (C=1) | 64 | 0.700 | 0.663 | 0.695 [0.59, 0.79] |
| codon_frequency + FBA | SVC (C=1) | 68 | 0.849 | 0.772 | 0.818 [0.73, 0.90] |
| normalized_chrom_pathways | logistic (C=10) | 197 | 0.705 | 0.722 | 0.775 [0.68, 0.86] |

| encoding | best model (chosen) | dim | CV | val | held-out [95% CI] |
|---|---|---|---|---|---|
| normalized_chrom_pathways + FBA | logistic (C=0.01) | 201 | 0.802 | 0.791 | 0.830 [0.75, 0.90] |
| nt_window_5979 | logistic (C=0.01) | 2,560 | 0.579 | 0.532 | 0.570 [0.46, 0.67] |
| nt_window_5979 + FBA | logistic (C=0.01) | 2,564 | 0.743 | 0.705 | 0.669 [0.55, 0.79] |
| nt_window_5979_max | logistic (C=0.001) | 2,560 | 0.626 | 0.608 | 0.636 [0.54, 0.73] |
| nt_window_5979_max + FBA | logistic (C=0.01) | 2,564 | 0.777 | 0.786 | 0.663 [0.54, 0.78] |
| nt_window_three_prime_5979 | logistic (C=0.001) | 2,560 | 0.555 | 0.580 | 0.526 [0.41, 0.64] |
| nt_window_three_prime_5979 + FBA | logistic (C=0.001) | 2,564 | 0.732 | 0.720 | 0.637 [0.51, 0.76] |
| nt_window_five_prime_5979 | cosine kNN (k=25) | 2,560 | 0.594 | 0.518 | 0.427 [0.32, 0.54] |
| nt_window_five_prime_5979 + FBA | logistic (C=0.01) | 2,564 | 0.733 | 0.657 | 0.693 [0.59, 0.79] |
| nt_window_three_prime_300 | random forest (max_depth=4) | 2,560 | 0.564 | 0.537 | 0.503 [0.39, 0.62] |
| nt_window_three_prime_300 + FBA | random forest (max_depth=None) | 2,564 | 0.752 | 0.757 | 0.659 [0.53, 0.78] |
| nt_window_five_prime_1003 | logistic (C=0.001) | 2,560 | 0.541 | 0.485 | 0.550 [0.43, 0.68] |
| nt_window_five_prime_1003 + FBA | logistic (C=0.01) | 2,564 | 0.727 | 0.665 | 0.636 [0.52, 0.75] |
| fudt_upstream | SVC (C=1) | 768 | 0.611 | 0.583 | 0.578 [0.47, 0.69] |
| fudt_upstream + FBA | random forest (max_depth=None) | 772 | 0.775 | 0.747 | 0.707 [0.60, 0.81] |
| fudt_downstream | SVC (C=10) | 768 | 0.589 | 0.553 | 0.469 [0.36, 0.58] |
| fudt_downstream + FBA | logistic (C=0.1) | 772 | 0.770 | 0.735 | 0.706 [0.59, 0.81] |
| species_lm_5p_3p | logistic (C=0.001) | 1,536 | 0.634 | 0.631 | 0.571 [0.47, 0.68] |
| species_lm_5p_3p + FBA | logistic (C=0.01) | 1,540 | 0.763 | 0.733 | 0.679 [0.56, 0.79] |
| calm | SVC (C=1) | 768 | 0.755 | 0.777 | 0.734 [0.64, 0.82] |
| calm + FBA | random forest (max_depth=8) | 772 | 0.850 | 0.833 | 0.806 [0.72, 0.89] |
| prot_T5_all | SVC (C=10) | 1,024 | 0.781 | 0.767 | 0.732 [0.62, 0.83] |
| prot_T5_all + FBA | logistic (C=0.01) | 1,028 | 0.842 | 0.826 | 0.821 [0.74, 0.90] |
| prot_T5_no_dubious | SVC (C=10) | 1,024 | 0.781 | 0.767 | 0.732 [0.62, 0.83] |
| prot_T5_no_dubious + FBA | logistic (C=0.01) | 1,028 | 0.842 | 0.826 | 0.821 [0.74, 0.90] |
| esm2_t33_650M_UR50D_all | random forest (max_depth=8) | 1,280 | 0.731 | 0.694 | 0.708 [0.61, 0.80] |
| esm2_t33_650M_UR50D_all + FBA | random forest (max_depth=8) | 1,284 | 0.815 | 0.794 | 0.758 [0.66, 0.86] |
| esm2_t33_650M_UR50D_no_dubious | random forest (max_depth=8) | 1,280 | 0.731 | 0.694 | 0.708 [0.61, 0.80] |
| esm2_t33_650M_UR50D_no_dubious + FBA | random forest (max_depth=8) | 1,284 | 0.815 | 0.794 | 0.758 [0.66, 0.86] |
| mixed | logistic (C=10) | 6,149 | 0.788 | 0.794 | 0.768 [0.66, 0.87] |
| mixed + FBA | logistic (C=10) | 6,153 | 0.833 | 0.837 | 0.792 [0.69, 0.89] |
| pfba_deletion_ypd | random forest (max_depth=None) | 4,131 | 0.770 | 0.723 | 0.756 [0.65, 0.86] |
| pfba_deletion_ypd + FBA | random forest (max_depth=None) | 4,135 | 0.776 | 0.736 | 0.764 [0.66, 0.86] |
| pfba_deletion_delta_ypd | random forest (max_depth=None) | 4,131 | 0.769 | 0.723 | 0.756 [0.65, 0.86] |
| pfba_deletion_delta_ypd + FBA | random forest (max_depth=None) | 4,135 | 0.776 | 0.736 | 0.764 [0.66, 0.86] |
| gene_flux_stats_ypd | random forest (max_depth=None) | 8 | 0.837 | 0.760 | 0.797 [0.70, 0.88] |
| gene_flux_stats_ypd + FBA | random forest (max_depth=None) | 12 | 0.847 | 0.777 | 0.817 [0.73, 0.90] |
| pathways | logistic (C=0.01) | 173 | 0.673 | 0.708 | 0.771 [0.68, 0.85] |
| pathways + FBA | random forest (max_depth=8) | 177 | 0.808 | 0.776 | 0.814 [0.73, 0.89] |
| chromosome | cosine kNN (k=10) | 17 | 0.523 | 0.537 | 0.532 [0.44, 0.63] |
| chromosome + FBA | logistic (C=0.1) | 21 | 0.758 | 0.768 | 0.692 [0.58, 0.80] |
| go | random forest (max_depth=None) | 2,362 | 0.852 | 0.895 | 0.807 [0.72, 0.88] |
| go + FBA | random forest (max_depth=None) | 2,366 | 0.877 | **0.915** | **0.844** [0.77, 0.91] |
| go_direct | logistic (C=0.01) | 2,887 | 0.827 | 0.770 | 0.739 [0.63, 0.84] |
| go_direct + FBA | logistic (C=0.01) | 2,891 | **0.877** | 0.805 | 0.787 [0.68, 0.88] |
| go_bp | random forest (max_depth=8) | 1,580 | 0.828 | 0.862 | 0.762 [0.66, 0.86] |
| go_bp + FBA | random forest (max_depth=8) | 1,584 | 0.856 | 0.884 | 0.815 [0.71, 0.90] |

The controls behave. The one-hot rung scores 0.506 and the six random widths score 0.45 to 0.59, chance within the 195-gene noise, and with FBA appended they score 0.56 to 0.69, FBA's own number, so no rung reads anything but its features. Without FBA four encodings are clear of 0.70: ProtT5 0.732, CaLM 0.734, the concatenation 0.768, and the chromosome-pathway indicator 0.775 [0.68, 0.86], each level with Flux Cone Learning's 0.742 on the same genes. FBA lifts every one of them: logistic regression on ProtT5 with the ratios scores 0.821 [0.74, 0.90] from 1,029 parameters, the support vector classifier on 64 codon frequencies 0.818 [0.73, 0.90], the random forest on CaLM 0.806 [0.72, 0.89], and logistic regression on the 197-dimensional chromosome-pathway indicator 0.830 [0.75, 0.90]. The rung with the highest cross-validation AUROC, the random forest on CaLM with FBA at 0.850, scores 0.806 on the held-out genes; the highest held-out number in the figure, 0.841 for cosine nearest neighbors on codon frequency with FBA, is the maximum over 216 rungs and is not a selected model. Neither an embedding nor the solver reaches 0.80 alone, and their sum does on four encodings. FBA's ratio is a feature the ontology computes for every gene at no training cost, so the transformer is measured against 0.80 to 0.83 from here on, not against the published 0.73.

The annotation vectors on their own. Logistic regression on the 173-column pathway multi-hot alone scores 0.771 [0.68, 0.85] with a cross-validation AUROC of 0.673, and with the four ratios 0.822, the level ProtT5 with the ratios reaches from 1,029 parameters; the chromosome one-hot alone is below chance (0.421) and 0.692 with the ratios. The GO multi-hot is the strongest single encoding on the ladder: logistic regression on the propagated vector scores 0.808 [0.72, 0.89] with no solve at all, above every other encoding without FBA, and 0.833 with the ratios; the random forest on it with the ratios scores 0.844 [0.77, 0.91] at a cross-validation AUROC of 0.877, the highest of any rung. The propagated form beats the direct annotations under every model (0.808 against 0.739 linear without the ratios), and

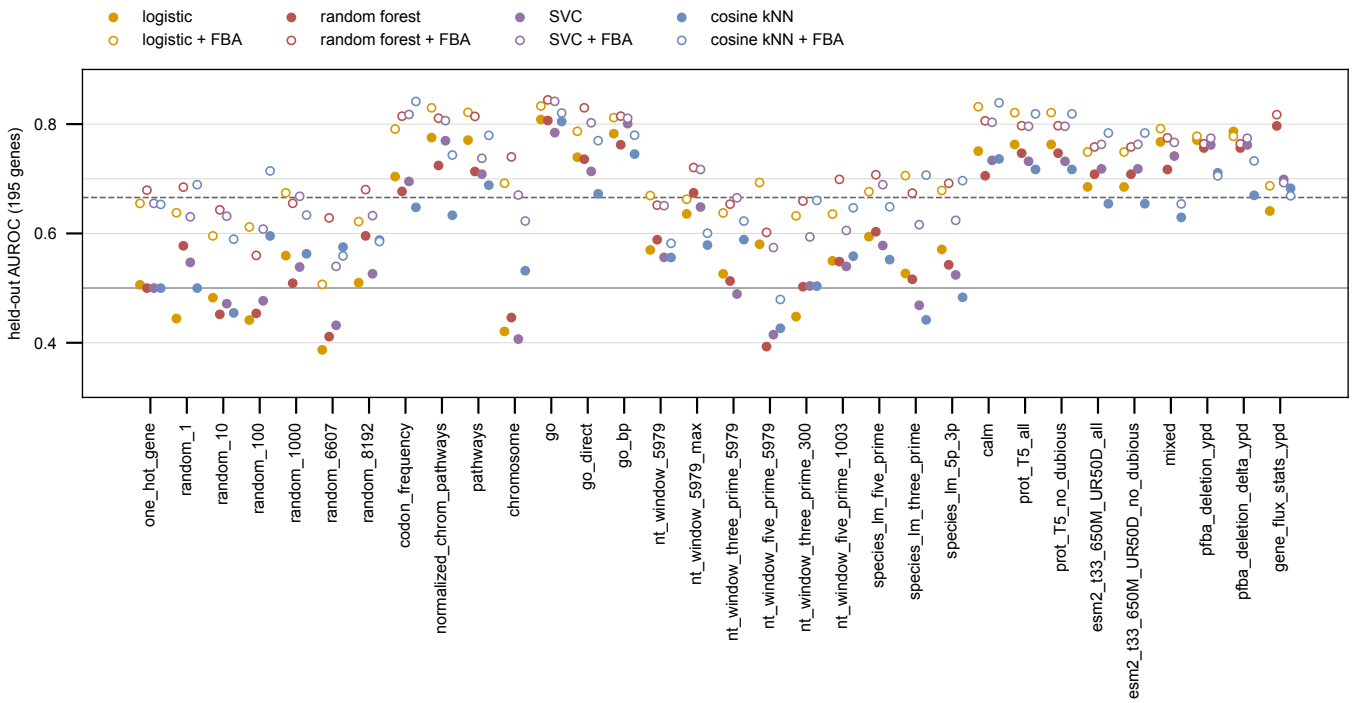



**Figure 1: The classical ladder on the released-label split.** Held-out AUROC on the 195 genes for every model class (color) on every encoding (column), filled without and open with the four FBA growth ratios appended; the dashed line is FBA alone on the shipped medium (0.654), the solid line chance. The last three columns are the solve-derived encodings. The hyperparameter of each mark was chosen by five-fold cross-validation inside the 629 training genes. Written by `experiments/028-gene-essentiality/scripts/logistic_baselines.py`.

the biological-process namespace alone reaches 0.812 with them.

The solve carries more than its growth ratio. The eight reaction statistics under a random forest score 0.797 [0.70, 0.88] with no sequence information, and 0.817 [0.73, 0.90] with the four ratios appended, at the highest cross-validation AUROC in the ladder (0.847); the growth ratio alone, which is one of the eight, scores 0.645 on the same genes. The full deletion flux vector scores 0.756 under its selected model and 0.771 under logistic regression, level with ProtT5 and with the pooled transformer, so a gene's position in the wild-type flux solution and how much the solve moves when it is deleted rank essential genes about as well as its protein sequence does, and the two sources are different: no other encoding reaches 0.80 without the solve, and the eight statistics come within 0.003 of it with nothing else.

## 2.5 Graph-only baselines: message passing over the gene graph ✗

The Gene_Graph project (2021 to 2023) classified essential genes by two-layer graph convolution and graph attention over YeastMine's physical, genetic and regulatory interaction graphs with all-ones node inputs, and saved no number. The same design is rerun here on the released split (`graph_baselines.py`): transductive node classification on one gene graph, where every gene's logit is a function of its neighborhood and its own features and no genotype is perturbed, on the physical interaction graph and on the union of the nine graphs the transformer reads, with all ones, with log degree and the clustering coefficient, or with ProtT5 as inputs, beside an MLP on ProtT5 with no graph (Table 5).

**Table 5:** Graph-only baselines on the released split: transductive node classification on one gene graph with no genotype perturbed, two layers of width 64, class-weighted binary cross entropy, 200 epochs, the epoch selected by a five-epoch moving average of the validation AUROC, scored once on the 195 held-out genes at seeds 2700 to 2702. The graph is the physical interaction graph or the union of the nine gene graphs the transformer reads; the inputs are all ones (the Gene_Graph design), log degree with the clustering coefficient, ProtT5, or ProtT5 with the chromosome-pathway indicator and the four FBA growth ratios as node features; two arms run 400 epochs. The MLP rows have no graph. Bold marks the highest value in a column.

| model | graph | inputs | dim | params | mean | sd | seeds | selected epochs |
|---|---|---|---|---|---|---|---|---|
| MLP | none | ProtT5 | 1,024 | 69,825 | 0.754 | 0.003 | 0.756, 0.755, 0.750 | 62, 58, 56 |
| GCN | physical | ones | 1 | 4,353 | 0.579 | 0.003 | 0.577, 0.577, 0.583 | 88, 100, 49 |
| GCN | physical | degree + clustering | 2 | 4,417 | 0.582 | 0.009 | 0.592, 0.571, 0.583 | 29, 26, 3 |
| GCN | physical | ProtT5 | 1,024 | 69,825 | 0.769 | 0.002 | 0.770, 0.766, 0.771 | 194, 193, 197 |
| GATv2 | physical | ones | 1 | 8,897 | 0.495 | 0.034 | 0.455, 0.493, 0.538 | 137, 148, 50 |
| GATv2 | physical | degree + clustering | 2 | 9,025 | 0.592 | 0.003 | 0.592, 0.588, 0.596 | 160, 192, 116 |
| GATv2 | physical | ProtT5 | 1,024 | 139,841 | 0.760 | 0.018 | 0.747, 0.747, 0.786 | 101, 93, 79 |
| GCN | union of nine | ones | 1 | 4,353 | 0.613 | 0.000 | 0.613, 0.613, 0.613 | 8, 5, 3 |
| GCN | union of nine | degree + clustering | 2 | 4,417 | 0.638 | 0.021 | 0.661, 0.644, 0.610 | 197, 197, 2 |
| GCN | union of nine | ProtT5 | 1,024 | 69,825 | 0.834 | 0.003 | 0.836, 0.829, 0.836 | 197, 196, 197 |
| GATv2 | union of nine | ones | 1 | 8,897 | 0.560 | 0.014 | 0.578, 0.556, 0.545 | 38, 107, 125 |
| GATv2 | union of nine | degree + clustering | 2 | 9,025 | 0.702 | 0.025 | 0.699, 0.672, 0.733 | 197, 191, 197 |
| GATv2 | union of nine | ProtT5 | 1,024 | 139,841 | 0.796 | 0.020 | 0.809, 0.768, 0.811 | 118, 197, 119 |
| MLP | none | ProtT5 + chrom-pathway | 1,221 | 82,433 | 0.817 | 0.000 | 0.817, 0.817, 0.818 | 70, 61, 60 |
| GCN | union of nine | ProtT5 + chrom-pathway | 1,221 | 82,433 | 0.850 | 0.003 | 0.849, 0.854, 0.847 | 197, 196, 197 |
| MLP | none | ProtT5 + FBA ratios | 1,028 | 70,081 | 0.803 | 0.013 | 0.814, 0.785, 0.810 | 71, 197, 76 |
| GCN | union of nine | ProtT5 + FBA ratios | 1,028 | 70,081 | 0.832 | 0.002 | 0.829, 0.833, 0.834 | 195, 197, 197 |
| MLP | none | ProtT5 + chrom-pathway + FBA ratios | 1,225 | 82,689 | 0.870 | 0.001 | 0.870, 0.872, 0.869 | 55, 58, 55 |
| GCN | union of nine | ProtT5 + chrom-pathway + FBA ratios | 1,225 | 82,689 | 0.847 | 0.005 | 0.843, 0.844, 0.855 | 195, 197, 196 |
| GCN | union of nine | ProtT5 (400 epochs) | 1,024 | 69,825 | 0.857 | 0.001 | 0.859, 0.856, 0.857 | 392, 397, 394 |
| GCN | union of nine | ProtT5 + chrom-pathway + FBA ratios (400 epochs) | 1,225 | 82,689 | **0.877** | 0.004 | 0.874, 0.874, 0.883 | 381, 396, 392 |

Structure alone carries little and structure with sequence carries a lot. With all-ones inputs the models read 0.50 to 0.61, chance to the ladder's random-vector level, and with degree and clustering 0.58 to 0.70. With ProtT5 the MLP reads 0.754, the ladder's logistic rung plus 0.02, and graph convolution over the union of the nine graphs reads 0.836, 0.829 and 0.836, a seed mean of 0.834 with a spread of 0.003 from 69,825 parameters: the edges add 0.08 to the protein embedding, the physical graph alone adds 0.015, and attention adds nothing over convolution on either graph. That cell is above the pooled transformer (0.774) and level with the transformer with FBA's four ratios (0.821) with no solve at all, and its three seeds select epochs 196 to 197 of 200 with the validation curve still rising; at 400 epochs the same cell reads 0.857 and selects epochs 392 to 397, so the budget is what bounds it.

The transformer's inputs move these cells the way they move the transformer, and the comparison says where the transformer's number comes from. Given the chromosome-pathway indicator and the four FBA ratios as node features beside ProtT5, the MLP with no graph reads 0.870, 0.872 and 0.869, a seed mean of 0.870 with a spread of 0.001 in six seconds of CPU, which is the round-9 transformer's 0.871 from the same three inputs; graph convolution on the union reads 0.847 at 200 epochs and 0.877 at 400, still rising. The indicator alone beside ProtT5 lifts the MLP to 0.817 and the convolution to 0.850, the ratios alone to 0.803 and 0.832. On this endpoint, then, the three inputs carry the number and the architectures on top of them are level: a two-layer MLP, graph convolution over the nine graphs, and the perturbed-graph transformer read 0.87 to 0.88 per cell from the same inputs. Given the transformer's own averaging, five fold models on the cross-validation files with their logits averaged on the held-out genes, the MLP reads 0.886 [0.82, 0.94] from fold models of 0.866 to 0.892, and the convolution at 400 epochs 0.888 [0.83, 0.93] from fold models of 0.807 to 0.897 (`graph_baselines.py -arms folds`), against the transformer's 0.893 [0.84, 0.94]. Three ensembles within 0.007 of one another, each inside the others' intervals, are one number at the resolution of 195 genes, and the MLP's costs thirty seconds.

## 2.6 The message-passing architectures, as equations and as baselines ✗

Every graph model in the comparator set is an instance of message passing: a node's representation is updated from an aggregate of its neighbors' representations,

$$h_i^{(k)} = \gamma^{(k)}\Big(h_i^{(k-1)}, \bigoplus_{j \in \mathcal{N}(i)} \phi^{(k)}\big(h_i^{(k-1)}, h_j^{(k-1)}, e_{ji}\big)\Big), \tag{1}$$

with $\phi$ the message, $\bigoplus$ a permutation-invariant aggregation and $\gamma$ the update, which is the form the MATH-490 project set out from and the form the Gene_Graph project's models take. The architectures differ in the three functions, and the differences are what the 1-WL result of the next subsection bears on.

**GCN** the symmetric-normalized mean with self-loops, $\hat{A} = A + I$, $\hat{d}_i = \sum_j \hat{A}_{ij}$: $h'_i = \Theta^\top \sum_{j \in \mathcal{N}(i) \cup \{i\}} \hat{A}_{ji}\,(\hat{d}_j \hat{d}_i)^{-1/2}\, h_j$.

**GAT and GATv2** attention over the neighborhood, $h'_i = \alpha_{ii}\Theta h_i + \sum_{j \in \mathcal{N}(i)} \alpha_{ij} \Theta h_j$, with

$$\alpha_{ij} = \mathrm{softmax}_j\big(a^\top \mathrm{LeakyReLU}(\Theta[h_i \| h_j])\big)$$

in the original form and the nonlinearity applied before the scoring vector in the v2 form, which is what makes the attention a function of both endpoints rather than a ranking of the neighbors alone; four heads here.

**GraphSAGE** the mean of the neighbors concatenated with the node's own representation, $h'_i = \Theta_1 h_i + \Theta_2 \operatorname{mean}_{j \in \mathcal{N}(i)} h_j$.

**GIN** the injective sum with a learned self-weight and a two-layer MLP, $h'_i = \operatorname{MLP}\big((1+\epsilon)\, h_i + \sum_{j \in \mathcal{N}(i)} h_j\big)$, the aggregator Xu et al. (2019) show to be as expressive as the 1-WL test, which the mean and the attention are not.

**RGCN** one weight per relation over a multigraph, $\mathcal{R}$ the relation set, $h'_i = \Theta_0 h_i + \sum_{r \in \mathcal{R}} \sum_{j \in \mathcal{N}_r(i)} \frac{1}{c_{i,r}} \Theta_r h_j$, with $\Theta_r$ a combination of $B$ shared bases so the parameter count does not grow with the relation count ($B = 4$ here).

**RGAT** the same relation-typed neighborhoods with attention within each relation, $h'_i = \sum_{r \in \mathcal{R}} \sum_{j \in \mathcal{N}_r(i)} \alpha_{ij}^{(r)} \Theta_r h_j$, the relational form Gene_Graph ran on its two-relation graph; here one GATv2 attention per relation on its own edge subset, summed, with no basis sharing, since the basis-shared implementation materializes a weight matrix per edge and 4.87 million typed edges times $64 \times 64$ is 80 GB.

The nine gene graphs the transformer reads enter the relational models as nine edge types on one node set, which is the multigraph the cell graph holds; the union of the nine loses the type. Gene_Graph's dual-primal convolution, which passes messages on the line graph as well as the graph, is described there and not rerun, since its attention on the primal graph reduces to GAT with self-loops. Table 6 runs GIN, GraphSAGE, RGCN and RGAT under the protocol of Table 5, with ProtT5 alone and with the transformer's three inputs, and Table 7 runs the MLP and GCN with the chromosome-pathway indicator split into its blocks, the cheap half of the attribution that Section 3 completes on the transformer.

**Table 6:** The message-passing architectures of the MATH-490 and Gene_Graph projects on the released split, the protocol of Table 5: GIN and GraphSAGE on the union of the nine graphs, and RGCN and RGAT with the nine graphs as nine edge types (RGCN with a basis decomposition of four bases; RGAT as one GATv2 attention per relation, summed, since the basis-shared form holds a weight matrix per edge), with ProtT5 alone and with the transformer's three inputs as node features, 200 epochs, seeds 2700 to 2702. Bold marks the highest value in a column.

| model | graph | inputs | dim | params | mean | sd | seeds | selected epochs |
|---|---|---|---|---|---|---|---|---|
| GIN | union of nine | ProtT5 | 1,024 | 78,147 | 0.665 | 0.059 | 0.696, 0.583, 0.718 | 186, 171, 195 |
| GraphSAGE | union of nine | ProtT5 | 1,024 | 139,457 | 0.783 | 0.002 | 0.784, 0.785, 0.780 | 87, 77, 82 |
| RGCN | nine typed | ProtT5 | 1,024 | 348,425 | 0.836 | 0.007 | 0.844, 0.828, 0.836 | 119, 87, 81 |
| RGAT | nine typed | ProtT5 | 1,024 | 1,258,049 | 0.852 | 0.011 | 0.853, 0.865, 0.839 | 196, 111, 113 |
| GIN | union of nine | ProtT5 + chrom-pathway + FBA ratios | 1,225 | 91,011 | 0.705 | 0.033 | 0.664, 0.744, 0.706 | 195, 182, 177 |
| GraphSAGE | union of nine | ProtT5 + chrom-pathway + FBA ratios | 1,225 | 165,185 | 0.887 | 0.002 | 0.886, 0.890, 0.885 | 98, 93, 92 |
| RGCN | nine typed | ProtT5 + chrom-pathway + FBA ratios | 1,225 | 412,745 | 0.860 | 0.009 | 0.847, 0.868, 0.866 | 181, 139, 134 |
| RGAT | nine typed | ProtT5 + chrom-pathway + FBA ratios | 1,225 | 1,489,601 | **0.889** | 0.004 | 0.887, 0.885, 0.895 | 153, 84, 118 |

Keeping the edge type is worth more than any choice of aggregator. On ProtT5 alone the relational models read 0.852 (RGAT) and 0.836 (RGCN) against 0.834 for graph convolution over the untyped union at the same 200 epochs, 0.783 for GraphSAGE and 0.665 for GIN; on the transformer's three inputs RGAT reads 0.887, 0.885 and 0.895 (mean 0.889), GraphSAGE 0.887, RGCN 0.860, GIN 0.705, against 0.847 for graph convolution. RGAT on the three inputs is the strongest of the cheap models in this document, level with the transformer's best single cell on the pathway input (Section 3.13) and 0.019 above the MLP on the same three inputs. GIN, the aggregator as expressive as the 1-WL test, is the weakest on both inputs and selects epochs 171 to 195 on five seeds of six, the sign of a model still moving at the end of its budget rather than of a converged one; what the next subsection measures is that 1-WL expressiveness is not the axis this endpoint turns on, so the ranking of the aggregators here says nothing about it. GraphSAGE, whose only difference from graph convolution is the separate weight on the node's own representation, gains 0.04 on the three inputs and loses 0.05 on ProtT5 alone.

**Table 7:** The chromosome-pathway indicator split into its three blocks (the 173-column pathway multi-hot, the 17-column chromosome one-hot, the seven scaled gene properties), each beside ProtT5 and the four FBA ratios, and the pathway block with the ratios and alone, on the MLP with no graph and on graph convolution over the union of the nine graphs, the protocol of Table 5, seeds 2700 to 2702. Bold marks the highest value in a column.

| model | graph | inputs | dim | params | mean | sd | seeds | selected epochs |
|---|---|---|---|---|---|---|---|---|
| MLP | none | ProtT5 + pathways + FBA ratios | 1,201 | 81,153 | **0.880** | 0.004 | 0.876, 0.885, 0.878 | 62, 56, 59 |
| GCN | union of nine | ProtT5 + pathways + FBA ratios | 1,201 | 81,153 | 0.846 | 0.001 | 0.845, 0.844, 0.847 | 197, 196, 197 |
| MLP | none | ProtT5 + chromosome + FBA ratios | 1,045 | 71,169 | 0.809 | 0.001 | 0.810, 0.809, 0.808 | 66, 64, 66 |
| GCN | union of nine | ProtT5 + chromosome + FBA ratios | 1,045 | 71,169 | 0.834 | 0.005 | 0.839, 0.837, 0.827 | 197, 197, 196 |
| MLP | none | ProtT5 + properties + FBA ratios | 1,035 | 70,529 | 0.798 | 0.012 | 0.788, 0.814, 0.791 | 197, 74, 196 |
| GCN | union of nine | ProtT5 + properties + FBA ratios | 1,035 | 70,529 | 0.834 | 0.003 | 0.834, 0.829, 0.838 | 196, 197, 197 |
| MLP | none | pathways + FBA ratios | 177 | 15,617 | 0.828 | 0.001 | 0.827, 0.828, 0.829 | 81, 72, 84 |
| GCN | union of nine | pathways + FBA ratios | 177 | 15,617 | 0.834 | 0.011 | 0.832, 0.847, 0.821 | 197, 197, 197 |
| MLP | none | pathways | 173 | 15,361 | 0.749 | 0.022 | 0.719, 0.761, 0.769 | 89, 81, 84 |
| GCN | union of nine | pathways | 173 | 15,361 | 0.871 | 0.000 | 0.871, 0.872, 0.871 | 197, 197, 197 |

The pathway block is the indicator. On the MLP, ProtT5 with the FBA ratios reads 0.803 (Table 5); adding the 173-column pathway multi-hot lifts it to 0.880, adding the chromosome one-hot leaves it at 0.809 and adding the seven gene properties at 0.798, both within one seed spread of the base, and the full 197-column indicator reads 0.870, below the pathway block alone. The pathway block with the four ratios and no ProtT5 reads 0.828, above ProtT5 with the ratios, so 177 columns carry more of the label than the 1,024-dimensional sequence embedding does. On graph convolution at 200 epochs the ordering is the same at a smaller spread (0.846 with the pathway block against 0.834 with either other block), with the selected epoch at 196 or 197 on every seed, the sign that the 200-epoch GCN is still climbing, as the 400-epoch rows of Table 5 showed. The one row where the graph matters is the pathway multi-hot alone: 0.749 on the MLP and 0.871 with convolution over the union of the nine graphs, the largest gain the graph gives any input in this document. The Gene Ontology multi-hot, the larger annotation of the same kind, runs under the same protocol (Table 8); its transformer round is Section 3.14.

**Table 8:** The Gene Ontology multi-hot on the MLP with no graph and on graph convolution over the union of the nine graphs, the protocol of Table 5: the propagated vector (2,362 columns), the biological-process namespace (1,580) and the direct annotations (2,887), each beside ProtT5 and the four FBA ratios; the propagated vector with the ratios and alone; the propagated vector with the pathway multi-hot beside ProtT5 and the ratios; and, as the control for the pathway block, one column that marks a gene annotated to any pathway (517 of 6,607 genes) beside ProtT5 and the ratios. Seeds 2700 to 2702. Bold marks the highest value in a column.

| model | graph | inputs | dim | params | mean | sd | seeds | selected epochs |
|---|---|---|---|---|---|---|---|---|
| MLP | none | ProtT5 + pathway any + FBA ratios | 1,029 | 70,145 | 0.774 | 0.019 | 0.761, 0.761, 0.801 | 197, 194, 75 |
| GCN | union of nine | ProtT5 + pathway any + FBA ratios | 1,029 | 70,145 | 0.832 | 0.003 | 0.831, 0.836, 0.829 | 197, 197, 197 |
| MLP | none | ProtT5 + GO + FBA ratios | 3,390 | 221,249 | 0.855 | 0.001 | 0.854, 0.854, 0.856 | 28, 27, 27 |
| GCN | union of nine | ProtT5 + GO + FBA ratios | 3,390 | 221,249 | 0.887 | 0.002 | 0.889, 0.888, 0.884 | 193, 195, 197 |
| MLP | none | ProtT5 + GO BP + FBA ratios | 2,608 | 171,201 | 0.819 | 0.003 | 0.822, 0.815, 0.820 | 191, 197, 178 |
| GCN | union of nine | ProtT5 + GO BP + FBA ratios | 2,608 | 171,201 | 0.885 | 0.003 | 0.882, 0.886, 0.887 | 196, 195, 197 |
| MLP | none | ProtT5 + GO direct + FBA ratios | 3,915 | 254,849 | 0.848 | 0.004 | 0.842, 0.851, 0.849 | 44, 16, 44 |
| GCN | union of nine | ProtT5 + GO direct + FBA ratios | 3,915 | 254,849 | 0.877 | 0.003 | 0.873, 0.881, 0.878 | 195, 197, 196 |
| MLP | none | GO + FBA ratios | 2,366 | 155,713 | 0.818 | 0.012 | 0.806, 0.834, 0.813 | 2, 29, 2 |
| GCN | union of nine | GO + FBA ratios | 2,366 | 155,713 | **0.891** | 0.003 | 0.895, 0.888, 0.890 | 195, 197, 197 |
| MLP | none | GO | 2,362 | 155,457 | 0.806 | 0.008 | 0.795, 0.810, 0.812 | 2, 13, 10 |
| GCN | union of nine | GO | 2,362 | 155,457 | 0.888 | 0.001 | 0.887, 0.889, 0.887 | 194, 196, 183 |
| MLP | none | ProtT5 + GO + pathways + FBA ratios | 3,563 | 232,321 | 0.859 | 0.002 | 0.862, 0.857, 0.859 | 28, 24, 30 |
| GCN | union of nine | ProtT5 + GO + pathways + FBA ratios | 3,563 | 232,321 | 0.884 | 0.003 | 0.885, 0.887, 0.880 | 196, 196, 193 |

Graph convolution on the GO multi-hot alone reads 0.888, with no sequence embedding and no solve, and 0.891 with the four ratios: level with the transformer's best single cell and above the MLP on all three of its inputs (0.870). On the MLP the same vector beside ProtT5 and the ratios reads 0.855, 0.025 under the pathway block, so without the graph GO carries less than the pathways and with it more; the propagated form beats the direct annotations on both models (0.887 against 0.877 on convolution), and the biological-process namespace alone loses nothing on convolution (0.885) and 0.036 on the MLP. Every convolution cell on GO selects its epoch between 183 and 197, still rising at the end of the budget. The one-column control that marks a gene as annotated to any pathway reads 0.774 on the MLP and 0.832 on convolution, the level of ProtT5 with the ratios, so the pathway block's gain is which pathway rather than whether there is one.

Given its budget, graph convolution on GO alone is the highest single model in this document. At 400 epochs it reads 0.912, 0.910 and 0.904 (mean 0.909, sd 0.003), with the four ratios 0.908, and beside ProtT5 with the ratios 0.901 (Table 9): two layers of convolution over the union of the nine gene graphs on 2,362 annotation columns, no sequence embedding and no solve, 43 seconds per seed on one card, above the transformer's best single cell (0.895) and above the reported fold ensemble's point (0.893) inside its interval. The three seeds still select epochs 382 to 397, but at 800

epochs the number does not move: 0.905, 0.911 and 0.901 (mean 0.906), 0.907 with the ratios, 0.903 beside ProtT5 (Table 10), the selected epochs again near the end of the budget and the held-out number the same, so the curve is flat past 400 and the selection rule is reading noise on its plateau. The sequence embedding and the solve add nothing to it at either budget, where on every other input they were the gain.

**Table 9:** Graph convolution over the union of the nine graphs on the propagated GO multi-hot at a 400-epoch budget, since every 200-epoch cell on it selected its epoch between 183 and 197: GO alone, GO with the four FBA ratios, and GO beside ProtT5 with the ratios. The protocol of Table 5 otherwise; seeds 2700 to 2702. Bold marks the highest value in a column.

| model | graph | inputs | dim | params | mean | sd | seeds | selected epochs |
|---|---|---|---|---|---|---|---|---|
| GCN | union of nine | GO (400 epochs) | 2,362 | 155,457 | 0.909 | 0.004 | 0.913, 0.910, 0.903 | 397, 395, 379 |
| GCN | union of nine | GO + FBA ratios (400 epochs) | 2,366 | 155,713 | **0.909** | 0.005 | 0.905, 0.906, 0.915 | 396, 392, 393 |
| GCN | union of nine | ProtT5 + GO + FBA ratios (400 epochs) | 3,390 | 221,249 | 0.901 | 0.003 | 0.900, 0.898, 0.905 | 397, 368, 391 |

**Table 10:** The three rows of Table 9 at an 800-epoch budget, since at 400 the three seeds still selected epochs 382 to 397. Seeds 2700 to 2702. Bold marks the highest value in a column.

| model | graph | inputs | dim | params | mean | sd | seeds | selected epochs |
|---|---|---|---|---|---|---|---|---|
| GCN | union of nine | GO (800 epochs) | 2,362 | 155,457 | 0.906 | 0.004 | 0.905, 0.911, 0.901 | 777, 784, 788 |
| GCN | union of nine | GO + FBA ratios (800 epochs) | 2,366 | 155,713 | **0.908** | 0.002 | 0.905, 0.911, 0.908 | 768, 790, 769 |
| GCN | union of nine | ProtT5 + GO + FBA ratios (800 epochs) | 3,390 | 221,249 | 0.903 | 0.001 | 0.905, 0.902, 0.903 | 769, 751, 792 |

On six seeds the number holds. Rerun on seeds 2700 to 2705 so its count matches the six-seed module arms (Table 11), the 400-epoch convolution on GO alone reads 0.912, 0.909, 0.902, 0.898, 0.901 and 0.905 (mean 0.905, sd 0.005), with the ratios 0.899 to 0.913 (mean 0.907) and beside ProtT5 with the ratios 0.895 to 0.905 (mean 0.900); the first three seeds differ from their earlier runs in the third decimal, and every one of the eighteen cells is above the transformer's seed mean on any input. The spread over seeds is 0.005, against 0.055 for the module's scalable form on the same six seeds (Section 4.22), whose one unfit run is the difference.

**Table 11:** The three rows of Table 9 on seeds 2700 to 2705, so the graph convolution's count matches the six-seed module arms of Section 4.22. The first three seeds are reruns of the Table 9 cells and differ from them in the third decimal. Bold marks the highest value in a column.

| model | graph | inputs | dim | params | mean | sd | seeds | selected epochs |
|---|---|---|---|---|---|---|---|---|
| GCN | union of nine | GO (400 epochs) | 2,362 | 155,457 | 0.905 | 0.005 | 0.912, 0.909, 0.902, 0.898, 0.901, 0.905 | 397, 397, 379, 387, 397, 396 |
| GCN | union of nine | GO + FBA ratios (400 epochs) | 2,366 | 155,713 | **0.907** | 0.004 | 0.904, 0.906, 0.913, 0.908, 0.899, 0.909 | 396, 392, 397, 397, 391, 396 |
| GCN | union of nine | ProtT5 + GO + FBA ratios (400 epochs) | 3,390 | 221,249 | 0.900 | 0.003 | 0.902, 0.898, 0.905, 0.898, 0.900, 0.895 | 397, 368, 391, 391, 386, 397 |

Under the reported protocol it is the highest number in this document. Five fold models of the 400-epoch convolution on GO alone score 0.910, 0.893, 0.918, 0.916 and 0.907 on the held-out genes, a fold mean of 0.909, and their averaged logits score 0.917 [0.87, 0.96]; at 800 epochs 0.915 [0.87, 0.95]. Against the transformer's reported ensemble (folds 0.860, 0.899, 0.889, 0.876, 0.884; 0.893 [0.84, 0.94]) the folds are above on four of five, the fold mean by 0.028 and the ensemble by 0.024, with the two intervals overlapping (Table 48). The five folds take two minutes on one card. With three seeds per fold, each fold's logit the mean of its seeds before the folds are averaged, the ensemble reads 0.917 [0.87, 0.95] at 400 epochs and 0.915 [0.87, 0.95] at 800, the same numbers as one seed per fold (folds 0.899 to 0.922, mean 0.910), so the ensemble is not sitting on a lucky seed. So the three architectures on the three inputs that were one number (0.886, 0.888, 0.893) now have a fourth point above them, and what moved it is the input and the graph together, not the architecture: the same convolution on the three inputs reads 0.888 as an ensemble, and the GO vector without the graph reads 0.855 on the MLP.

It is the convolution, not message passing in general. On GO alone at 400 epochs relational convolution reads 0.843, relational attention 0.844 and GraphSAGE 0.823 (Table 12), 0.07 to 0.09 under the symmetric-normalized mean, where on the three inputs those same forms were level with or above it. The pathway multi-hot added to GO reads 0.905, nothing on top of GO alone. What the normalized mean does that the others do not, on this input, is not measured; hypothesis, untested: the signal is the neighborhood's annotations smoothed over the graph, and a form with a separate path for the node's own 2,362 columns fits those instead.

**Table 12:** Relational convolution and relational attention over the nine graphs as nine edge types on the propagated GO multi-hot alone, GraphSAGE over the union on it, and graph convolution over the union on GO with the pathway multi-hot, all at 400 epochs, the protocol of Table 5 otherwise; seeds 2700 to 2702. Bold marks the highest value in a column.

| model | graph | inputs | dim | params | mean | sd | seeds | selected epochs |
|---|---|---|---|---|---|---|---|---|
| RGCN | nine typed | GO (400 epochs) | 2,362 | 776,585 | 0.843 | 0.019 | 0.831, 0.829, 0.869 | 101, 149, 91 |
| RGAT | nine typed | GO (400 epochs) | 2,362 | 2,799,425 | 0.845 | 0.009 | 0.836, 0.843, 0.858 | 34, 56, 25 |
| GCN | union of nine | GO + pathways (400 epochs) | 2,535 | 166,529 | **0.904** | 0.002 | 0.906, 0.901, 0.905 | 388, 396, 397 |
| GraphSAGE | union of nine | GO (400 epochs) | 2,362 | 310,721 | 0.823 | 0.001 | 0.822, 0.824, 0.824 | 307, 380, 394 |

## 2.7 The Weisfeiler-Leman test on the deletion graphs ✗

A message-passing model over a gene graph is bounded by what color refinement can separate: two graphs that the 1-WL test assigns the same signature are the same graph to it, whatever the model's weights (Xu et al. 2019). The transformer's perturbed graph is the gene graph with the deleted gene's token removed, so the first question is whether single deletions are even distinguishable by structure, and the second is whether structure alone ranks them. `wl_knockout_test.py` answers both on the physical interaction graph and on the union of the nine graphs the transformer reads (Table 13): for every labeled gene it hashes the graph with that gene's node removed by color refinement at one, two and three iterations with degree as the initial color, counts collisions, and fits logistic regression on the WL subtree kernel's feature map of each gene's 2-hop ego graph. Figure 2 illustrates the refinement and the kernel on a toy pair of six-node graphs, not the yeast graphs.

**Table 13:** The 1-WL test on the single-deletion graphs. For every labeled gene the gene graph with that gene's node removed is hashed by Weisfeiler-Leman color refinement with degree as the initial color; a collision is a set of deletions with one hash, which no 1-WL-bounded message-passing model can tell apart by structure. The WL kernel rung is logistic regression on the bag of colors of each gene's 2-hop ego graph refined for three iterations, hashed to 1,024 bins, with C chosen by five-fold cross-validation inside the training genes: physical: C = 0.001, CV 0.578, validation 0.626, held-out 0.490; union of nine: C = 0.01, CV 0.443, validation 0.478, held-out 0.564.

| graph | edges | $k$ | deletions | distinct | in collision | largest | essential | min |
|---|---|---|---|---|---|---|---|---|
| physical | 139,463 | 1 | 1,121 | 1,118 | 6 | 2 | 0 | 11 |
| physical | 139,463 | 2 | 1,121 | 1,118 | 6 | 2 | 0 | 10 |
| physical | 139,463 | 3 | 1,121 | 1,118 | 6 | 2 | 0 | 18 |
| union of nine | 1,516,034 | 1 | 1,121 | 1,121 | 0 | 1 | 0 | 130 |
| union of nine | 1,516,034 | 2 | 1,121 | 1,121 | 0 | 1 | 0 | 91 |
| union of nine | 1,516,034 | 3 | 1,121 | 1,121 | 0 | 1 | 0 | 89 |

The test separates the deletions and the kernel does not rank them. On the physical graph, one iteration already gives 1,118 distinct signatures among 1,121 deletions, three pairs collide, none of the six genes in them is essential, and two more iterations change nothing; on the union every deletion has its own signature at one iteration. A single deletion changes the degree of every neighbor, and on graphs this dense (139,463 and 1,516,034 edges) that is enough for the test, so 1-WL expressivity is not what bounds a message-passing model on this endpoint. The kernel rung reads 0.490 on the physical graph and 0.564 on the union, chance on 195 genes, which is the same reading the graph-only baselines gave with all-ones and degree inputs: a gene's neighborhood structure, however finely refined, does not say whether the gene is essential, and every number above chance in this document comes from what is written on the nodes.

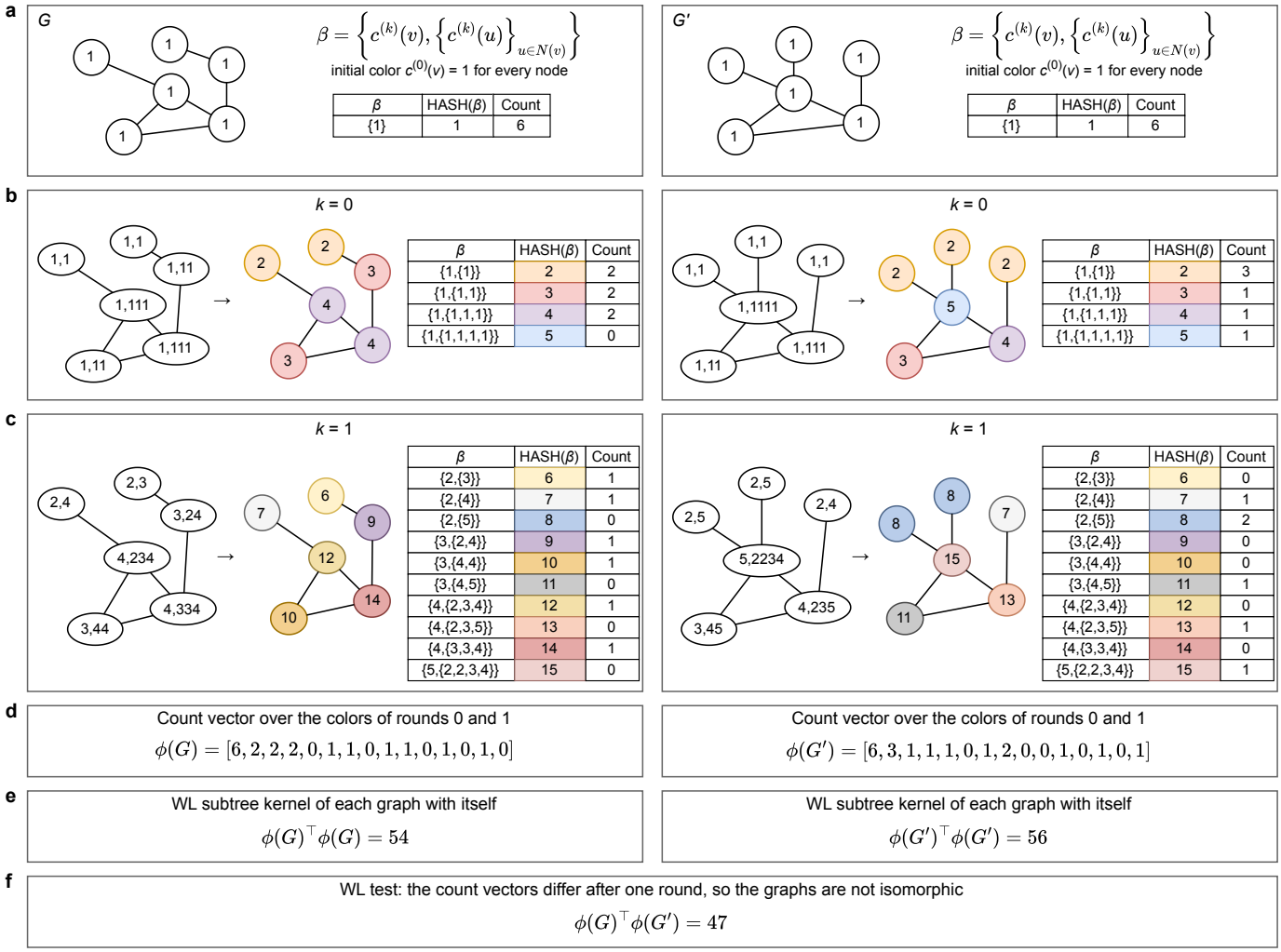



**Figure 2: The 1-WL test and the WL subtree kernel on a toy pair of graphs.** **a**, Two six-node graphs $G$ and $G'$ with six edges each, every node given the initial color 1; each round of color refinement replaces a node's color by the hash of the multiset $\beta$ of its own color and its neighbors' colors, and the table counts the nodes of each color. **b**, Round $k = 0$: the multiset at each node (left), the hashed colors (right) and the count of each color; a node's new color is its degree in disguise, and $G'$ alone holds a degree-4 node. **c**, Round $k = 1$: the same refinement on the round-0 colors, where the two graphs share one of the ten colors. **d**, The count vector $\phi$ over the colors of rounds 0 and 1, the WL subtree kernel's feature map. **e**, The kernel of each graph with itself, $\phi^\top \phi$. **f**, The kernel across the pair; the count vectors differ after one round, so the test declares the graphs non-isomorphic. Drawn in draw.io from `notes/assets/drawio/028-wl-test.drawio`; the yeast deletion graphs are hashed the same way by `wl_knockout_test.py`.

# 3 The backbone: splits, labels, weight decay and the shape of the curve ✗

Every cell in this section is the pooled backbone at width 12 with one transformer block and a one-layer preprocessor (16,874 parameters), trained for 80 epochs at three initialization seeds with the split pinned. Round 1 (jobs 2149 to 2181) moves one lever at a time away from the cell that trains on the released-label split file (`ess_fcl_val0.2_s0`, 629 training genes, 158 validation genes): the old 75-gene validation draw, the genome-wide labels, weight decay at 0, $10^{-2}$ and $10^{-1}$, batch 32, learning rate at $3 \times 10^{-4}$ and $3 \times 10^{-3}$, the class weight 772/126, and graph regularization off. Table 14 carries every number; Figure 3 the curves.

**Table 14:** Round 1, the backbone. Width 12, one block, one-layer preprocessor, 80 epochs, seeds 2700 to 2702. Held-out AUROC on the 195 genes: the seed mean with its standard deviation, the three seeds, the gene bootstrap of the pooled per-gene logits, the epochs selected on validation, and the median absolute epoch-to-epoch step of the validation curve. Bold marks the highest value in a column. ↑ (↓) marks an arm whose held-out AUROC is above (below) the first arm's on every one of the same three seeds.

| arm | mean | sd | seeds | pooled [95% CI] | selected epochs | val step |
|---|---|---|---|---|---|---|
| wave-11 protocol (75-gene val) | 0.759 | 0.020 | 0.781, 0.755, 0.741 | 0.760 [0.67, 0.84] | 18, 38, 14 | 0.0056 |
| released labels, 158-gene val | 0.725 | 0.019 | 0.745, 0.722, 0.708 ↓ | 0.738 [0.65, 0.83] | 66, 77, 77 | 0.0026 |
| weight decay 0 | 0.725 | 0.019 | 0.745, 0.720, 0.709 ↓ | 0.735 [0.64, 0.82] | 66, 77, 77 | 0.0027 |
| weight decay 1e-2 | 0.726 | 0.017 | 0.745, 0.722, 0.711 ↓ | 0.735 [0.64, 0.83] | 65, 77, 77 | 0.0029 |
| weight decay 1e-1 | 0.725 | 0.018 | 0.746, 0.721, 0.709 ↓ | 0.735 [0.64, 0.82] | 66, 77, 77 | 0.0025 |
| batch 32 | **0.765** | 0.005 | 0.766, 0.770, 0.759 | 0.764 [0.67, 0.85] | 74, 73, 77 | 0.0032 |
| genome labels (5,795 genes) | 0.704 | 0.024 | 0.706, 0.726, 0.679 ↓ | 0.705 [0.61, 0.80] | 77, 77, 69 | 0.0020 |
| lr 3e-4 | 0.658 | 0.093 | 0.737, 0.682, 0.556 ↓ | 0.732 [0.64, 0.82] | 77, 77, 74 | 0.0023 |
| lr 3e-3 | 0.686 | 0.043 | 0.718, 0.704, 0.637 ↓ | 0.713 [0.62, 0.81] | 77, 77, 50 | 0.0024 |
| class weight 772/126 | 0.759 | 0.016 | 0.764, 0.772, 0.741 | **0.769** [0.68, 0.86] | 77, 75, 53 | 0.0034 |
| graph regularization off | 0.725 | 0.019 | 0.745, 0.722, 0.709 ↓ | 0.738 [0.65, 0.83] | 66, 77, 77 | 0.0026 |

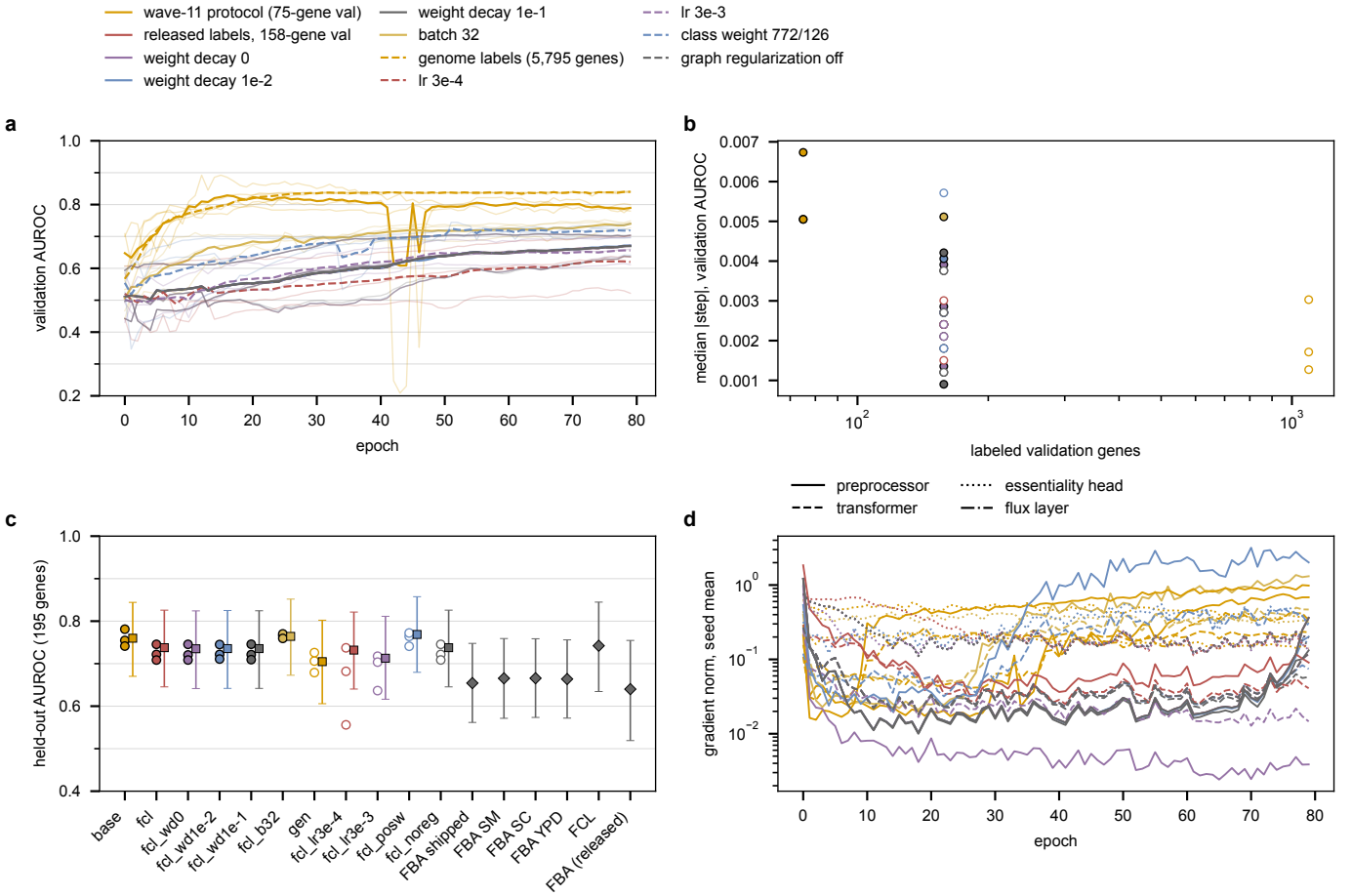



**Figure 3: Round 1, the backbone.** **a**, Validation AUROC per epoch, thin lines per seed and the seed mean per arm. **b**, The median absolute epoch-to-epoch step of that curve against the number of labeled validation genes it was read on. **c**, Held-out AUROC on the 195 genes per cell, the seed mean, and the gene bootstrap of the pooled per-gene logits, beside FBA on each medium. **d**, Per-module gradient norm per epoch, seed mean, on the arms that logged it. Generated by `scripts/wave12_readout.py -round w12`.

### 3.1 The curve was noisy because the validation set was small ✗

The median epoch-to-epoch step of the validation AUROC is 0.0056 on the 75-gene draw, 0.0026 on the 158-gene split and 0.0020 on the 1,093-gene genome-wide split (Figure 3b). Two of the three width-24 cells of the previous protocol selected epochs 4 and 5 on validation peaks of 0.82 and 0.86 and scored 0.74 on the held-out genes; no cell on a split file selects before epoch 50. The old protocol's curve reads near 0.8 on its own 75 genes and the split cells read 0.65 to 0.72 on theirs, so the levels are not comparable, only the steps are.

### 3.2 The split file costs training genes, and the cells are under-trained ✗

The split-file cell scores 0.725 against the old protocol's 0.759 on the same held-out genes. It trains on 629 labeled genes where the old draw trained on roughly 740, and every one of its seeds selected epoch 66 to 77 of 80 with the validation curve still rising (Figure 3a). At batch 128 an epoch is five optimizer steps, so 80 epochs is 400 updates. The two levers that move all three seeds in the same direction are both more optimization: batch 32, four times the steps per epoch, lifts the seeds by 0.020, 0.048 and 0.051 and collapses their spread to 0.005; the class weight, which steepens the curve, lifts them by 0.019, 0.051 and 0.033. Round 3 combines the two and raises the epoch budget.

### 3.3 Weight decay and graph regularization do nothing here ✗

Weight decay at 0, $10^{-2}$ and $10^{-1}$ reproduces the base cell to the third decimal on every seed, and so does removing graph regularization. At 400 updates with learning rate $10^{-3}$, a decay of $10^{-1}$ shrinks a weight by four percent, so it is not a lever at this budget. Graph regularization is a different case: its configuration pins the nine gene graphs to attention heads 0 to 8 of the *second* transformer block, so on a one-block cell the term is never evaluated, and the arm with it off reproduces the arm with it on because both are the same cell. The term was inert on every one-block cell in this document; a two-block cell with four heads indexes a head it does not have and fails (round 8, job 2261), so the two-block cells run with it off. The learning rate at $3 \times 10^{-4}$ and $3 \times 10^{-3}$ is worse than $10^{-3}$ on the seed mean, with seed spreads of 0.09 and 0.04 that put the pooled numbers inside the base interval.

### 3.4 More labels from the genome make the held-out ranking worse ✗

The genome-wide file adds SGD's essential list and the viable single-mutant fitness records as nonessential, 4,368 training genes against 629, and scores 0.704 against 0.725, three of three seeds down. Its validation AUROC on its own 1,093 genes is 0.84 while the released held-out genes read 0.70, the widest validation-to-held-out gap of any arm. The viable-fitness negatives are a different population from the released split's nonessential genes, and a model that ranks them well does not rank the released genes better. Hypothesis, untested: the released split's nonessential class is enriched for genes with a measurable fitness defect, which the genome-wide negatives are not.

### 3.5 Gradient norms ✗

On the split-file cells the preprocessor's gradient norm starts near $10^{-2}$ and rises about thirty-fold over 80 epochs while the head's stays flat near 0.4 (Figure 3d); the clip at 10 never binds. The rise tracks the validation curve's climb, which is the same under-training reading from the gradient side.

### 3.6 Round 3: the two levers combined, a longer budget, and the fold ensemble ✗

Round 3 (jobs 2209 and 2224 to 2236) combines the two round-1 levers that moved three of three seeds, batch 32 and the class weight, at 80 and at 200 epochs, runs batch 32 alone at 200 epochs as the class weight's control, and trains five fold models on the cross-validation files, which partition the 787 labeled non-test genes five ways with the held-out genes fixed; their logits are averaged on the held-out genes, the protocol behind the published Flux Cone Learning number (Table 15, Figure 4). With the loader workers removed a 200-epoch cell takes 11 minutes.

**Table 15:** Round 3, more optimization and the fold ensemble. Batch 32 with the class weight at 80 and 200 epochs, batch 32 alone at 200 epochs, and the five fold models of the cross-validation files, each on its own fold. Bold marks the highest value in a column. ↑ (↓) marks an arm whose held-out AUROC is above (below) the first arm's on every one of the same three seeds.

| arm | mean | sd | seeds | pooled [95% CI] | selected epochs | val step |
|---|---|---|---|---|---|---|
| batch 32 + class weight, 80 epochs | 0.750 | 0.043 | 0.780, 0.769, 0.700 | 0.774 [0.69, 0.86] | 75, 68, 31 | 0.0026 |
| batch 32 + class weight, 200 epochs | 0.774 | 0.006 | 0.780, 0.769, 0.773 | 0.780 [0.69, 0.87] | 75, 68, 148 | 0.0031 |
| batch 32, 200 epochs | 0.761 | 0.005 | 0.766, 0.761, 0.755 | 0.764 [0.67, 0.85] | 74, 100, 181 | 0.0026 |
| 5-fold, fold 0 | 0.780 | 0.000 | 0.780 | 0.780 [0.69, 0.86] | 75 | 0.0033 |
| 5-fold, fold 1 | **0.806** | 0.000 | 0.806 | **0.806** [0.72, 0.89] | 50 | 0.0036 |
| 5-fold, fold 2 | 0.797 | 0.000 | 0.797 | 0.797 [0.72, 0.87] | 14 | 0.0033 |
| 5-fold, fold 3 | 0.751 | 0.000 | 0.751 | 0.751 [0.66, 0.84] | 184 | 0.0021 |
| 5-fold, fold 4 | 0.789 | 0.000 | 0.789 | 0.789 [0.70, 0.87] | 108 | 0.0039 |

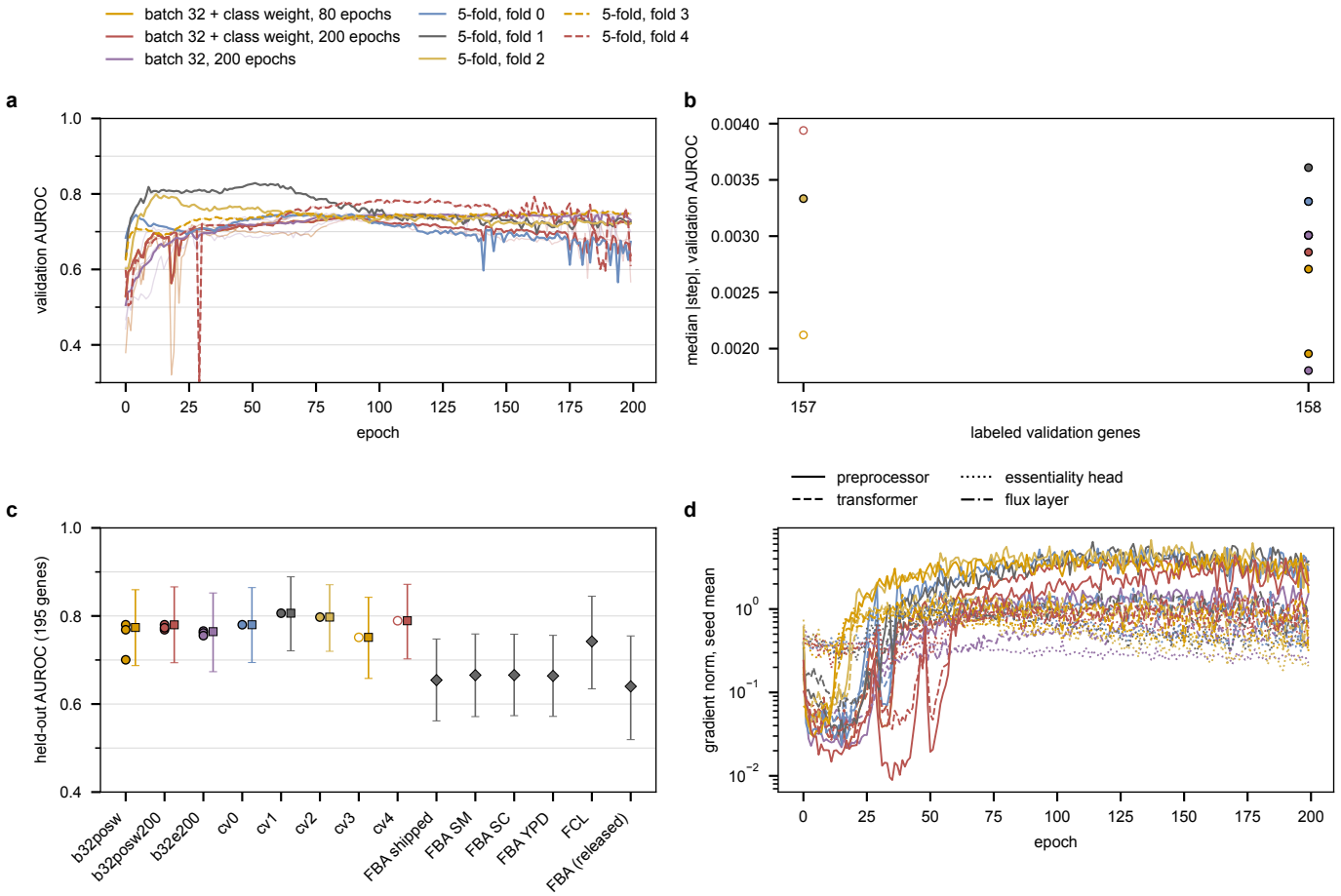



**Figure 4: Round 3, the levers combined and the fold models. a**, Validation AUROC per epoch, seed mean per arm; each fold model reads its own validation fold. **b**, The median absolute epoch-to-epoch step per cell against the validation size, 157 or 158 genes. **c**, Held-out AUROC per cell (points), the seed mean and the gene bootstrap of the pooled per-gene logits (square with interval), beside FBA on four media, Flux Cone Learning and the released FBA predictions on the same 195 genes. **d**, Gradient norm per module per epoch, seed mean. Written by `experiments/028-gene-essentiality/scripts/wave12_readout.py -round w14`.

Batch 32 with the class weight at 200 epochs scores 0.780, 0.769 and 0.773, a seed mean of 0.774 with a spread of 0.006, the tightest cell of any round; the pooled bootstrap is 0.780 [0.69, 0.87]. Batch 32 alone at 200 epochs scores 0.766, 0.761 and 0.755, so the class weight adds 0.01 to 0.02 on three of three seeds at the longer budget as it did at 80. The longer budget itself moves one seed: the 80- and 200-epoch cells share a trajectory through epoch 80 and select the same epoch on seeds 2700 and 2701 (75 and 68), while seed 2702 selects epoch 31 at 80 epochs and epoch 148 at 200, 0.700 against 0.773. Fold 0 of the cross-validation files is the released-label split file exactly (same 629 training, 158 validation and 195 held-out genes, both drawn at seed 0), so its fold model is the 200-epoch cell's first seed. The five fold models score 0.780, 0.806, 0.797, 0.751 and 0.789 on the held-out genes, a fold mean of 0.785, and their averaged logits score 0.795 [0.71, 0.88] (`cv_ensemble_readout.py`). That is the number to set beside the published 0.73 and beside 0.742 on these genes: the backbone under Flux Cone Learning's own protocol is 0.05 above it, and 0.01 to 0.04 below the ladder's rungs with FBA appended.

## 3.7 Round 5: FBA's growth ratios as an input ✗

The ladder's rungs cross 0.80 only with FBA's four growth ratios appended to the encoding, and no transformer cell has seen those numbers. Round 5 (jobs 2249 and 2250) gives the round-3 cell the same complement: the `fba` readout adds an affine of the deleted gene's knockout-to-wild-type growth ratio on the four media to the context head's logit, five parameters on top of the 16,561, with the four weights written into every epoch row. It runs at the three seeds, paired against the round-3 cell, and on the five cross-validation folds for the ensemble (Table 16, Figure 5). The read was fixed first: the transformer gains the ladder's complement if its seed mean moves from 0.774 toward 0.82 on three of three seeds and the fold ensemble from 0.795 toward the ladder's 0.81 to 0.83.

**Table 16:** Round 5, FBA's growth ratios as an input. The round-3 pooled cell and the same cell with an affine of FBA's four per-medium growth ratios for the deleted gene added to the context head, at three seeds, and that readout on the five cross-validation folds whose logits are averaged on the held-out genes. Bold marks the highest value in a column. ↑ (↓) marks an arm whose held-out AUROC is above (below) the first arm's on every one of the same three seeds.

| arm | mean | sd | seeds | pooled [95% CI] | selected epochs | val step |
|---|---|---|---|---|---|---|
| batch 32 + class weight, 200 epochs | 0.774 | 0.006 | 0.780, 0.769, 0.773 | 0.780 [0.69, 0.87] | 75, 68, 148 | 0.0031 |
| context head + FBA growth ratios (round-3 levers) | 0.821 | 0.020 | 0.814, 0.805, 0.844 ↑ | 0.838 [0.76, 0.91] | 28, 70, 61 | 0.0036 |
| context head + FBA ratios, fold 0 | 0.814 | 0.000 | 0.814 | 0.814 [0.73, 0.89] | 28 | 0.0033 |
| context head + FBA ratios, fold 1 | 0.824 | 0.000 | 0.824 | 0.824 [0.74, 0.90] | 42 | 0.0042 |
| context head + FBA ratios, fold 2 | 0.839 | 0.000 | 0.839 | 0.839 [0.77, 0.91] | 37 | 0.0048 |
| context head + FBA ratios, fold 3 | 0.775 | 0.000 | 0.775 | 0.775 [0.69, 0.86] | 122 | 0.0042 |
| context head + FBA ratios, fold 4 | **0.853** | 0.000 | 0.853 | **0.853** [0.78, 0.92] | 54 | 0.0036 |

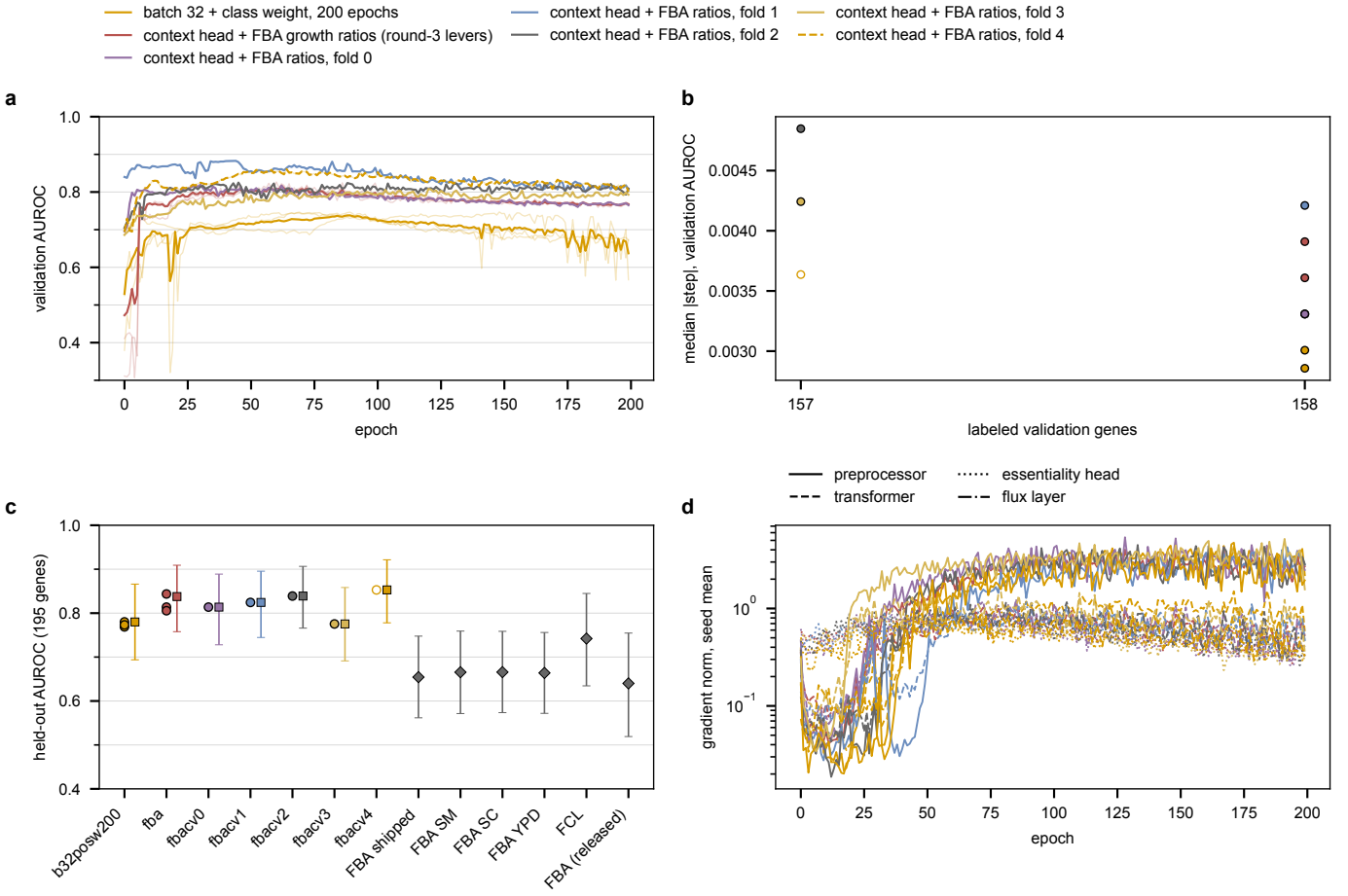



**Figure 5: Round 5, FBA's growth ratios as an input.** **a**, Validation AUROC per epoch, one thin line per seed and the seed mean per arm; each fold model reads its own fold. **b**, The median absolute epoch-to-epoch step per cell against the validation size. **c**, Held-out AUROC per cell (points), the seed mean and the gene bootstrap of the pooled per-gene logits (square with interval), beside FBA on four media, Flux Cone Learning and the released FBA predictions on the same 195 genes. **d**, Gradient norm per module per epoch, seed mean. Written by `experiments/028-gene-essentiality/scripts/wave12_readout.py -round w16`.

Both halves of the read hold. The cell scores 0.814, 0.805 and 0.844 against 0.780, 0.769 and 0.773 on the same seeds, up by 0.034, 0.037 and 0.071, a seed mean of 0.821 and a pooled bootstrap of 0.838 [0.76, 0.91]. The five fold models score 0.814, 0.824, 0.839, 0.775 and 0.853, a fold mean of 0.821, and their averaged logits 0.822 [0.74, 0.90] on the held-out

genes, against 0.795 without the ratios. The transformer with the solver's four numbers sits where the ladder's best rungs sit (0.806 to 0.830), 0.08 above Flux Cone Learning on the same genes and 0.09 above its published number. The four weights end negative on every cell, largest on the shipped-bound and YPD columns (about −0.5 to −0.7 at the selected epoch against −0.3 to −0.5 on SM and near zero to −0.4 on SC), which is the direction a growth ratio must carry (a deletion that stops growth is essential), and the cells select earlier, at epochs 28 to 70 against 68 to 148: the affine is learned in the first epochs and the context head fills in around it.

The two sources are complementary and the complement is the same for the transformer as for a logistic regression on a frozen embedding: the backbone alone reads 0.77, the solver alone 0.66, and the sum 0.82. What the flux layer could not supply in rounds 2 and 4, where its growth column matched the head and its $k_{cat}$ content went unread, the FBA solve supplies as four numbers, which locates the module's value on this endpoint in the solve itself rather than in the learned layer.

### 3.8 Round 8: width and depth on the FBA cell ✗

The round-5 cell ties the ladder's best rungs and does not pass them, and it is the smallest transformer in the document. Round 8 (jobs 2259, 2260, 2262 and 2263) moves its size one axis at a time with the readout, batch, class weight and budget fixed: width 24 and width 48 at one block (40,902 and 112,902 parameters against 16,566), two blocks at width 12 (18,450), and the two-layer preprocessor (542,480, most of it in the preprocessor's first layer), each at the three seeds, three cells packed on one card (Table 17, Figure 6). The read was fixed first: a cell passes the ladder if its seed mean clears 0.83 with three of three seeds above the round-5 cell.

**Table 17:** Round 8, width and depth on the FBA cell. The round-5 cell (width 12, one block, one-layer preprocessor, context head plus the FBA affine) against the same readout at width 24 and 48, at two blocks, and with the two-layer preprocessor. Batch 32, class weight, 200 epochs, seeds 2700 to 2702. Bold marks the highest value in a column. ↑ (↓) marks an arm whose held-out AUROC is above (below) the first arm's on every one of the same three seeds.

| arm | mean | sd | seeds | pooled [95% CI] | selected epochs | val step |
|---|---|---|---|---|---|---|
| context head + FBA growth ratios (round-3 levers) | 0.821 | 0.020 | 0.814, 0.805, 0.844 | **0.838** [0.76, 0.91] | 28, 70, 61 | 0.0036 |
| context head + FBA ratios, width 24 | 0.817 | 0.005 | 0.818, 0.822, 0.812 | 0.823 [0.74, 0.90] | 50, 41, 66 | 0.0045 |
| context head + FBA ratios, width 48 | **0.824** | 0.004 | 0.827, 0.819, 0.826 | 0.831 [0.74, 0.91] | 58, 86, 64 | 0.0037 |
| context head + FBA ratios, two blocks | 0.812 | 0.007 | 0.810, 0.820, 0.806 | 0.827 [0.74, 0.90] | 57, 144, 41 | 0.0032 |
| context head + FBA ratios, two-layer preprocessor | 0.821 | 0.014 | 0.804, 0.829, 0.828 | 0.820 [0.74, 0.90] | 82, 73, 103 | 0.0029 |

No cell passes. Width 48 has the highest seed mean of the four, 0.824 against 0.821, with a seed spread of 0.004, and it is up on seeds 2700 and 2701 by 0.013 and 0.014 and down on seed 2702 by 0.018; width 24 scores 0.817, the two-layer preprocessor 0.821, and two blocks 0.812, the last down on two of three seeds. Seed 2702 of the round-5 cell, 0.844, is the highest single seed in the document and every round-8 cell is below it on that seed, so no arm clears the bar on all three. The seed spread narrows with width, from 0.020 at width 12 to 0.005 at 24 and 0.004 at 48, and every round-8 seed mean lies inside the round-5 cell's pooled bootstrap of 0.84 [0.76, 0.91]. The validation curves peak at 0.80 to 0.82 between epochs 40 and 90 on every arm and drift down from there, most at width 24, and the cells select epochs 41 to 144. The two-block cells run with graph regularization off, since the term indexes a second-block head that a four-head block does not have (the failed job 2261); it was inert on every one-block cell, so the comparison is like for like.

A width-12 transformer with FBA's four numbers reads 0.82 on these genes and neither four times the width, a second block, nor a wider preprocessor moves it by more than the seed noise. Capacity is not the lever on this endpoint and this split; the levers left untested are the gene encoding the transformer reads and what the solve supplies beyond four growth ratios.

### 3.9 Round 9: the FBA cell on the ladder's other encodings ✗

Every transformer cell so far reads ProtT5, and the ladder's best selected rung reads something else: logistic regression on the 197-dimensional chromosome-pathway indicator with the four ratios scores 0.830. Round 9 (jobs 2280 to 2283) opens the round-5 cell on the ladder's other encodings through `-node-embeddings`, with the readout, width, batch, class weight and budget fixed: the chromosome-pathway indicator alone, the indicator concatenated with ProtT5, ESM2, and the ladder's seven-encoding concatenation (ProtT5, CaLM, the two species-LM embeddings, the 5,979-base window, codon frequency and the indicator), each at the three seeds (Table 18, Figure 7; *chrom-pathway* in the figure is the chromosome-pathway indicator). The read is the round-8 one: a cell passes the ladder if its seed mean clears 0.83 with three of three seeds above the round-5 cell.

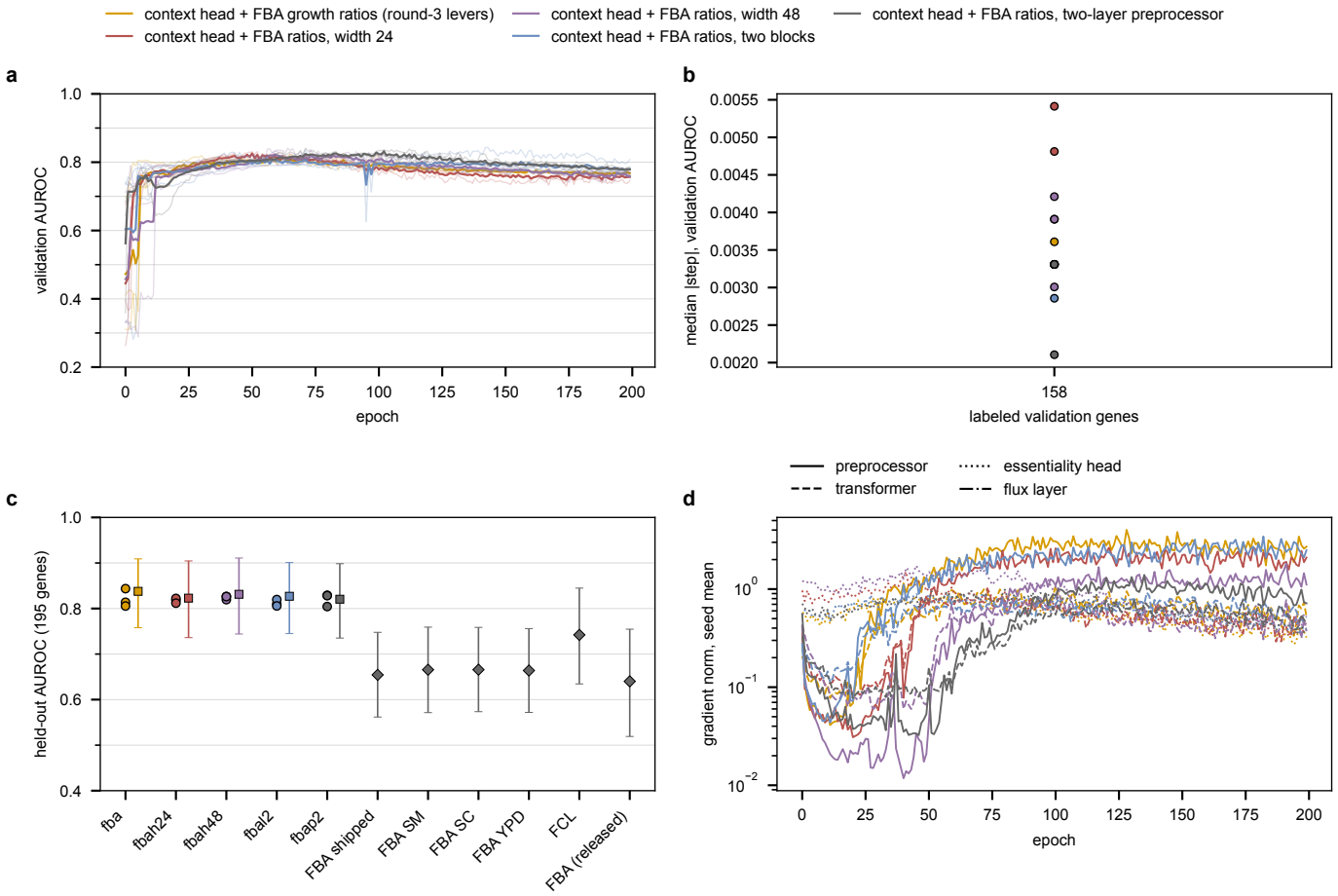



**Figure 6: Round 8, width and depth on the FBA cell.** **a**, Validation AUROC per epoch, one thin line per seed and the seed mean per arm. **b**, The median absolute epoch-to-epoch step per cell against the validation size. **c**, Held-out AUROC per cell (points), the seed mean and the gene bootstrap of the pooled per-gene logits (square with interval), beside FBA on four media, Flux Cone Learning and the released FBA predictions on the same 195 genes. **d**, Gradient norm per module per epoch, seed mean. Written by `experiments/028-gene-essentiality/scripts/wave12_readout.py -round w19`.

**Table 18:** Round 9, the FBA affine on the ladder's other encodings. The round-5 cell (ProtT5 backbone, context head plus the FBA affine) against the same cell opened on the chromosome-pathway indicator, on the indicator concatenated with ProtT5, on ESM2, and on the ladder's seven-encoding concatenation. Width 12, one block, batch 32, class weight, 200 epochs, seeds 2700 to 2702. Bold marks the highest value in a column. ↑ (↓) marks an arm whose held-out AUROC is above (below) the first arm's on every one of the same three seeds.

| arm | mean | sd | seeds | pooled [95% CI] | selected epochs | val step |
|---|---|---|---|---|---|---|
| context head + FBA growth ratios (round-3 levers) | 0.821 | 0.020 | 0.814, 0.805, 0.844 | 0.838 [0.76, 0.91] | 28, 70, 61 | 0.0036 |
| chrom-pathway backbone + FBA ratios | 0.781 | 0.027 | 0.810, 0.759, 0.774 ↓ | 0.823 [0.75, 0.90] | 21, 2, 2 | 0.0083 |
| chrom-pathway + ProtT5 backbone + FBA ratios | **0.871** | 0.015 | 0.860, 0.866, 0.888 ↑ | **0.868** [0.80, 0.93] | 53, 51, 38 | 0.0030 |
| ESM2 backbone + FBA ratios | 0.740 | 0.008 | 0.734, 0.736, 0.749 ↓ | 0.737 [0.64, 0.83] | 133, 192, 64 | 0.0060 |
| seven-encoding backbone + FBA ratios | 0.820 | 0.009 | 0.822, 0.810, 0.827 | 0.823 [0.74, 0.90] | 91, 92, 55 | 0.0026 |

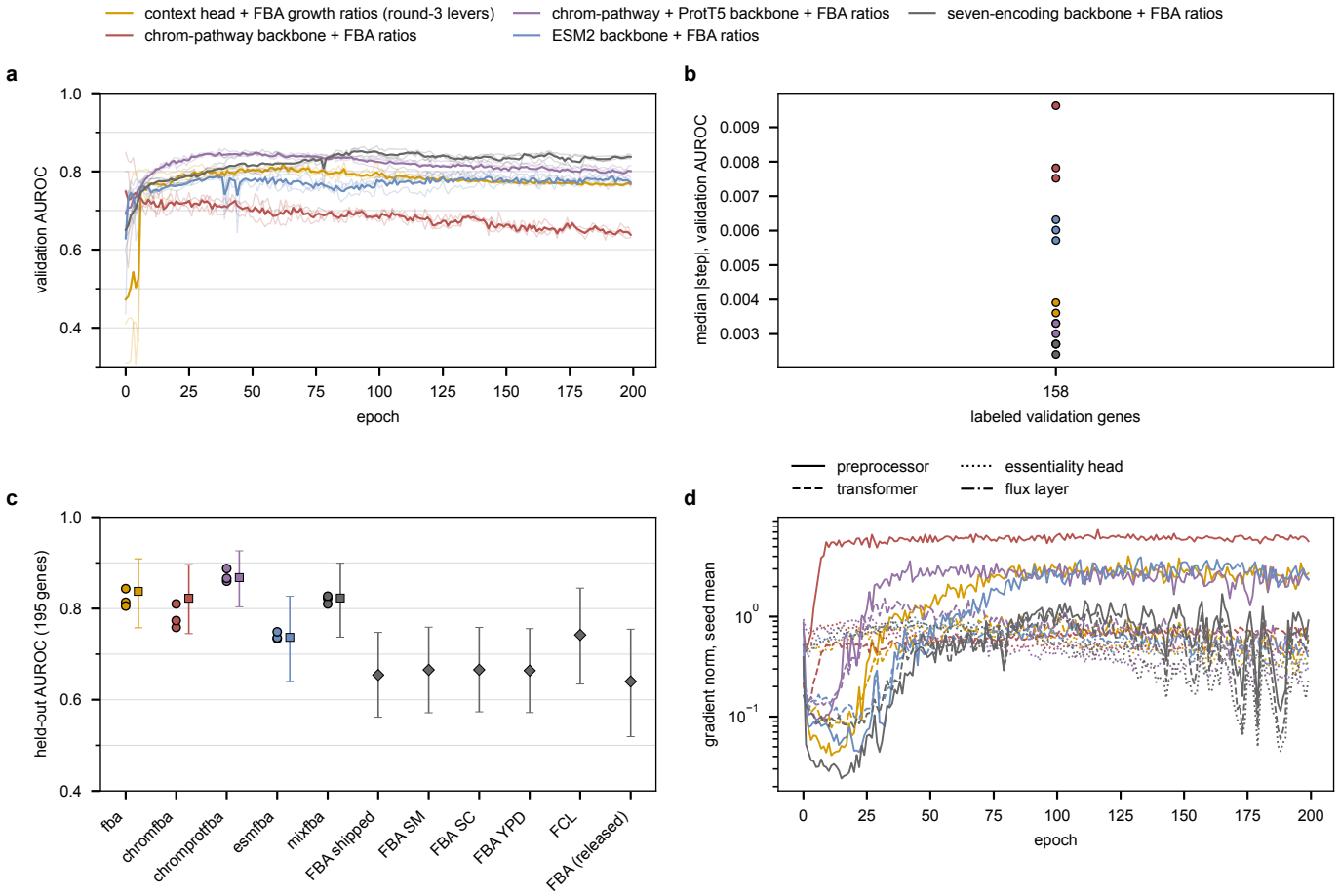



**Figure 7: Round 9, the FBA cell on the ladder's other encodings.** **a**, Validation AUROC per epoch, one thin line per seed and the seed mean per arm. **b**, The median absolute epoch-to-epoch step per cell against the validation size. **c**, Held-out AUROC per cell (points), the seed mean and the gene bootstrap of the pooled per-gene logits (square with interval), beside FBA on four media, Flux Cone Learning and the released FBA predictions on the same 195 genes. **d**, Gradient norm per module per epoch, seed mean. Written by `experiments/028-gene-essentiality/scripts/wave12_readout.py -round w20`.

The indicator concatenated with ProtT5 passes. It scores 0.860, 0.866 and 0.888 against the round-5 cell's 0.814, 0.805 and 0.844 on the same seeds, up by 0.046, 0.061 and 0.044, a seed mean of 0.871 with a spread of 0.015 and a pooled bootstrap of 0.868 [0.80, 0.93]; the lower end of that interval sits at the ladder's best rung. It is the first cell in this document whose seed mean clears 0.83 and whose weakest seed clears the round-5 cell's strongest. Its cells select epochs 38 to 53 on a validation curve that peaks near 0.85 between epochs 25 and 50 and drifts to 0.80 by epoch 200 (Figure 7a), and the cell has 18,930 parameters, 2,364 more than the round-5 cell for the wider input. The indicator alone does not: 0.810, 0.759 and 0.774, a mean of 0.781, with two seeds selecting epoch 2 and a validation curve that falls from 0.73 from the first epochs while its preprocessor's gradient norm sits near 5 for the whole run (Figure 7d), against 0.1 to 3 on the other arms. The two encodings add through the transformer, 0.87 against 0.83 for the indicator's rung and 0.82 for ProtT5's; the ladder never fit the pair on its own. Beside the chromosome and pathway indicators the encoding holds five continuous gene properties, two of them the gene's expression level and dispersion, and round 14 below says which columns carry the gain. ESM2 with the affine scores 0.740, below ProtT5 with the affine by 0.08 and level with ESM2's own ladder rung, and the seven-encoding concatenation 0.820, the round-5 cell's number from 78,066 parameters: adding five encodings to the two that pass returns the concatenation to where ProtT5 alone sits. Round 12 confirms the passing cell under the fold protocol and on four more seeds before the number is reported, since 0.87

on 195 genes is one cell's draw.

### 3.10 Round 12: the passing cell confirmed, and the number to report ✗

Round 12 (jobs 2291 to 2293) runs the round-9 cell at seeds 2703 to 2706 on the split file and as five fold models on the cross-validation files, the protocol behind the published Flux Cone Learning number, beside the round-5 cell's own fold models (Table 19, Figure 8). The four new seeds score 0.846, 0.877, 0.868 and 0.878, so over seven seeds the cell reads 0.869 with a spread of 0.014 and a pooled bootstrap of 0.870 [0.81, 0.93]; the weakest of the seven is above the round-5 cell's strongest. The five fold models score 0.860, 0.899, 0.889, 0.876 and 0.884 on the held-out genes, a fold mean of 0.881, every fold above the best of the round-5 folds (0.853), and their averaged logits score 0.893 [0.84, 0.94] (`cv_ensemble_readout.py -prefix w23_chromprotcv`), against 0.822 for the round-5 folds and 0.795 for the backbone alone. The fold models select epochs 25 to 53.

**Table 19:** Round 12, the passing cell confirmed. The round-5 cell (ProtT5 backbone, context head plus the FBA affine) and its five fold models, against the round-9 cell (chromosome-pathway indicator concatenated with ProtT5, same readout) on seven seeds, 2700 to 2706, and on the five cross-validation folds whose logits are averaged on the held-out genes. Width 12, one block, batch 32, class weight, 200 epochs. Bold marks the highest value in a column. ↑ (↓) marks an arm whose held-out AUROC is above (below) the first arm's on every one of the same three seeds.

| arm | mean | sd | seeds | pooled [95% CI] | selected epochs | val step |
|---|---|---|---|---|---|---|
| context head + FBA growth ratios (round-3 levers) | 0.821 | 0.020 | 0.814, 0.805, 0.844 | 0.838 [0.76, 0.91] | 28, 70, 61 | 0.0036 |
| chrom-pathway + ProtT5 backbone + FBA ratios | 0.869 | 0.014 | 0.846 to 0.888 (7) | 0.870 [0.81, 0.93] | 38 to 53 | 0.0027 |
| context head + FBA ratios, fold 0 | 0.814 | 0.000 | 0.814 | 0.814 [0.73, 0.89] | 28 | 0.0033 |
| context head + FBA ratios, fold 1 | 0.824 | 0.000 | 0.824 | 0.824 [0.74, 0.90] | 42 | 0.0042 |
| context head + FBA ratios, fold 2 | 0.839 | 0.000 | 0.839 | 0.839 [0.77, 0.91] | 37 | 0.0048 |
| context head + FBA ratios, fold 3 | 0.775 | 0.000 | 0.775 | 0.775 [0.69, 0.86] | 122 | 0.0042 |
| context head + FBA ratios, fold 4 | 0.853 | 0.000 | 0.853 | 0.853 [0.78, 0.92] | 54 | 0.0036 |
| chrom-pathway + ProtT5 + FBA ratios, fold 0 | 0.860 | 0.000 | 0.860 | 0.860 [0.79, 0.92] | 53 | 0.0030 |
| chrom-pathway + ProtT5 + FBA ratios, fold 1 | **0.899** | 0.000 | 0.899 | **0.899** [0.84, 0.95] | 32 | 0.0030 |
| chrom-pathway + ProtT5 + FBA ratios, fold 2 | 0.889 | 0.000 | 0.889 | 0.889 [0.83, 0.94] | 25 | 0.0055 |
| chrom-pathway + ProtT5 + FBA ratios, fold 3 | 0.875 | 0.000 | 0.875 | 0.875 [0.81, 0.93] | 34 | 0.0045 |
| chrom-pathway + ProtT5 + FBA ratios, fold 4 | 0.884 | 0.000 | 0.884 | 0.884 [0.82, 0.94] | 27 | 0.0055 |

The number to set beside the published 0.73 and beside 0.742 on these genes is the fold ensemble, 0.893 [0.84, 0.94]: a width-12 transformer reading the chromosome-pathway indicator and ProtT5, with FBA's four growth ratios on its logit, scored under Flux Cone Learning's own protocol on its own held-out genes, is 0.15 above it, and its interval excludes the ladder's best rung (0.830). The cell has 18,930 parameters. Every part of the gain is accounted for by an input the ontology computes for every gene at no training cost: the four solver numbers (0.774 to 0.821 on the seed mean) and the pathway indicator concatenated with the protein embedding (0.821 to 0.869). The inputs are the number: a two-layer MLP on the same three inputs, with no graph and no perturbation, reads 0.870 per cell in six seconds of CPU and 0.886 [0.82, 0.94] as a fold ensemble (Section 2), so what the transformer's architecture adds on this endpoint is not measurable on 195 genes. The 0.893 is reported as the document's number because it is the cell the rounds selected under the fixed reads; the MLP's 0.886 and the graph convolution's 0.888 are the same number.

### 3.11 Round 11: the affine widened with the solve ✗

The ladder's solve-derived encodings (Section 2) carry 0.15 AUROC beyond the growth ratio. Round 11 (jobs 2294 to 2296) appends them to the round-5 cell's affine, standardized column by column over the labeled genes: the eight reaction statistics (12 columns with the ratios), the deletion's pFBA flux vector (4,135), and both (4,139), each at the three seeds on the ProtT5 backbone (Table 20, Figure 9), under the round-8 read.

**Table 20:** Round 11, the affine widened with the solve. The round-5 cell (context head plus an affine of the four FBA growth ratios) against the same cell whose affine also reads the eight statistics of the deleted gene's reactions under the wild-type pFBA solve and its variability, the deletion's own pFBA flux vector (4,131 columns), or both; every column standardized over the labeled genes. Width 12, one block, batch 32, class weight, 200 epochs, seeds 2700 to 2702. Bold marks the highest value in a column. ↑ (↓) marks an arm whose held-out AUROC is above (below) the first arm's on every one of the same three seeds.

| arm | mean | sd | seeds | pooled [95% CI] | selected epochs | val step |
|---|---|---|---|---|---|---|
| context head + FBA growth ratios (round-3 levers) | 0.821 | 0.020 | 0.814, 0.805, 0.844 | 0.838 [0.76, 0.91] | 28, 70, 61 | 0.0036 |
| context head + FBA ratios + eight reaction statistics | 0.795 | 0.019 | 0.816, 0.790, 0.779 | 0.809 [0.70, 0.90] | 55, 71, 124 | 0.0049 |
| context head + FBA ratios + deletion pFBA vector | 0.831 | 0.076 | 0.876, 0.743, 0.874 | 0.853 [0.76, 0.93] | 70, 7, 99 | 0.0064 |
| context head + FBA ratios + statistics + pFBA vector | **0.858** | 0.008 | 0.850, 0.858, 0.867 ↑ | **0.858** [0.77, 0.93] | 188, 153, 162 | 0.0056 |

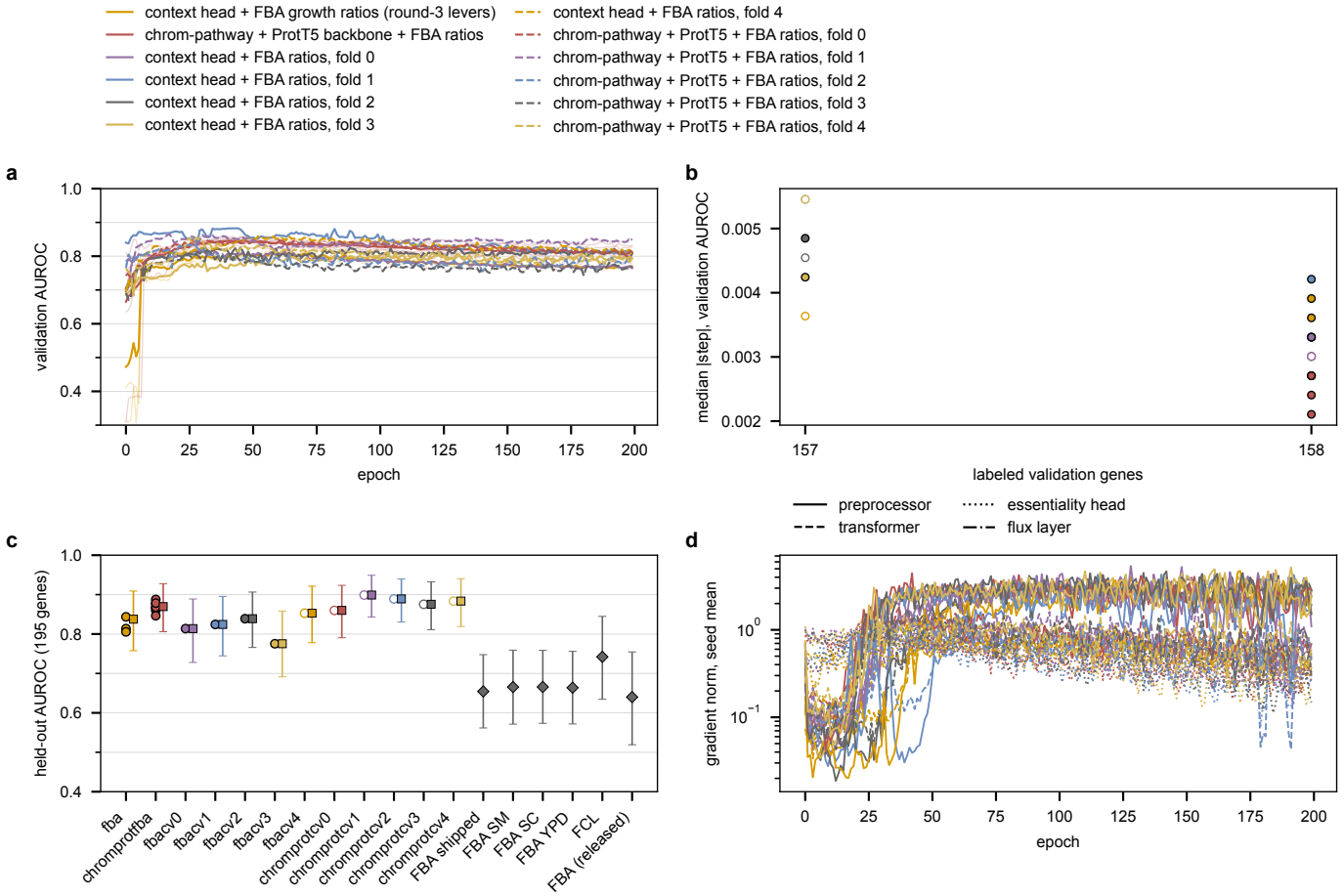



**Figure 8: Round 12, the passing cell confirmed.** **a**, Validation AUROC per epoch, one thin line per seed and the seed mean per arm; each fold model reads its own fold. **b**, The median absolute epoch-to-epoch step per cell against the validation size, 157 or 158 genes. **c**, Held-out AUROC per cell (points), the seed mean and the gene bootstrap of the pooled per-gene logits (square with interval), beside FBA on four media, Flux Cone Learning and the released FBA predictions on the same 195 genes. **d**, Gradient norm per module per epoch, seed mean. Written by `experiments/028-gene-essentiality/scripts/wave12_readout.py -round w23`.

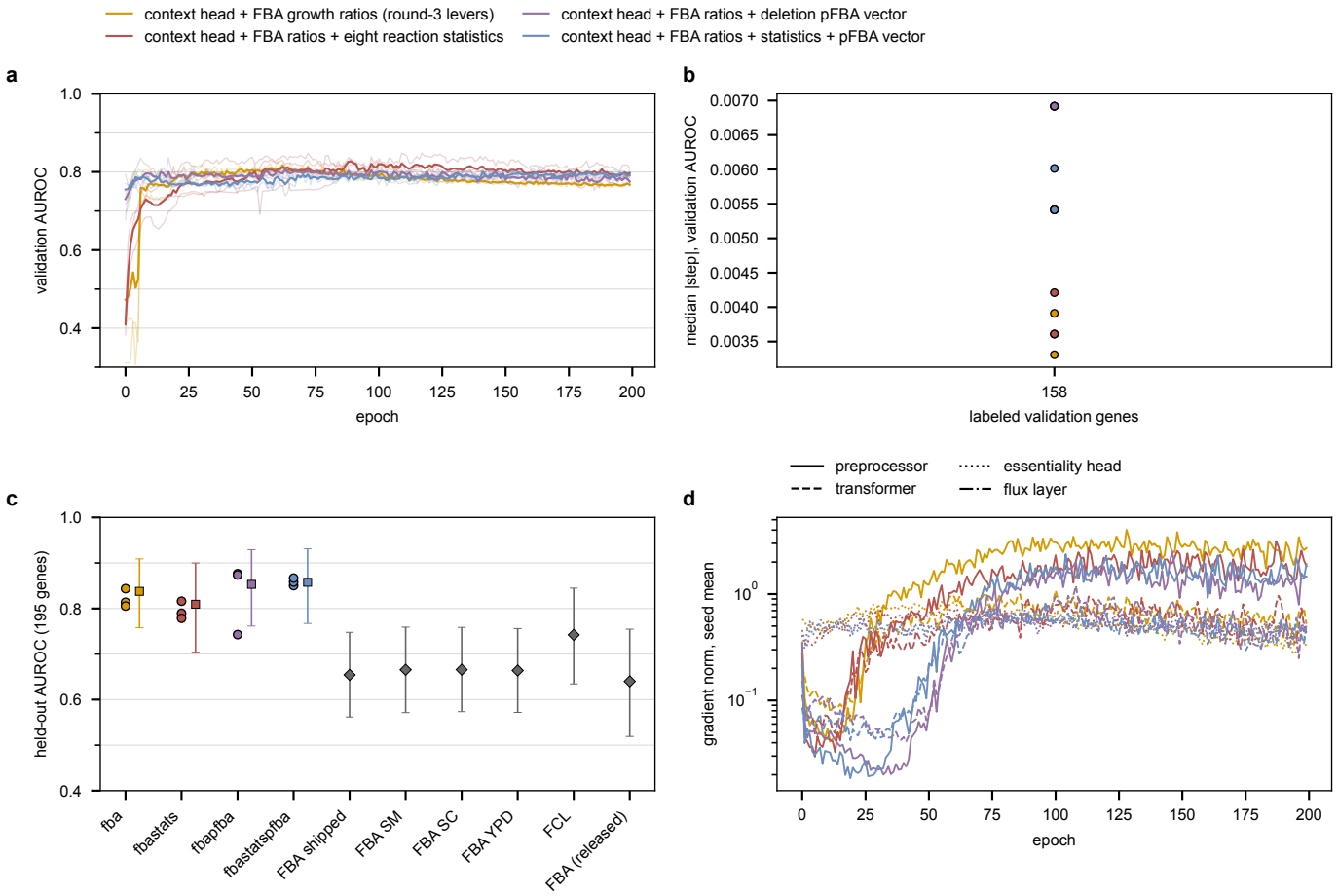



**Figure 9: Round 11, the affine widened with the solve.** **a**, Validation AUROC per epoch, one thin line per seed and the seed mean per arm. **b**, The median absolute epoch-to-epoch step per cell against the validation size. **c**, Held-out AUROC per cell (points), the seed mean and the gene bootstrap of the pooled per-gene logits (square with interval), beside FBA on four media, Flux Cone Learning and the released FBA predictions on the same 195 genes. **d**, Gradient norm per module per epoch, seed mean. Written by `experiments/028-gene-essentiality/scripts/wave12_readout.py -round w22`.

The statistics and the vector together pass; neither does alone. With both, the cell scores 0.850, 0.858 and 0.867 against 0.814, 0.805 and 0.844, up on three of three seeds by 0.036, 0.053 and 0.023, a seed mean of 0.858 with a spread of 0.008, and it selects late, at epochs 153 to 188, on a validation curve that is still level at 200. The eight statistics alone read 0.816, 0.790 and 0.779, down on three of three seeds, though a random forest on the same twelve numbers scores 0.817 on the ladder: through a linear affine the statistics do not add to what the context head already carries, and what the random forest reads in them the affine cannot. The vector alone reads 0.877, 0.743 and 0.874, two seeds at the level of the pair and one collapsed at epoch 7, a collapse no seed shows with the eight statistics beside the vector. Round 16 puts the passing affine on the round-9 backbone.

## 3.12 Round 14: which columns of the indicator carry the gain ✗

The chromosome-pathway indicator is three things: five scaled gene properties, a chromosome one-hot, and a multi-hot over 173 SGD pathways. Round 14 (jobs 2307 and 2308) drops the pathway columns (`normalized_chrom`, 24 dimensions) and runs the round-9 cell with and without ProtT5 beside it (Table 21, Figure 10).

**Table 21:** Round 14, attribution inside the chromosome-pathway indicator. The round-5 cell, the round-9 cells on the indicator alone and with ProtT5, and the same two cells on the indicator without its pathway columns (`normalized_chrom`: the five gene properties and the chromosome one-hot), all with the FBA affine. Width 12, one block, batch 32, class weight, 200 epochs, seeds 2700 to 2702. Bold marks the highest value in a column. ↑ (↓) marks an arm whose held-out AUROC is above (below) the first arm's on every one of the same three seeds.

| arm | mean | sd | seeds | pooled [95% CI] | selected epochs | val step |
|---|---|---|---|---|---|---|
| context head + FBA growth ratios (round-3 levers) | 0.821 | 0.020 | 0.814, 0.805, 0.844 | 0.838 [0.76, 0.91] | 28, 70, 61 | 0.0036 |
| chrom-pathway backbone + FBA ratios | 0.781 | 0.027 | 0.810, 0.759, 0.774 ↓ | 0.823 [0.75, 0.90] | 21, 2, 2 | 0.0083 |
| chrom-pathway + ProtT5 backbone + FBA ratios | **0.871** | 0.015 | 0.860, 0.866, 0.888 ↑ | **0.868** [0.80, 0.93] | 53, 51, 38 | 0.0030 |
| gene properties + chromosome backbone + FBA ratios | 0.669 | 0.030 | 0.636, 0.676, 0.694 ↓ | 0.663 [0.54, 0.77] | 32, 69, 12 | 0.0110 |
| gene properties + chromosome + ProtT5 backbone + FBA ratios | 0.812 | 0.003 | 0.811, 0.816, 0.810 | 0.812 [0.72, 0.89] | 65, 55, 53 | 0.0039 |

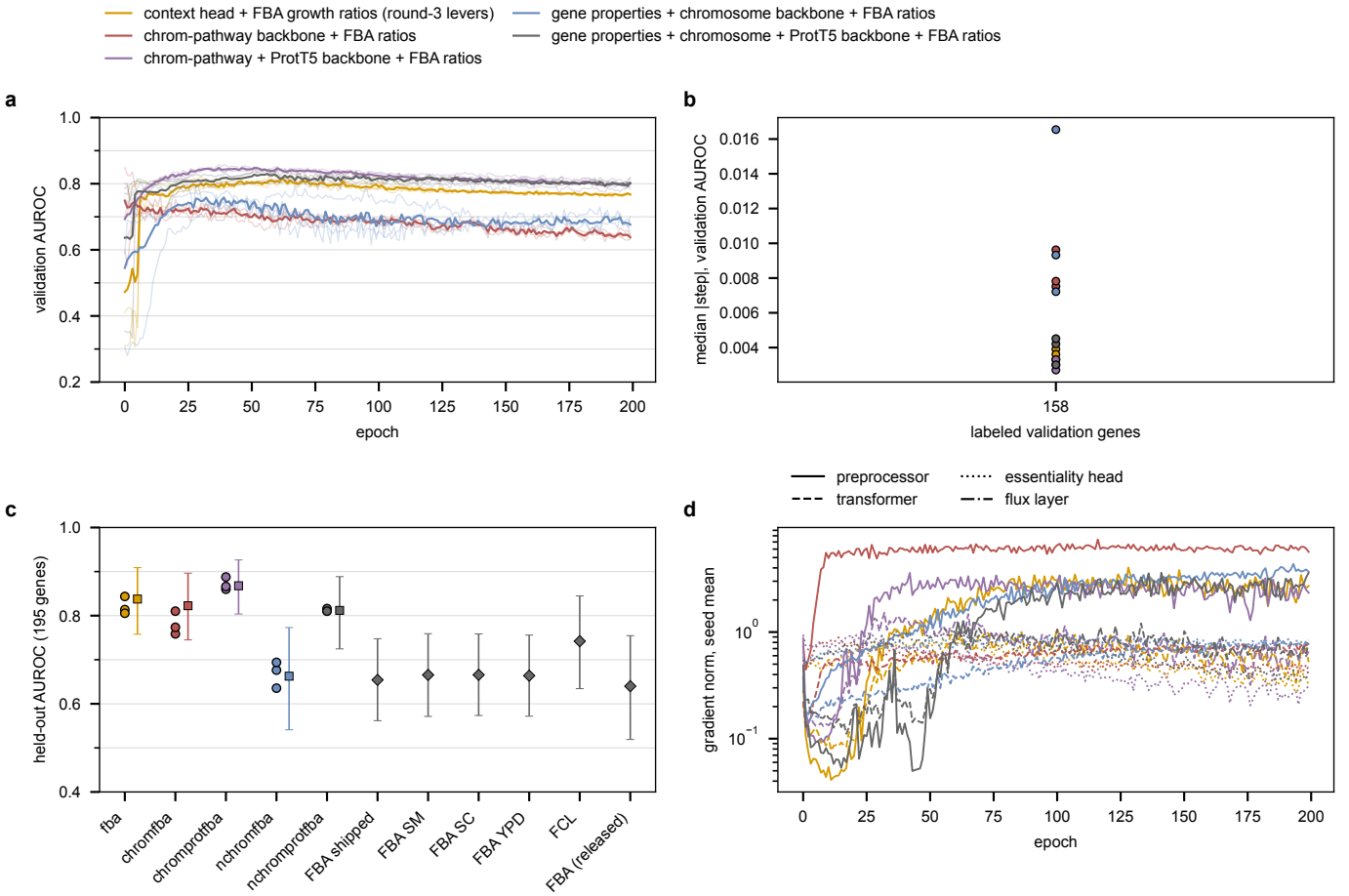



**Figure 10: Round 14, attribution inside the chromosome-pathway indicator.** **a**, Validation AUROC per epoch, one thin line per seed and the seed mean per arm. **b**, The median absolute epoch-to-epoch step per cell against the validation size. **c**, Held-out AUROC per cell (points), the seed mean and the gene bootstrap of the pooled per-gene logits (square with interval), beside FBA on four media, Flux Cone Learning and the released FBA predictions on the same 195 genes. **d**, Gradient norm per module per epoch, seed mean. Written by `experiments/028-gene-essentiality/scripts/wave12_readout.py -round w25`.

The pathways carry it. Without them, the properties and the chromosome beside ProtT5 read 0.811, 0.816 and 0.810, a seed mean of 0.813 against 0.871 with the pathways and 0.821 for ProtT5 alone, so the five properties and the chromosome add nothing to the protein embedding, expression included; on their own they read 0.669, the solver's level. The 173 pathway columns are what lifts the cell from 0.82 to 0.87, and they do it only with ProtT5 beside them: the full indicator alone reads 0.78 (round 9). A gene's pathway membership and its protein sequence are complementary inputs on this endpoint, and each is cheap.

## 3.13 Round 22: the indicator split into its blocks ✗

Round 14 attributes the gain by subtraction, the indicator without its pathway columns; round 22 (job 2338) attributes it by addition, one block at a time on the round-9 recipe. The 197-column indicator is three blocks, the 173-column pathway multi-hot, the 17-column chromosome one-hot and the seven scaled gene properties, and each is built as an embedding of its own (`pathways`, `chromosome`, `normalized_props` in the embedding registry) and run beside ProtT5 with the FBA affine, three seeds each, with the pathway block also run without ProtT5. The read is paired against round 9's cell: the block whose three seeds match round 9's is the block that carries the gain, and a block that reads at the ProtT5-alone level (0.821) carries none. On the cheap models the same split runs on the MLP and on graph convolution (Table 7), where the pathway block beside ProtT5 and the ratios reads 0.876, 0.885 and 0.878 on the MLP, above the full indicator's 0.870.

**Table 22:** Round 22, the chromosome-pathway indicator split into its blocks. The round-9 cell (the paired comparator: the 197-column indicator beside ProtT5 with the FBA affine), the round-5 affine on ProtT5 alone, and the round-14 cell (properties and chromosome, no pathways), against the same recipe with one block at a time: the 173-column pathway multi-hot, the 17-column chromosome one-hot, the seven scaled properties, and the pathway block alone without ProtT5. Batch 32, class weight, 200 epochs, seeds 2700 to 2702. Bold marks the highest value in a column. ↑ (↓) marks an arm whose held-out AUROC is above (below) the first arm's on every one of the same three seeds.

| arm | mean | sd | seeds | pooled [95% CI] | selected epochs | val step |
|---|---|---|---|---|---|---|
| chrom-pathway + ProtT5 backbone + FBA ratios | 0.871 | 0.015 | 0.860, 0.866, 0.888 | 0.868 [0.80, 0.93] | 53, 51, 38 | 0.0030 |
| context head + FBA growth ratios (round-3 levers) | 0.821 | 0.020 | 0.814, 0.805, 0.844 ↓ | 0.838 [0.76, 0.91] | 28, 70, 61 | 0.0036 |
| gene properties + chromosome + ProtT5 backbone + FBA ratios | 0.812 | 0.003 | 0.811, 0.816, 0.810 ↓ | 0.812 [0.72, 0.89] | 65, 55, 53 | 0.0039 |
| pathway multi-hot + ProtT5 backbone + FBA ratios | **0.890** | 0.005 | 0.889, 0.895, 0.885 | **0.890** [0.83, 0.94] | 31, 45, 55 | 0.0054 |
| chromosome one-hot + ProtT5 backbone + FBA ratios | 0.811 | 0.019 | 0.824, 0.821, 0.789 ↓ | 0.818 [0.73, 0.90] | 40, 44, 55 | 0.0039 |
| seven gene properties + ProtT5 backbone + FBA ratios | 0.824 | 0.020 | 0.816, 0.846, 0.809 ↓ | 0.829 [0.75, 0.90] | 71, 63, 38 | 0.0035 |
| pathway multi-hot backbone + FBA ratios | 0.766 | 0.007 | 0.773, 0.760, 0.766 ↓ | 0.774 [0.68, 0.86] | 123, 140, 96 | 0.0018 |

The transformer says what the MLP said. ProtT5 with the pathway block and the affine reads 0.889, 0.895 and 0.885 (mean 0.890; pooled 0.890 with a gene bootstrap of 0.834 to 0.942), against round 9's full indicator at 0.860, 0.866 and 0.888: above by 0.03 on two seeds and under by 0.003 on the third, the seed round 9 selected at its best. ProtT5 with the chromosome block reads 0.824, 0.821 and 0.789 and with the seven properties 0.816, 0.846 and 0.809, both at the level of ProtT5 with the affine alone (0.821). The pathway block without ProtT5 reads 0.773, 0.760 and 0.766 on the transformer where the MLP had it at 0.828, the one place the two models part: the transformer needs the sequence embedding to use the pathway columns, and the MLP does not. The block that carries the gain is the pathway multi-hot, and the 24 columns beside it carry none of it. The best single cell in this section is now the pathway block with ProtT5 and the affine at 0.895, which puts the input, not the architecture, 0.02 above the ensemble the document reports.

## 3.14 Round 26: the Gene Ontology multi-hot as a backbone input ✗

If 173 pathway memberships carry the gain over the protein embedding, the larger annotation of the same kind is the Gene Ontology, and the same question applies to it. The GO multi-hot (Section 1) is built as its own dataset beside the chromosome, pathway and property vectors (Table 23), so each is a starting vector any model can take. Round 26 (jobs 2363 and 2364) runs the round-9 recipe with it: GO beside ProtT5 with the affine, the biological-process namespace alone beside ProtT5, GO alone with the affine, and GO with the pathway multi-hot beside ProtT5, three seeds each (Table 24, Figure 12). The read pairs each cell against the round-22 pathway cell by seed: GO beside ProtT5 at or above it on three seeds of three says GO carries at least what the pathways carry; GO with the pathways above it on three of three says the two annotations add. The cheap models run the same inputs on the MLP and on graph convolution (Table 8), and the ladder gains a linear rung on each annotation vector alone.

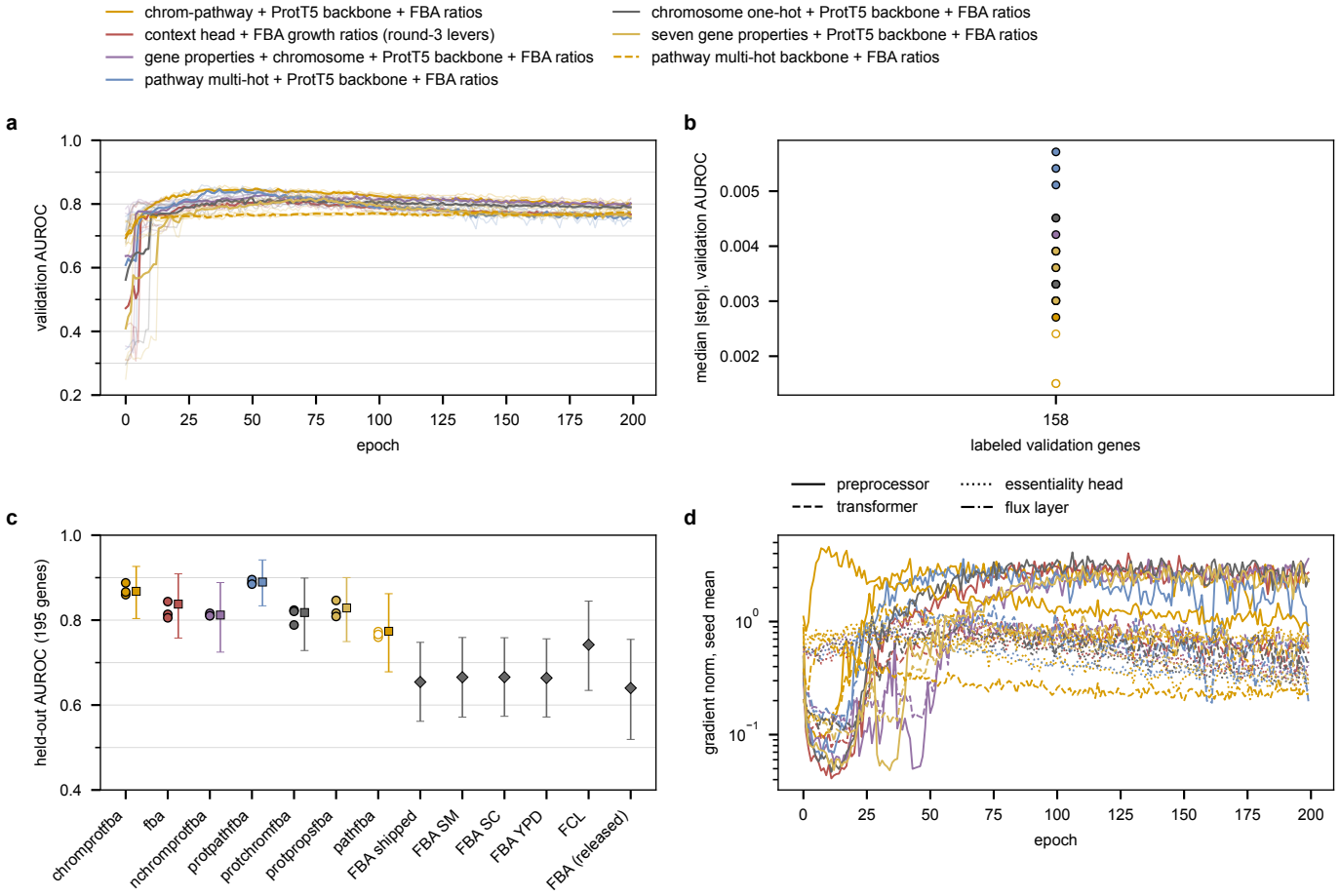



**Figure 11: Round 22, the chromosome-pathway indicator split into its blocks on the transformer.** **a**, Validation AUROC per epoch, one thin line per seed and the seed mean per arm. **b**, The median absolute epoch-to-epoch step per cell against the validation size. **c**, Held-out AUROC per cell (points), the seed mean and the gene bootstrap of the pooled per-gene logits (square with interval), beside FBA on four media, Flux Cone Learning and the released FBA predictions on the same 195 genes. **d**, Gradient norm per module per epoch, seed mean. Written by `experiments/028-gene-essentiality/scripts/wave12_readout.py -round w38`.

**Table 23:** The per-gene starting vectors used in this document, each a dataset in the source tree registered under its name in `NodeEmbeddingBuilder` so any model opens it by name. Width is the column count of the built tensor; covered is the number of genes with a nonzero row over the genes the dataset holds. Written by `notes_tex_tables.py` from the built tensors.

| name | what | width | covered | construction | module |
|---|---|---|---|---|---|
| `prot_T5_all` | protein language model embedding | 1,024 | 6,607 / 6,607 | ProtT5 mean-pooled over the protein sequence | `protT5.py` |
| `pathways` | pathway multi-hot | 173 | 517 / 6,607 | one column per SGD pathway a gene is annotated to | `sgd_gene_graph_hot.py` |
| `normalized_props` | scaled gene properties | 7 | 6,607 / 6,607 | protein length, molecular weight, isoelectric point, expression median and median absolute deviation, scaled to [0, 1] | `sgd_gene_graph_hot.py` |
| `normalized_chrom_pathways` | chromosome-pathway indicator | 197 | 6,607 / 6,607 | the three blocks above concatenated | `sgd_gene_graph_hot.py` |
| `go` | GO multi-hot, propagated | 2,362 | 5,501 / 6,607 | SGD GO annotations propagated to ancestors (is_a, part_of), roots dropped, terms with ten or more genes | `go_term_hot.py` |
| `go_direct` | GO multi-hot, direct | 2,887 | 5,492 / 6,607 | the annotated terms only, terms with two or more genes | `go_term_hot.py` |
| `go_bp` | GO biological process | 1,580 | 4,909 / 6,607 | the propagated vector restricted to one namespace | `go_term_hot.py` |
| `go_mf` | GO molecular function | 430 | 4,146 / 6,607 | as above | `go_term_hot.py` |
| `go_cc` | GO cellular component | 352 | 5,256 / 6,607 | as above | `go_term_hot.py` |

**Table 24:** Round 26, the Gene Ontology multi-hot as a backbone input. The round-5 affine on ProtT5, the round-9 indicator cell and the round-22 pathway cell (the comparators), against the round-9 recipe with the GO multi-hot beside ProtT5 (the SGD annotations propagated to their ancestors over terms with ten or more genes, 2,362 columns), the biological-process namespace alone beside ProtT5 (1,580 columns), the GO multi-hot alone, and GO with the pathway multi-hot beside ProtT5. Batch 32, class weight, 200 epochs, seeds 2700 to 2702. Bold marks the highest value in a column. ↑ (↓) marks an arm whose held-out AUROC is above (below) the first arm's on every one of the same three seeds.

| arm | mean | sd | seeds | pooled [95% CI] | selected epochs | val step |
|---|---|---|---|---|---|---|
| context head + FBA growth ratios (round-3 levers) | 0.821 | 0.020 | 0.814, 0.805, 0.844 | 0.838 [0.76, 0.91] | 28, 70, 61 | 0.0036 |
| chrom-pathway + ProtT5 + FBA ratios | 0.871 | 0.015 | 0.860, 0.866, 0.888 ↑ | 0.868 [0.80, 0.93] | 53, 51, 38 | 0.0030 |
| pathway multi-hot + ProtT5 + FBA ratios | **0.890** | 0.005 | 0.889, 0.895, 0.885 ↑ | **0.890** [0.83, 0.94] | 31, 45, 55 | 0.0054 |
| GO multi-hot (propagated, 2,362) + ProtT5 + FBA ratios | 0.833 | 0.008 | 0.830, 0.827, 0.842 | 0.839 [0.76, 0.91] | 123, 48, 62 | 0.0044 |
| GO biological process (1,580) + ProtT5 + FBA ratios | 0.821 | 0.014 | 0.805, 0.832, 0.826 | 0.827 [0.73, 0.91] | 180, 160, 104 | 0.0034 |
| GO multi-hot backbone + FBA ratios | 0.800 | 0.012 | 0.810, 0.787, 0.804 ↓ | 0.810 [0.72, 0.89] | 28, 11, 139 | 0.0048 |
| GO + pathway multi-hots + ProtT5 + FBA ratios | 0.849 | 0.005 | 0.852, 0.852, 0.844 | 0.851 [0.78, 0.92] | 131, 48, 126 | 0.0019 |

The transformer takes nothing from GO that it did not have from ProtT5, and beside the pathway block GO costs it. GO with ProtT5 and the affine reads 0.830, 0.827 and 0.842 (mean 0.833), level with ProtT5 and the affine alone (0.821) and 0.05 to 0.07 under the pathway cell on every seed; the biological-process namespace alone reads 0.821, and GO alone with the affine 0.800. GO added to the pathway cell reads 0.852, 0.852 and 0.844 against that cell's 0.889, 0.895 and 0.885, a cost of 0.04 on three seeds of three. Both clauses of the read fail. The same vector is the strongest single encoding on the ladder (0.808 linear with no solve, 0.844 under a random forest with the ratios) and, on the cheap models under the same protocol, lifts the MLP to 0.855 and graph convolution to 0.887 (Table 8), so what fails is not the vector but the transformer's use of it; why a width-12 cell reads 173 sparse pathway columns and not 2,362 dense GO columns is not measured. The one-column control that marks a gene as annotated to any pathway reads 0.774 on the MLP and 0.832 on graph convolution, the ProtT5-with-ratios level, so the pathway block's gain is which pathway, not whether there is one.

## 3.15 Round 28: the GO cells with a wider cell ✗

If the width-12 preprocessor is what keeps the transformer from reading 2,362 GO columns, a wider one reads them. Round 28 (job 2378) runs the two round-26 GO cells, GO beside ProtT5 and GO with the pathways beside ProtT5, at width 24 with a two-layer preprocessor (the round-8 cell's shape), three seeds each, against the width-12 cells and the pathway cell (Table 25). The read pairs each wide cell against its width-12 cell by seed: above on three of three and the preprocessor is the limit; level and it is not.

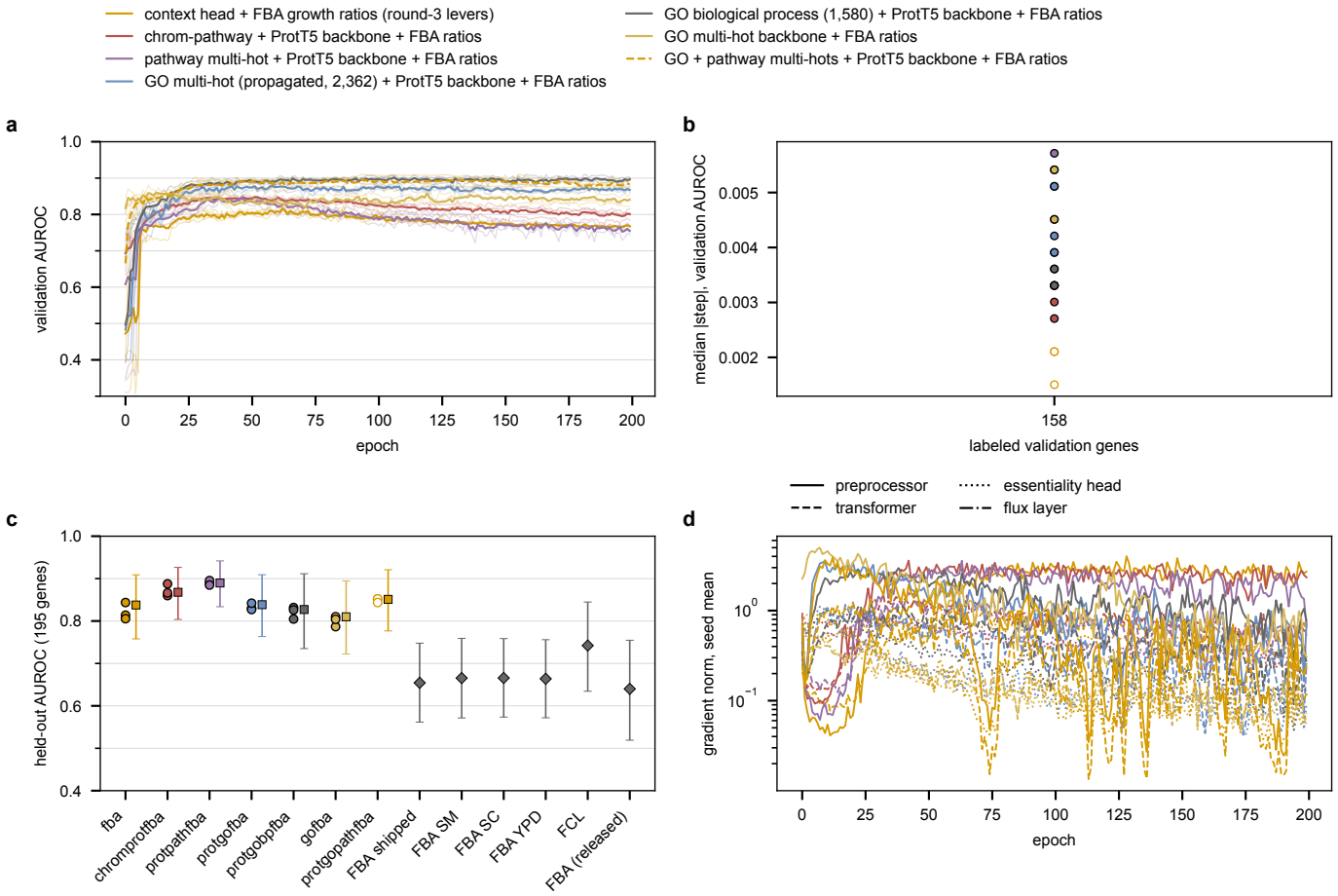



**Figure 12: Round 26, the Gene Ontology multi-hot as a backbone input. a**, Validation AUROC per epoch, one thin line per seed and the seed mean per arm. **b**, The median absolute epoch-to-epoch step per cell against the validation size. **c**, Held-out AUROC per cell (points), the seed mean and the gene bootstrap of the pooled per-gene logits (square with interval), beside FBA on four media, Flux Cone Learning and the released FBA predictions on the same 195 genes. **d**, Gradient norm per module per epoch, seed mean. Written by `experiments/028-gene-essentiality/scripts/wave12_readout.py -round w42`.

**Table 25:** Round 28, the GO cells with a wider cell. The round-5 affine on ProtT5, the round-22 pathway cell and the two round-26 GO cells at width 12 with a one-layer preprocessor (the comparators), against the same GO cells at width 24 with a two-layer preprocessor. Batch 32, class weight, 200 epochs, seeds 2700 to 2702. Bold marks the highest value in a column. ↑ (↓) marks an arm whose held-out AUROC is above (below) the first arm's on every one of the same three seeds.

| arm | mean | sd | seeds | pooled [95% CI] | selected epochs | val step |
|---|---|---|---|---|---|---|
| context head + FBA growth ratios (round-3 levers) | 0.821 | 0.020 | 0.814, 0.805, 0.844 | 0.838 [0.76, 0.91] | 28, 70, 61 | 0.0036 |
| pathway + ProtT5 + FBA ratios | **0.890** | 0.005 | 0.889, 0.895, 0.885 ↑ | **0.890** [0.83, 0.94] | 31, 45, 55 | 0.0054 |
| GO multi-hot (propagated, 2,362) + ProtT5 + FBA ratios | 0.833 | 0.008 | 0.830, 0.827, 0.842 | 0.839 [0.76, 0.91] | 123, 48, 62 | 0.0044 |
| GO + pathway multi-hots + ProtT5 + FBA ratios | 0.849 | 0.005 | 0.852, 0.852, 0.844 | 0.851 [0.78, 0.92] | 131, 48, 126 | 0.0019 |
| GO + ProtT5 + FBA ratios, width 24, two-layer preprocessor | 0.842 | 0.005 | 0.847, 0.841, 0.838 | 0.847 [0.77, 0.92] | 152, 65, 129 | 0.0037 |
| GO + pathways + ProtT5 + FBA ratios, width 24, two-layer preprocessor | 0.852 | 0.003 | 0.849, 0.855, 0.853 ↑ | 0.858 [0.78, 0.93] | 42, 182, 76 | 0.0028 |

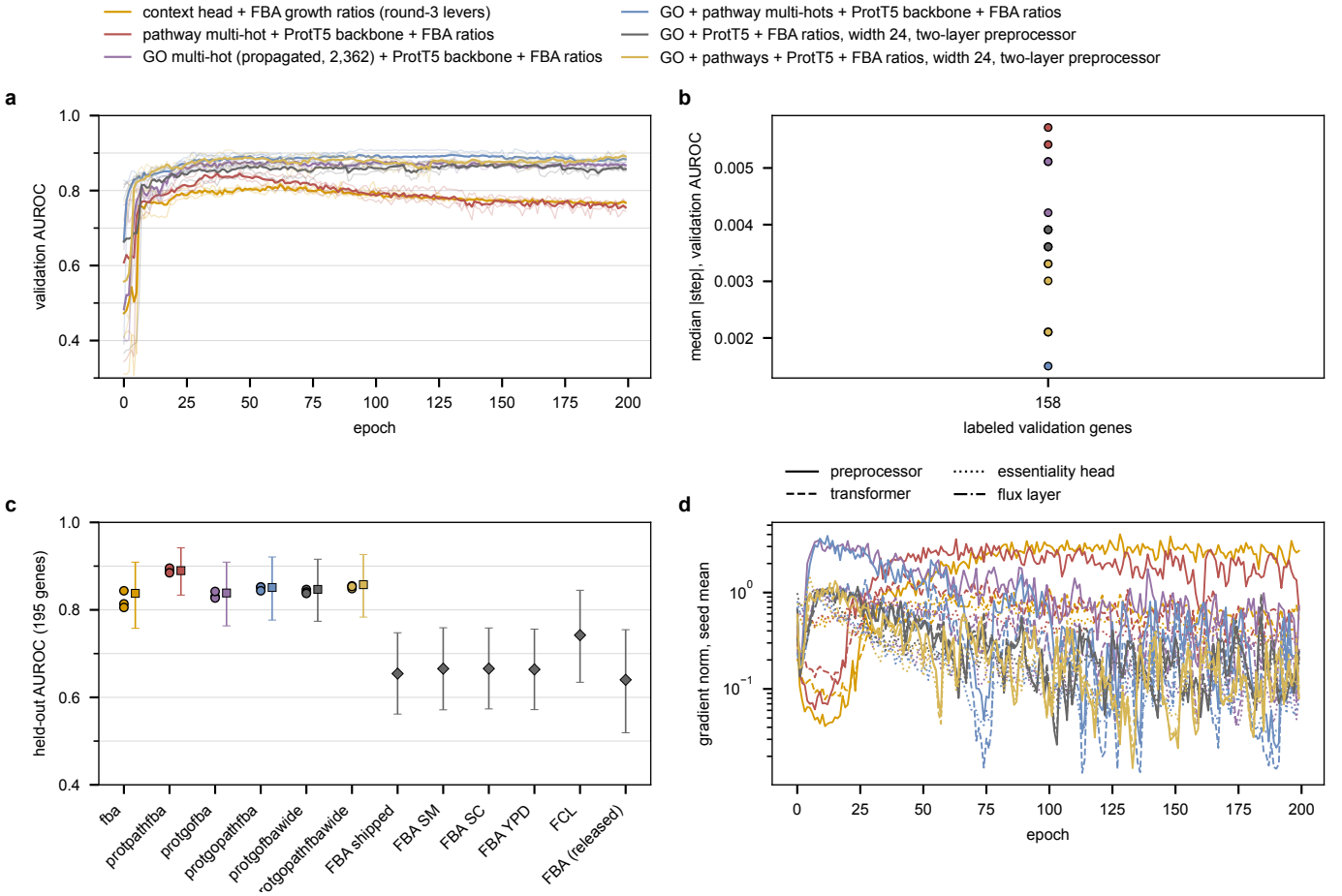



**Figure 13: Round 28, the GO cells with a wider cell. a**, Validation AUROC per epoch, one thin line per seed and the seed mean per arm. **b**, The median absolute epoch-to-epoch step per cell against the validation size. **c**, Held-out AUROC per cell (points), the seed mean and the gene bootstrap of the pooled per-gene logits (square with interval), beside FBA on four media, Flux Cone Learning and the released FBA predictions on the same 195 genes. **d**, Gradient norm per module per epoch, seed mean. Written by `experiments/028-gene-essentiality/scripts/wave12_readout.py -round w44`.

The preprocessor is not the limit. At width 24 with a two-layer preprocessor, 5.8 million parameters against 17 thousand, GO beside ProtT5 reads 0.847, 0.841 and 0.838 against the width-12 cell's 0.830, 0.827 and 0.842, above on two seeds by 0.01 to 0.02 and level on the third; GO with the pathways beside ProtT5 reads 0.849, 0.855 and 0.853 against 0.852, 0.852 and 0.844, level. A cell 350 times larger reads the GO vector the same, so whatever keeps the transformer at 0.84 on an input that gives graph convolution 0.91 is not capacity. What it is, is not measured; the pooled readout, which averages the perturbed tokens, and the density of the GO rows against the pathway rows are the untested candidates.

## 3.16 Round 16: the round-9 backbone with the round-11 affine ✗

Round 16 (jobs 2309 and 2310) puts the widened affine on the round-9 backbone: the chromosome-pathway indicator with ProtT5, and the four ratios with the eight statistics and the deletion flux vector on the logit, and the same without the statistics, three seeds each, against round 9 (Table 26, Figure 14).

**Table 26:** Round 16, the round-9 backbone with the round-11 affine. The round-5 cell, the round-9 cell (chromosome-pathway indicator with ProtT5), the round-11 cell (ProtT5 with the widened affine), and the round-9 backbone with the widened affine: the four ratios and the pFBA vector, and those plus the eight statistics. Width 12, one block, batch 32, class weight, 200 epochs, seeds 2700 to 2702. Bold marks the highest value in a column. ↑ (↓) marks an arm whose held-out AUROC is above (below) the first arm's on every one of the same three seeds.

| arm | mean | sd | seeds | pooled [95% CI] | selected epochs | val step |
|---|---|---|---|---|---|---|
| context head + FBA growth ratios (round-3 levers) | 0.821 | 0.020 | 0.814, 0.805, 0.844 | 0.838 [0.76, 0.91] | 28, 70, 61 | 0.0036 |
| chrom-pathway + ProtT5 + FBA ratios | 0.871 | 0.015 | 0.860, 0.866, 0.888 ↑ | 0.868 [0.80, 0.93] | 53, 51, 38 | 0.0030 |
| context head + FBA ratios + statistics + pFBA vector | 0.858 | 0.008 | 0.850, 0.858, 0.867 ↑ | 0.858 [0.77, 0.93] | 188, 153, 162 | 0.0056 |
| chrom-pathway + ProtT5 + FBA ratios + pFBA vector | 0.828 | 0.068 | 0.769, 0.812, 0.902 | 0.867 [0.79, 0.93] | 6, 13, 68 | 0.0027 |
| chrom-pathway + ProtT5 + ratios + statistics + pFBA | **0.880** | 0.018 | 0.863, 0.878, 0.898 ↑ | **0.889** [0.83, 0.94] | 60, 38, 42 | 0.0029 |

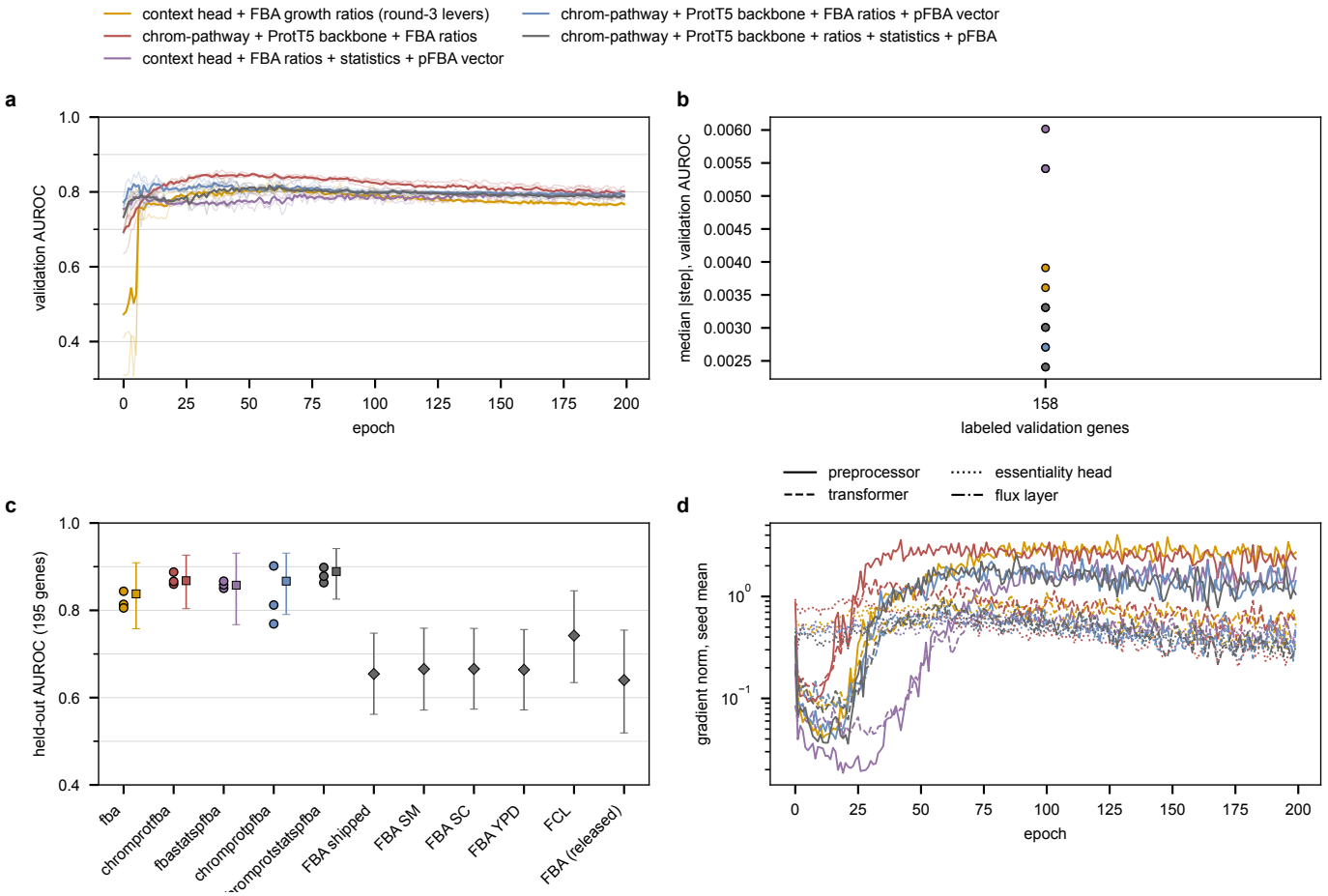



**Figure 14: Round 16, the round-9 backbone with the round-11 affine.** **a**, Validation AUROC per epoch, one thin line per seed and the seed mean per arm. **b**, The median absolute epoch-to-epoch step per cell against the validation size. **c**, Held-out AUROC per cell (points), the seed mean and the gene bootstrap of the pooled per-gene logits (square with interval), beside FBA on four media, Flux Cone Learning and the released FBA predictions on the same 195 genes. **d**, Gradient norm per module per epoch, seed mean. Written by `experiments/028-gene-essentiality/scripts/wave12_readout.py -round w27`.

The two gains add, by a little. With the statistics and the vector the cell reads 0.863, 0.878 and 0.898 against round 9's 0.860, 0.866 and 0.888, up by 0.003, 0.012 and 0.010 on three of three seeds, a seed mean of 0.880 with a spread of 0.018 and a pooled bootstrap of 0.889 [0.83, 0.94]. The step is a fifth of the one the indicator gave and inside the seed spread, so it is the direction rather than the size that holds. Without the statistics the vector alone reads 0.769, 0.812 and 0.902, two seeds selecting epochs 6 and 13, the instability round 11 showed on the ProtT5 backbone, and the same cure. Round 17 runs the passing cell on the five folds, with the read fixed first: its ensemble replaces the round-12 ensemble's 0.893 as the reported number only if its fold models are above round 12's on at least four of the five folds; otherwise 0.893 stands.

## 3.17 Round 17: the round-16 cell on the five folds ✗

Round 17 (jobs 2312 and 2313) trains the round-16 cell on the five cross-validation files (Table 27, Figure 15). Its fold models score 0.863, 0.860, 0.887, 0.841 and 0.882 on the held-out genes, a fold mean of 0.867, against round 12's 0.860,

0.899, 0.889, 0.876 and 0.884: above on fold 0 by 0.003 and below on the other four, by 0.002 to 0.039. Their averaged logits score 0.901 [0.85, 0.95], above the round-12 ensemble's 0.893 by 0.008, with the two intervals almost coincident.

**Table 27:** Round 17, the round-16 cell on the five folds. The round-9 and round-16 cells on the split file, the round-12 fold models of the round-9 cell, and the five fold models of the round-16 cell (chromosome-pathway indicator with ProtT5, the affine reading the four ratios, the eight statistics and the pFBA vector), whose logits are averaged on the held-out genes. Width 12, one block, batch 32, class weight, 200 epochs. Bold marks the highest value in a column. ↑ (↓) marks an arm whose held-out AUROC is above (below) the first arm's on every one of the same three seeds.

| arm | mean | sd | seeds | pooled [95% CI] | selected epochs | val step |
|---|---|---|---|---|---|---|
| chrom-pathway + ProtT5 + FBA ratios | 0.871 | 0.015 | 0.860, 0.866, 0.888 | 0.868 [0.80, 0.93] | 53, 51, 38 | 0.0030 |
| chrom-pathway + ProtT5 + ratios + statistics + pFBA | 0.880 | 0.018 | 0.863, 0.878, 0.898 ↑ | 0.889 [0.83, 0.94] | 60, 38, 42 | 0.0029 |
| chrom-pathway + ProtT5 + FBA ratios, fold 0 | 0.860 | 0.000 | 0.860 | 0.860 [0.79, 0.92] | 53 | 0.0030 |
| chrom-pathway + ProtT5 + FBA ratios, fold 1 | **0.899** | 0.000 | 0.899 | **0.899** [0.84, 0.95] | 32 | 0.0030 |
| chrom-pathway + ProtT5 + FBA ratios, fold 2 | 0.889 | 0.000 | 0.889 | 0.889 [0.83, 0.94] | 25 | 0.0055 |
| chrom-pathway + ProtT5 + FBA ratios, fold 3 | 0.875 | 0.000 | 0.875 | 0.875 [0.81, 0.93] | 34 | 0.0045 |
| chrom-pathway + ProtT5 + FBA ratios, fold 4 | 0.884 | 0.000 | 0.884 | 0.884 [0.82, 0.94] | 27 | 0.0055 |
| chrom-pathway + ProtT5 + widened affine, fold 0 | 0.863 | 0.000 | 0.863 | 0.863 [0.79, 0.93] | 60 | 0.0024 |
| chrom-pathway + ProtT5 + widened affine, fold 1 | 0.860 | 0.000 | 0.860 | 0.860 [0.77, 0.93] | 135 | 0.0078 |
| chrom-pathway + ProtT5 + widened affine, fold 2 | 0.887 | 0.000 | 0.887 | 0.887 [0.81, 0.95] | 26 | 0.0085 |
| chrom-pathway + ProtT5 + widened affine, fold 3 | 0.840 | 0.000 | 0.840 | 0.840 [0.75, 0.92] | 38 | 0.0052 |
| chrom-pathway + ProtT5 + widened affine, fold 4 | 0.882 | 0.000 | 0.882 | 0.882 [0.82, 0.94] | 42 | 0.0082 |

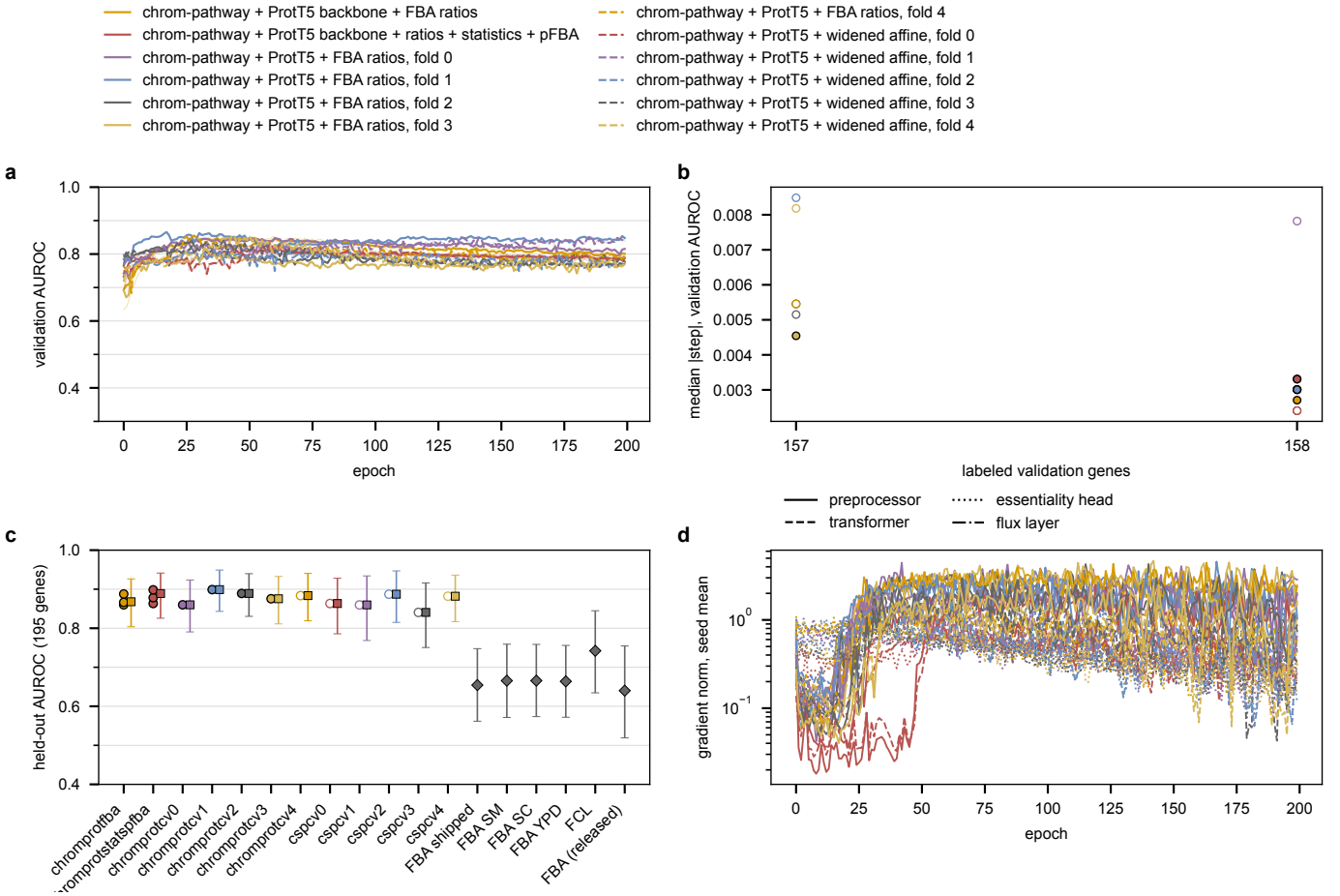



**Figure 15: Round 17, the round-16 cell on the five folds.** **a**, Validation AUROC per epoch, one thin line per seed and the seed mean per arm; each fold model reads its own fold. **b**, The median absolute epoch-to-epoch step per cell against the validation size, 157 or 158 genes. **c**, Held-out AUROC per cell (points), the seed mean and the gene bootstrap of the pooled per-gene logits (square with interval), beside FBA on four media, Flux Cone Learning and the released FBA predictions on the same 195 genes. **d**, Gradient norm per module per epoch, seed mean. Written by `experiments/028-gene-essentiality/scripts/wave12_readout.py -round w28`.

By the rule fixed before the round, 0.893 stands: the widened affine's fold models are below round 12's on four of five folds, and an ensemble 0.008 higher from fold models 0.014 lower on average is the kind of draw the rule exists to refuse. The two cells are one cell to the resolution of 195 genes, and the reported number for this document is the round-12 ensemble, 0.893 [0.84, 0.94], from the chromosome-pathway indicator with ProtT5 and the four FBA growth ratios.

# 4 The metabolic module: the solve helps, so why not the layer? ✗

One linear program per deletion on yeast-GEM ranks essential genes at 0.65 on its own and lifts every model in this document by 0.05 when its four growth ratios are appended (Sections 2 and 3). The metabolic module is built on the same genome-scale model: the same stoichiometry, the same bounds, the same gene-protein-reaction rules closing the deleted gene's reactions. The question of this section is why routing the prediction through it does not do what the solve does. The rounds answer it in steps: the module neither helps nor hurts on the backbone (rounds 2 and 4); it adds nothing on top of the solve (round 6); its growth column is not growth but a free coordinate, because the layer never reaches the polytope the solve optimizes over (the probe, round 7); handed the solve's own point on that polytope, the anchored layer's constraint set discards it (round 10) while the GEM's own box keeps it, and the gated module cell then reaches the level of the solve appended as an input (round 13); a layer that solves the program instead of projecting toward it is infeasible on the anchored box, whose capacity ceilings exclude growth (round 15), and on the GEM's box it is feasible, ranks above FBA before training, and gated reaches the same level (round 15b); with the ceilings calibrated to admit growth, the growth column alone ranks the held-out genes at 0.84 on one seed and the program's infeasible region absorbs the other two (round 15c). The gradient reaches the alternating layer throughout, so its null is not a gradient-flow failure; the layer that solves has the linear program's own gradient, the dual of the binding bounds. The section closes with what the comparison says a layer has to compute to help.

The pooled backbone reads its logit from an MLP on the perturbed context (Figure 17). A flux arm builds the enzyme-constrained thermodynamic flux layer on the same context and exposes one physically named quantity, the flux through the growth reaction `r_2111`. Three readouts put that quantity to different uses, and they differ in what the backbone's gradient has to pass through.

**biomass (forced)** $\ell = a\, v_{\text{growth}} + b$. The growth column is the only path from the context to the logit, so the module carries the whole prediction or the cell scores chance (Figure 18). This is FBA's own reading of essentiality with a learned threshold, and it is the wave-9 design.

**context + growth** $\ell = \text{MLP}(h) + a\, v_{\text{growth}} + b$. Both paths from the first step; the per-module gradient norms say how much of each update the module carries when it is not forced to carry all of it.

**gated** $\ell = \text{MLP}(h) + g\,(a\, v_{\text{growth}} + b)$ with $g_0 = 0$ (Figure 19). At initialization the cell is the pooled model exactly; the gate's trajectory, logged every epoch, says whether the optimizer chose the flux path, and the `flux_layer` gradient line says whether gradient reached the layer, which at $g = 0$ it cannot.




**Figure 16: Color key for the readout diagrams.** White is the input; orange a learned part of the backbone (the context, the abundance head); yellow the flux layer and its capacity terms, dashed where the box is a fixed table (the $k_{cat}$ table); purple the fixed stoichiometry of the GEM; red the prediction and the loss. The key holds for every readout diagram in this section, Figures 17 to 87. Rendered from `notes/experiments.028-gene-essentiality.mermaid.readouts.md`.

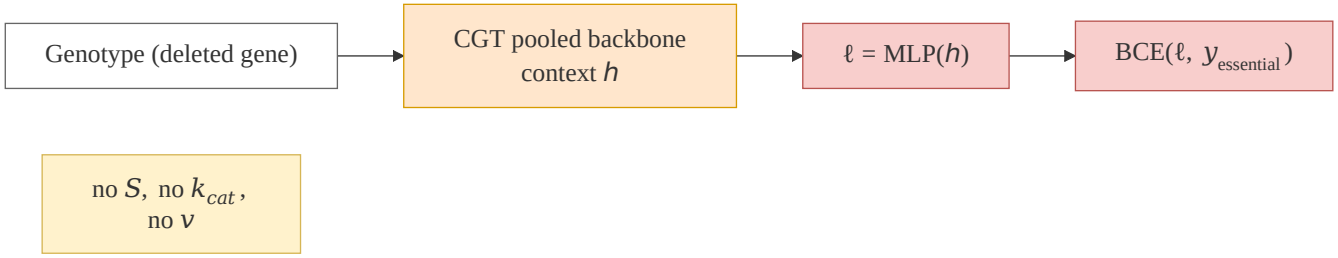



**Figure 17: The pooled readout.** The backbone alone: one MLP from the perturbed context to a logit, binary cross entropy against the released label. No stoichiometry, no kinetic parameters, no flux. Colors as in Figure 16.

## 4.1 What the layer's flux violates ✗

Every flux arm in this document builds the anchored layer: the predicted $k_{cat}$ tier (97 measured, 782 homolog, 2,831 predicted, 18 default catalytic units), the union-external reaction energies (3,779 of 4,131 reactions with a $\Delta_r G'$), the

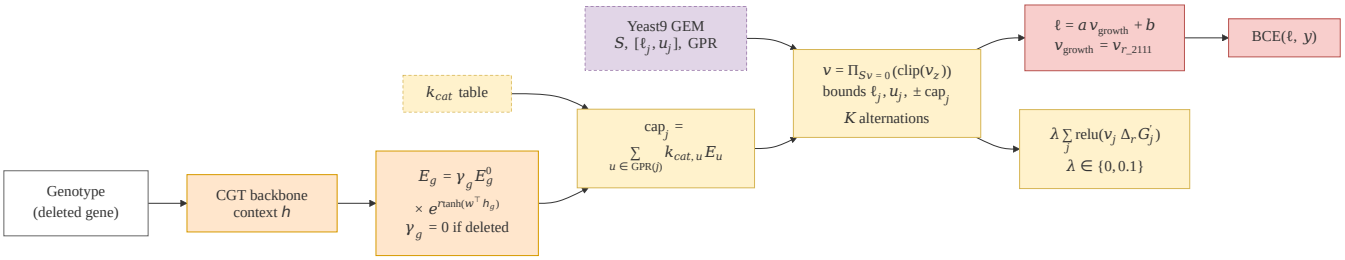



**Figure 18: The forced biomass readout.** The context sets each enzyme's abundance (a deleted gene's is zero), the $k_{cat}$ table turns abundances into capacities, and the flux is the projection onto $Sv = 0$ of the capacity-clipped proposal; the logit is an affine of the growth column alone. The thermodynamic penalty is a loss on the flux with no path to the prediction, at scale 0.1 or 0. Colors as in Figure 16.

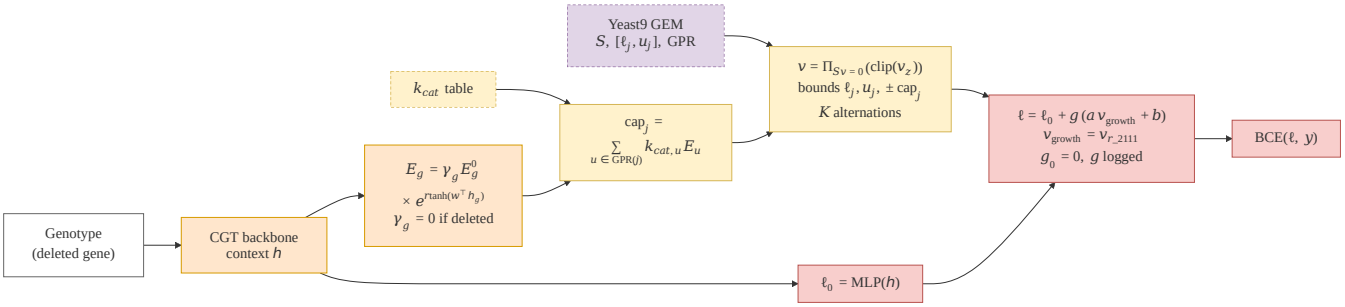



**Figure 19: The gated readout.** The pooled MLP's logit plus the growth affine behind a gate initialized at zero. The gate is a parameter of the readout; its value is written into every epoch row as `ess_gate`. Colors as in Figure 16.

enzyme-capacity ceiling from the abundance head in the box, and the second law, the protein budget and the Gibbs dissipation as penalty terms at the constraint scale. The layer logs every one of those residuals each epoch, and Table 28 reads them at the selected epoch across the module arms. With the balance exact, the flux is outside its bounds on 58 to 63 percent of reactions, runs against its reaction energy on 44 to 51 percent, uses 0.2 to 7 times the protein pool, and dissipates 5,000 to 12,000 J/gDW/h against the 3,700 limit, with the penalty on or off; with the bounds exact (round 7) the second-law violation falls to a third and the dissipation rises to 100,000. No arm's flux satisfies the constraints the layer was built to hold, so on this endpoint the $k_{cat}$ table and the reaction energies are inputs to a projection that does not honor them, which is the precise sense in which the module is not the solve.

The Gibbs dissipation is reported here as a diagnostic and nothing else. The Niebel limit is an empirical plateau of one organism's physiology, not a constraint of the network, and a model that stays under it because it was penalized for exceeding it has been told the answer. The anchored arm's registry entry carries the dissipation hinge, so the round-2 and round-4 cells at constraint scale 0.1 trained against it; every cell of the layer that solves ran at constraint scale 0, where the hinge has weight zero, and from round 18 the module cells run on the `nodiss` entry of the same arm, where the term is absent by construction and the rate is logged only.

**Table 28:** What the layer's flux violates at the selected epoch, seed mean per module arm: the median mass-balance residual $|Sv|$ over metabolites; the fraction of reactions outside their bounds; the fraction of reactions running against their reaction energy ($v_j \Delta_r G_j' > 0$); the enzyme budget used as a multiple of the protein pool; and the Gibbs dissipation in J/gDW/h against the 3,700 limit. Rounds are abbreviated r2 to r13. Every arm builds the anchored layer (predicted $k_{cat}$ tier, union-external $\Delta G$) unless it names the GEM box. Constraint scale 0 means no penalty on any of these terms; 0.1 penalizes all of them.

| arm | cells | $\|Sv\|$ median | box | second law | budget | dissipation |
|---|---|---|---|---|---|---|
| r2 forced, scale 0.1 | 3 | 1.4e-08 | 0.63 | 0.48 | 0.2 | -8,375 |
| r2 forced, scale 0 | 3 | 1.3e-08 | 0.62 | 0.51 | 0.8 | 7,099 |
| r2 gated, scale 0.1 | 3 | 1.1e-08 | 0.63 | 0.44 | 0.3 | -9,854 |
| r4 forced, scale 0 | 3 | 7.9e-09 | 0.59 | 0.51 | 2.4 | 10,264 |
| r4 gated, scale 0 | 3 | 1.1e-08 | 0.58 | 0.51 | 3.6 | 12,277 |
| r6 forced + FBA affine | 3 | 7.5e-09 | 0.59 | 0.51 | 2.2 | 10,115 |
| r7 forced, clip last | 3 | 9.0e-01 | 0.00 | 0.32 | 1.3 | 102,065 |
| r7 forced, box form | 3 | 2.0e+00 | 0.00 | 0.38 | 1.4 | 120,815 |
| r10 forced, warm start | 3 | 1.3e-09 | 0.33 | 0.49 | 6.9 | 5,815 |
| r10 gated, warm start | 3 | 9.3e-10 | 0.36 | 0.49 | 1.3 | 5,104 |
| r13 forced, GEM box, warm start | 3 | 2.8e-09 | 0.26 | 0.00 | nan | nan |
| r13 gated, GEM box, warm start | 3 | 2.5e-09 | 0.27 | 0.00 | nan | nan |

## 4.2 What holds by construction, and what does not ✗

Three constructions of the module appear in this section, and they differ in which constraints their flux satisfies exactly, which it satisfies through a penalty, and which it satisfies only when the backbone finds a way to. The alternating layer (Figure 20) writes a proposal in the null space of $S$, so mass balance holds at the start, and alternates the balance projection with the box clip $K$ times, so whichever of the two runs last holds exactly and the other is a residual; the second law, the protein pool and the capacity ceilings are hinges at the constraint scale, and Table 28 measures what the hinges leave. The layer that solves (Figure 32) returns the growth optimum of the linear program under the layer's bounds, so mass balance, the box and every capacity ceiling hold exactly at once, with no penalty in the loss; what it does not guarantee is that a program exists, and rounds 15 to 15d measure what happens when it does not. The elastic layer that solves (Figure 38) keeps all of that and adds two things by construction: the protein pool, by projecting the abundance head's output onto the budget before the ceilings are formed, and feasibility, by letting the one forced flux in the model, the ATP maintenance reaction's lower bound of 0.7 mmol/gDW/h, go unmet at a price. Its zero flux with the maintenance fully unmet satisfies every constraint, so the program solves for every genotype at every ceiling, and a genotype that cannot meet its maintenance is not absent from the gradient but priced into it. Under every construction the second law enters only through the anchored reaction energies and the dissipation only as a logged rate. What "the enzyme-capacity box contains no growing flux" means, stated exactly with the elastic program in hand: at the ceilings as parameterized the maintenance flux the model forces cannot be carried, the shortfall is 0.54 of 0.70 with zero growth (`capacity_calibration.py -lp-slack-penalty 1`, 32 held-out genotypes), the hard program's feasible set is therefore empty, and the solver returns nothing; from a ceiling scale of 10 the shortfall is zero and growth is 0.01, and from 30 the box admits the growth the rounds train on.

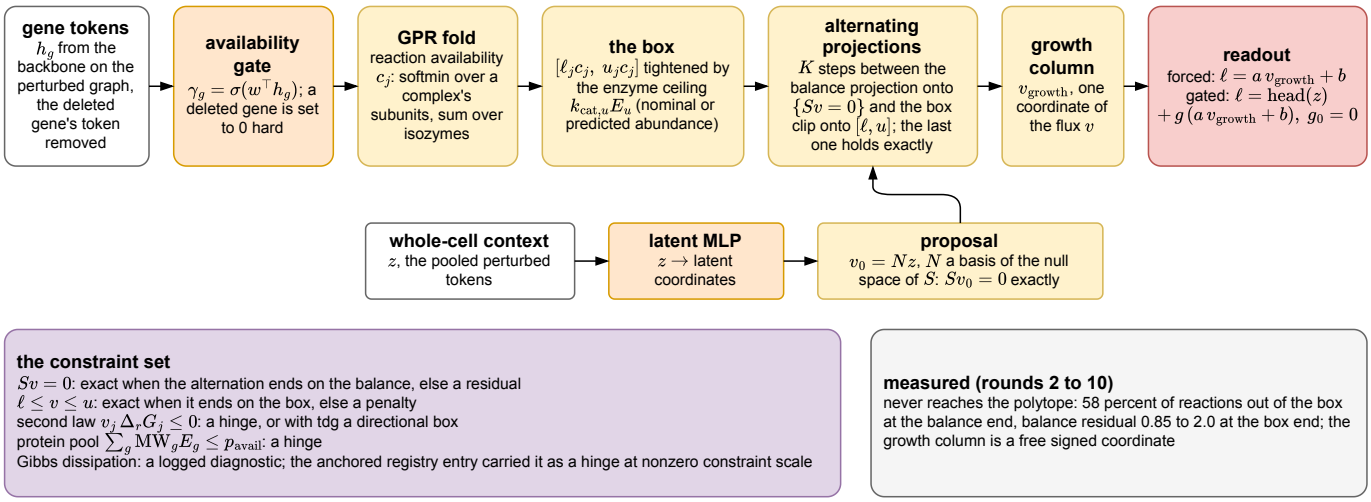



**Figure 20: The alternating layer, rounds 2 to 13.** The backbone's gene tokens set a learned availability gate, which the gene-protein-reaction rule folds to a per-reaction availability and a box tightened by the enzyme-capacity ceiling; a proposal written in the null space of $S$ from the whole-cell context is alternately projected onto the balance and clipped onto the box, and the growth column of the result enters the readout, alone or behind a zero-initialized gate. Of the constraint set only the projection the alternation ends on holds exactly; the second law and the protein pool are hinges, and the Gibbs dissipation is a logged diagnostic. Colors as in Figure 16, the same in Figures 32 and 38. Drawn in draw.io from `notes/assets/drawio/028-module.gen.py`.

## 4.3 Round 2: the module on the stable backbone, with and without the penalties ✗

Round 2 (jobs 2182 to 2208) puts the three readouts on the width-12 cell of Section 3, on the released-label split, at three seeds each: the forced and gated readouts at constraint scale 0.1 and 0, the context readout at both scales, the forced readout at scale 0 with the $k_{cat}$ vector permuted across catalytic units (same architecture, no unit-to-value correspondence), and the stoichiometry-only layer (`flux_off`: no thermodynamics, no capacity) with the forced and context readouts. Every flux arm carries the wave-9 flux configuration otherwise: the predicted $k_{cat}$ tier, the union-external thermodynamic source, twenty warmup epochs, the dataset medium, the mixture concentration space. The paired comparator is the round-1 pooled cell on the same split at the same batch, weight decay and budget (Table 29, Figure 21).

The reads were fixed before the numbers: *not hurt* is the gated arm against the pooled cell paired by seed, with the gate's trajectory beside it; *help* is the forced arm above the pooled cell on three of three seeds and above its shuffled-$k_{cat}$ control; the *collapse's cause* is the forced arm at scale 0.1 against scale 0, since on the betaxanthin targets the penalty scale was the dose that starved the layer's input.

**Table 29:** Round 2, the metabolic module. The round-1 pooled cell on the released-label split (`fcl`) and the same cell with the prediction routed through the flux layer: the growth column alone (forced), the context head plus the growth affine, and that affine behind a zero-initialized gate, at constraint scale 0.1 and 0, with the shuffled-kcat control and the stoichiometry-only layer. Bold marks the highest value in a column. ↑ (↓) marks an arm whose held-out AUROC is above (below) the first arm's on every one of the same three seeds.

| arm | mean | sd | seeds | pooled [95% CI] | selected epochs | val step |
|---|---|---|---|---|---|---|
| released labels, 158-gene val | 0.725 | 0.019 | 0.745, 0.722, 0.708 | 0.738 [0.65, 0.83] | 66, 77, 77 | 0.0026 |
| growth column only, constraints 0.1 | 0.706 | 0.105 | 0.759, 0.585, 0.774 | 0.765 [0.67, 0.85] | 77, 26, 76 | 0.0037 |
| growth column only, constraints 0 | 0.732 | 0.068 | 0.770, 0.653, 0.772 | **0.773** [0.68, 0.86] | 72, 72, 72 | 0.0047 |
| growth column only, kcat shuffled, constraints 0 | **0.751** | 0.029 | 0.771, 0.718, 0.765 | 0.769 [0.68, 0.86] | 74, 77, 75 | 0.0029 |
| context + growth column, constraints 0.1 | 0.666 | 0.086 | 0.566, 0.709, 0.722 | 0.717 [0.61, 0.82] | 49, 77, 77 | 0.0053 |
| context + growth column, constraints 0 | 0.667 | 0.096 | 0.556, 0.727, 0.718 | 0.724 [0.62, 0.82] | 49, 73, 77 | 0.0068 |
| gated growth column, constraints 0.1 | 0.719 | 0.029 | 0.721, 0.747, 0.690 | 0.738 [0.64, 0.83] | 41, 77, 77 | 0.0027 |
| gated growth column, constraints 0 | 0.735 | 0.018 | 0.714, 0.744, 0.747 | 0.747 [0.65, 0.84] | 40, 77, 77 | 0.0031 |
| growth column only, stoichiometry only | 0.606 | 0.072 | 0.524, 0.631, 0.661 ↓ | 0.664 [0.56, 0.77] | 19, 73, 65 | 0.0033 |
| context + growth column, stoichiometry only | 0.705 | 0.023 | 0.719, 0.717, 0.678 ↓ | 0.755 [0.67, 0.84] | 75, 76, 61 | 0.0042 |

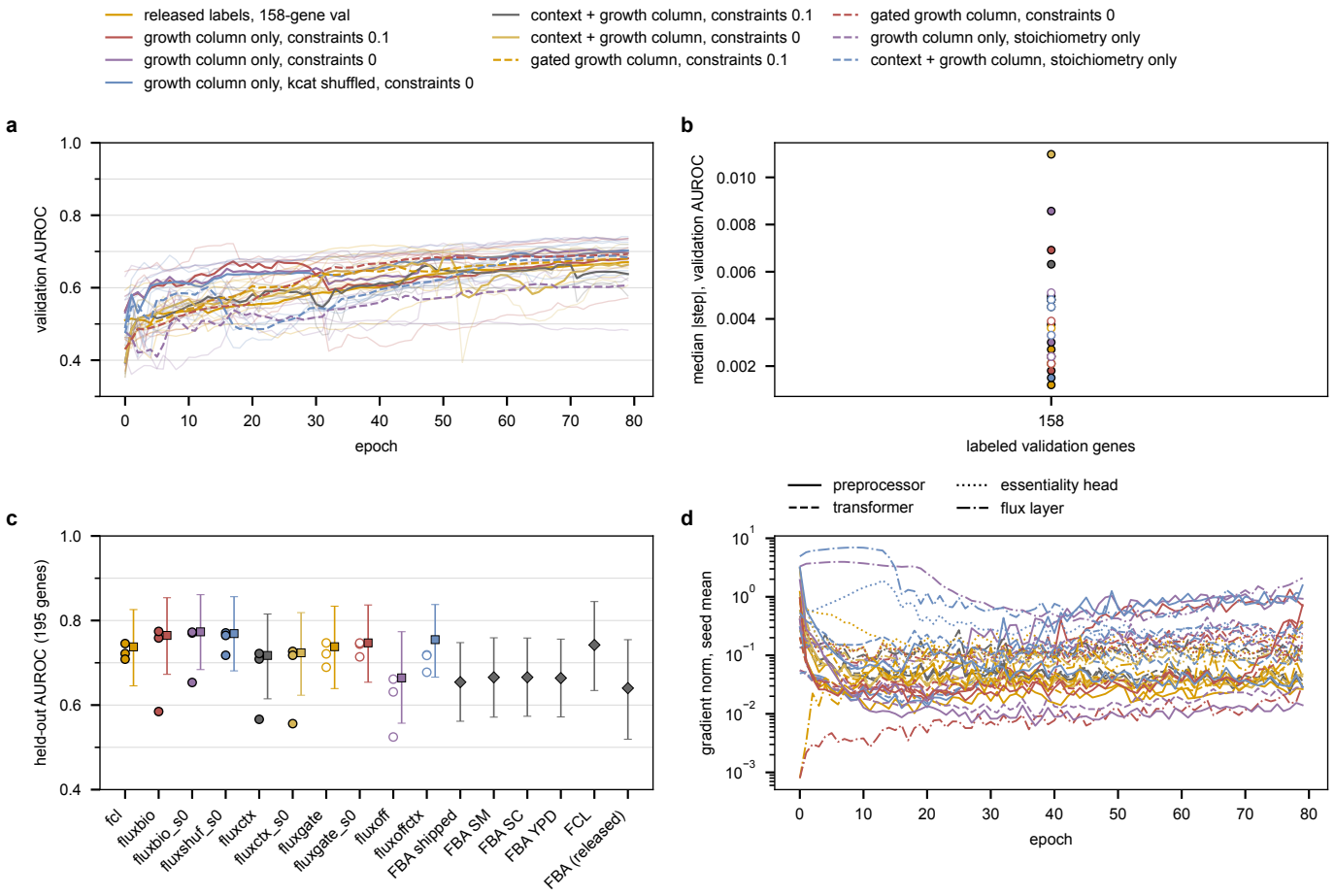



**Figure 21: Round 2, the metabolic module. a**, Validation AUROC per epoch, one thin line per seed and the seed mean per arm. **b**, The median absolute epoch-to-epoch step of that curve per cell, against the number of labeled validation genes, 158 for every cell in this round. **c**, Held-out AUROC per cell (points), the seed mean and the gene bootstrap of the pooled per-gene logits (square with interval), beside FBA on four media, Flux Cone Learning and the released FBA predictions on the same 195 genes. **d**, Gradient norm per module per epoch, seed mean; the dash-dot line is the flux layer. Written by `experiments/028-gene-essentiality/scripts/wave12_readout.py -round w13`.

*Not hurt* holds. The gated arm at scale 0 scores 0.714, 0.744 and 0.747 against the pooled cell's 0.745, 0.722 and 0.708 on the same seeds, two of three above by 0.02 to 0.04 and one below by 0.03, a seed mean of 0.735 against 0.725; at scale 0.1 the same comparison is one of three above. The gate leaves zero and stops: at epoch 80 it reads −0.056 to −0.063 on all six gated cells, and the flux layer's gradient norm sits at 0.01 to 0.07 against 0.22 to 0.36 on the forced arm's two seeds that train. The optimizer takes a small negative weight on the growth affine and the layer receives a tenth of the forced arm's gradient, so the gated cell is a pooled cell with a small correction, and it scores like one.

*Help* does not hold. The forced arm at scale 0 scores 0.770, 0.653 and 0.772, two of three seeds above the pooled cell and one seed 0.07 below; at scale 0.1 the split is the same and the low seed reads 0.585. The shuffled-$k_{cat}$ control, the same layer with the catalytic constants permuted across units, scores 0.771, 0.718 and 0.765, a seed mean of 0.751 against the

real table's 0.732, so whatever the growth column carries here does not come from the correspondence between units and constants. The forced arm's pooled bootstrap, 0.773 [0.68, 0.86], is above Flux Cone Learning's 0.742 and FBA's 0.640 to 0.666 on the same genes, level with the pooled cell's 0.738, and below the ladder's 0.80 line.

The collapse on this endpoint is not the penalty's. Seed 2701 falls at both scales, to 0.585 at 0.1 and 0.653 at 0, and the two runs are the same run: the gradient norms start where the other seeds' do, decay within ten epochs to a hundredth of them (preprocessor 0.002, transformer 0.003 at epoch 80, against 1.0 and 0.5 on seed 2700) and stay there, with the training loss flat at 0.46 against 0.34, which is a stalled seed rather than a starved layer. The stoichiometry-only layer collapses on all three seeds (0.524, 0.631, 0.661), so the capacity bound and the thermodynamic term are what keep the growth column informative rather than what starve it, the opposite of the betaxanthin reading. Adding the context head beside the growth affine without a gate costs 0.06 on the seed mean (0.666 and 0.667 against 0.725) and puts one seed at 0.56; the same sum behind the zero-initialized gate loses nothing.

## 4.4 Round 4: the module on the best backbone ✗

Round 4 (jobs 2246 to 2248, three cells per GPU) repeats the gated readout, the forced readout and the shuffled-$k_{cat}$ control at constraint scale 0 on the round-3 cell, batch 32 with the class weight at 200 epochs, paired by seed against it (Table 30, Figure 22). The reads are the round-2 reads.

**Table 30:** Round 4, the module on the round-3 levers. The round-3 pooled cell (batch 32, class weight, 200 epochs) and the same cell with the prediction routed through the flux layer at constraint scale 0: the gated readout, the forced growth column, and the forced growth column with the kcat vector shuffled. Bold marks the highest value in a column. ↑ (↓) marks an arm whose held-out AUROC is above (below) the first arm's on every one of the same three seeds.

| arm | mean | sd | seeds | pooled [95% CI] | selected epochs | val step |
|---|---|---|---|---|---|---|
| batch 32 + class weight, 200 epochs | 0.774 | 0.006 | 0.780, 0.769, 0.773 | 0.780 [0.69, 0.87] | 75, 68, 148 | 0.0031 |
| gated growth column, constraints 0 (round-3 levers) | 0.771 | 0.010 | 0.766, 0.782, 0.765 | 0.768 [0.68, 0.85] | 67, 71, 77 | 0.0040 |
| growth column only, constraints 0 (round-3 levers) | 0.777 | 0.005 | 0.772, 0.781, 0.778 | 0.778 [0.69, 0.86] | 108, 74, 84 | 0.0057 |
| growth column only, kcat shuffled (round-3 levers) | **0.779** | 0.006 | 0.785, 0.781, 0.772 | **0.781** [0.69, 0.87] | 66, 71, 74 | 0.0048 |

Every arm lands within 0.015 of the pooled cell on every seed. The gated readout scores 0.766, 0.782 and 0.765 against 0.780, 0.769 and 0.773, one of three above; the forced readout 0.772, 0.781 and 0.778, two of three above by 0.005 and 0.012 and one below by 0.008; the shuffled-$k_{cat}$ control 0.785, 0.781 and 0.772, two of three above, a seed mean of 0.779 against the real table's 0.777. The gate now moves: it leaves zero after epoch 10 and climbs steadily to $-0.17$, $+0.20$ and $+0.11$ at epoch 200, of either sign, with the flux layer's gradient at 0.06 to 0.12 against 0.22 to 0.65 in the forced arm. The stalled seed of round 2 does not recur under batch 32 with the class weight. On every arm the validation curve peaks between epochs 60 and 110 and falls afterwards (Figure 22a), so the 200-epoch budget is more than the cell uses and the smoothed selection rule is what the number rests on.

The two rounds together answer the two questions. The module does not hurt: on the stable backbone and on the best one, the gated cell is the pooled cell to within seed noise. The module does not help: the forced growth column, which carries the whole prediction, matches the pooled MLP head to within 0.01, and it matches it equally well with the catalytic constants permuted, so the growth column is as expressive a readout as the head but its $k_{cat}$ content is not what it reads. On this endpoint, at this width, the enzyme-constrained flux is a learned nonlinearity over the context rather than a physical prior that ranks genes.

## 4.5 Round 6: one medium against four, the layer on top of the solve, and two other embeddings ✗

Round 5 (Section 3) put FBA's four growth ratios beside the context head and reached the ladder's level. Round 6 (jobs 2251 to 2256, three cells per GPU) asks three questions of that result on the same cell, batch 32 with the class weight at 200 epochs, three seeds each, paired against the round-5 cell (Table 31, Figure 25): whether one medium's ratio carries what four do; whether the flux layer adds anything once the solve is present, with the forced growth column plus the affine (no context head) and the gated readout plus the affine, both at constraint scale 0; and whether the backbone on another node embedding does better, on codon frequency (64 dimensions, the encoding that reaches 0.82 with FBA on the ladder from a support vector classifier) and on the species-LM composite the expression line uses (the 5′ and 3′ species language model, ProtT5 and CaLM, 3,332 dimensions), with and without the affine.

Figure 23 draws the round-5 affine and Figure 24 the gated readout with that affine beside it.

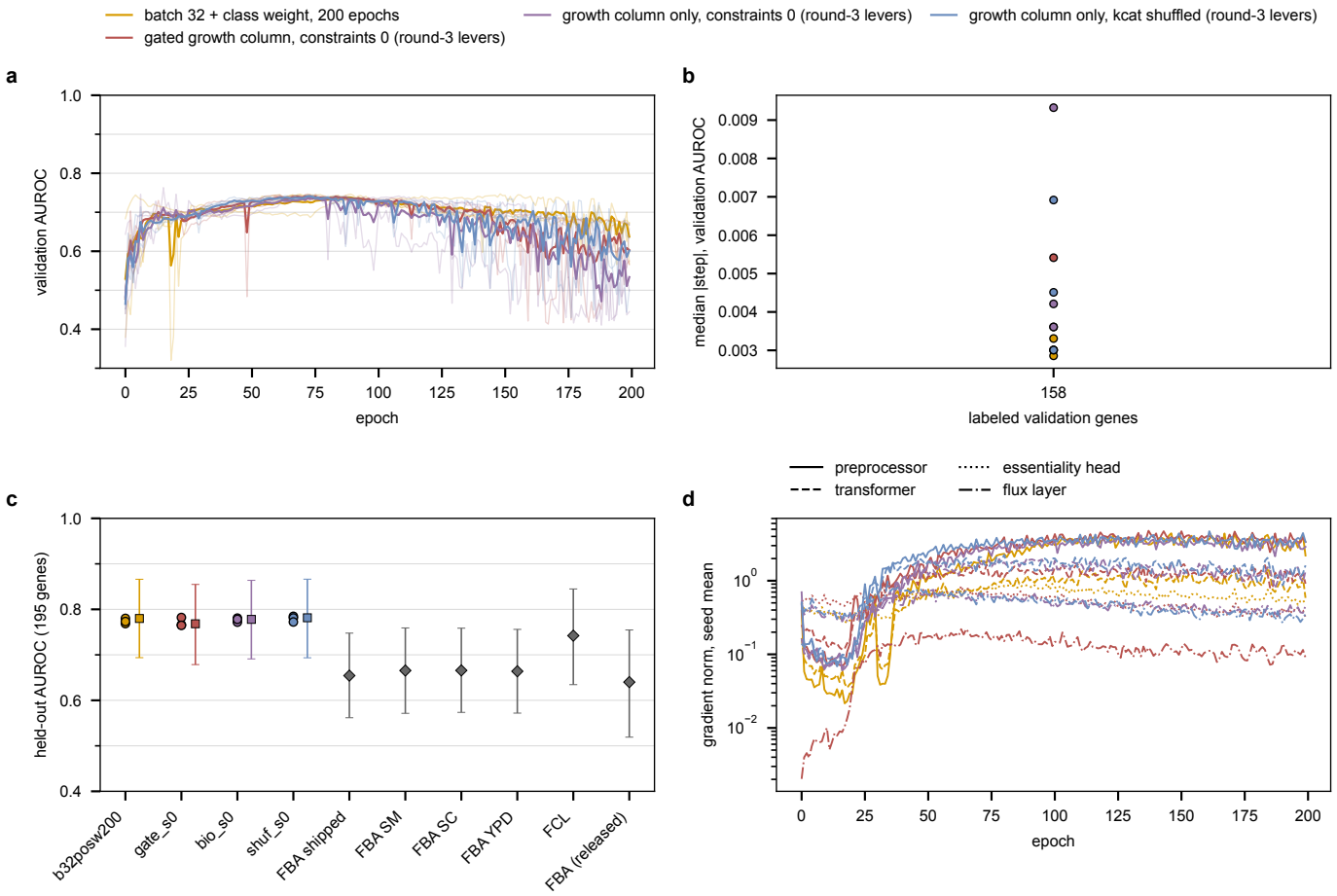



**Figure 22: Round 4, the module on the round-3 cell. a**, Validation AUROC per epoch, one thin line per seed and the seed mean per arm. **b**, The median absolute epoch-to-epoch step per cell against the validation size, 158 for every cell. **c**, Held-out AUROC per cell (points), the seed mean and the gene bootstrap of the pooled per-gene logits (square with interval), beside FBA on four media, Flux Cone Learning and the released FBA predictions on the same 195 genes. **d**, Gradient norm per module per epoch, seed mean; the dash-dot line is the flux layer. Written by `experiments/028-gene-essentiality/scripts/wave12_readout.py -round w15`.

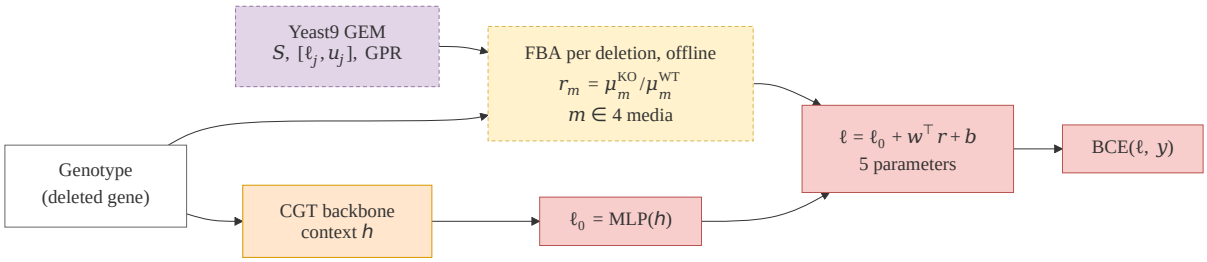



**Figure 23: The FBA affine, round 5.** The pooled readout (Figure 17) plus an affine of the deleted gene's knockout-to-wild-type growth ratio on four media, solved offline by FBA and read as a fixed table. No flux layer is in the graph, and the five added parameters are the four weights and the intercept. Colors as in Figure 16.

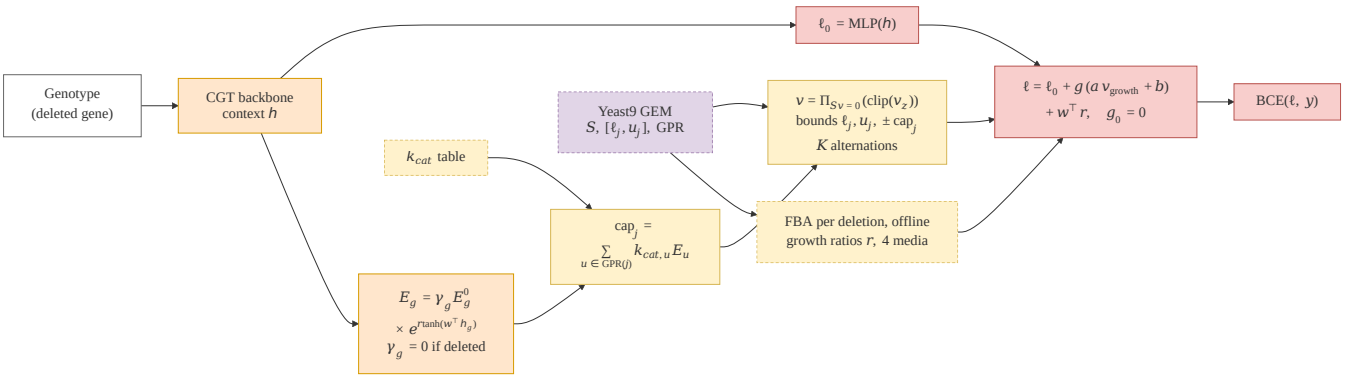



**Figure 24: The gated readout beside the FBA affine, round 6.** The gated readout of Figure 19 with the affine of Figure 23 added to the same logit, so the alternating layer's growth column and the solve's growth ratios reach the prediction side by side. The forced arm of the round drops the context head $\ell_0$ and keeps the other two terms. Colors as in Figure 16.

**Table 31:** Round 6, one medium, the layer on top of the solve, and two other embeddings. The round-3 pooled cell and the round-5 FBA-readout cell, then: the FBA affine on the YPD ratio alone; the forced and the gated growth column of the flux layer with the FBA affine beside them (constraint scale 0); and the pooled backbone opened on codon frequency, and on the species-LM composite (5' and 3' species LM, ProtT5, CaLM), with and without the FBA affine. Batch 32, class weight, 200 epochs, seeds 2700 to 2702. Bold marks the highest value in a column. ↑ (↓) marks an arm whose held-out AUROC is above (below) the first arm's on every one of the same three seeds.

| arm | mean | sd | seeds | pooled [95% CI] | selected epochs | val step |
|---|---|---|---|---|---|---|
| batch 32 + class weight, 200 epochs | 0.774 | 0.006 | 0.780, 0.769, 0.773 | 0.780 [0.69, 0.87] | 75, 68, 148 | 0.0031 |
| context head + FBA growth ratios (round-3 levers) | **0.821** | 0.020 | 0.814, 0.805, 0.844 ↑ | **0.838** [0.76, 0.91] | 28, 70, 61 | 0.0036 |
| context head + FBA ratio, YPD only | 0.795 | 0.022 | 0.819, 0.775, 0.789 ↑ | 0.808 [0.73, 0.89] | 18, 98, 61 | 0.0034 |
| growth column + FBA ratios, no context head | 0.818 | 0.014 | 0.834, 0.811, 0.809 ↑ | 0.820 [0.74, 0.90] | 74, 83, 57 | 0.0052 |
| gated growth column + FBA ratios | **0.821** | 0.011 | 0.810, 0.821, 0.832 ↑ | 0.827 [0.75, 0.90] | 116, 43, 92 | 0.0041 |
| codon-frequency backbone + FBA ratios | 0.731 | 0.043 | 0.747, 0.682, 0.764 ↓ | 0.721 [0.62, 0.81] | 16, 172, 14 | 0.0130 |
| species-LM composite backbone + FBA ratios | 0.816 | 0.013 | 0.802, 0.825, 0.822 ↑ | 0.819 [0.73, 0.90] | 149, 65, 65 | 0.0031 |
| species-LM composite backbone | 0.778 | 0.005 | 0.783, 0.779, 0.773 | 0.776 [0.69, 0.85] | 30, 46, 23 | 0.0035 |

Four media carry more than one. The YPD ratio alone scores 0.819, 0.775 and 0.789 against 0.814, 0.805 and 0.844 with all four, two of three seeds below by 0.03 to 0.05, a seed mean of 0.795 against 0.821; the one column still lifts the pooled cell by 0.02.

The learned layer adds nothing on top of the solve. The forced growth column with the affine beside it and no context head scores 0.834, 0.811 and 0.809, a mean of 0.818, and the gated readout with the affine 0.810, 0.821 and 0.832, a mean of 0.821, each one of three seeds above the round-5 cell and each within its seed spread. The growth column is as good a companion to the solve as the MLP head, and no better; its gradient norm, 0.4 to 1.1 in the forced cells, says the layer trains rather than idles. Put beside rounds 2, 4 and 5, the pattern is one statement: the module's ranking is what the backbone gives it, and the physics that ranks genes on this endpoint is the solve's optimum, which the projection layer does not compute.

The embeddings do not move the ceiling. The species-LM composite scores 0.783, 0.779 and 0.773 without the affine, against ProtT5's 0.780, 0.769 and 0.773 at 2.7 times the parameters, and 0.802, 0.825 and 0.822 with it, against 0.814, 0.805 and 0.844: within noise both times. The backbone on codon frequency with the affine scores 0.747, 0.682 and 0.764, a mean of 0.731, below the pooled ProtT5 cell and 0.09 below the ladder's codon rung with FBA (0.818), so the 64 codon numbers that a support vector classifier reads well are not read well through a 12-wide transformer; the selected epochs (16, 172, 14) and the step of 0.013 in panel b say the cell never settled.

## 4.6 The growth column is a signed free coordinate ✗

Four rounds put the layer's growth column level with the pooled head and blind to its own $k_{cat}$ table while one linear program per deletion lifts every model by 0.05. The probe in Table 32 reads the column itself. It rebuilds a forced cell from its result file, deletes each of the 982 labeled genes through the same batches training used, and reads $v_{\text{growth}}$ at initialization and under the trained weights, scoring the negative column against the label (a deletion that stops growth is essential) and the column against FBA's ratio.

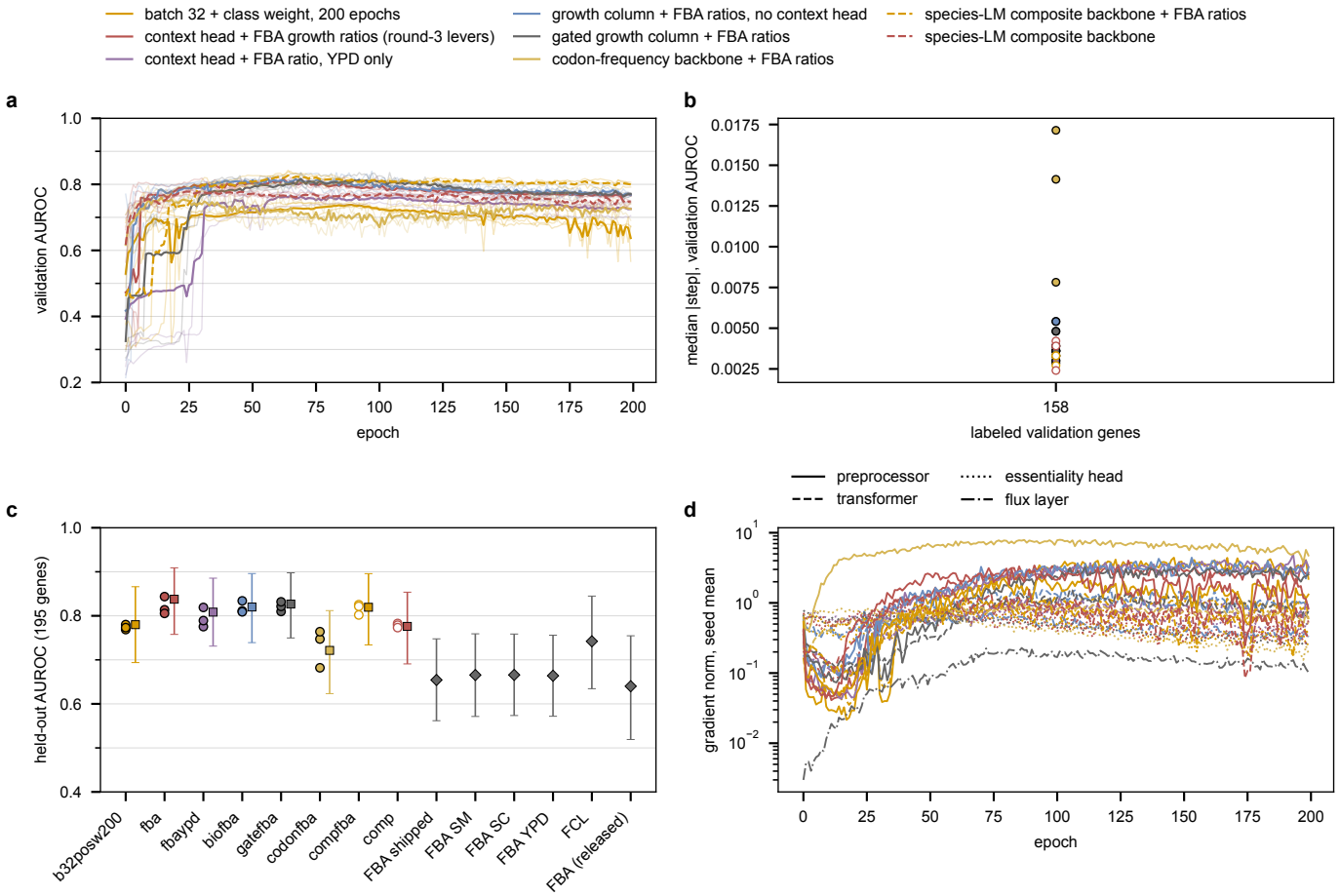



**Figure 25: Round 6.** **a**, Validation AUROC per epoch, one thin line per seed and the seed mean per arm; the second tier of arms is dashed. **b**, The median absolute epoch-to-epoch step per cell against the validation size. **c**, Held-out AUROC per cell (points, hollow for the second tier), the seed mean and the gene bootstrap of the pooled per-gene logits (square with interval), beside FBA on four media, Flux Cone Learning and the released FBA predictions on the same 195 genes. **d**, Gradient norm per module per epoch, seed mean; the dash-dot line is the flux layer. Written by `experiments/028-gene-essentiality/scripts/wave12_readout.py -round w17`.

**Table 32:** The flux layer's growth column, read for every labeled deletion at initialization and under the cell's trained weights (`flux_growth_probe.py`). Median $v_{\text{growth}}$ over the 982 labeled deletions (mmol/gDW/h; the wild type's FVA range on YPD is 0.48 to 0.54); AUROC of the NEGATIVE growth column against the released label on the train, validation and held-out genes, so 0.5 is chance and a value below 0.5 is a ranking with the sign of growth inverted; Spearman of the column against FBA's knockout-to-wild-type ratio on the shipped bounds and on YPD.

| cell | weights | median $v$ | train | val | held-out | $\rho$ shipped | $\rho$ YPD |
|---|---|---|---|---|---|---|---|
| `w13_fluxbio_e80_s2700` | initialization | -0.27 | 0.618 | 0.555 | 0.547 | +0.12 | +0.11 |
| `w13_fluxbio_e80_s2700` | trained (epoch 77) | -2.41 | 0.091 | 0.266 | 0.241 | -0.26 | -0.22 |
| `w15_bio_s0_e200_s2700` | initialization | -0.27 | 0.618 | 0.555 | 0.547 | +0.12 | +0.11 |
| `w15_bio_s0_e200_s2700` | trained (epoch 108) | -1.43 | 0.048 | 0.258 | 0.228 | -0.27 | -0.24 |
| `w18_biobox_e200_s2700` | initialization | 45.08 | 0.402 | 0.389 | 0.326 | -0.09 | -0.05 |
| `w18_biobox_e200_s2700` | trained (epoch 180) | 0.08 | 0.103 | 0.250 | 0.233 | -0.25 | -0.22 |
| `w18_biobox_e200_s2701` | initialization | 50.26 | 0.512 | 0.503 | 0.603 | -0.14 | -0.08 |
| `w18_biobox_e200_s2701` | trained (epoch 95) | 0.02 | 0.637 | 0.696 | 0.609 | +0.10 | +0.05 |
| `w18_biobox_e200_s2702` | initialization | 53.07 | 0.572 | 0.602 | 0.506 | +0.17 | +0.15 |
| `w18_biobox_e200_s2702` | trained (epoch 140) | 0.27 | 0.125 | 0.265 | 0.215 | -0.28 | -0.23 |
| `w18_bioprojbox_e200_s2700` | initialization | 0.00 | 0.500 | 0.500 | 0.500 | – | – |
| `w18_bioprojbox_e200_s2700` | trained (epoch 2) | 0.00 | 0.500 | 0.500 | 0.500 | – | – |
| `w18_bioprojbox_e200_s2701` | initialization | 0.00 | 0.500 | 0.500 | 0.500 | – | – |
| `w18_bioprojbox_e200_s2701` | trained (epoch 2) | 0.00 | 0.500 | 0.500 | 0.500 | – | – |
| `w18_bioprojbox_e200_s2702` | initialization | 0.00 | 0.500 | 0.500 | 0.500 | – | – |
| `w18_bioprojbox_e200_s2702` | trained (epoch 2) | 0.00 | 0.500 | 0.500 | 0.500 | – | – |
| `w21_warmbio_e200_s2700` | initialization | -0.07 | 0.247 | 0.259 | 0.337 | -0.35 | -0.43 |
| `w21_warmbio_e200_s2700` | trained (epoch 112) | -0.10 | 0.272 | 0.215 | 0.383 | -0.38 | -0.38 |
| `w21_warmbio_e200_s2702` | initialization | -0.10 | 0.304 | 0.305 | 0.392 | -0.40 | -0.44 |
| `w21_warmbio_e200_s2702` | trained (epoch 157) | -0.20 | 0.070 | 0.114 | 0.215 | -0.37 | -0.39 |
| `w21_warmgate_e200_s2700` | initialization | -0.07 | 0.247 | 0.259 | 0.337 | -0.35 | -0.43 |
| `w21_warmgate_e200_s2700` | trained (epoch 181) | -0.10 | 0.422 | 0.328 | 0.410 | -0.39 | -0.45 |
| `w24_warmoffbio_e200_s2700` | initialization | 0.38 | 0.798 | 0.698 | 0.683 | +0.47 | +0.57 |
| `w24_warmoffbio_e200_s2700` | trained (epoch 53) | 0.55 | 0.689 | 0.638 | 0.623 | +0.54 | +0.64 |
| `w24_warmoffgate_e200_s2700` | initialization | 0.38 | 0.798 | 0.698 | 0.683 | +0.47 | +0.57 |
| `w24_warmoffgate_e200_s2700` | trained (epoch 85) | 0.55 | 0.735 | 0.748 | 0.670 | +0.56 | +0.64 |
| `w26_lpbio_e200_s2700` | initialization | 0.00 | 0.500 | 0.500 | 0.500 | – | – |
| `w26_lpbio_e200_s2700` | trained (epoch 2) | 0.00 | 0.500 | 0.500 | 0.500 | – | – |
| `w26_lpgate_e200_s2700` | initialization | 0.00 | 0.500 | 0.500 | 0.500 | – | – |
| `w26_lpgate_e200_s2700` | trained (epoch 78) | 0.00 | 0.500 | 0.500 | 0.500 | – | – |
| `w29_lpoffbio_e200_s2700` | initialization | 0.19 | 0.744 | 0.767 | 0.734 | +0.40 | +0.50 |
| `w29_lpoffbio_e200_s2700` | trained (epoch 11) | 0.00 | 0.500 | 0.500 | 0.500 | – | – |
| `w29_lpoffgate_e200_s2700` | initialization | 0.19 | 0.744 | 0.767 | 0.734 | +0.40 | +0.50 |
| `w29_lpoffgate_e200_s2700` | trained (epoch 2) | 0.42 | 0.779 | 0.775 | 0.768 | +0.43 | +0.54 |
| `w30_lpcapbio100_e200_s2700` | initialization | 0.10 | 0.765 | 0.682 | 0.730 | +0.49 | +0.55 |
| `w30_lpcapbio100_e200_s2700` | trained (epoch 2) | 0.00 | 0.501 | 0.500 | 0.500 | +0.02 | +0.01 |
| `w30_lpcapgate100_e200_s2700` | initialization | 0.10 | 0.765 | 0.682 | 0.730 | +0.49 | +0.55 |
| `w30_lpcapgate100_e200_s2700` | trained (epoch 6) | 0.33 | 0.685 | 0.635 | 0.624 | +0.41 | +0.59 |

At initialization the column carries a weak signal in the right direction: 0.55 on the held-out genes and a Spearman of +0.11 with the solve, nothing like FBA's 0.65, and no deletion reaches zero growth. Training inverts it. Under the round-4 forced cell's weights the negative column reads 0.23 on the held-out genes, which is that cell's 0.77 with the sign of growth reversed: the affine learned a positive slope, an essential deletion gets the *higher* growth, and the column is anticorrelated with the solve (−0.25). The round-2 forced cell does the same (0.24, −0.26). The median growth under the trained weights is −1.4 and −2.4 mmol/gDW/h against a wild-type FVA range of 0.48 to 0.54 on YPD: for most deletions the column is negative, which growth cannot be.

The column can do this because the arm's alternations end on the balance projection, so $Sv = 0$ holds to machine precision and the box does not: 58 percent of reactions sit outside their bounds at the selected epoch (0.64 on the round-2 cell, 0.59 on the round-6 cell), the growth reaction's floor of zero among them. $v_{\text{growth}}$ is then a signed, unbounded coordinate the backbone writes through the abundance head, and the readout uses it as a free latent. That is the mechanism behind the shuffled-$k_{cat}$ parity and behind the layer matching the head: the quantity the forced readout reads is not growth.

## 4.7 Round 7: the growth column with the bounds enforced ✗

Round 7 (jobs 2257 and 2258, three cells per GPU) puts the forced readout on two layers whose bounds hold exactly: the same alternations ending on the clip instead of the balance, and the pure box form, $v = \ell + (u - \ell)\,\sigma(z)$, with no projection at all. Same cell otherwise, batch 32 with the class weight at 200 epochs, constraint scale 0, three seeds, paired against the round-4 forced cell and the pooled cell (Table 33, Figure 27); the probe runs on every cell (Table 32).

Figure 26 draws the box form.

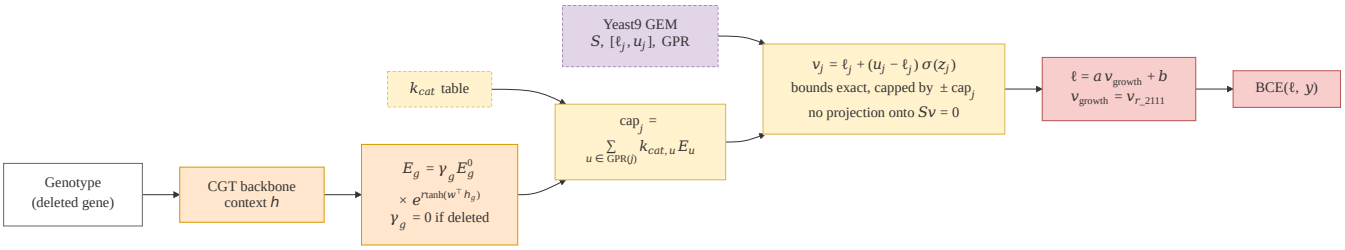



**Figure 26: The box form, round 7.** The forced readout of Figure 18 with the alternating projections replaced by a sigmoid that places every flux inside its bounds, so the box holds exactly and mass balance is never imposed. The round's other arm keeps the alternations of Figure 18 and ends them on the clip instead of on the balance projection. Colors as in Figure 16.

**Table 33:** Round 7, the growth column with the bounds enforced. The round-3 pooled cell, the round-4 forced growth column (projection ending on the balance, box open), and the forced growth column on two layers whose bounds hold exactly: the projection ending on the box, and the pure box parameterization. Constraint scale 0, batch 32, class weight, 200 epochs, seeds 2700 to 2702. Bold marks the highest value in a column. ↑ (↓) marks an arm whose held-out AUROC is above (below) the first arm's on every one of the same three seeds.

| arm | mean | sd | seeds | pooled [95% CI] | selected epochs | val step |
|---|---|---|---|---|---|---|
| batch 32 + class weight, 200 epochs | 0.774 | 0.006 | 0.780, 0.769, 0.773 | 0.780 [0.69, 0.87] | 75, 68, 148 | 0.0031 |
| growth column only, constraints 0 (round-3 levers) | **0.777** | 0.005 | 0.772, 0.781, 0.778 | 0.778 [0.69, 0.86] | 108, 74, 84 | 0.0057 |
| growth column only, projection ending on the box | 0.500 | 0.000 | 0.500, 0.500, 0.500 ↓ | 0.500 [0.50, 0.50] | 2, 2, 2 | 0.0000 |
| growth column only, box parameterization | 0.720 | 0.097 | 0.767, 0.609, 0.785 | **0.781** [0.69, 0.87] | 180, 95, 140 | 0.0074 |

With the clip last, the growth column is zero for every one of the 982 deletions, at initialization and after training: the clip pins the growth reaction at its floor, the cells score 0.500 exactly on three of three seeds and select epoch 2, and the balance residual, the loose end now, is 0.85 to 1.0 of the layer's own turnover. With the box form the column lives in $[0, u]$ and starts near $u/2$ (median 45 to 53 mmol/gDW/h against a wild-type 0.5); $Sv = 0$ is then never approached, the balance residual reads 2.0 at every epoch, and the cells score 0.767, 0.609 and 0.785, a seed mean of 0.720 below the pooled cell's 0.774. The probe on the trained box cells inverts the sign on two of three seeds as before (0.23 and 0.22 on the held-out genes, Spearman $-0.25$ and $-0.28$ with the solve) and reads 0.61 with $+0.10$ on the third.

The rounds close the module question at the layer rather than at the readout. Whichever constraint the layer holds exactly, the other is violated by an amount comparable to the fluxes themselves: 58 percent of reactions out of bounds when the balance is exact, a balance residual of 0.85 to 2.0 when the bounds are. The alternations never reach the polytope that FBA optimizes over, so the growth column is either a free signed coordinate, a constant zero, or an unbalanced sigmoid, and none of them is the quantity whose optimum over that polytope ranks essential genes at 0.65 on its own and lifts the transformer by 0.05. On this endpoint the physics that helps is the solve, and the layer would have to reach feasibility before its growth column could carry it.

## 4.8 Round 10: the layer warm-started from the deletion's own solve ✗

If the layer cannot reach the polytope from a free proposal, it can be handed a point on it. One parsimonious FBA solve per labeled gene on YPD, the medium the essentiality cell's box is built on, gives a reference flux per deletion (`ess_reference_flux_table.py`, 1,121 genes, 104 of them with zero growth, about eight minutes of CPU), and under the warm-start parameterization the layer proposes that reference plus a null-space residual of relative size $\alpha$ before its alternations. The growth column then starts at the deletion's own pFBA growth rather than at a free coordinate, and the negative of that growth ranks the 195 held-out genes at 0.645 on its own. Round 10 (jobs 2284 to 2287) runs the forced growth column and the gated column with the context head on the anchored layer at constraint scale 0, with $\alpha$ frozen at 0.01, learned from 0.01, and frozen at 0.1, each at the three seeds, paired against the round-4 forced and gated cells and the round-5 affine (Table 34, Figure 29). Two reads were fixed first: the probe at initialization must find the growth column at the solve's level, about 0.65 on the held-out genes, or the warm start did not reach the readout; and the gated cell passes if it sits at the round-5 affine's 0.82 with three of three seeds above the round-4 gated cell.

Figure 28 draws the warm-started path.

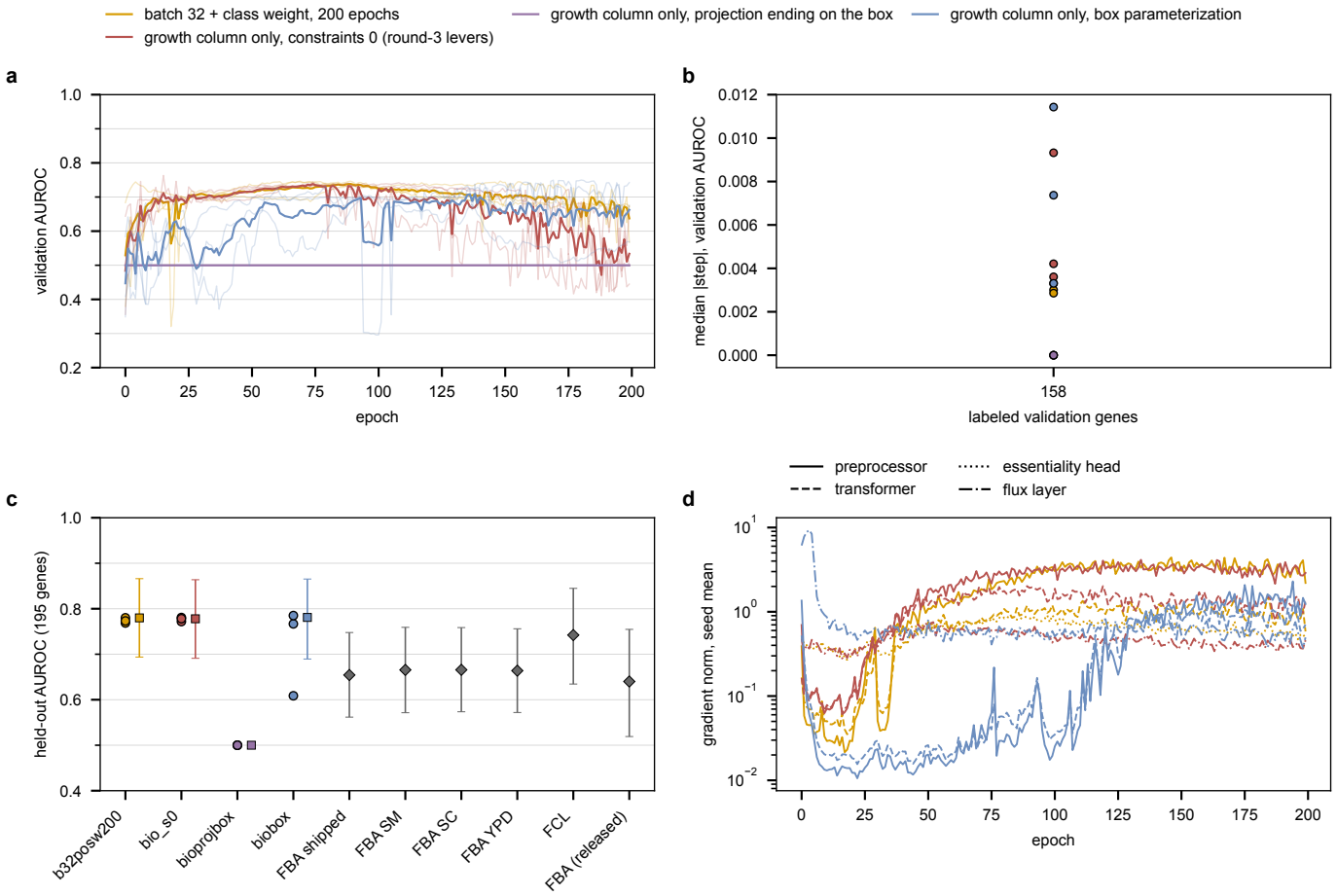



**Figure 27: Round 7, the growth column with the bounds enforced.** **a**, Validation AUROC per epoch, one thin line per seed and the seed mean per arm. **b**, The median absolute epoch-to-epoch step per cell against the validation size. **c**, Held-out AUROC per cell (points), the seed mean and the gene bootstrap of the pooled per-gene logits (square with interval), beside FBA on four media, Flux Cone Learning and the released FBA predictions on the same 195 genes. **d**, Gradient norm per module per epoch, seed mean; the dash-dot line is the flux layer. Written by `experiments/028-gene-essentiality/scripts/wave12_readout.py -round w18`.

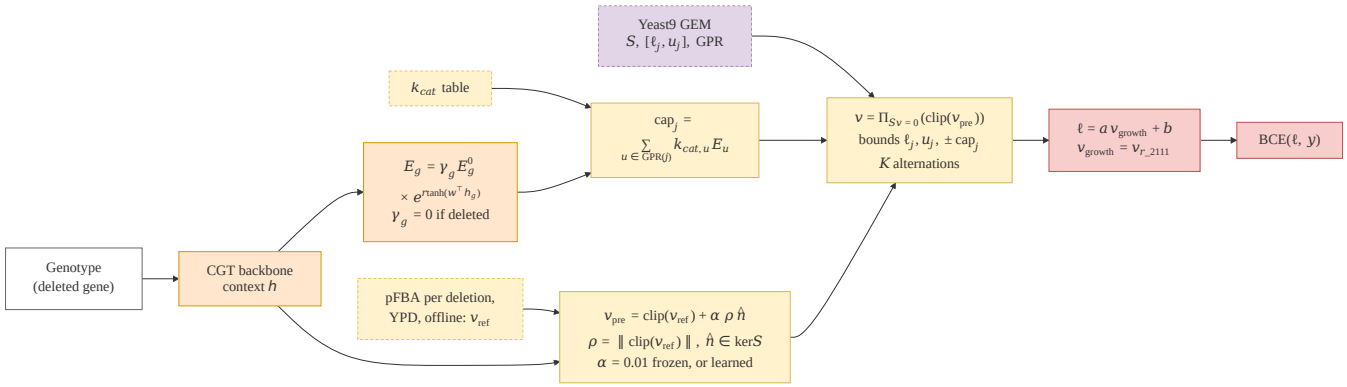



**Figure 28: The warm-started forced readout on the anchored layer, round 10.** The forced readout of Figure 18 with the proposal changed: the alternations start from the deletion's own parsimonious FBA flux $v_{\text{ref}}$, a fixed table solved offline on YPD, plus a null-space step of relative size $\alpha$ whose direction the context sets. The box is still the GEM's bounds tightened by the capacity ceilings. The gated arm of the round puts the same growth column behind the gate of Figure 19. Colors as in Figure 16.

**Table 34:** Round 10, the flux layer warm-started from the deletion's own solve. The pooled cell, the round-4 forced and gated growth columns, and the round-5 FBA affine, against the forced and gated growth columns whose layer proposes the deletion's own parsimonious FBA flux on YPD plus a null-space residual of relative size alpha, with alpha frozen at 0.01, learned, or frozen at 0.1. Constraint scale 0, batch 32, class weight, 200 epochs, seeds 2700 to 2702. Bold marks the highest value in a column. ↑ (↓) marks an arm whose held-out AUROC is above (below) the first arm's on every one of the same three seeds.

| arm | mean | sd | seeds | pooled [95% CI] | selected epochs | val step |
|---|---|---|---|---|---|---|
| batch 32 + class weight, 200 epochs | 0.774 | 0.006 | 0.780, 0.769, 0.773 | 0.780 [0.69, 0.87] | 75, 68, 148 | 0.0031 |
| growth column only, constraints 0 (round-3 levers) | 0.777 | 0.005 | 0.772, 0.781, 0.778 | 0.778 [0.69, 0.86] | 108, 74, 84 | 0.0057 |
| gated growth column, constraints 0 (round-3 levers) | 0.771 | 0.010 | 0.766, 0.782, 0.765 | 0.768 [0.68, 0.85] | 67, 71, 77 | 0.0040 |
| context head + FBA growth ratios (round-3 levers) | **0.821** | 0.020 | 0.814, 0.805, 0.844 ↑ | **0.838** [0.76, 0.91] | 28, 70, 61 | 0.0036 |
| growth column, pFBA warm start, alpha 0.01 frozen | 0.704 | 0.073 | 0.629, 0.706, 0.776 | 0.801 [0.71, 0.90] | 112, 144, 157 | 0.0222 |
| gated column + context, pFBA warm start, alpha 0.01 frozen | 0.776 | 0.034 | 0.810, 0.776, 0.742 | 0.784 [0.70, 0.87] | 181, 70, 4 | 0.0037 |
| gated column + context, pFBA warm start, alpha learned | 0.775 | 0.035 | 0.812, 0.772, 0.741 | 0.796 [0.72, 0.87] | 146, 71, 4 | 0.0036 |
| gated column + context, pFBA warm start, alpha 0.1 frozen | 0.751 | 0.044 | 0.784, 0.767, 0.701 | 0.777 [0.69, 0.86] | 79, 83, 3 | 0.0035 |

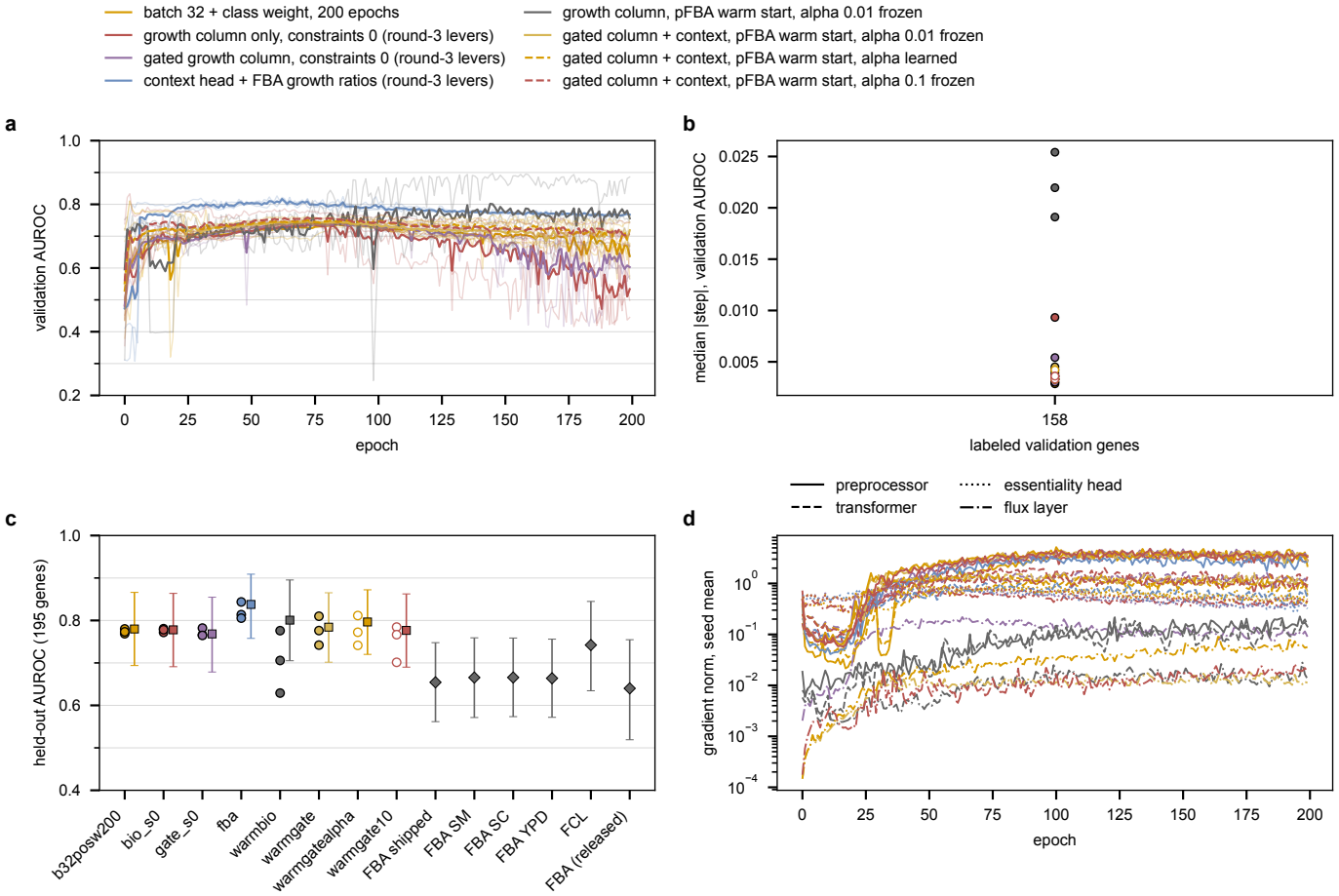



**Figure 29: Round 10, the layer warm-started from the deletion's own solve.** **a**, Validation AUROC per epoch, one thin line per seed and the seed mean per arm. **b**, The median absolute epoch-to-epoch step per cell against the validation size. **c**, Held-out AUROC per cell (points), the seed mean and the gene bootstrap of the pooled per-gene logits (square with interval), beside FBA on four media, Flux Cone Learning and the released FBA predictions on the same 195 genes. **d**, Gradient norm per module per epoch, seed mean; the dash-dot line is the flux layer. Written by `experiments/028-gene-essentiality/scripts/wave12_readout.py -round w21`.

Neither read holds. The probe at initialization finds the growth column negative (median −0.07 to −0.10 mmol/gDW/h against references of 0 to 0.5) and anti-correlated with the solve it was started from: Spearman −0.43 to −0.44 against the YPD ratio, and the negative column ranks the held-out genes at 0.34 to 0.39, which is the solve's ranking with its sign reversed. The layer's own diagnostics say where the reference goes: the anchored layer's box is the GEM's bounds tightened by the enzyme-capacity ceilings and the second-law directions, the reference violates it on 28 to 39 percent of reactions at initialization, and the proposal the alternations return sits 0.45 to 0.57 of the reference's norm away from it at $\alpha = 0.01$. The projections that make the flux balanced move it half its length off the point it was handed and reverse the growth column's correlation with the solve on the way. Trained, the forced cells read 0.629, 0.706 and 0.776 against the round-4 forced cell's 0.772, 0.781 and 0.778, down on three of three seeds, and the probe on their trained weights inverts further (0.22 to 0.41); the gated cell reads 0.810, 0.776 and 0.742 against 0.766, 0.782 and 0.765, one seed up and two down, with seed 2702 selecting epoch 4 and its gate ending at −2.0. Letting $\alpha$ learn changes nothing on the held-out genes (0.812, 0.772, 0.741) while $\alpha$ grows to 0.19, 0.02 and 0.11 and the box violation on one seed rises from 0.30 to 0.45, the job-1946 failure mode in miniature; $\alpha$ frozen at 0.1 reads 0.751. The warm start hands

the layer the solve and the anchored layer's constraint set does not keep it. Round 13 repeats the warm start on the stoichiometry-only layer, whose box is the GEM's own bounds, so the reference is inside it by construction.

## 4.9 Round 13: the warm start on the stoichiometry-only layer ✗

Round 13 (jobs 2297 and 2298) repeats the warm start on the stoichiometry-only layer, whose box is the GEM's own bounds scaled by the availability gate, with no capacity ceiling and no second-law direction, forced and gated, $\alpha$ frozen at 0.01, three seeds each (Table 35, Figure 31). The reads are round 10's: the probe at initialization at the solve's level, and the gated cell at the round-5 affine's 0.82 with three of three seeds above the round-4 gated cell.

Figure 30 draws the same warm start on this layer.

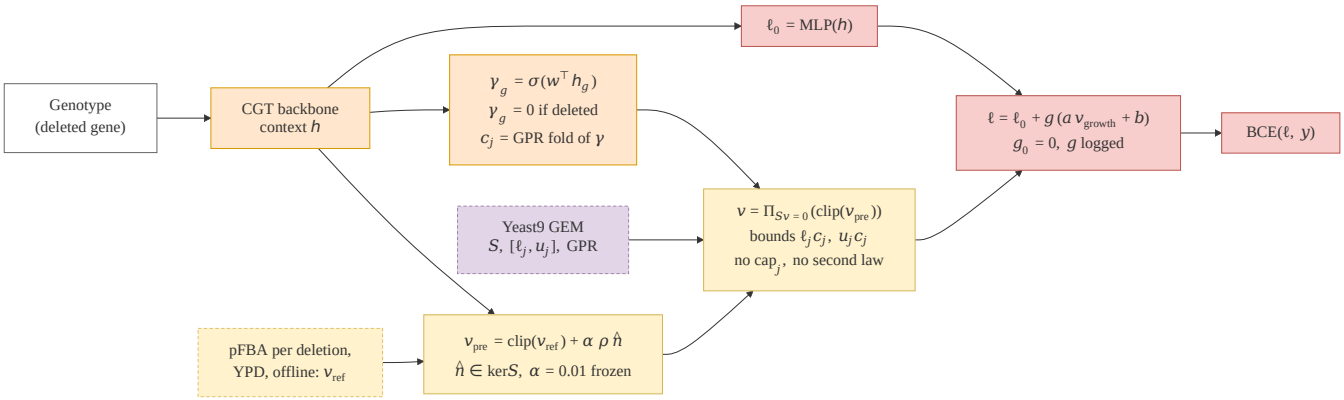



**Figure 30: The warm-started gated readout on the stoichiometry-only layer, round 13.** The proposal of Figure 28 on a layer with no abundance head, no $k_{cat}$ table and no capacity ceiling: a learned availability gate, zero for the deleted gene, scales the GEM's own bounds. The growth column enters behind the zero-initialized gate of Figure 19, and the forced arm reads it alone. Colors as in Figure 16.

**Table 35:** Round 13, the warm start on the stoichiometry-only layer. The pooled cell, the round-4 forced and gated columns and the round-10 warm-started columns on the anchored layer, against the same warm start on the stoichiometry-only layer, whose box is the GEM's own bounds rather than the capacity- and second-law-tightened one, so the deletion's pFBA reference sits inside it. Alpha frozen at 0.01, constraint scale 0, batch 32, class weight, 200 epochs, seeds 2700 to 2702. Bold marks the highest value in a column. ↑ (↓) marks an arm whose held-out AUROC is above (below) the first arm's on every one of the same three seeds.

| arm | mean | sd | seeds | pooled [95% CI] | selected epochs | val step |
|---|---|---|---|---|---|---|
| batch 32 + class weight, 200 epochs | 0.774 | 0.006 | 0.780, 0.769, 0.773 | 0.780 [0.69, 0.87] | 75, 68, 148 | 0.0031 |
| growth column only, constraints 0 (round-3 levers) | 0.777 | 0.005 | 0.772, 0.781, 0.778 | 0.778 [0.69, 0.86] | 108, 74, 84 | 0.0057 |
| gated growth column, constraints 0 (round-3 levers) | 0.771 | 0.010 | 0.766, 0.782, 0.765 | 0.768 [0.68, 0.85] | 67, 71, 77 | 0.0040 |
| growth column, pFBA warm start, alpha 0.01 frozen | 0.704 | 0.073 | 0.629, 0.706, 0.776 | 0.801 [0.71, 0.90] | 112, 144, 157 | 0.0222 |
| gated column + context, pFBA warm start, alpha 0.01 frozen | 0.776 | 0.034 | 0.810, 0.776, 0.742 | 0.784 [0.70, 0.87] | 181, 70, 4 | 0.0037 |
| growth column, GEM box only, pFBA warm start, alpha 0.01 frozen | 0.664 | 0.058 | 0.654, 0.726, 0.612 ↓ | 0.676 [0.55, 0.79] | 53, 2, 23 | 0.0236 |
| gated column + context, GEM box only, pFBA warm start, alpha 0.01 frozen | **0.816** | 0.019 | 0.794, 0.829, 0.825 ↑ | **0.802** [0.72, 0.88] | 85, 5, 118 | 0.0042 |

Both reads hold, and this is the first module cell in the document that carries the solve. At initialization the growth column is positive (median 0.38 mmol/gDW/h), its Spearman with the solve is +0.47 to +0.57 across the four media, and its negative ranks the held-out genes at 0.683, above FBA's own 0.654 to 0.666 on the same genes; the alternations leave the reference where it was, 0.3 percent of its norm away, with a box violation of 0.24 to 0.29 that is the availability gate's untrained scaling of the bounds rather than a displacement of the flux. The forced cells read 0.654, 0.726 and 0.612, a mean of 0.664, which is the solve's number: trained, the column keeps its sign (Spearman +0.54 to +0.64) and its ranking (0.62 to 0.67), and the layer's learned parts add nothing to it. The gated cells read 0.794, 0.829 and 0.825 against the round-4 gated cell's 0.766, 0.782 and 0.765, up by 0.028, 0.047 and 0.060 on three of three seeds, a seed mean of 0.816 against the affine's 0.821, and on the two seeds that train past epoch 5 the gate ends at +0.97 and +0.90, the first positive gates in the document: the head adds the growth column with its physical sign. The third seed selects epoch 5 with its gate at zero and still reads 0.829, the seed-to-seed spread the round-9 cells also show.

Put beside round 10, the comparison isolates the loss. The same reference on the anchored layer arrived sign-reversed and half its norm away; on the GEM's own box it arrives intact. The capacity ceilings and the second-law directions, tightened onto a flux that was solved without them, are what the projections could not reconcile, and the layer that solves (round 15) applies those constraints inside the solve instead of after it.

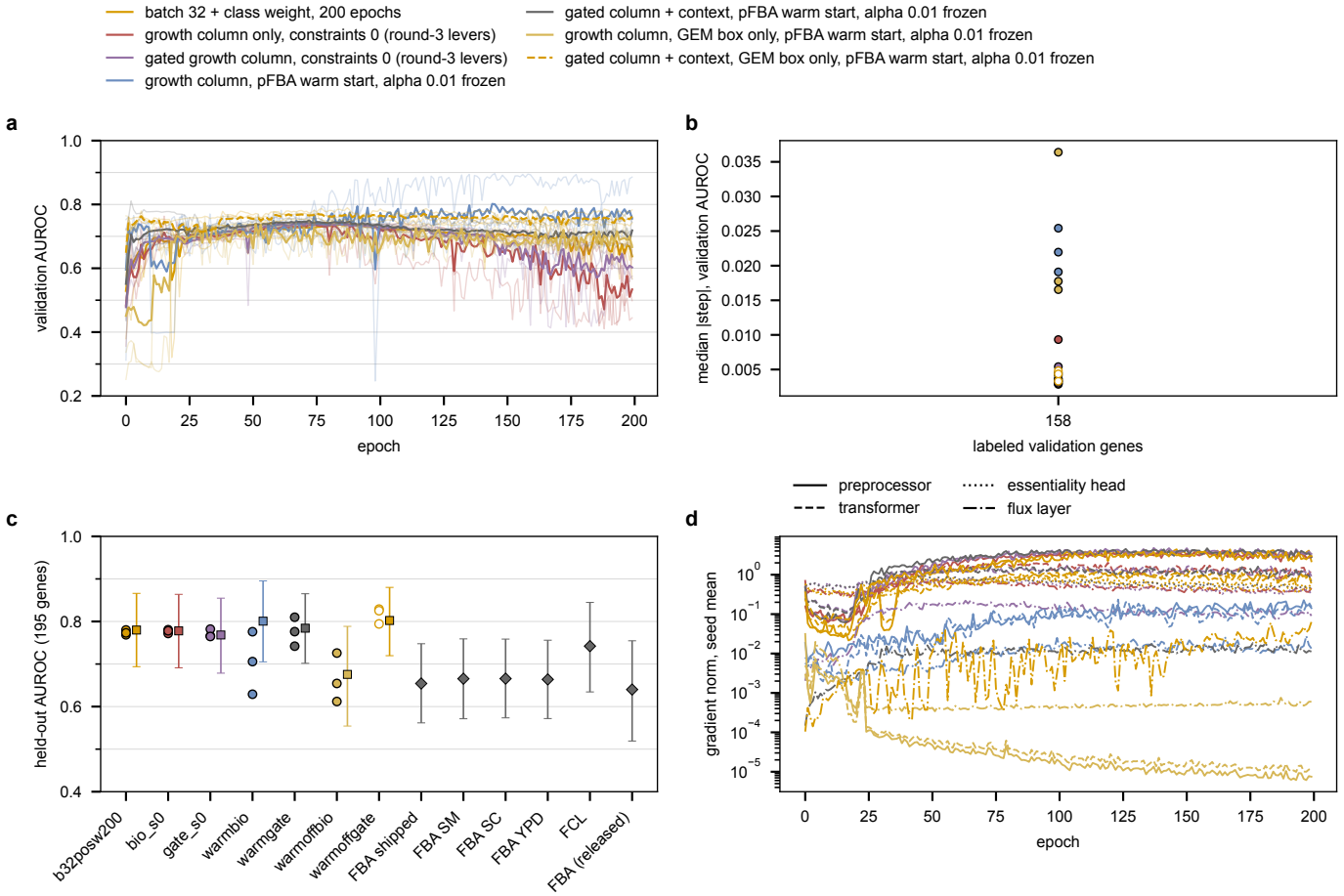



**Figure 31: Round 13, the warm start on the stoichiometry-only layer.** **a**, Validation AUROC per epoch, one thin line per seed and the seed mean per arm. **b**, The median absolute epoch-to-epoch step per cell against the validation size. **c**, Held-out AUROC per cell (points), the seed mean and the gene bootstrap of the pooled per-gene logits (square with interval), beside FBA on four media, Flux Cone Learning and the released FBA predictions on the same 195 genes. **d**, Gradient norm per module per epoch, seed mean; the dash-dot line is the flux layer. Written by `experiments/028-gene-essentiality/scripts/wave12_readout.py -round w24`.

## 4.10 Round 15: the layer that solves ✗

The projection route carries the solve only when handed it. Round 15 replaces the alternations with the solve: under the `lp` parameterization the layer's flux is the growth optimum of the linear program $\max\{v_{\text{growth}} : Sv = 0,\ \ell \le v \le u\}$ over the layer's own bounds, one HiGHS solve per genotype in a forked pool, and the backbone reaches the prediction through the bounds alone, with the gradient the bound duals give (`torchcell.metabolism.lp_layer`; the envelope theorem, checked against finite differences on the GEM). On the anchored layer the bounds are the GEM's scaled by availability and capped by the enzyme-capacity ceilings from the abundance head, so this is the GECKO construction with a learned enzyme pool. Round 15 (jobs 2304 to 2306) runs it forced, forced with the second-law box, and gated, three seeds each (Table 36, Figure 33); Figure 32 draws the construction.

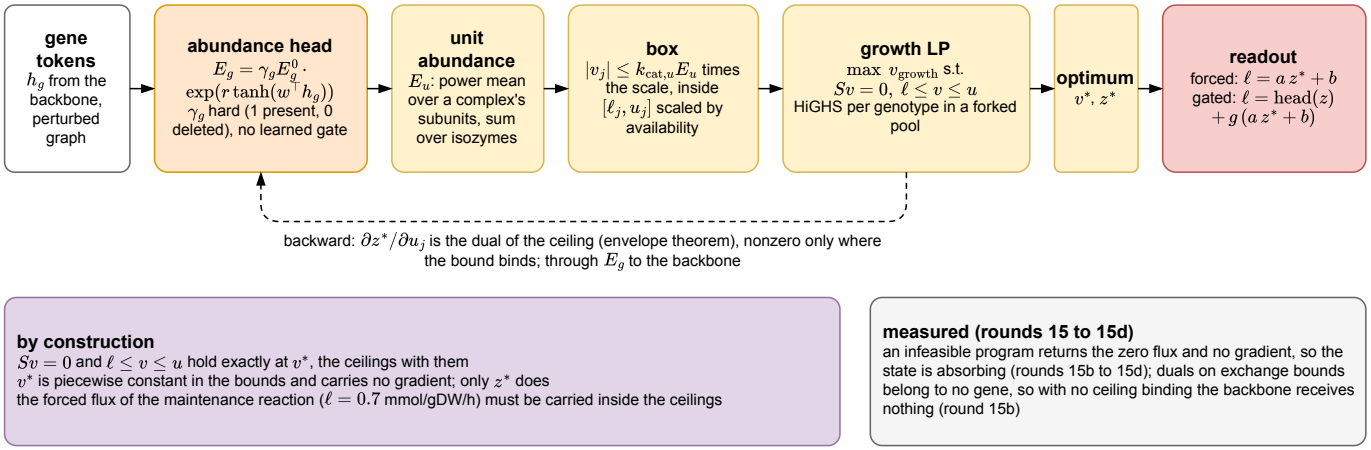



**Figure 32: The layer that solves, rounds 15 to 15d.** The abundance head sets each gene's enzyme abundance with availability a hard fact, the unit abundance and the $k_{cat}$ ceilings form the box, and the flux is the growth optimum of the linear program over that box, one HiGHS solve per genotype; the program's value $z^*$ enters the readout. Mass balance and the box hold exactly at the optimum. The dashed path is the backward pass: the derivative of $z^*$ in a bound is that bound's dual, nonzero only where the bound binds, so the backbone is trained through the ceilings and only through them; an infeasible program returns no flux and no gradient. Colors as in Figure 16. Drawn in draw.io from `notes/assets/drawio/028-module.gen.py`.

**Table 36:** Round 15, the layer that solves. The pooled cell, the round-4 forced and gated growth columns and the round-5 FBA affine, against a flux layer whose flux is the growth optimum of a linear program under the layer's own bounds (availability and the enzyme-capacity ceiling from the abundance head), with the bound duals as its gradient: the forced growth column, the same with the box tightened to the second-law direction before the solve, and the gated column with the context head. Constraint scale 0, batch 32, class weight, 200 epochs, seeds 2700 to 2702. Bold marks the highest value in a column. ↑ (↓) marks an arm whose held-out AUROC is above (below) the first arm's on every one of the same three seeds.

| arm | mean | sd | seeds | pooled [95% CI] | selected epochs | val step |
|---|---|---|---|---|---|---|
| batch 32 + class weight, 200 epochs | 0.774 | 0.006 | 0.780, 0.769, 0.773 | 0.780 [0.69, 0.87] | 75, 68, 148 | 0.0031 |
| growth column only, constraints 0 (round-3 levers) | 0.777 | 0.005 | 0.772, 0.781, 0.778 | 0.778 [0.69, 0.86] | 108, 74, 84 | 0.0057 |
| gated growth column, constraints 0 (round-3 levers) | 0.771 | 0.010 | 0.766, 0.782, 0.765 | 0.768 [0.68, 0.85] | 67, 71, 77 | 0.0040 |
| context head + FBA growth ratios (round-3 levers) | **0.821** | 0.020 | 0.814, 0.805, 0.844 ↑ | **0.838** [0.76, 0.91] | 28, 70, 61 | 0.0036 |
| growth column, the layer that solves | 0.500 | 0.000 | 0.500, 0.500, 0.500 ↓ | 0.500 [0.50, 0.50] | 2, 2, 2 | 0.0000 |
| growth column, the layer that solves, second-law box | 0.500 | 0.000 | 0.500, 0.500, 0.500 ↓ | 0.500 [0.50, 0.50] | 2, 2, 2 | 0.0000 |
| gated column + context, the layer that solves | 0.773 | 0.012 | 0.771, 0.786, 0.763 | 0.770 [0.68, 0.86] | 78, 81, 63 | 0.0028 |

Every forced cell scores 0.500 exactly and selects epoch 2, because the program is infeasible for every genotype at every epoch: the probe finds the growth column at zero for all 982 deletions at initialization and under the trained weights. The gated cells read 0.771, 0.786 and 0.763, the pooled cell's 0.774, with gates within 0.07 of zero, which is the head working alone beside a dead path. The cause is measured on one genotype's bounds taken from the layer: under the GEM's own bounds the program solves at the wild-type growth of 0.0858, and under the layer's bounds it has no feasible point, with or without the maintenance flux released and with the deleted gene's reactions reopened. What differs is the capacity ceiling: the layer's upper bounds have a median of 0.084 mmol/gDW/h and a tenth of the reactions below 0.007, against 100 on the GEM, because the predicted $k_{cat}$ tier multiplied by the abundance head's nominal abundances gives ceilings two to four orders below the fluxes growth needs. The alternating layer never noticed, because it never satisfied the box; the solve cannot proceed without it. That is the same fact as round 10's, read exactly: the enzyme-capacity box as parameterized contains no flux that meets the model's maintenance requirement, let alone one that grows (Section 4.2 puts the shortfall at 0.54 of 0.70), and every arm built on it has been reading a flux off the polytope. Round 15b runs the solve on the stoichiometry-only box, where round 13 showed the polytope holds growth. The capacity ceiling that admits growth is then calibrated the way GECKO calibrates its pool, by one scale on

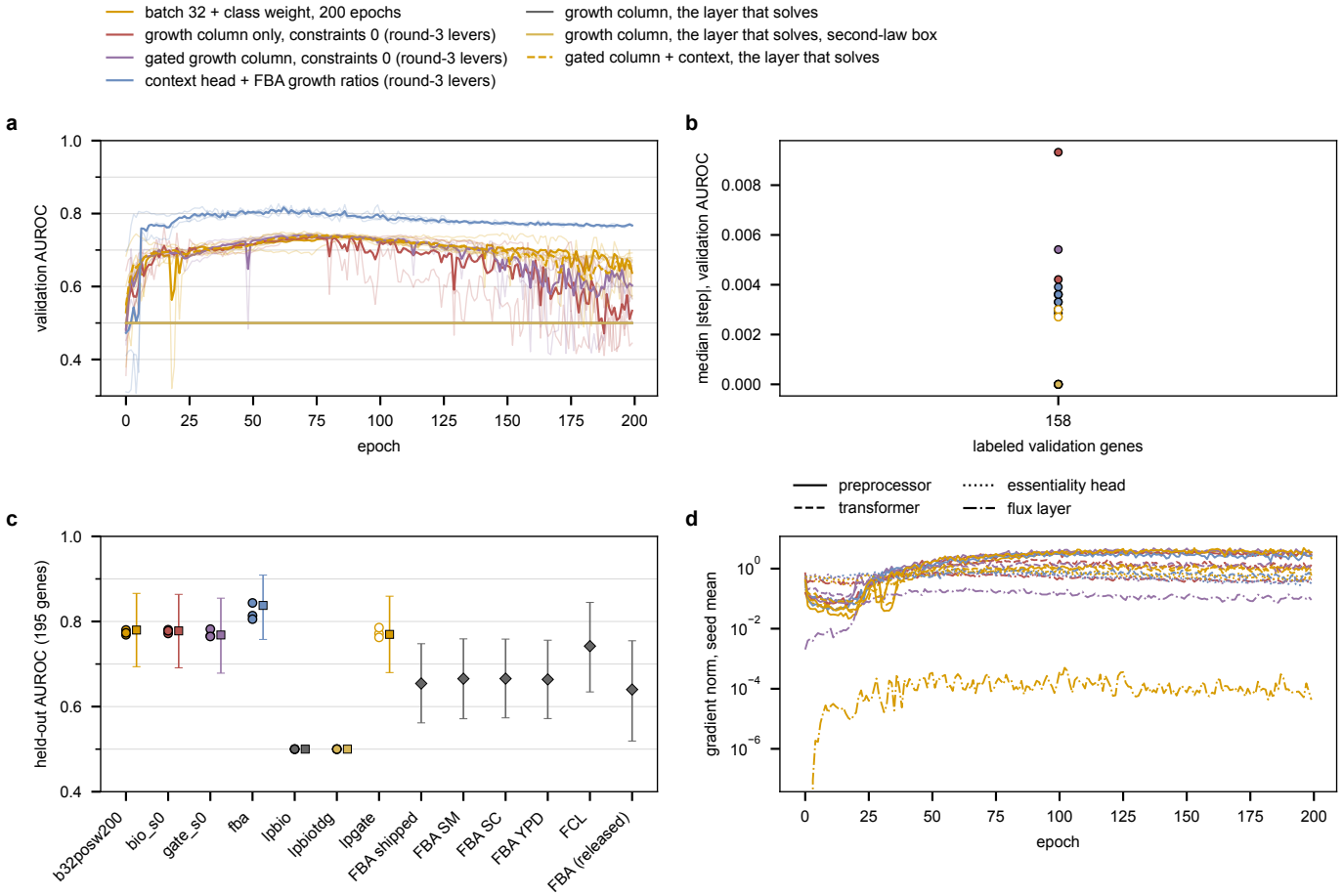



**Figure 33: Round 15, the layer that solves on the anchored box.** **a**, Validation AUROC per epoch, one thin line per seed and the seed mean per arm. **b**, The median absolute epoch-to-epoch step per cell against the validation size. **c**, Held-out AUROC per cell (points), the seed mean and the gene bootstrap of the pooled per-gene logits (square with interval), beside FBA on four media, Flux Cone Learning and the released FBA predictions on the same 195 genes. **d**, Gradient norm per module per epoch, seed mean; the dash-dot line is the flux layer. Written by `experiments/028-gene-essentiality/scripts/wave12_readout.py -round w26`.

every ceiling (`capacity_calibration.py`, through the layer's own forward pass on a batch of 32 held-out genotypes at initialization): the program is infeasible up to a scale of 10, feasible for 31 of the 32 from a scale of 30 (the thirty-second is lethal) with a median growth of 0.042, reaches 0.10 at 100 and 0.20 at 1,000, and at 10,000 returns the unconstrained wild type's 0.535, where the ceilings no longer bind. Round 15c runs the layer that solves at 100 and 1,000, where bounds the genes own bind at the optimum.

## 4.11 Round 15b: the layer that solves on the stoichiometry-only box ✗

Round 15b (jobs 2314 and 2315) runs the same solve on the stoichiometry-only layer, whose bounds are the GEM's scaled by the availability gate, forced and gated, three seeds each, beside the round-13 warm-started columns on the same box and the round-5 affine (Table 37, Figure 35); the read is round 10's. Figure 34 draws the layer on this box.

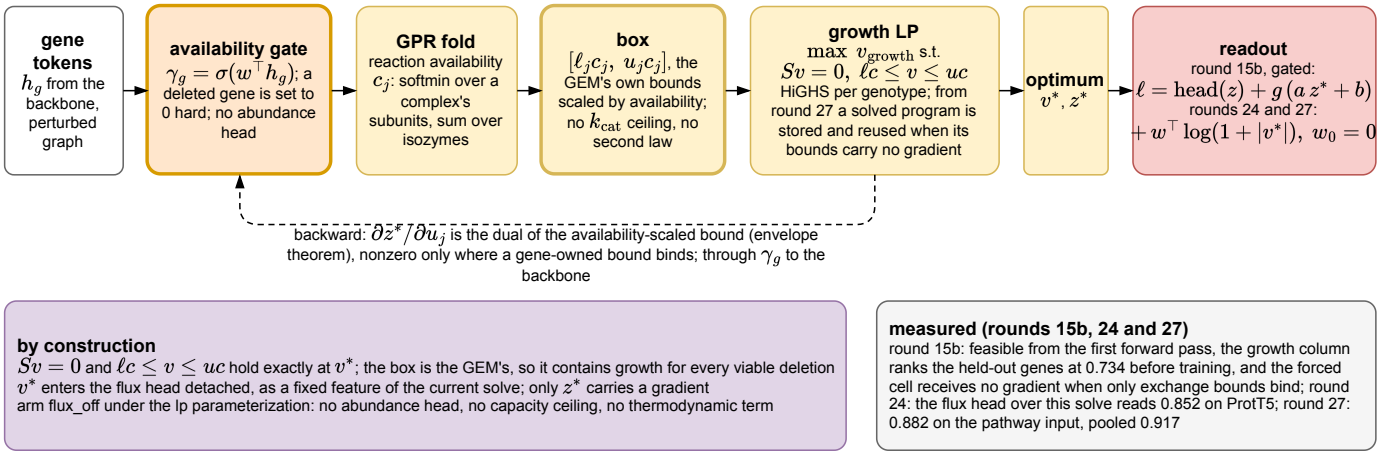



**Figure 34: The layer that solves on the GEM's own box, rounds 15b, 24 and 27.** A learned availability gate, zero for a deleted gene, is folded through the gene-protein-reaction rules into a reaction availability that scales the GEM's own bounds; there is no abundance head, no $k_{cat}$ ceiling and no thermodynamic term, and the flux is the growth optimum of the linear program over that box. The dashed path is the backward pass, the dual of a gene-owned bound where one binds. Round 15b reads the program's value through the forced or the gated readout, and rounds 24 and 27 add a zero-initialized head over $\log(1 + |v^*|)$ of the detached optimum; from round 27 the solver stores a solved program and reuses it when its bounds carry no gradient. Thick borders mark the blocks that differ from Figure 32. Colors as in Figure 16. Drawn in draw.io from `notes/assets/drawio/028-module.gen.py`.

**Table 37:** Round 15b, the layer that solves on the stoichiometry-only box. The pooled cell, the round-5 FBA affine and the round-13 warm-started columns on the stoichiometry-only layer, against the growth LP under that layer's box (the GEM's bounds scaled by availability, no capacity ceiling): the forced growth column and the gated column with the context head. Constraint scale 0, batch 32, class weight, 200 epochs, seeds 2700 to 2702. Bold marks the highest value in a column. ↑ (↓) marks an arm whose held-out AUROC is above (below) the first arm's on every one of the same three seeds.

| arm | mean | sd | seeds | pooled [95% CI] | selected epochs | val step |
|---|---|---|---|---|---|---|
| batch 32 + class weight, 200 epochs | 0.774 | 0.006 | 0.780, 0.769, 0.773 | 0.780 [0.69, 0.87] | 75, 68, 148 | 0.0031 |
| context head + FBA growth ratios (round-3 levers) | 0.821 | 0.020 | 0.814, 0.805, 0.844 ↑ | **0.838** [0.76, 0.91] | 28, 70, 61 | 0.0036 |
| growth column, GEM box only, pFBA warm start, alpha 0.01 frozen | 0.664 | 0.058 | 0.654, 0.726, 0.612 ↓ | 0.676 [0.55, 0.79] | 53, 2, 23 | 0.0236 |
| gated column + context, GEM box only, pFBA warm start, alpha 0.01 frozen | 0.816 | 0.019 | 0.794, 0.829, 0.825 ↑ | 0.802 [0.72, 0.88] | 85, 5, 118 | 0.0042 |
| growth column, GEM-box solve | 0.613 | 0.098 | 0.500, 0.669, 0.669 ↓ | 0.669 [0.58, 0.76] | 11, 51, 2 | 0.0000 |
| gated column + context, GEM-box solve | **0.824** | 0.021 | 0.800, 0.837, 0.835 ↑ | 0.837 [0.76, 0.91] | 2, 103, 4 | 0.0046 |

The program is feasible here and its growth column is growth from the first forward pass: the probe at initialization finds a positive column with 10 percent of the 982 deletions at exactly zero growth, the lethal ones, a Spearman of +0.40 to +0.50 with FBA's ratio across the four media, and a ranking of 0.734 on the held-out genes, above FBA's own 0.654 to 0.666 on the same genes. The layer that solves under the availability-scaled box is a better ranker than the unscaled solve before any training, which is the uniform enzyme ceiling the gate imposes doing GECKO's work in miniature. The gated cells read 0.800, 0.837 and 0.835 against the round-4 gated cell's 0.766, 0.782 and 0.765, up by 0.034, 0.055 and 0.070 on three of three seeds, a seed mean of 0.824 against the affine's 0.821 and a pooled bootstrap of 0.837 [0.76, 0.91]: both halves of the read hold, and this is a module cell at the affine's level with the solve inside the forward pass and the gradient through it. The seed that trains longest (2701, epoch 103) ends with its gate at +1.42, the largest positive gate in the document, and its growth column ranks the held-out genes at 0.768 on its own after training, above its 0.734 at initialization; the other two seeds select epochs 2 and 4 with gates near zero, so on those the number is the head's early ranking and the column's contribution is not read.

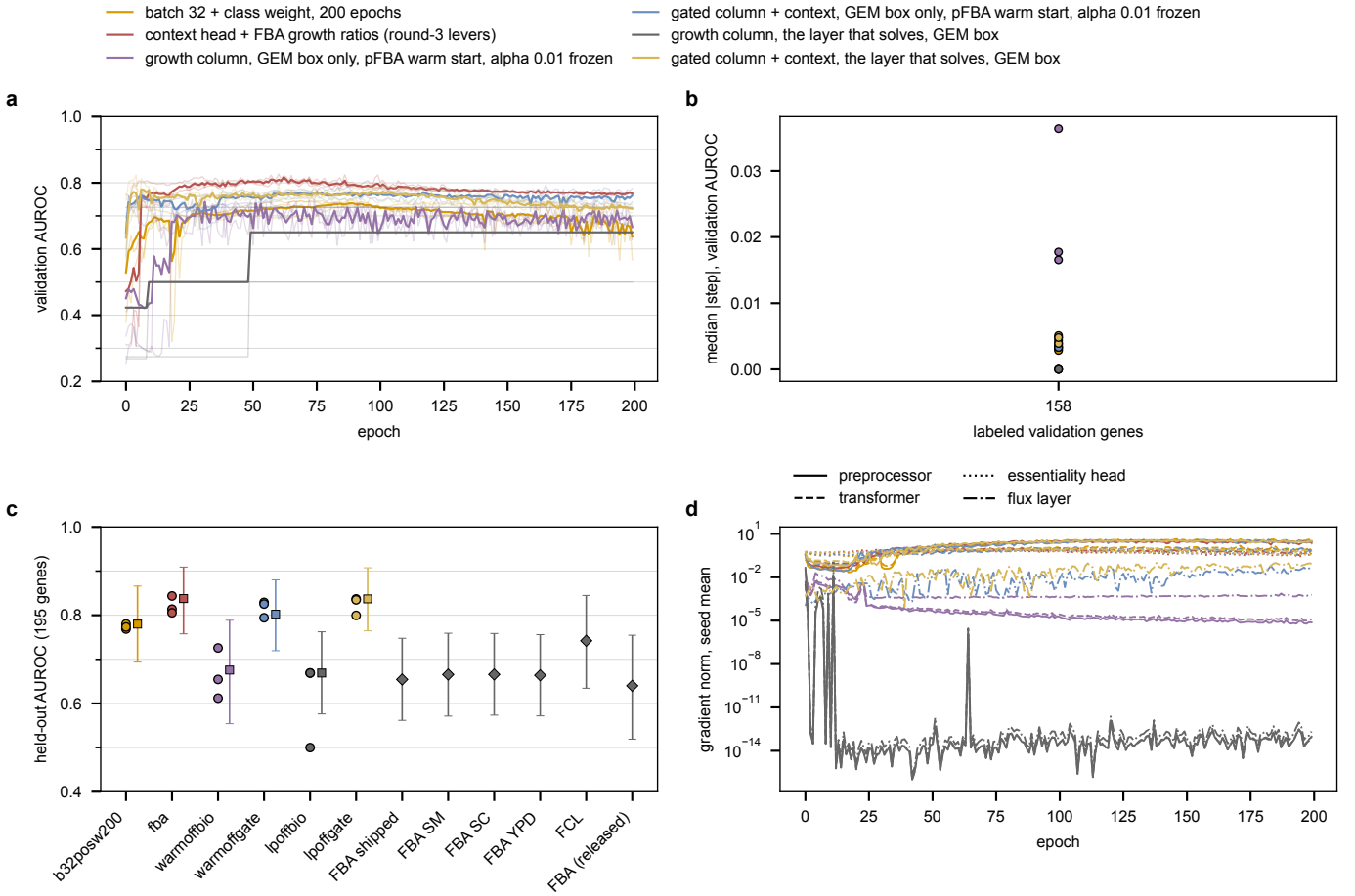



**Figure 35: Round 15b, the layer that solves on the stoichiometry-only box.** **a**, Validation AUROC per epoch, one thin line per seed and the seed mean per arm. **b**, The median absolute epoch-to-epoch step per cell against the validation size. **c**, Held-out AUROC per cell (points), the seed mean and the gene bootstrap of the pooled per-gene logits (square with interval), beside FBA on four media, Flux Cone Learning and the released FBA predictions on the same 195 genes. **d**, Gradient norm per module per epoch, seed mean; the dash-dot line is the flux layer. Written by `experiments/028-gene-essentiality/scripts/wave12_readout.py -round w29`.

The forced cells fail in two ways that are the linear program's own. On two seeds the gradient into the backbone is $10^{-14}$ for the whole run and the cell reads 0.669, the solve's number with a learned threshold: at the optimum the binding constraints are the exchange bounds, which no gene owns, so the duals on the learnable bounds are zero and the backbone has nothing to move. On the third seed the early gradient moves the availability gate until the program is infeasible for every genotype (epoch 11), the column is zero from then on, and the cell reads 0.500; an infeasible program returns no gradient, so the state is absorbing. The gated cell avoids both because the head carries the loss while the column contributes where its duals are live. A layer that solves therefore needs two more things the rounds name: bounds the genes own that bind at the optimum (the capacity ceilings, once calibrated to admit growth), and a barrier that keeps the program feasible under training.

## 4.12 Round 15c: the layer that solves with calibrated capacity ceilings ✗

Round 15c (jobs 2316 to 2318) runs the solve on the anchored box with the capacity ceilings scaled by 100 and by 1,000, the range the calibration put growth in with the ceilings still binding, gated at both scales and forced at 100, three seeds each (Table 38, Figure 36). Bounds the genes own now bind at the optimum: the layer reports 3 to 8 percent of reactions at their capacity ceiling at the selected epoch, against zero on the GEM box.

**Table 38:** Round 15c, the layer that solves with calibrated capacity ceilings. The pooled cell, the round-5 FBA affine and the round-15b cells on the GEM box, against the growth LP on the anchored box with the enzyme-capacity ceilings scaled by 100 and by 1,000 (capacity_calibration.py: the program admits growth from a scale of 30 and the ceilings stop binding near 10,000), forced and gated. Constraint scale 0, batch 32, class weight, 200 epochs, seeds 2700 to 2702. Bold marks the highest value in a column. ↑ (↓) marks an arm whose held-out AUROC is above (below) the first arm's on every one of the same three seeds.

| arm | mean | sd | seeds | pooled [95% CI] | selected epochs | val step |
|---|---|---|---|---|---|---|
| batch 32 + class weight, 200 epochs | 0.774 | 0.006 | 0.780, 0.769, 0.773 | 0.780 [0.69, 0.87] | 75, 68, 148 | 0.0031 |
| context head + FBA growth ratios (round-3 levers) | 0.821 | 0.020 | 0.814, 0.805, 0.844 ↑ | 0.838 [0.76, 0.91] | 28, 70, 61 | 0.0036 |
| growth column, the layer that solves, GEM box | 0.613 | 0.098 | 0.500, 0.669, 0.669 ↓ | 0.669 [0.58, 0.76] | 11, 51, 2 | 0.0000 |
| gated column + context, the layer that solves, GEM box | **0.824** | 0.021 | 0.800, 0.837, 0.835 ↑ | 0.837 [0.76, 0.91] | 2, 103, 4 | 0.0046 |
| growth column, the layer that solves, capacity x100 | 0.614 | 0.198 | 0.500, 0.843, 0.500 | **0.843** [0.76, 0.91] | 2, 29, 2 | 0.0001 |
| gated column + context, the layer that solves, capacity x100 | 0.800 | 0.017 | 0.820, 0.795, 0.786 ↑ | 0.790 [0.71, 0.87] | 6, 88, 63 | 0.0037 |
| gated column + context, the layer that solves, capacity x1000 | 0.795 | 0.012 | 0.809, 0.786, 0.791 ↑ | 0.793 [0.71, 0.87] | 6, 51, 75 | 0.0039 |

The forced column carries the prediction alone on one seed. At scale 100 seed 2701 starts with 87 percent of its genotypes infeasible, is pulled back to none by epoch 50 by the gradient from the feasible rows, selects epoch 29 and reads 0.8425 on the held-out genes: the growth column of a linear program under learned enzyme ceilings, with no context head and no appended ratio, at the level of the affine (0.821) and of the round-9 backbone's seeds, 0.18 above FBA's own ranking on the same genes. The other two seeds enter the absorbing state within five epochs, 97 percent of genotypes infeasible from then on, and read 0.500. The probe at initialization reads the column at 0.730 with a Spearman of +0.49 to +0.55 against the solve, the same start as the GEM box's. The gated cells stay feasible for every genotype throughout and read 0.820, 0.795 and 0.786 at scale 100 and 0.809, 0.786 and 0.791 at 1,000, three of three seeds above the round-4 gated cell at both scales, with gates ending between +0.05 and +1.04, but below the GEM box's 0.824 and the affine's 0.821, so the calibrated ceilings cost the gated cell 0.02 to 0.03 against the plain box while binding.

What the one seed shows is the module's upper bound so far: the solve with learned capacity ceilings is a 0.84 ranker on its own, and the physics is doing it, since the column is growth (10 percent of deletions at exactly zero, the lethal ones) and the ceilings bind. What the other two show is that the linear program's infeasible region is absorbing under gradient descent on its bounds, which round 15d tests with more slack and a smaller step.

## 4.13 Round 15d: the forced layer that solves against the absorbing state ✗

Round 15d (jobs 2319 and 2320) reruns the forced cell at capacity scale 1,000, where the calibration leaves ten times the slack, and at scale 100 with the learning rate at $3 \times 10^{-4}$, three seeds each (Table 39, Figure 37).

**Table 39:** Round 15d, the forced layer that solves against the absorbing state. The pooled cell, the round-5 FBA affine and the round-15c forced cell at capacity scale 100, against the same cell at scale 1,000 and at scale 100 with the learning rate at $3 \times 10^{-4}$. Constraint scale 0, batch 32, class weight, 200 epochs, seeds 2700 to 2702. Bold marks the highest value in a column. ↑ (↓) marks an arm whose held-out AUROC is above (below) the first arm's on every one of the same three seeds.

| arm | mean | sd | seeds | pooled [95% CI] | selected epochs | val step |
|---|---|---|---|---|---|---|
| batch 32 + class weight, 200 epochs | 0.774 | 0.006 | 0.780, 0.769, 0.773 | 0.780 [0.69, 0.87] | 75, 68, 148 | 0.0031 |
| context head + FBA growth ratios (round-3 levers) | **0.821** | 0.020 | 0.814, 0.805, 0.844 ↑ | 0.838 [0.76, 0.91] | 28, 70, 61 | 0.0036 |
| growth column, the layer that solves, capacity x100 | 0.614 | 0.198 | 0.500, 0.843, 0.500 | **0.843** [0.76, 0.91] | 2, 29, 2 | 0.0001 |
| growth column, the layer that solves, capacity x1000 | 0.714 | 0.106 | 0.615, 0.826, 0.702 | 0.819 [0.73, 0.90] | 73, 30, 45 | 0.0000 |
| growth column, the layer that solves, capacity x100, lr 3e-4 | 0.582 | 0.142 | 0.500, 0.745, 0.500 ↓ | 0.745 [0.64, 0.84] | 2, 119, 2 | 0.0000 |

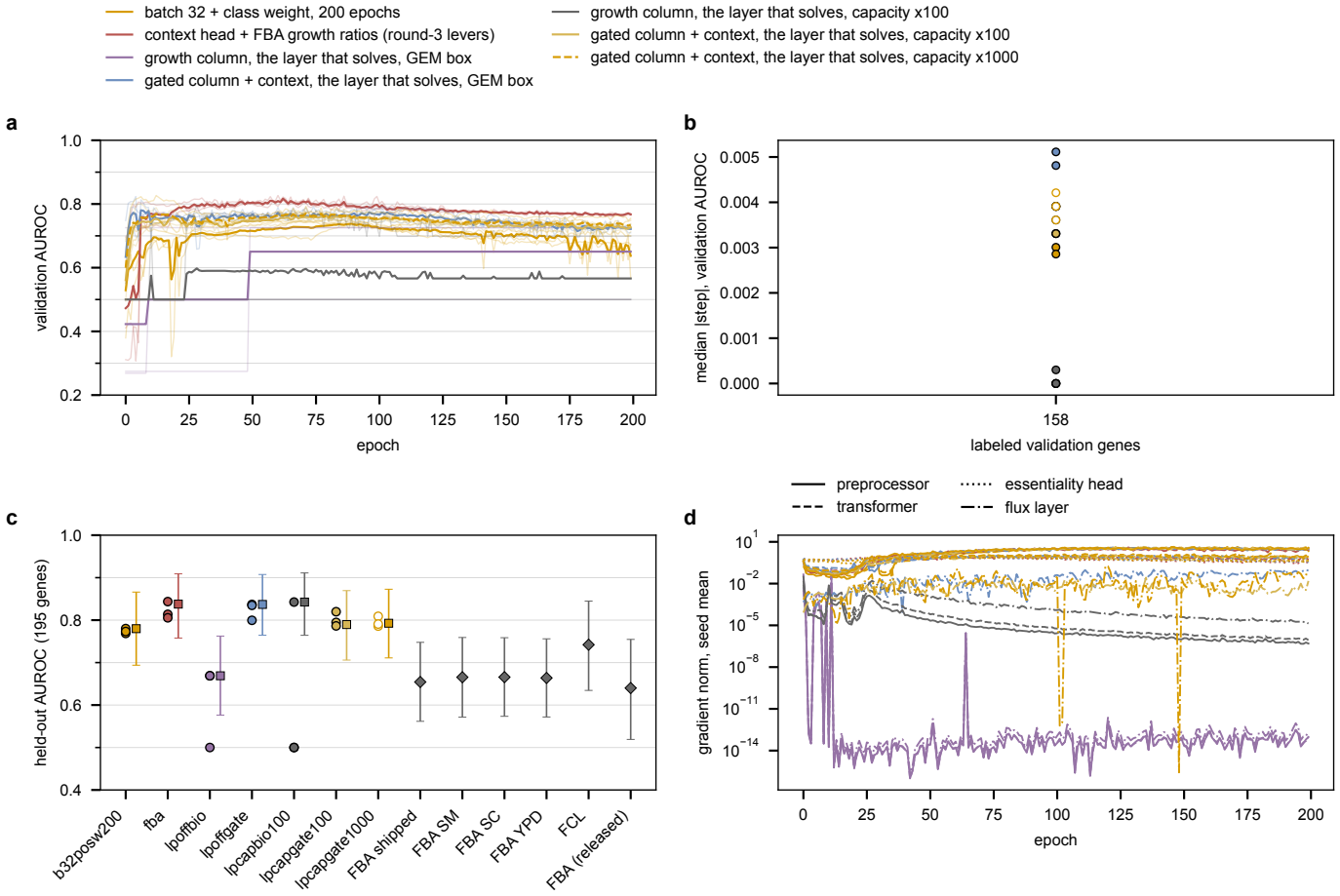



**Figure 36: Round 15c, the layer that solves with calibrated capacity ceilings.** **a**, Validation AUROC per epoch, one thin line per seed and the seed mean per arm. **b**, The median absolute epoch-to-epoch step per cell against the validation size. **c**, Held-out AUROC per cell (points), the seed mean and the gene bootstrap of the pooled per-gene logits (square with interval), beside FBA on four media, Flux Cone Learning and the released FBA predictions on the same 195 genes. **d**, Gradient norm per module per epoch, seed mean; the dash-dot line is the flux layer. Written by `experiments/028-gene-essentiality/scripts/wave12_readout.py -round w30`.

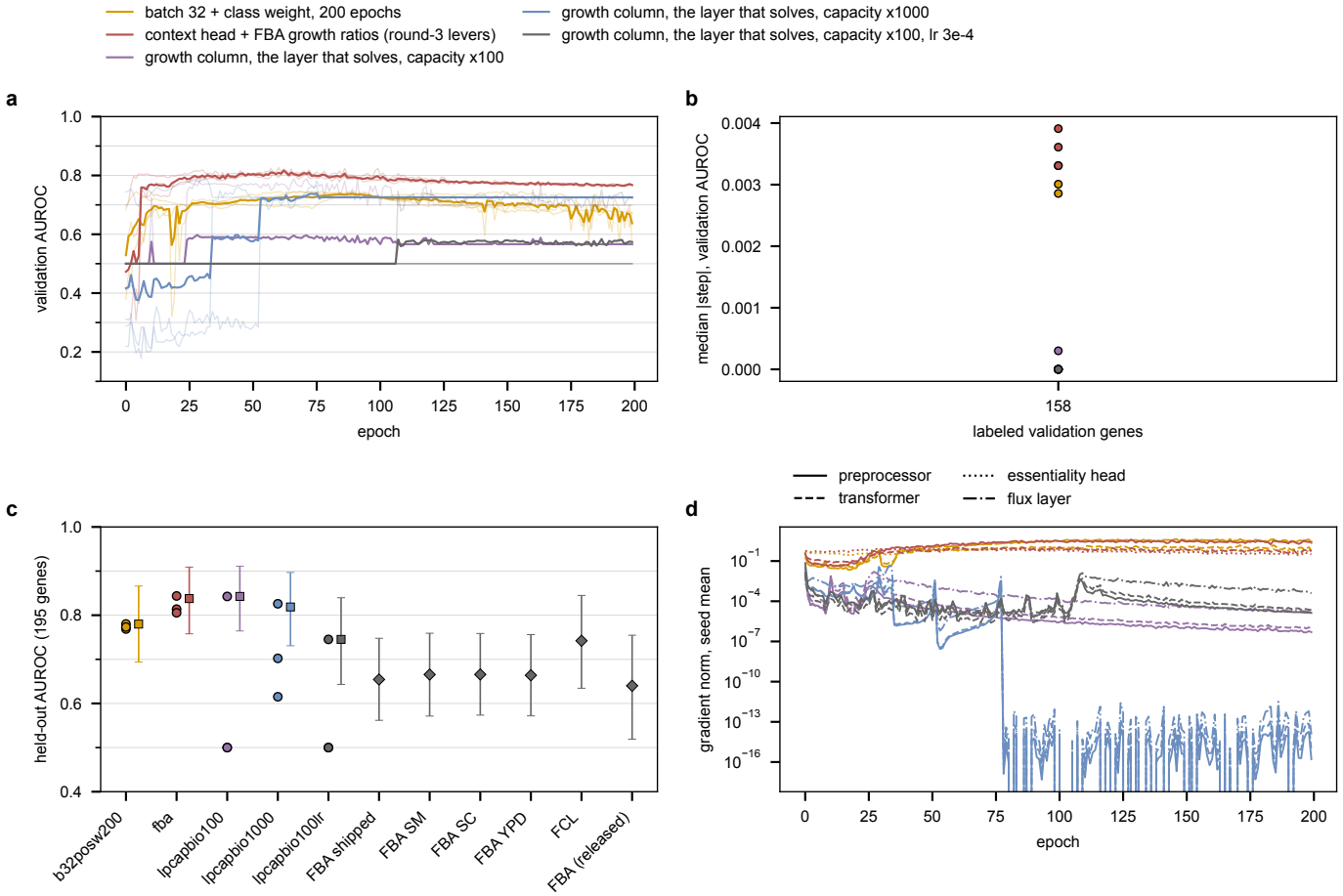



**Figure 37: Round 15d, the forced layer that solves against the absorbing state.** **a**, Validation AUROC per epoch, one thin line per seed and the seed mean per arm. **b**, The median absolute epoch-to-epoch step per cell against the validation size. **c**, Held-out AUROC per cell (points), the seed mean and the gene bootstrap of the pooled per-gene logits (square with interval), beside FBA on four media, Flux Cone Learning and the released FBA predictions on the same 195 genes. **d**, Gradient norm per module per epoch, seed mean; the dash-dot line is the flux layer. Written by `experiments/028-gene-essentiality/scripts/wave12_readout.py -round w31`.

Slack removes the absorbing state and the step size does not. At scale 1,000 no genotype's program is infeasible at any epoch on any seed, and the forced column reads 0.615, 0.826 and 0.702, a mean of 0.714 with a spread of 0.106: one seed near the affine's level and two near the solve's, from a column that is growth throughout. At scale 100 with the smaller step two seeds are absorbed by epoch 5 as before and the third is infeasible on 73 to 97 percent of genotypes until epoch 100, recovers, and selects epoch 119 at 0.745. The absorbing state is a property of how much room the ceilings leave, and with room the forced column is a 0.62 to 0.83 ranker whose spread across seeds is the largest in the document. Across rounds 15c and 15d the forced layer that solves has reached 0.84 and 0.83 on two of nine seeds, so the module can carry the whole prediction at the affine's level, and does so unreliably; the gated form is the reliable one, and it reads the affine's level on the GEM box.

## 4.14 Round 18: the elastic layer that solves, feasible by construction ✗

Rounds 15b to 15d leave the forced layer that solves with one failure mode, the absorbing infeasible state, and one unmeasured question, whether the growth column alone is a seed-robust 0.83 ranker once that state cannot occur. Round 18 (jobs 2321 to 2324) removes the state by construction rather than by slack in the ceilings: the elastic program of Section 4.2, every forced flux priced at $M = 1$ per mmol/gDW/h of shortfall, so that the value the readout sees is growth where maintenance is met and a negative number where it is not, and the gradient on the ceilings that block maintenance is the dual of the elastic row, of magnitude $M$, in place of no gradient at all. The program is solved in micromole units, because HiGHS's presolve at its absolute tolerances reads ceilings of $10^{-6}$ as an empty box and reports the feasible program infeasible; in those units it solves at the hard program's cost. Four cells, three seeds each, all on the `nodiss` arm: the forced column at ceiling scale 100 and 1,000 (the two scales of rounds 15c and 15d), the forced column at 100 with the protein pool held by construction, and the gated column at 100. The reads were fixed before launch: the forced cells pass if all three seeds are above the round-4 forced column and none is below 0.80, the level the hard program reached on two of nine seeds; the pool cell is read paired against the elastic cell without it; the gated cell passes if all three seeds are above the round-4 gated cell with a seed mean at or above the GEM box's 0.824. The infeasible fraction reading zero at every epoch is a check on the construction, not a result. Beside the score, every cell logs what the program did: the fraction of genotypes with unmet maintenance, the fraction on which a gene-owned bound carries a dual (the fraction on which the backbone receives any gradient through the solve), and the gradient norm of each part of the layer (Figure 39). Figure 38 draws the construction.

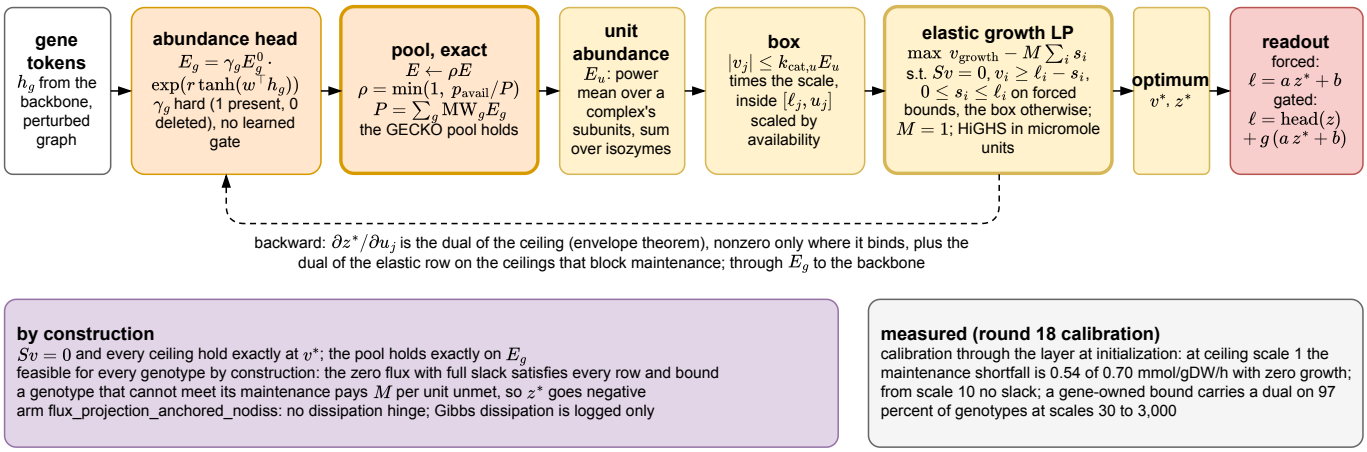



**Figure 38: The elastic layer that solves with the pool by construction, round 18.** Two blocks change against Figure 32, drawn with a heavier border: the abundance vector is projected onto the protein budget before it sets the ceilings, so the GECKO pool constraint holds exactly, and the growth program is elastic, every forced flux (the ATP maintenance reaction, lower bound 0.7 mmol/gDW/h) carrying a priced slack so that the zero flux with full slack is feasible for every genotype and the value $z^*$ falls by $M$ per unit of maintenance the ceilings cannot carry. The backward pass gains the dual of the elastic row on the ceilings that block maintenance. The arm carries no dissipation hinge; the dissipation is logged only. Colors as in Figure 16. Drawn in draw.io from `notes/assets/drawio/028-module.gen.py`.

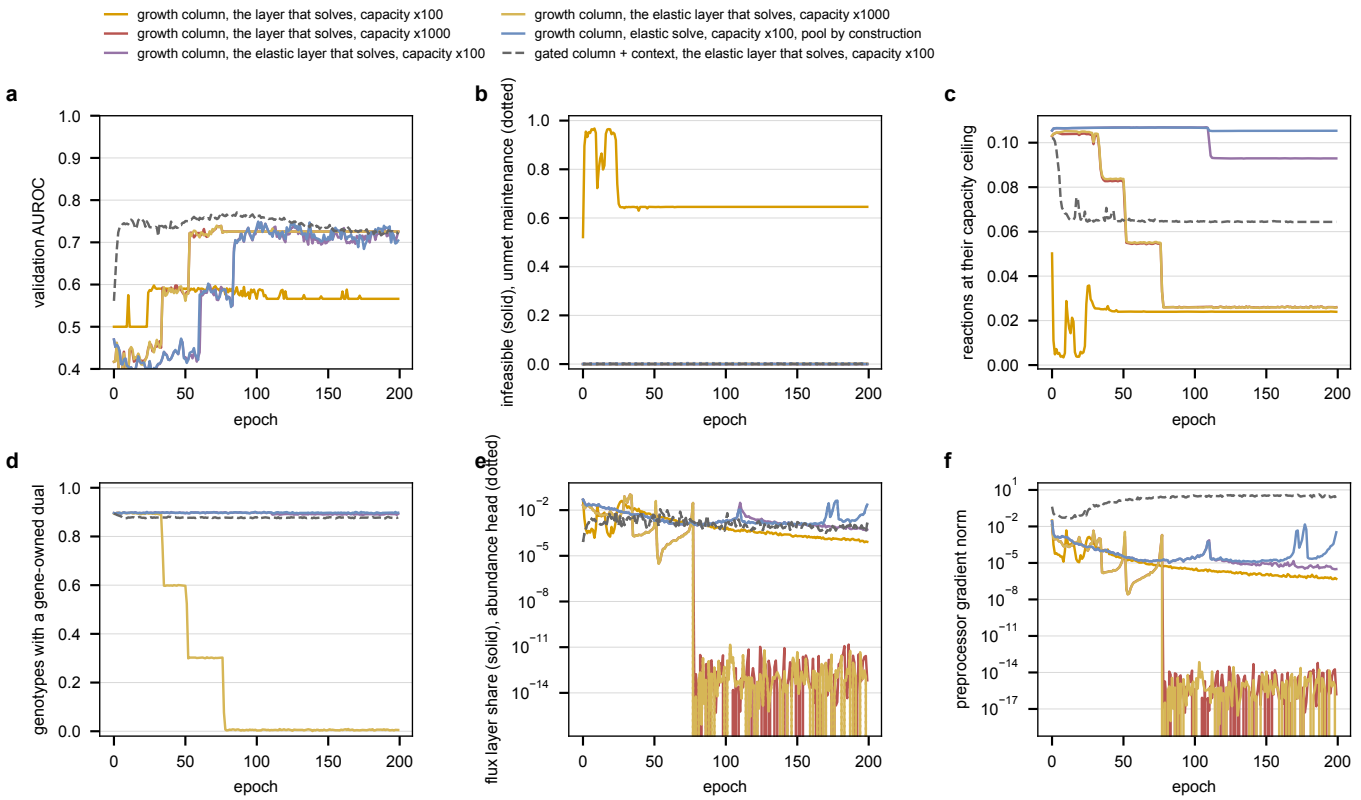



**Figure 39: The layer that solves, probed per epoch.** Seed mean per arm on the cells whose flux is a linear program. **a**, Validation AUROC. **b**, The fraction of genotypes whose program was infeasible (solid) and, on the elastic program, the fraction with unmet maintenance (dotted). **c**, The fraction of gene-owned reactions at their capacity ceiling. **d**, The fraction of genotypes on which a gene-owned bound carries a dual, logged from round 18. **e**, The flux layer's share of the summed per-module gradient norm (solid) and the abundance head's share (dotted, round 18). **f**, The preprocessor's gradient norm. Written by `experiments/028-gene-essentiality/scripts/lp_probe_readout.py`.

**Table 40:** Round 18, the forced column of the elastic layer that solves. The round-4 forced column of the alternating layer (the paired comparator), the round-5 FBA affine and the hard program's forced cells at capacity scale 100 and 1,000 (rounds 15c and 15d), against the elastic program (every forced flux priced at 1 per mmol/gDW/h of shortfall, feasible for every genotype by construction) at scale 100 and 1,000, and at 100 with the protein pool held by construction. Arm `flux_projection_anchored_nodiss`, constraint scale 0, batch 32, class weight, 200 epochs, seeds 2700 to 2702. Bold marks the highest value in a column. ↑ (↓) marks an arm whose held-out AUROC is above (below) the first arm's on every one of the same three seeds.

| arm | mean | sd | seeds | pooled [95% CI] | selected epochs | val step |
|---|---|---|---|---|---|---|
| growth column only, constraints 0 (round-3 levers) | 0.777 | 0.005 | 0.772, 0.781, 0.778 | 0.778 [0.69, 0.86] | 108, 74, 84 | 0.0057 |
| context head + FBA growth ratios (round-3 levers) | **0.821** | 0.020 | 0.814, 0.805, 0.844 ↑ | 0.838 [0.76, 0.91] | 28, 70, 61 | 0.0036 |
| growth column, the layer that solves, capacity x100 | 0.614 | 0.198 | 0.500, 0.843, 0.500 | **0.843** [0.76, 0.91] | 2, 29, 2 | 0.0001 |
| growth column, the layer that solves, capacity x1000 | 0.714 | 0.106 | 0.615, 0.826, 0.702 | 0.819 [0.73, 0.90] | 73, 30, 45 | 0.0000 |
| growth column, the elastic layer that solves, capacity x100 | 0.714 | 0.024 | 0.704, 0.741, 0.697 ↓ | 0.697 [0.56, 0.82] | 106, 132, 180 | 0.0103 |
| growth column, the elastic layer that solves, capacity x1000 | 0.713 | 0.110 | 0.607, 0.826, 0.707 | 0.819 [0.73, 0.90] | 73, 30, 45 | 0.0000 |
| growth column, elastic solve, capacity x100, pool by construction | 0.706 | 0.037 | 0.704, 0.743, 0.670 ↓ | 0.697 [0.57, 0.82] | 106, 126, 187 | 0.0097 |

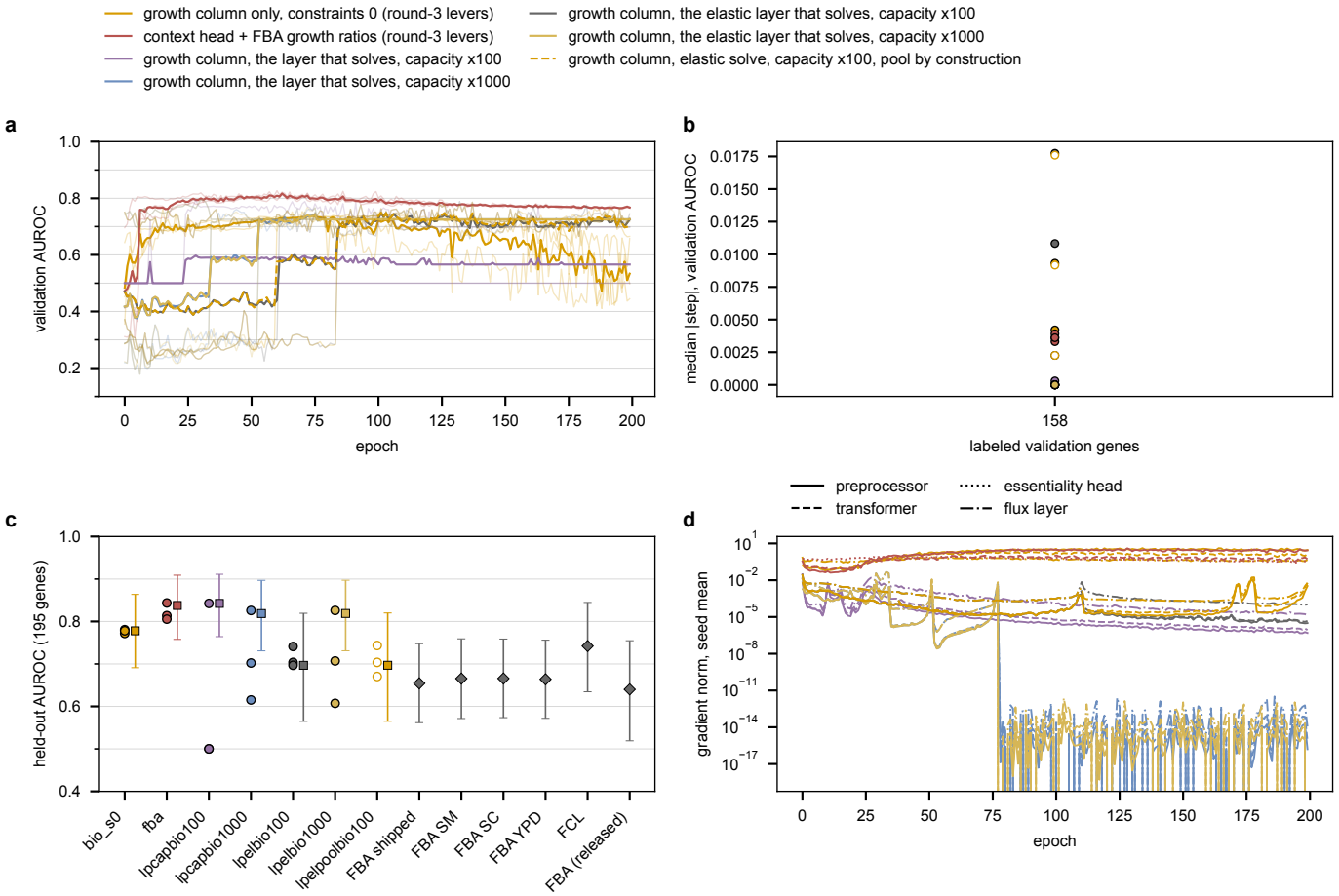



**Figure 40: Round 18, the forced column of the elastic layer that solves. a**, Validation AUROC per epoch, one thin line per seed and the seed mean per arm. **b**, The median absolute epoch-to-epoch step per cell against the validation size. **c**, Held-out AUROC per cell (points), the seed mean and the gene bootstrap of the pooled per-gene logits (square with interval), beside FBA on four media, Flux Cone Learning and the released FBA predictions on the same 195 genes. **d**, Gradient norm per module per epoch, seed mean; the dash-dot line is the flux layer. Written by `experiments/028-gene-essentiality/scripts/wave12_readout.py -round w32`.

**Table 41:** Round 18, the gated column of the elastic layer that solves. The round-4 gated cell (the paired comparator), the round-5 FBA affine, the gated hard program on the GEM box (round 15b) and at capacity scale 100 (round 15c), against the gated elastic program at scale 100. Arm `flux_projection_anchored_nodiss`, constraint scale 0, batch 32, class weight, 200 epochs, seeds 2700 to 2702. Bold marks the highest value in a column. ↑ (↓) marks an arm whose held-out AUROC is above (below) the first arm's on every one of the same three seeds.

| arm | mean | sd | seeds | pooled [95% CI] | selected epochs | val step |
|---|---|---|---|---|---|---|
| gated growth column, constraints 0 (round-3 levers) | 0.771 | 0.010 | 0.766, 0.782, 0.765 | 0.768 [0.68, 0.85] | 67, 71, 77 | 0.0040 |
| context head + FBA growth ratios (round-3 levers) | 0.821 | 0.020 | 0.814, 0.805, 0.844 ↑ | **0.838** [0.76, 0.91] | 28, 70, 61 | 0.0036 |
| gated column + context, GEM-box solve | **0.824** | 0.021 | 0.800, 0.837, 0.835 ↑ | 0.837 [0.76, 0.91] | 2, 103, 4 | 0.0046 |
| gated column + context, the layer that solves, capacity x100 | 0.800 | 0.017 | 0.820, 0.795, 0.786 ↑ | 0.790 [0.71, 0.87] | 6, 88, 63 | 0.0037 |
| gated column + context, elastic solve, x100 | 0.803 | 0.018 | 0.816, 0.810, 0.783 ↑ | 0.800 [0.72, 0.88] | 6, 95, 63 | 0.0038 |

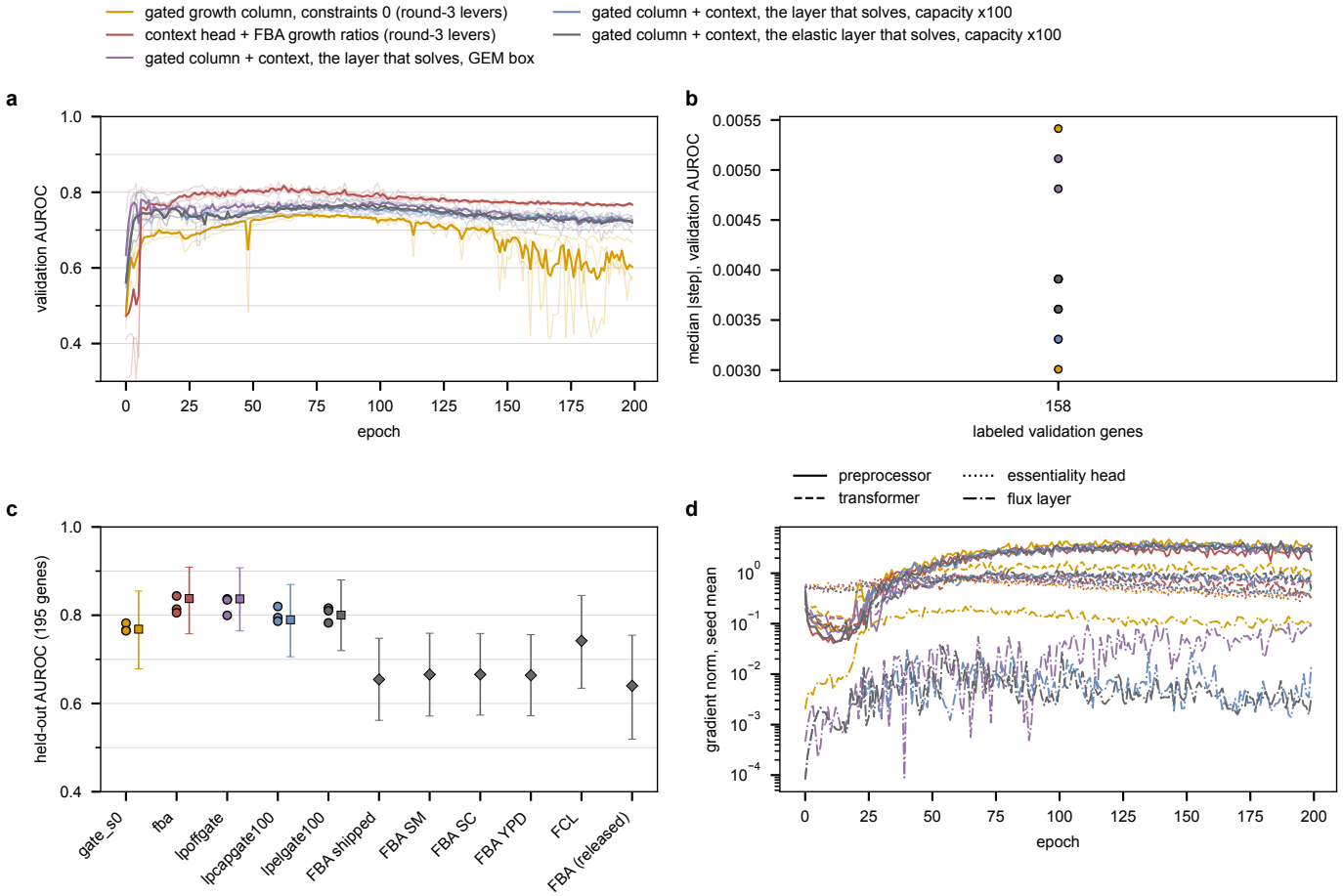



**Figure 41: Round 18, the gated column of the elastic layer that solves.** **a**, Validation AUROC per epoch, one thin line per seed and the seed mean per arm. **b**, The median absolute epoch-to-epoch step per cell against the validation size. **c**, Held-out AUROC per cell (points), the seed mean and the gene bootstrap of the pooled per-gene logits (square with interval), beside FBA on four media, Flux Cone Learning and the released FBA predictions on the same 195 genes. **d**, Gradient norm per module per epoch, seed mean; the dash-dot line is the flux layer. Written by `experiments/028-gene-essentiality/scripts/wave12_readout.py -round w33`.

The construction did what it was built to do and the score did not follow. On every cell no genotype's program is infeasible at any epoch, the maintenance flux is met on every genotype at the selected epoch, and a gene-owned bound carries a dual on 88 to 90 percent of genotypes, so the two seeds that the hard program absorbed at scale 100 now train from the first step (Figure 39b, d). The forced column reads 0.704, 0.741 and 0.697 at scale 100, a seed mean of 0.714 with a spread of 0.02 where the hard program's was 0.20, three of three seeds below the round-4 forced column, and none at 0.80: the read fails on both clauses. At scale 1,000 the elastic cell reads 0.607, 0.826 and 0.707, the hard program's 0.615, 0.826 and 0.702 to within a seed's noise, which is what the construction promises when no program is infeasible, since the two programs are then the same program. With the pool held by construction the column reads 0.704, 0.743 and 0.670: the projection never engaged (the factor stayed at 1 on every batch), so the abundance head never asked for more protein than the budget and the cell is the one without it. The gated cell reads 0.816, 0.810 and 0.783, three of three above the round-4 gated cell and level with the hard gated cell at the same scale (0.800), below the GEM box's 0.824, so its read fails on the second clause.

What the histories add is the diagnosis the score alone would not give. The forced column is not overfitting through the abundance head: its training loss barely moves, from 1.41 to 1.25 on the seed that trains best and from 1.26 to 1.25 on another, against 1.33 to 0.31 for the alternating layer's forced coordinate (round 4) and 1.26 to 0.17 for the gated cell beside it, and its validation curve sits at 0.72 to 0.79, the solve's own ranking at initialization (0.73) with a learned sign: two seeds spend their first sixty epochs at 0.23 to 0.30, the inverted affine, before the sign flips (Figure 40a). The gradient is present, since the duals reach the abundance head on nine genotypes in ten and it carries $10^{-3}$ to $10^{-2}$ of the summed norm (Figure 39e), but the path it travels is stiff: the head is one linear map of the width-12 token, twelve weights shared by every gene, trained at the backbone's rate, and each step moves ceilings whose duals are of order 0.01 per mmol/gDW/h. Under the hard program at scale 1,000 the same probe shows the other failure, the backbone's gradient collapsing to $10^{-13}$ after epoch 78 when no gene-owned bound binds any longer (Figure 39d, e). Round 19 tests the stiffness directly, with the layer's parameters at ten times the learning rate and with a two-layer abundance head, and reads the training loss at the selected epoch against round 18's 1.25 beside the score.

## 4.15 Round 19: the step and the head's capacity ✗

Round 19 (jobs 2326 to 2329) moves the two things that make the path through the solve stiff: the flux layer's parameters at ten times the learning rate (a second optimizer group), a two-layer abundance head in place of the twelve-weight linear map, and both, on the forced elastic column at ceiling scale 100, with both changes behind the gate as well, three seeds each (Tables 42 and 43, Figures 42 and 43); the read is round 18's, paired against the round-18 elastic cells.

**Table 42:** Round 19, the forced elastic column with a faster layer and a deeper head. The round-18 forced elastic cell at capacity scale 100 (the paired comparator), the round-4 forced column and the round-5 FBA affine, against the same elastic cell with the flux layer's parameters at ten times the learning rate, with a two-layer abundance head, and with both. Arm `flux_projection_anchored_nodiss`, constraint scale 0, batch 32, class weight, 200 epochs, seeds 2700 to 2702. Bold marks the highest value in a column. ↑ (↓) marks an arm whose held-out AUROC is above (below) the first arm's on every one of the same three seeds.

| arm | mean | sd | seeds | pooled [95% CI] | selected epochs | val step |
|---|---|---|---|---|---|---|
| growth column, the elastic layer that solves, capacity x100 | 0.714 | 0.024 | 0.704, 0.741, 0.697 | 0.697 [0.56, 0.82] | 106, 132, 180 | 0.0103 |
| growth column only, constraints 0 (round-3 levers) | 0.777 | 0.005 | 0.772, 0.781, 0.778 ↑ | 0.778 [0.69, 0.86] | 108, 74, 84 | 0.0057 |
| context head + FBA growth ratios (round-3 levers) | **0.821** | 0.020 | 0.814, 0.805, 0.844 ↑ | **0.838** [0.76, 0.91] | 28, 70, 61 | 0.0036 |
| growth column, elastic solve, x100, flux layer lr x10 | 0.677 | 0.042 | 0.704, 0.629, 0.697 | 0.699 [0.57, 0.82] | 115, 64, 182 | 0.0044 |
| growth column, elastic solve, x100, two-layer abundance head | 0.752 | 0.072 | 0.823, 0.754, 0.678 | 0.818 [0.74, 0.89] | 2, 10, 85 | 0.0048 |
| growth column, elastic solve, x100, two-layer head + lr x10 | 0.720 | 0.070 | 0.799, 0.691, 0.669 | 0.787 [0.69, 0.87] | 9, 16, 191 | 0.0000 |

**Table 43:** Round 19, the gated elastic column with a faster layer and a deeper head. The round-18 gated elastic cell at capacity scale 100 (the paired comparator), the round-4 gated cell and the round-5 FBA affine, against the gated elastic cell with the two-layer abundance head and the flux layer at ten times the learning rate. Arm `flux_projection_anchored_nodiss`, constraint scale 0, batch 32, class weight, 200 epochs, seeds 2700 to 2702. Bold marks the highest value in a column. ↑ (↓) marks an arm whose held-out AUROC is above (below) the first arm's on every one of the same three seeds.

| arm | mean | sd | seeds | pooled [95% CI] | selected epochs | val step |
|---|---|---|---|---|---|---|
| gated column + context, elastic solve, x100 | 0.803 | 0.018 | 0.816, 0.810, 0.783 | 0.800 [0.72, 0.88] | 6, 95, 63 | 0.0038 |
| gated growth column, constraints 0 (round-3 levers) | 0.771 | 0.010 | 0.766, 0.782, 0.765 ↓ | 0.768 [0.68, 0.85] | 67, 71, 77 | 0.0040 |
| context head + FBA growth ratios (round-3 levers) | **0.821** | 0.020 | 0.814, 0.805, 0.844 | **0.838** [0.76, 0.91] | 28, 70, 61 | 0.0036 |
| gated column + context, elastic solve, x100, two-layer head + lr x10 | 0.774 | 0.025 | 0.786, 0.790, 0.745 ↓ | 0.791 [0.71, 0.87] | 85, 66, 10 | 0.0021 |

Neither lever is the limit. The faster layer reads 0.705, 0.629 and 0.697, below round 18 on two seeds; the deeper head 0.823, 0.754 and 0.678, above on two, with its best seed selected at epoch 2; both together 0.799, 0.691 and 0.669. The training loss at the selected epoch is 1.24 to 1.26 in every cell, round 18's number, and where the deeper head does fit further later in the run (to 1.07 on two seeds) the validation curve has already peaked and decays, so the extra

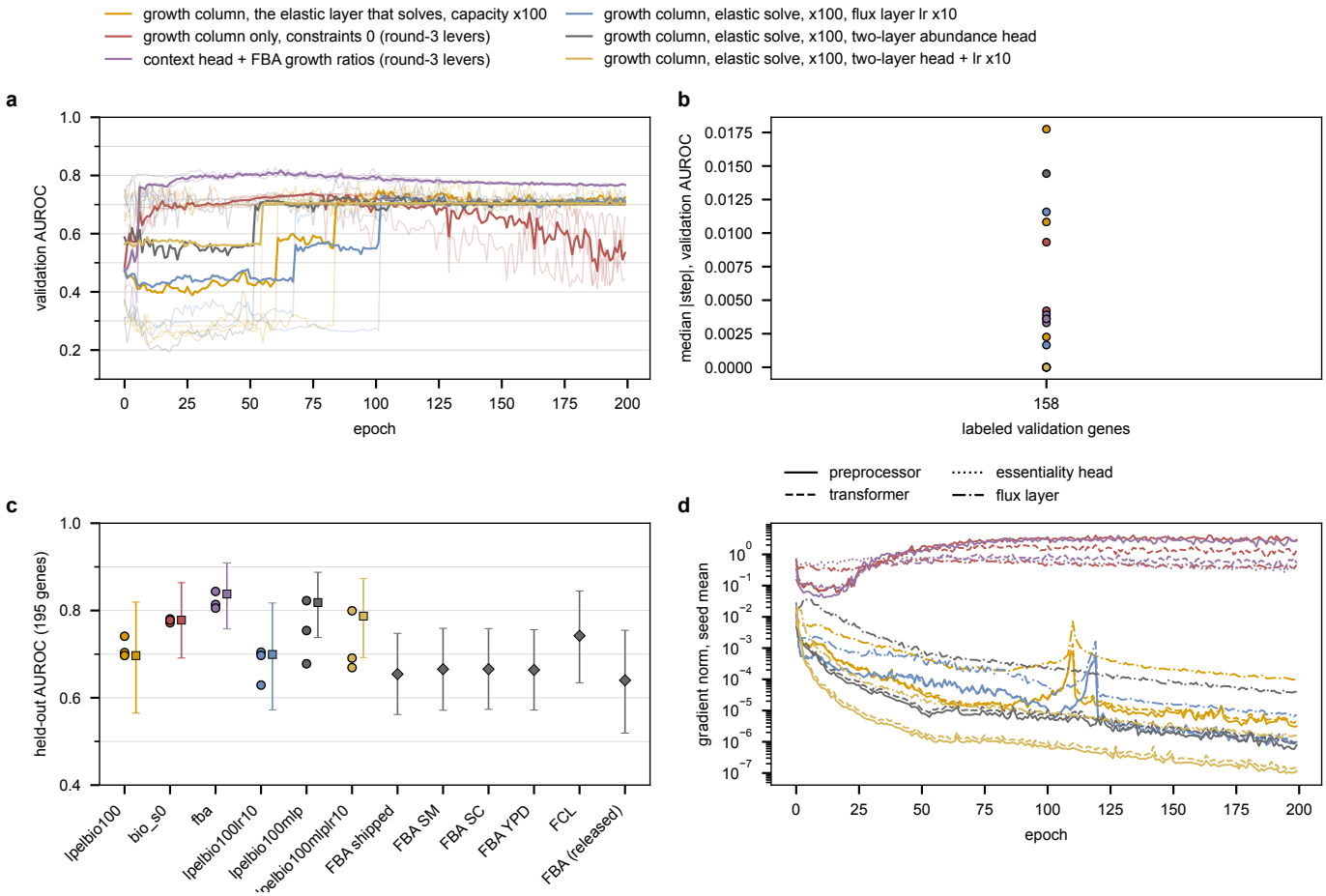



**Figure 42: Round 19, the forced elastic column with a faster layer and a deeper head. a**, Validation AUROC per epoch, one thin line per seed and the seed mean per arm. **b**, The median absolute epoch-to-epoch step per cell against the validation size. **c**, Held-out AUROC per cell (points), the seed mean and the gene bootstrap of the pooled per-gene logits (square with interval), beside FBA on four media, Flux Cone Learning and the released FBA predictions on the same 195 genes. **d**, Gradient norm per module per epoch, seed mean; the dash-dot line is the flux layer. Written by `experiments/028-gene-essentiality/scripts/wave12_readout.py -round w34`.

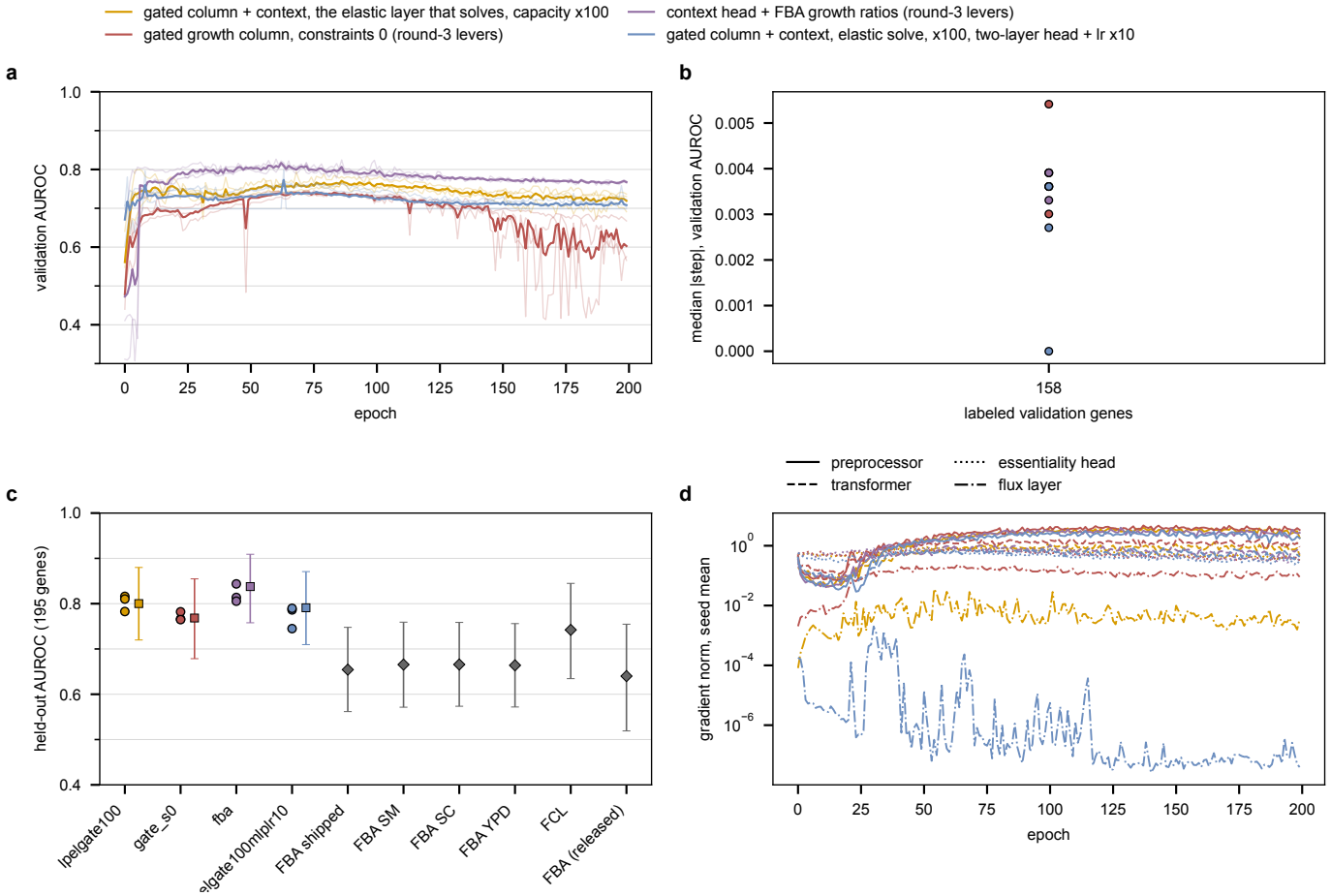



**Figure 43: Round 19, the gated elastic column with a faster layer and a deeper head.** **a**, Validation AUROC per epoch, one thin line per seed and the seed mean per arm. **b**, The median absolute epoch-to-epoch step per cell against the validation size. **c**, Held-out AUROC per cell (points), the seed mean and the gene bootstrap of the pooled per-gene logits (square with interval), beside FBA on four media, Flux Cone Learning and the released FBA predictions on the same 195 genes. **d**, Gradient norm per module per epoch, seed mean; the dash-dot line is the flux layer. Written by `experiments/028-gene-essentiality/scripts/wave12_readout.py -round w35`.

capacity memorizes and does not rank. Behind the gate the two changes cost the cell: 0.786, 0.790 and 0.745 against the round-18 gated cell's 0.816, 0.810 and 0.783, with the gate ending negative on every seed ($-0.51$, $-0.45$, $-0.09$), the head learning to subtract a column the faster layer has bent toward the training set. No arm passes either clause. Rounds 18 and 19 together say that the forced column cannot be fit through the ceilings by any of feasibility, slack, step size or head capacity, and the next subsection measures why.

## 4.16 Why the ceilings cannot carry the ranking ✗

The gradient the layer that solves hands the backbone is the dual of the bounds that bind at the optimum, so what the backbone can learn through the solve is bounded by which bounds those are and whether they differ between genotypes. `lp_dual_overlap.py` reads them off the elastic cell at initialization on 64 held-out genotypes at three ceiling scales (`results/lp_dual_overlap.json`). At scale 100 the median genotype has six gene-owned reactions at their ceiling, of 2,709 that carry a gene, and the median Jaccard overlap between two genotypes' binding sets is 1.0: the same six reactions bind whichever gene is deleted, with a union of 96 over the 64 genotypes carried by one outlier genotype with 90 and by the lethal deletions, whose programs bind nothing. At scale 30 the median is eight reactions with a union of 22, at 1,000 two with a union of four, and the median dual is 0.005 to 0.009 per mmol/gDW/h throughout. The deleted gene's own reactions never bind, because the deletion has already closed them: the gene's identity reaches the value through the shape of the polytope, which the head cannot move, and never through a bound the head owns. The abundance head can therefore raise or lower one growth level for every genotype at once, which the readout's intercept absorbs, and it cannot move one genotype's value relative to another's, which is what the labels ask for. That is the mechanism behind rounds 18 and 19: the elastic program removes the absorbing state and the gradient arrives on nine genotypes in ten, and it arrives on the same six ceilings for all of them. What the growth column carries is the solve's own ranking of the deletions, 0.73 at initialization, with a learned sign and threshold; the constructions that let the module reach the affine's level are the ones that put a learned head beside the column (the gated cells, 0.80 to 0.82) rather than behind it.

## 4.17 Round 20: the gated forms on six seeds, and the last forced probes ✗

Round 20 (jobs 2332 to 2335) does two things the dual-overlap probe asks for. It puts the two gated forms of the layer that solves, the GEM-box cell of round 15b and the elastic cell of round 18, on three more seeds (2703 to 2705) so the reliable form is read on six, and it runs the two probes of the forced column the probe leaves open: the elastic column at ceiling scale 30, where the median genotype has eight binding reactions rather than six, with the flux layer at ten times the learning rate, and the scale-100 column with the abundance head's reach raised from $e^3$ to $e^6$, so a head that wants to move a ceiling by a decade can (Table 44, Figure 44). The gated read is round 18's, extended to six seeds: every seed above its paired gated cell without the solve (round 4, 0.771), and a seed mean at or above the GEM-box gated cell's 0.824. The forced read is unchanged.

**Table 44:** Round 20, the gated forms on six seeds and the last forced probes. The round-4 gated cell (the paired comparator) and the round-5 affine, against the gated layer that solves on the GEM box (round 15b, seeds 2700 to 2702) and on seeds 2703 to 2705, the gated elastic cell at capacity scale 100 (round 18) and on seeds 2703 to 2705, and two last probes of the forced elastic column: at ceiling scale 30 with the flux layer at ten times the learning rate, and at 100 with the abundance head's reach doubled to $e^6$. Constraint scale 0, batch 32, class weight, 200 epochs. Bold marks the highest value in a column. ↑ (↓) marks an arm whose held-out AUROC is above (below) the first arm's on every one of the same three seeds.

| arm | mean | sd | seeds | pooled [95% CI] | selected epochs | val step |
|---|---|---|---|---|---|---|
| gated growth column, constraints 0 (round-3 levers) | 0.771 | 0.010 | 0.766, 0.782, 0.765 | 0.768 [0.68, 0.85] | 67, 71, 77 | 0.0040 |
| context head + FBA growth ratios (round-3 levers) | **0.821** | 0.020 | 0.814, 0.805, 0.844 ↑ | **0.838** [0.76, 0.91] | 28, 70, 61 | 0.0036 |
| gated column + context, GEM-box solve | 0.814 | 0.020 | 0.787 to 0.837 (6) | 0.816 [0.74, 0.89] | 2 to 103 | 0.0045 |
| gated column + context, elastic solve, x100 | 0.804 | 0.013 | 0.783 to 0.816 (6) | 0.803 [0.72, 0.88] | 6 to 95 | 0.0038 |
| growth column, elastic solve, x100 | 0.714 | 0.024 | 0.704, 0.741, 0.697 ↓ | 0.697 [0.56, 0.82] | 106, 132, 180 | 0.0103 |
| growth column, elastic solve, x30, flux layer lr x10 | 0.630 | 0.074 | 0.653, 0.548, 0.690 ↓ | 0.722 [0.61, 0.83] | 110, 2, 187 | 0.0092 |
| growth column, elastic solve, x100, reach $e^6$, flux layer lr x10 | 0.652 | 0.029 | 0.669, 0.619, 0.669 ↓ | 0.741 [0.63, 0.84] | 154, 177, 159 | 0.0032 |

The gated forms are reliable and stop short of the affine. On seeds 2703 to 2705 the GEM-box gated cell reads 0.818, 0.807 and 0.787 and the elastic gated cell 0.795, 0.808 and 0.814; over six seeds the two arms read 0.814 (sd 0.020) and 0.804 (sd 0.013), pooled 0.816 and 0.803, twelve seeds of twelve above the gated cell without the solve (0.771). Neither reaches the second clause: the GEM-box form's three-seed 0.824 was the top of its spread, and six seeds put both forms 0.01 to 0.02 under the affine's 0.821 (0.814, 0.805 and 0.844 by seed), a gap inside one seed standard deviation of either arm. The gate ends at $-0.95$, 0.68 and 0.80 on the GEM box and $-1.01$, $-1.41$ and $-0.79$ on the elastic program: the head uses the column with either sign, and the sign is not what the seeds agree on. The forced probes fail as the probe of Section 4.16 predicts. At scale 30 with the faster layer the column reads 0.653, 0.548 and 0.690, the middle seed selected at epoch 2 with the maintenance slack active on every genotype; at scale 100 with the head's reach at $e^6$ it reads 0.669, 0.619 and 0.669, and the two better seeds end with no gene-owned dual at all, the head having pushed

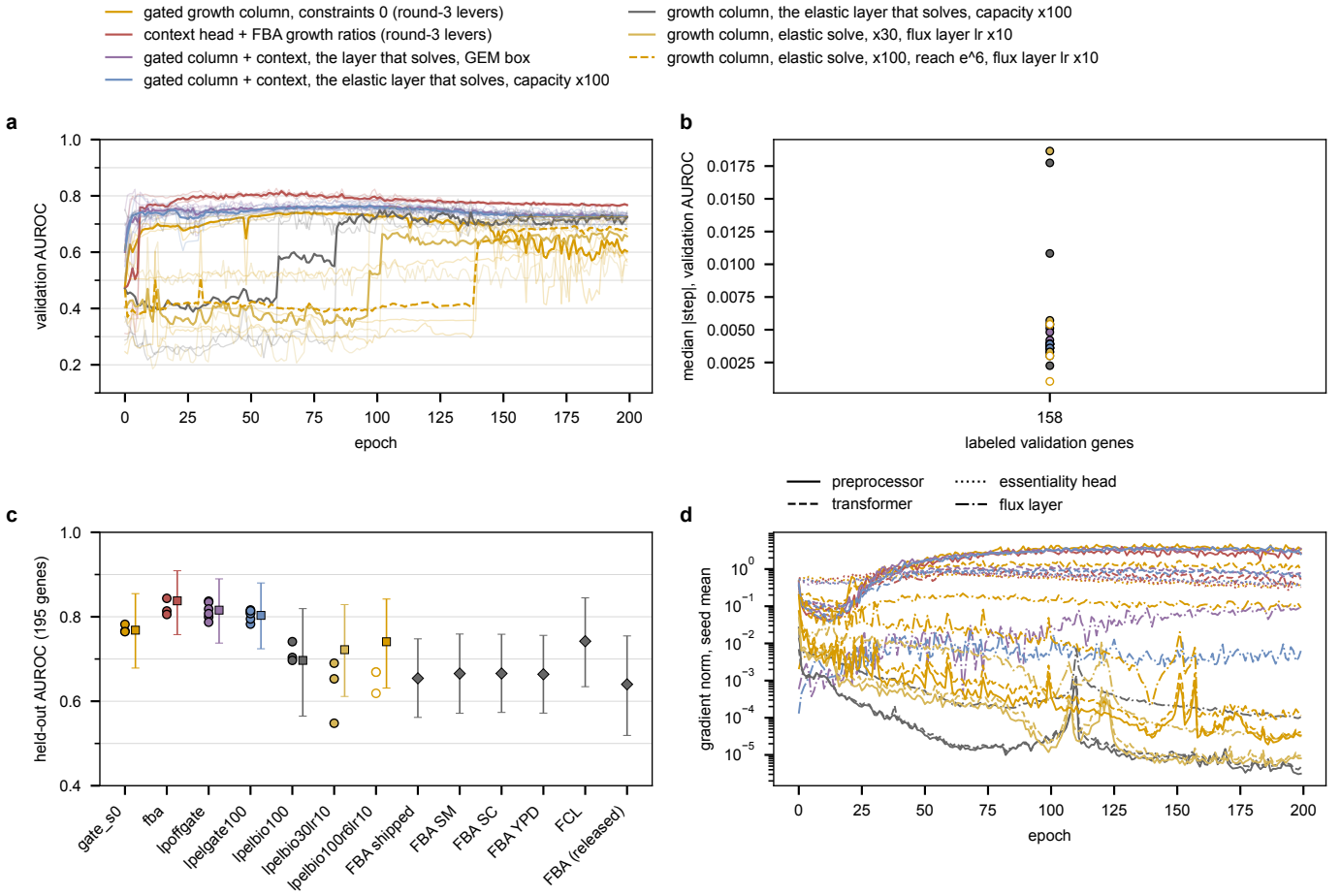



**Figure 44: Round 20, the gated forms on six seeds and the last forced probes.** **a**, Validation AUROC per epoch, one thin line per seed and the seed mean per arm. **b**, The median absolute epoch-to-epoch step per cell against the validation size. **c**, Held-out AUROC per cell (points), the seed mean and the gene bootstrap of the pooled per-gene logits (square with interval), beside FBA on four media, Flux Cone Learning and the released FBA predictions on the same 195 genes. **d**, Gradient norm per module per epoch, seed mean; the dash-dot line is the flux layer. Written by `experiments/028-gene-essentiality/scripts/wave12_readout.py -round w36`.

every ceiling off the optimum, while the third sits in full slack. A wider reach hands the head both exits from the ceilings, above the optimum and below the maintenance, and neither one ranks the deletions.

## 4.18 Round 21: the gated cell with a head over the solved flux ✗

If the ceilings cannot move one genotype's growth against another's, the thing the solve does hold per genotype is the flux vector it returns: round 11 measured the deletion's own pFBA vector at 0.77 as an encoding, above its growth ratio at 0.65. Round 21 (jobs 2336 and 2337) reads that vector from inside the cell. The `gated_flux` readout (Figure 45) is the round-18 gated elastic cell plus a zero-initialized linear head over $\log(1 + |v^*_{\mathcal{R}}|)$, $v^*$ the optimum the elastic program returns, detached, over all 4,131 reactions or over the 274 exchange reactions (the uptake and secretion profile), constraint scale 0, three seeds (Table 45, Figure 46). The head's weight norm is logged each epoch. The read is paired against the round-18 gated elastic cell by seed, the pass being three of three above it and a seed mean at or above the affine's 0.821.

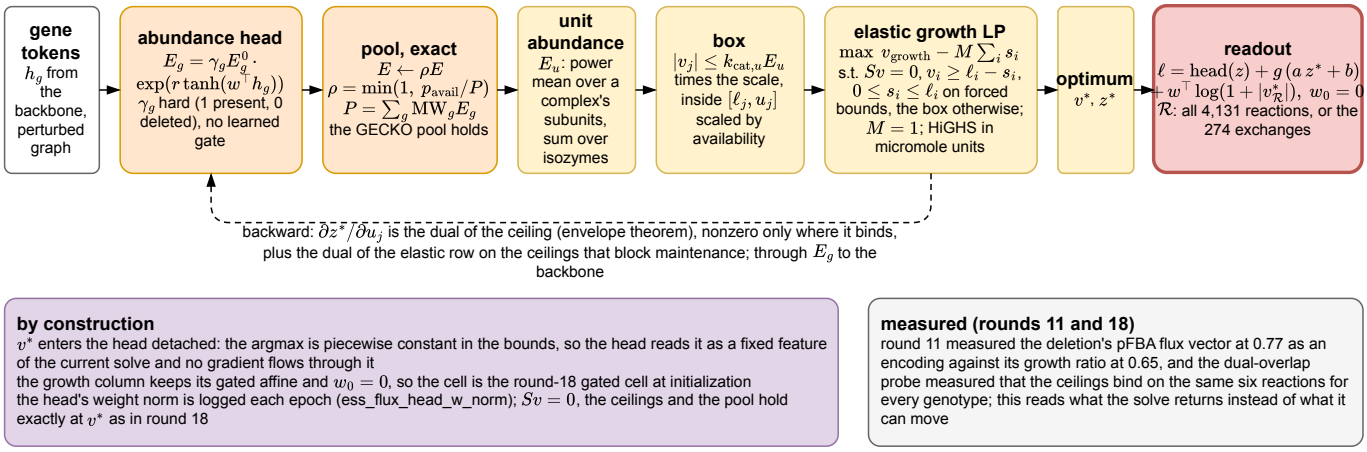



**Figure 45: The gated elastic cell with a head over the solved flux, round 21.** One block changes against Figure 38, drawn with a heavier border: the readout adds a zero-initialized linear head over $\log(1 + |v^*_{\mathcal{R}}|)$, the flux vector the solve returns restricted to a reaction set $\mathcal{R}$ (all 4,131 reactions, or the 274 exchange reactions). The vector enters detached, since the argmax is piecewise constant in the bounds, so the head reads the solve's answer as a feature and the backbone keeps receiving the duals through the growth column alone. The growth column keeps its gated affine, so the cell is the round-18 gated cell at initialization. Colors as in Figure 16. Drawn in draw.io from `notes/assets/drawio/028-module.gen.py`.

**Table 45:** Round 21, the gated elastic cell with a head over the solved flux. The round-18 gated elastic cell (the paired comparator) and the round-5 affine, against the same cell with a zero-initialized linear head over $\log(1 + |v^*|)$ of the solved flux, over all 4,131 reactions and over the 274 exchange reactions. Constraint scale 0, batch 32, class weight, 200 epochs, seeds 2700 to 2702. Bold marks the highest value in a column. ↑ (↓) marks an arm whose held-out AUROC is above (below) the first arm's on every one of the same three seeds.

| arm | mean | sd | seeds | pooled [95% CI] | selected epochs | val step |
|---|---|---|---|---|---|---|
| gated column + context, elastic solve, x100 | 0.803 | 0.018 | 0.816, 0.810, 0.783 | 0.800 [0.72, 0.88] | 6, 95, 63 | 0.0038 |
| context head + FBA growth ratios (round-3 levers) | 0.821 | 0.020 | 0.814, 0.805, 0.844 | 0.838 [0.76, 0.91] | 28, 70, 61 | 0.0036 |
| gated + head over all solved fluxes | **0.868** | 0.015 | 0.883, 0.853, 0.868 ↑ | **0.868** [0.79, 0.93] | 66, 83, 91 | 0.0158 |
| gated + head over the solved exchange fluxes | 0.835 | 0.020 | 0.831, 0.856, 0.817 ↑ | 0.843 [0.76, 0.92] | 83, 197, 89 | 0.0045 |

Both heads pass, and the head over every reaction passes by more. Over all 4,131 reactions the cell reads 0.883, 0.853 and 0.868 (mean 0.868, sd 0.015; pooled 0.868 with a gene bootstrap of 0.792 to 0.934), above the round-18 gated elastic cell on every seed (0.816, 0.810, 0.783) and 0.047 above the affine's 0.821, which is the first module cell to pass a pre-registered read in this document and the first to sit above the affine on the ProtT5 backbone. Over the 274 exchange reactions it reads 0.831, 0.856 and 0.817 (mean 0.835), also above its comparator on every seed and above the affine's mean, with one seed selected at epoch 197. Two logged quantities say what changed. The training loss at the selected epoch is 0.61 to 0.79, where every forced cell of rounds 15 to 20 sat at 1.24 to 1.26: the head fits, and the growth column never did. The gate on the growth column ends at 0.01, 0.05 and 0.09 under the full head and 0.24 to 0.42 under the exchange head, so the growth column is used little or not at all once the head can read the vector behind it, and the head's weight norm grows to 3.3 to 5.6 from zero. What is measured is that the flux vector the elastic program returns for a deletion, read linearly, ranks the held-out genes at 0.87 on a backbone whose own pooled cell ranks them at 0.77. Which part of the cell makes that vector informative, the deletion's closure of its reactions through the GPR rules, the learned ceilings that reshape the optimum per genotype, or the head's reach over the exchange and transport columns, is what Section 4.20 separates.

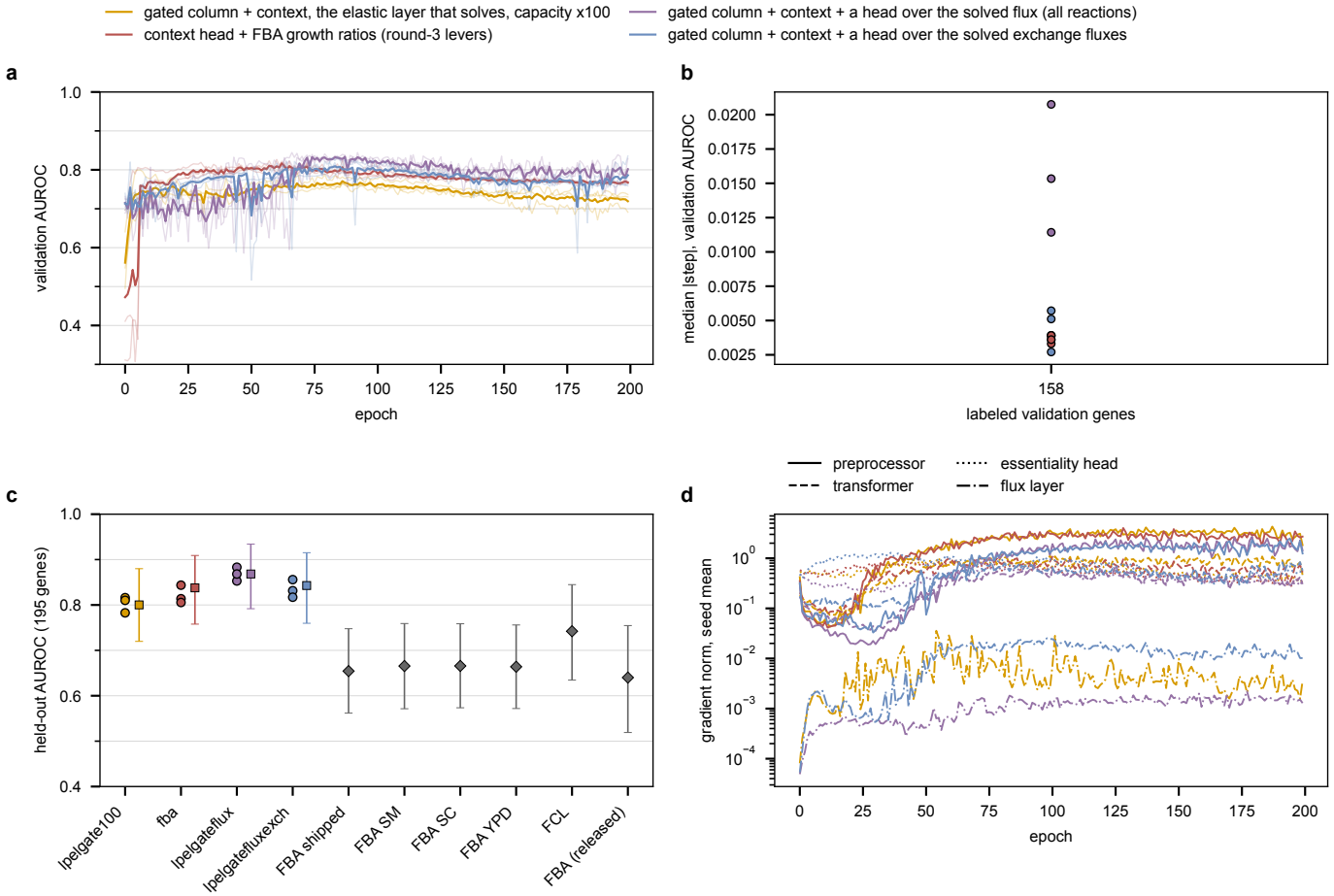



**Figure 46: Round 21, the gated elastic cell with a head over the solved flux.** **a**, Validation AUROC per epoch, one thin line per seed and the seed mean per arm. **b**, The median absolute epoch-to-epoch step per cell against the validation size. **c**, Held-out AUROC per cell (points), the seed mean and the gene bootstrap of the pooled per-gene logits (square with interval), beside FBA on four media, Flux Cone Learning and the released FBA predictions on the same 195 genes. **d**, Gradient norm per module per epoch, seed mean; the dash-dot line is the flux layer. Written by `experiments/028-gene-essentiality/scripts/wave12_readout.py -round w37`.

## 4.19 Round 23: the module on the ladder's top input ✗

Every module cell so far sits on the ProtT5 backbone, where the affine adds 0.05 to the pooled cell (0.771 to 0.821). Section 2.6 found the input that carries the rest of the ladder, the 173-column pathway multi-hot beside ProtT5, and round 22 (Section 3.13) runs the affine on it. Round 23 (jobs 2341 to 2344) asks whether the module delivers the same bump there: the three gated forms of the layer that solves, the GEM-box solve of round 15b, the elastic solve of round 18 and the elastic solve with the flux head of round 21, each on ProtT5 with the pathway multi-hot, beside the pooled cell on that input with no readout as the module-free control, three seeds (Table 46, Figure 47). The read pairs each form by seed against the control and against the round-22 affine on the same input: a form passes if it is above the control on three seeds of three and at or above the affine on three of three.

**Table 46:** Round 23, the gated module forms on the ladder's top input. The pooled cell on ProtT5 with the pathway multi-hot and no readout (the module-free control) and the round-22 FBA affine on that input (the paired comparator), against the three gated forms of the layer that solves on the same input: the GEM-box solve (round 15b), the elastic solve at capacity scale 100 (round 18) and the elastic solve with the head over the solved flux (round 21). Constraint scale 0, batch 32, class weight, 200 epochs, seeds 2700 to 2702. Bold marks the highest value in a column. ↑ (↓) marks an arm whose held-out AUROC is above (below) the first arm's on every one of the same three seeds.

| arm | mean | sd | seeds | pooled [95% CI] | selected epochs | val step |
|---|---|---|---|---|---|---|
| pathway + ProtT5, no readout | 0.860 | 0.002 | 0.859, 0.859, 0.862 | 0.861 [0.79, 0.92] | 36, 56, 46 | 0.0040 |
| pathway + ProtT5 + FBA ratios | **0.890** | 0.005 | 0.889, 0.895, 0.885 ↑ | 0.890 [0.83, 0.94] | 31, 45, 55 | 0.0054 |
| pathway + ProtT5; gated column + context, GEM-box solve | 0.868 | 0.005 | 0.862, 0.871, 0.869 ↑ | 0.867 [0.80, 0.93] | 47, 61, 40 | 0.0052 |
| pathway + ProtT5; gated column + context, elastic solve, x100 | 0.867 | 0.007 | 0.859, 0.872, 0.869 | 0.872 [0.81, 0.93] | 43, 58, 34 | 0.0048 |
| pathway + ProtT5; gated + head over all solved fluxes | 0.845 | 0.087 | 0.716 to 0.911 (6) | **0.910** [0.86, 0.95] | 11 to 150 | 0.0203 |

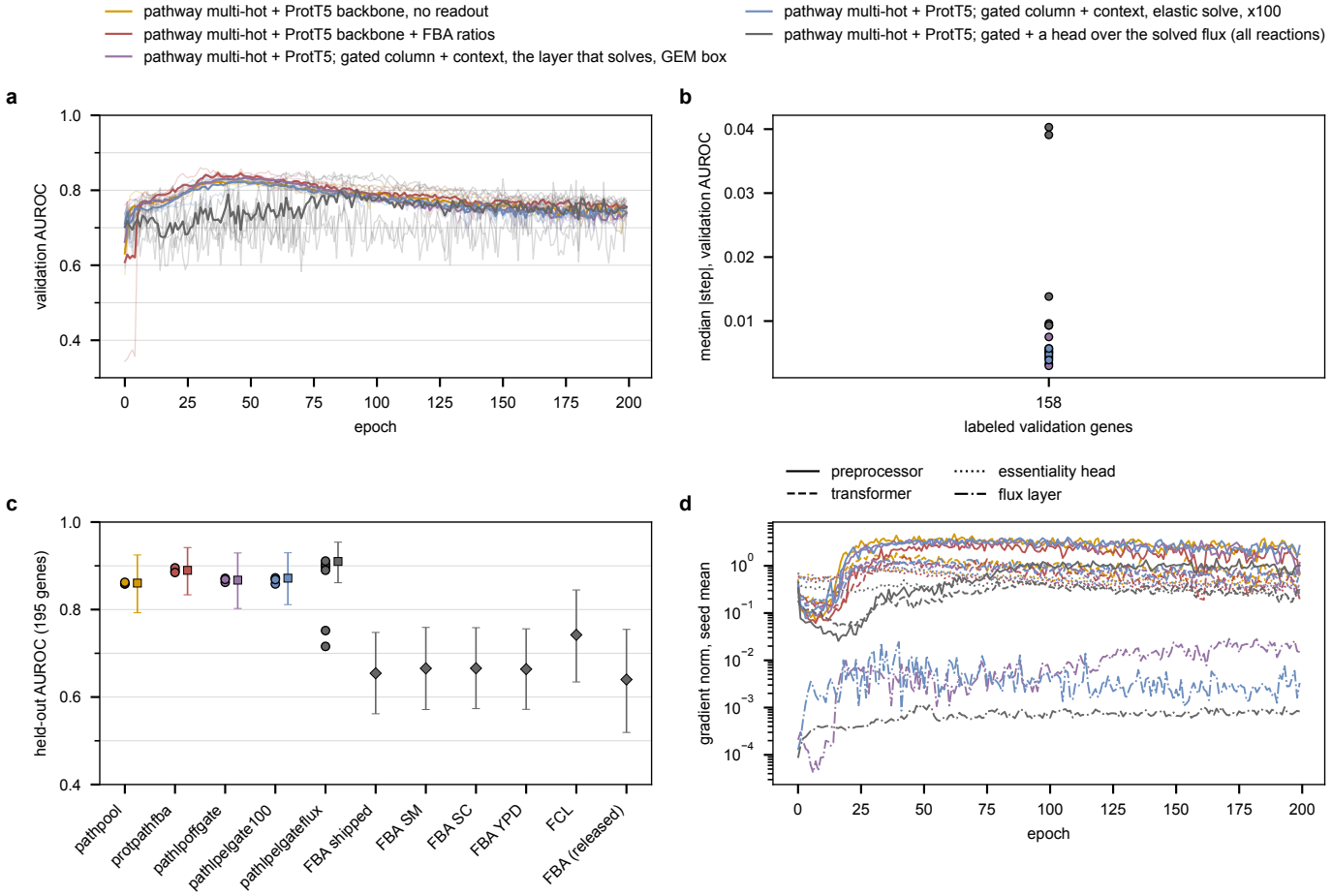



**Figure 47: Round 23, the gated module forms on ProtT5 with the pathway multi-hot. a**, Validation AUROC per epoch, one thin line per seed and the seed mean per arm. **b**, The median absolute epoch-to-epoch step per cell against the validation size. **c**, Held-out AUROC per cell (points), the seed mean and the gene bootstrap of the pooled per-gene logits (square with interval), beside FBA on four media, Flux Cone Learning and the released FBA predictions on the same 195 genes. **d**, Gradient norm per module per epoch, seed mean; the dash-dot line is the flux layer. Written by `experiments/028-gene-essentiality/scripts/wave12_readout.py -round w39`.

The input does most of the work, the affine the rest, and the flux head is the one form that reaches past it. The pooled cell on ProtT5 with the pathway multi-hot and no readout reads 0.859, 0.859 and 0.862 (mean 0.860), 0.09 above the same cell on ProtT5 alone, and the affine on that input reads 0.889, 0.895 and 0.885 (mean 0.890, Section 3.13). The

two gated forms without the head sit between: the GEM-box solve at 0.862, 0.871 and 0.869 and the elastic solve at 0.859, 0.872 and 0.869, above the control on three seeds and on two, and below the affine on three of three. On seed 2700 the elastic cell's gate ended at exactly zero and its held-out score is the control's to four decimals, which is what a closed gate means: the cell is the pooled model. The flux head on this input reads 0.897, 0.716 and 0.903. The two seeds that fit are the two highest single cells in this document, 0.008 and 0.018 above the affine on their seeds, with the pooled per-gene logits of all three at 0.907; the third seed never fit, its training loss 1.02 at the selected epoch against 0.55 and 0.74, its validation curve flat at 0.69 to 0.72 for 200 epochs where the other two pass through a late transition near epoch 80 to a loss of 0.2 to 0.3. So the form fails the read as written, above the control on two seeds of three, and its seed mean of 0.839 is set by the seed that did not fit. Three more seeds (round 25) read 0.890, 0.752 and 0.911, the middle one unfit again, so over six seeds the form reads 0.897, 0.716, 0.903, 0.890, 0.752 and 0.911: four cells at 0.890 to 0.911, the highest single transformer cells in the document, and two at 0.72 to 0.75 that never left a training loss of 1.0, a seed mean of 0.845 with a standard deviation of 0.087. The pooled per-gene logits of the six read 0.910 [0.86, 0.95], the highest bootstrap point in the document, the two unfit seeds diluted rather than decisive in the mean logit. Hypothesis, untested: the transition is delayed rather than absent on the unfit seeds, since the round-21 cells on ProtT5 alone fit by epoch 66 to 91 on every seed and the pathway input adds 173 columns the head has to learn around; a longer budget on an unfit seed is the direct test.

## 4.20 Round 24: what carries the head over the solved flux ✗

Round 21's first pack passed its read (Section 4.18), and three things could be carrying it: the flux vector the program returns under the deletion, which the ladder's pFBA-vector encoding put at 0.77 on its own; the learned ceilings, which shape that vector per genotype through the abundance head; and the linear head's choice of reactions. Round 24 (jobs 2347 to 2349) separates them. The same head over the flux of the GEM-box solve (round 15b's program, no ceilings and no abundance head) says whether the enzyme-constrained part is needed: at or above round 21 on three seeds of three and it is not, below on three of three and it is, anything else undecided at this size. The head over the 2,709 gene-owned reactions only says whether the exchange and transport columns matter. And the passing cell runs on seeds 2703 to 2705, so its read stands on six (Table 47, Figure 49).

Figure 48 draws the head over the GEM-box solve.

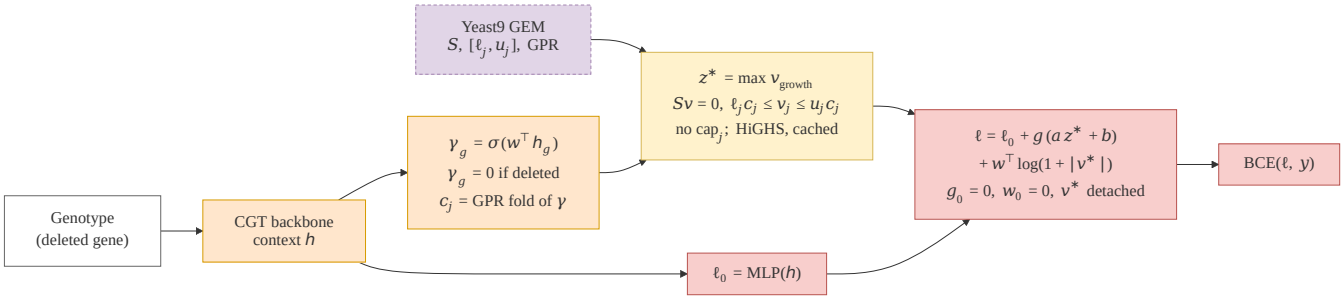



**Figure 48: The solve on the GEM's own box with a flux head, rounds 24 and 27.** The gated flux readout of Figure 45 with the abundance head, the $k_{cat}$ table and the capacity ceilings removed: the linear program runs on the GEM's bounds scaled by the availability gate (Figure 34), and a zero-initialized linear head reads $\log(1 + |v^*|)$ of the detached optimum. From round 27 the solver stores each solved program and reuses it when its bounds carry no gradient. Colors as in Figure 16.

**Table 47:** Round 24, what carries the head over the solved flux. The round-18 gated elastic cell, the round-5 affine on ProtT5 and the round-9 affine on the chromosome-pathway indicator with ProtT5 (the comparators), the round-21 cells (the head over all solved fluxes, on six seeds once seeds 2703 to 2705 land; the head over the 274 exchange fluxes), against the head over the 2,709 gene-owned fluxes, and the head over the flux of the GEM-box solve with no learned ceilings. Constraint scale 0, batch 32, class weight, 200 epochs. Bold marks the highest value in a column. ↑ (↓) marks an arm whose held-out AUROC is above (below) the first arm's on every one of the same three seeds.

| arm | mean | sd | seeds | pooled [95% CI] | selected epochs | val step |
|---|---|---|---|---|---|---|
| gated column + context, elastic solve, x100 | 0.803 | 0.018 | 0.816, 0.810, 0.783 | 0.800 [0.72, 0.88] | 6, 95, 63 | 0.0038 |
| context head + FBA growth ratios (round-3 levers) | 0.821 | 0.020 | 0.814, 0.805, 0.844 | 0.838 [0.76, 0.91] | 28, 70, 61 | 0.0036 |
| chrom-pathway + ProtT5 + FBA ratios | **0.871** | 0.015 | 0.860, 0.866, 0.888 ↑ | **0.868** [0.80, 0.93] | 53, 51, 38 | 0.0030 |
| gated + head over all solved fluxes | 0.862 | 0.013 | 0.847 to 0.883 (6) | 0.866 [0.79, 0.93] | 66 to 135 | 0.0179 |
| gated + head over the solved exchange fluxes | 0.835 | 0.020 | 0.831, 0.856, 0.817 ↑ | 0.843 [0.76, 0.92] | 83, 197, 89 | 0.0045 |
| gated + head over the solved gene-owned fluxes | 0.861 | 0.009 | 0.871, 0.860, 0.853 ↑ | 0.861 [0.78, 0.93] | 93, 86, 75 | 0.0197 |
| gated + head over the GEM-box solve's flux (all reactions) | 0.852 | 0.019 | 0.873, 0.836, 0.847 ↑ | 0.860 [0.78, 0.93] | 68, 164, 49 | 0.0184 |

The vector carries it, the ceilings sharpen it a little, and the gene-owned columns are where it lives. On six seeds the head over every solved flux reads 0.862 (sd 0.013, pooled 0.866 with a gene bootstrap of 0.789 to 0.933), every seed

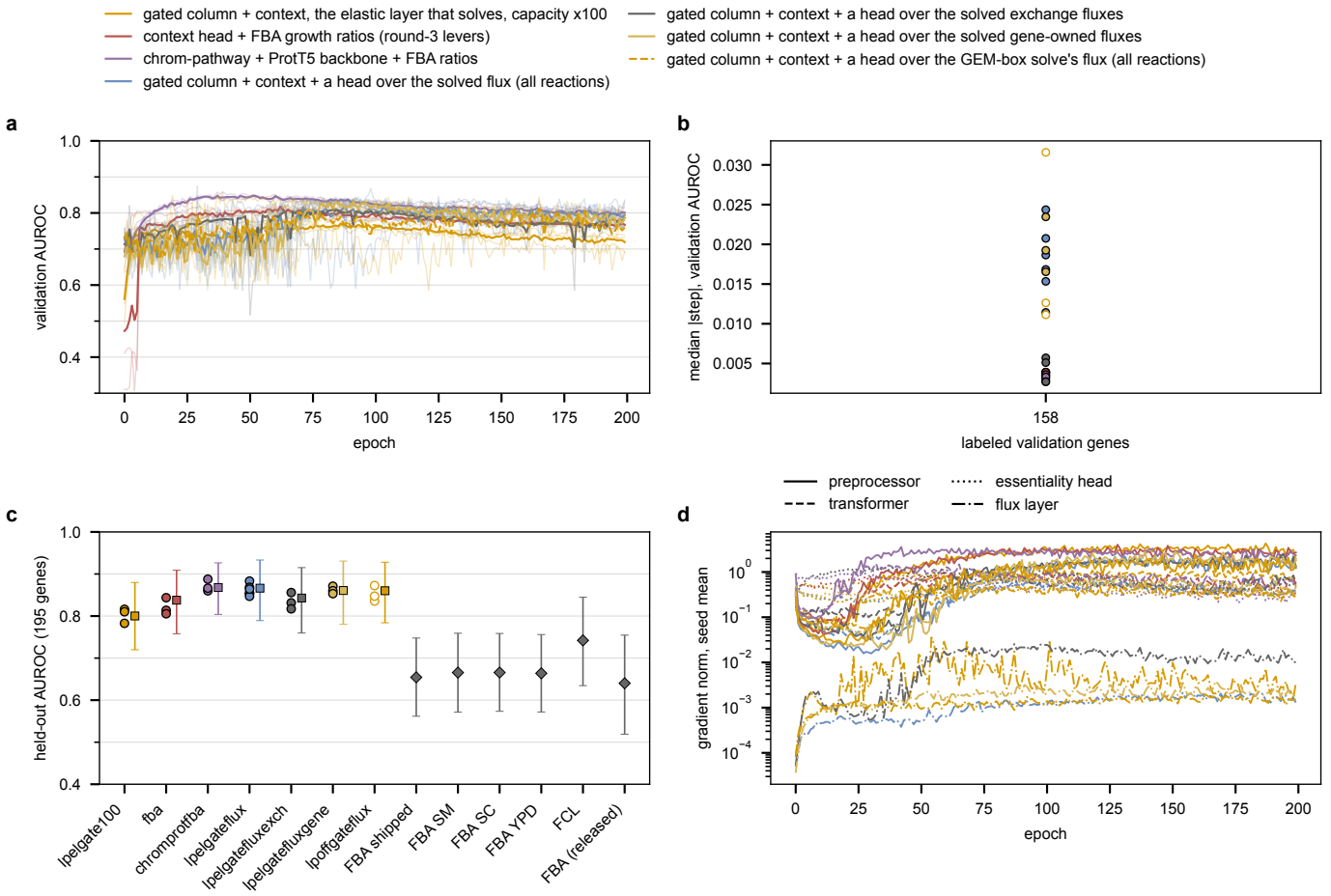



**Figure 49: Round 24, what carries the head over the solved flux.** **a**, Validation AUROC per epoch, one thin line per seed and the seed mean per arm. **b**, The median absolute epoch-to-epoch step per cell against the validation size. **c**, Held-out AUROC per cell (points), the seed mean and the gene bootstrap of the pooled per-gene logits (square with interval), beside FBA on four media, Flux Cone Learning and the released FBA predictions on the same 195 genes. **d**, Gradient norm per module per epoch, seed mean; the dash-dot line is the flux layer. Written by `experiments/028-gene-essentiality/scripts/wave12_readout.py -round w40`.

above its paired gated elastic cell, so the round-21 read holds on six. The same head over the GEM-box solve, with nothing learned in the layer, reads 0.873, 0.836 and 0.847 (mean 0.852), below the elastic cell on three seeds of three by 0.010, 0.017 and 0.021: by the read as written the learned ceilings are needed, and what they add is inside one seed standard deviation of either arm. With nothing learned the flux is a function of the genotype and the medium alone, and a linear head over it still sits 0.03 above the affine and 0.08 above the ladder's linear rung on the deletion pFBA vector (0.771); which of the transformer's context head, the class-weighted training, or the difference between the program's argmax and the parsimonious point accounts for that gap is not measured. The head over the 2,709 gene-owned reactions reads 0.871, 0.860 and 0.853 (mean 0.861), level with the full head (below on two seeds, above on one), where the 274 exchange fluxes alone read 0.835: the uptake and secretion profile is the part of the vector that carries least.

## 4.21 Round 25: the flux head under the reported protocol ✗

The document's number is a fold ensemble (Section 3): five models trained on the five cross-validation files, their logits averaged on the held-out genes. Round 25 (jobs 2352 to 2354) runs the round-23 flux-head cell, the gated elastic solve with the head over every solved flux on ProtT5 with the pathway multi-hot, under that protocol, and on seeds 2703 to 2705 of the split file so the single-split read stands on six seeds and the number of seeds that miss the fitting transition is counted out of six rather than three. The ensemble is read beside the round-12 ensemble's 0.893 with its gene bootstrap, with no pass rule (Table 48); the six-seed read is round 23's.

**Table 48:** Every fold ensemble in the document on the 195 held-out genes: five models trained on the five cross-validation files (each a different fifth of the 787 labeled non-test genes held for validation), their per-gene logits averaged and scored once, Flux Cone Learning's own test protocol. Fold AUROCs are the single models on the held-out genes; the ensemble carries a 2,000-draw gene bootstrap. Written by `cv_ensemble_readout.py`. Bold marks the highest value in a column.

| cell | folds | fold AUROCs | fold mean | ensemble [95% CI] |
|---|---|---|---|---|
| round 3: pooled transformer, ProtT5, no readout | 5 | 0.780, 0.806, 0.797, 0.751, 0.789 | 0.785 | 0.795 [0.71, 0.88] |
| round 5: ProtT5 + FBA affine | 5 | 0.814, 0.824, 0.839, 0.775, 0.853 | 0.821 | 0.822 [0.74, 0.90] |
| round 12: chromosome-pathway indicator + ProtT5 + FBA affine (the reported number) | 5 | 0.860, 0.899, 0.889, 0.875, 0.884 | 0.881 | 0.893 [0.84, 0.94] |
| round 17: chromosome-pathway indicator + ProtT5 + the widened affine (ratios, statistics, pFBA vector) | 5 | 0.863, 0.860, 0.887, 0.840, 0.882 | 0.867 | 0.901 [0.85, 0.95] |
| round 25: pathway multi-hot + ProtT5, gated elastic solve + the head over the solved flux | 5 | 0.901, 0.702, 0.859, 0.904, 0.687 | 0.811 | 0.897 [0.84, 0.95] |
| round 27: pathway multi-hot + ProtT5, the head over the cached GEM-box solve (nothing learned in the layer) | 5 | 0.912, 0.859, 0.918, 0.910, 0.901 | 0.900 | 0.912 [0.86, 0.95] |
| round 34: pathway multi-hot + ProtT5, the deletion's pFBA vector as $\log(1+\lvert v\rvert)$ appended to the affine (no layer) | 5 | 0.911, 0.901, 0.838, 0.911, 0.913 | 0.895 | 0.920 [0.87, 0.96] |
| round 34: the same, three seeds per fold (each fold model the seed-mean logit) | 5 | 0.910, 0.907, 0.913, 0.900, 0.915 | 0.909 | **0.921** [0.87, 0.96] |
| round 37: the same cell over the interior-point store (HiGHS ipm, no crossover, KKT $10^{-4}$) | 5 | 0.913, 0.836, 0.881, 0.897, 0.911 | 0.888 | 0.909 [0.86, 0.95] |
| round 37: the interior-point store, three seeds per fold | 5 | 0.915, 0.851, 0.887, 0.884, 0.926 | 0.893 | 0.912 [0.86, 0.95] |
| round 39: the same cell over the PDLP store (HiGHS PDLP, no crossover, KKT $10^{-4}$) | 5 | 0.921, 0.859, 0.887, 0.878, 0.907 | 0.890 | 0.909 [0.86, 0.95] |
| round 73: the same cell over the dual-simplex vertex store with its 429 clamp-carrying columns dropped | 5 | 0.876, 0.808, 0.913, 0.862, 0.898 | 0.871 | 0.901 [0.85, 0.95] |
| round 39: the PDLP store, three seeds per fold | 5 | 0.915, 0.881, 0.884, 0.880, 0.908 | 0.894 | 0.911 [0.86, 0.95] |
| round 54: the layer-free cell at width 48, one seed per fold | 5 | 0.881, 0.912, 0.927, 0.880, 0.908 | 0.902 | 0.917 [0.87, 0.96] |
| round 54: the width-48 cell, three seeds per fold | 5 | 0.917, 0.895, 0.919, 0.894, 0.916 | 0.908 | 0.920 [0.87, 0.96] |
| round 58: GO and the $\log(1+\lvert v\rvert)$ vector as affine columns, one seed per fold | 5 | 0.911, 0.907, 0.896, 0.886, 0.884 | 0.897 | 0.918 [0.86, 0.97] |
| round 58: the same, two seeds per fold | 5 | 0.907, 0.907, 0.894, 0.880, 0.886 | 0.895 | 0.917 [0.86, 0.97] |
| round 62: GO reduced to 64 components and the vector as affine columns, one seed per fold | 5 | 0.905, 0.880, 0.843, 0.873, 0.917 | 0.884 | 0.914 [0.86, 0.96] |
| round 62: the same, three seeds per fold | 5 | 0.917, 0.878, 0.843, 0.901, 0.911 | 0.890 | 0.917 [0.86, 0.96] |
| round 66: the round-33 cell on the fixed one-hot code, kernels pinned, one seed per fold | 5 | 0.912, 0.911, 0.830, 0.886, 0.914 | 0.891 | 0.917 [0.87, 0.96] |
| round 66: the same, three seeds per fold | 5 | 0.911, 0.909, 0.913, 0.894, 0.915 | 0.908 | 0.920 [0.87, 0.96] |
| MLP on the transformer's three inputs (no graph) | 5 | 0.870, 0.892, 0.880, 0.866, 0.884 | 0.878 | 0.886 [0.82, 0.94] |
| graph convolution on the three inputs, 400 epochs | 5 | 0.874, 0.807, 0.897, 0.888, 0.882 | 0.870 | 0.887 [0.83, 0.93] |
| graph convolution on the GO multi-hot alone, 400 epochs | 5 | 0.914, 0.893, 0.918, 0.916, 0.906 | 0.909 | 0.918 [0.87, 0.95] |
| graph convolution on the GO multi-hot alone, 800 epochs | 5 | 0.905, 0.900, 0.915, 0.911, 0.909 | 0.908 | 0.915 [0.87, 0.95] |
| graph convolution on GO alone, 400 epochs, three seeds per fold | 5 | 0.907, 0.899, 0.922, 0.916, 0.907 | 0.910 | 0.917 [0.87, 0.95] |
| graph convolution on GO alone, 800 epochs, three seeds per fold | 5 | 0.906, 0.901, 0.919, 0.912, 0.909 | 0.909 | 0.915 [0.87, 0.95] |

The ensemble is 0.897 [0.84, 0.95], one number with the reported 0.893, and it is made of three fold models at 0.90 and two that never fit. The folds read 0.901, 0.702, 0.859, 0.904 and 0.687; folds 1 and 4 stay at a training loss of 1.0

through their budget, the trace seed 2701 showed in round 23, so three of the eight cells of this form on the pathway input miss the fitting transition. The fold mean is 0.811 against round 12's 0.881, and by the rule round 17 fixed (fold models above round 12's on at least four folds of five, here two) the reported number stands. The fit folds are the highest fold models in the document; the form's problem is the third of its cells that do not fit, and the six-seed read of Table 46 says how often.

## 4.22 Round 27: the scalable form ✗

What the foundation-model setting can carry is a transform, not a layer. Round 24 measured that the head over the GEM-box solve, with nothing learned in the layer, reads 0.852 on ProtT5, 0.01 to 0.02 under the elastic cell with its ceilings and 0.03 above the affine; and with nothing learned the flux is a function of the genotype and the medium alone, so the program need only be solved once per genotype. Round 27 adds that cache to the solver (a solved program is reused whenever its bounds carry no gradient) and runs the cached GEM-box head on ProtT5 with the pathway multi-hot on six seeds (job 2368, then 2388 to 2390) and on the five folds (2369, then 2391), the cells of rounds 23 and 25 with the learned ceilings removed (Table 49, Figure 50; Table 48). The read is round 23's, and the wall time per cell is reported beside the number, since that is what the form is for.

**Table 49:** Round 27, the scalable form on the pathway input. The pooled control on ProtT5 with the pathway multi-hot, the round-22 affine on that input, and the round-23 flux head on the elastic solve (the comparators), against the same head over the GEM-box solve with nothing learned in the layer and the program cached per genotype, seeds 2700 to 2705. Constraint scale 0, batch 32, class weight, 200 epochs. Bold marks the highest value in a column. ↑ (↓) marks an arm whose held-out AUROC is above (below) the first arm's on every one of the same three seeds.

| arm | mean | sd | seeds | pooled [95% CI] | selected epochs | val step |
|---|---|---|---|---|---|---|
| pathway + ProtT5, no readout | 0.860 | 0.002 | 0.859, 0.859, 0.862 | 0.861 [0.79, 0.92] | 36, 56, 46 | 0.0040 |
| pathway + ProtT5 + FBA ratios | 0.890 | 0.005 | 0.889, 0.895, 0.885 ↑ | 0.890 [0.83, 0.94] | 31, 45, 55 | 0.0054 |
| pathway + ProtT5; gated + head over all solved fluxes | 0.845 | 0.087 | 0.716 to 0.911 (6) | 0.910 [0.86, 0.95] | 11 to 150 | 0.0203 |
| the same cell, first attempt (three survivors of the six-cell pack; per-gene dumps overwritten) | **0.905** | 0.005 | 0.911, 0.901, 0.903 ↑ | – | 55, 76, 78 | 0.0174 |
| pathway + ProtT5; gated + head over the GEM-box solve's flux (cached, nothing learned in the layer) | 0.882 | 0.055 | 0.772 to 0.917 (6) | **0.917** [0.87, 0.96] | 42 to 61 | 0.0176 |

The scalable form matches the best cells and, on nine runs, misses the fitting transition once. The six-cell pack (job 2368) lost three cells to the launcher's memory ceiling; its three survivors, seeds 2701 to 2703, read 0.911, 0.901 and 0.903 (mean 0.905, sd 0.005) with training losses of 0.53 to 0.77 and selected epochs 55 to 78, and their per-gene dumps were overwritten by the rerun, so the table carries their pinned numbers without a bootstrap. The rerun of all six seeds in pairs (jobs 2388 to 2390, 40 GB each) reads 0.902, 0.772, 0.897, 0.917, 0.903 and 0.901 on seeds 2700 to 2705 (mean 0.882, sd 0.055; pooled 0.917 with a gene bootstrap of 0.871 to 0.957, above the elastic form's 0.910 on the same six seeds), selected epochs 42 to 61, training losses 0.05 to 0.18 on the five that fit and 1.04 on seed 2701, which never fit. Seed 2701 is the seed the elastic form never fit at 200 epochs (0.716); the scalable form fit it in the first attempt (0.911) and not in the second (0.772), which is round 29's finding again, on a form with no learned ceilings in the layer: the miss is a chance event of the run and not of the seed or of the layer's ceilings. Over the nine runs of this form, eight fit and read 0.897 to 0.917; over the twelve runs of the elastic form on the same input (six seeds, five folds and the 400-epoch rerun), nine fit (Section 4.23). The wall time was 139 minutes per cell at six cells on one card and 83 minutes at two, against 86 for the uncached elastic form at three: at either packing the layer's own forward on the card, not the solve, is the cost. The cache's hit rate was not logged, and the epoch rows say it was probably never hit: the solver reuses a program only when its bounds carry no gradient, and this arm keeps a learned availability gate on the bounds whose gradient norm is nonzero in the epoch rows ($5 \times 10^{-4}$ at epoch 50 of seed 2700, with a gene-owned bound carrying a dual on 85 percent of genotypes). The name "cached" for this form in the tables and figures records the intent of the round, not a measured reuse; the form that does solve once per genotype is the layer-free one of round 33. The fold read of this form (job 2391, five cells at 120 GB, 144 minutes per cell at five per card) reads 0.912, 0.859, 0.918, 0.910 and 0.901 on the folds (mean 0.900) and 0.912 [0.86, 0.95] as the fold ensemble, one number with the reported 0.893 and the elastic form's 0.897 and under the GO convolution's 0.917 (Table 48); all five folds fit, with the training loss under 0.6 between epochs 51 and 61, and the fold at 0.859 is the one whose validation curve kept rising to its last epochs and selected epoch 192. Under the family protocol of Section 5.3, the fold models' standardized logits averaged, the form reads 0.922 [0.88, 0.96] alone, the highest single transformer family, and 0.954 [0.92, 0.98] with the GO convolution's fold models, the maximum of the nine fold combinations tried and reported as such.

## 4.23 Round 29: the unfit seed under a longer budget ✗

A third of the elastic flux-head cells never fit within 200 epochs (rounds 23 to 25), and the constant learning rate leaves two readings open: the fitting transition is delayed on those seeds, or it is absent. Round 29 reruns the round-23 cell on seed 2701, the seed whose 200-epoch run never fit, with the same program and a 400-epoch budget (job 2387; Table 50,

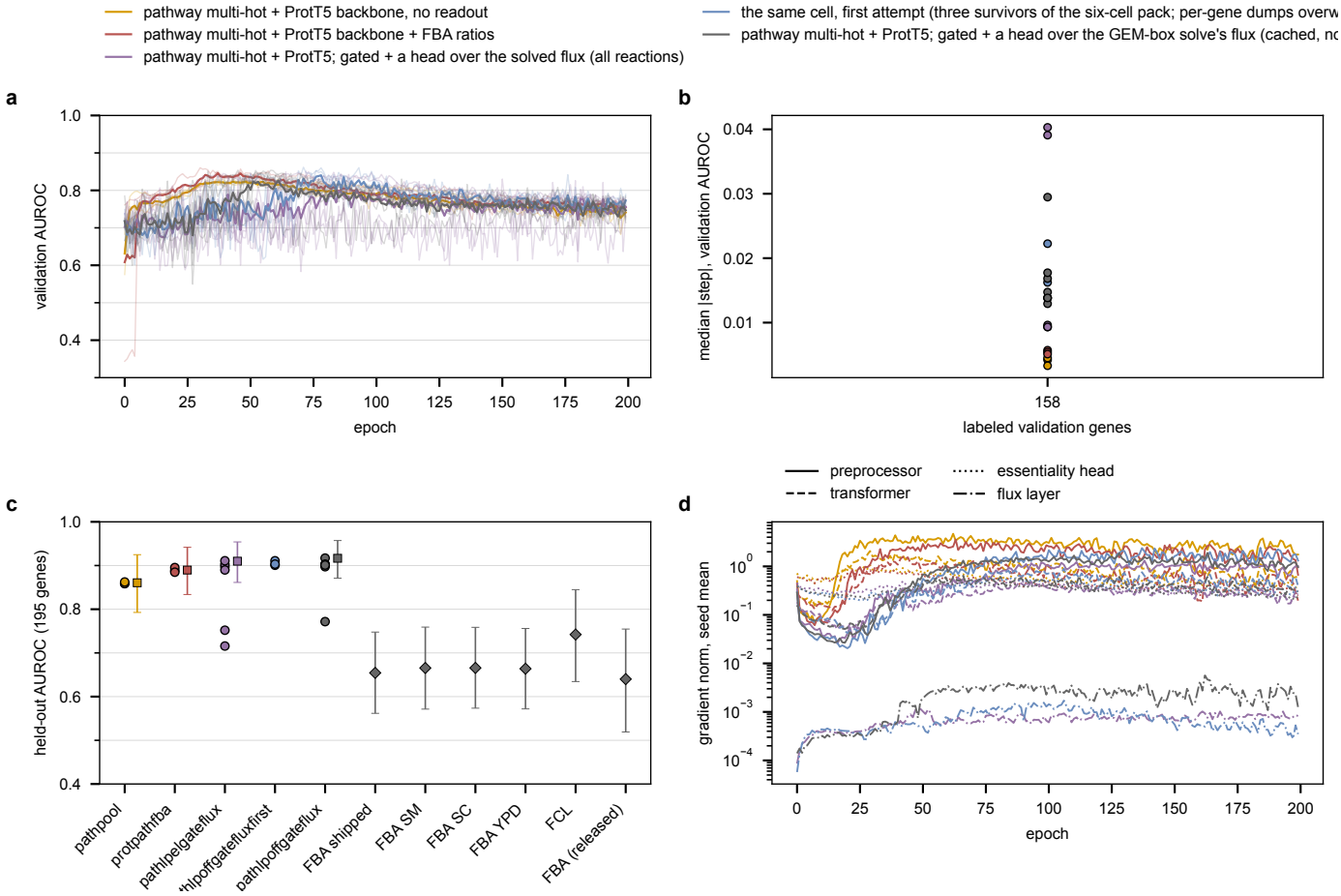



**Figure 50: Round 27, the scalable form on the pathway input.** **a**, Validation AUROC per epoch, one thin line per seed and the seed mean per arm. **b**, The median absolute epoch-to-epoch step per cell against the validation size. **c**, Held-out AUROC per cell (points), the seed mean and the gene bootstrap of the pooled per-gene logits (square with interval), beside FBA on four media, Flux Cone Learning and the released FBA predictions on the same 195 genes. **d**, Gradient norm per module per epoch, seed mean; the dash-dot line is the flux layer. Written by `experiments/028-gene-essentiality/scripts/wave12_readout.py -round w43`.

Figure 51). Because the learning rate is constant, the first 200 epochs of the two runs are the same program on the same seed, so the pair also reads whether a run is reproducible.

**Table 50:** Round 29, the unfit seed under a 400-epoch budget. The round-23 flux-head cell on ProtT5 with the pathway multi-hot, seed 2701, whose 200-epoch run never fit, beside the same program and seed run for 400 epochs. Batch 32, class weight, one seed; the two runs diverge from the first epoch, so the pair reads a chance event, not a delay. Bold marks the highest value in a column. ↑ (↓) marks an arm whose held-out AUROC is above (below) the first arm's on every one of the same three seeds.

| arm | mean | sd | seeds | pooled [95% CI] | selected epochs | val step |
|---|---|---|---|---|---|---|
| pathway multi-hot + ProtT5; gated + a head over the solved flux (all reactions) | 0.716 | 0.000 | 0.716 | 0.716 [0.60, 0.82] | 150 | 0.0391 |
| the same cell, 400 epochs | **0.889** | 0.000 | 0.889 | **0.889** [0.84, 0.94] | 82 | 0.0165 |

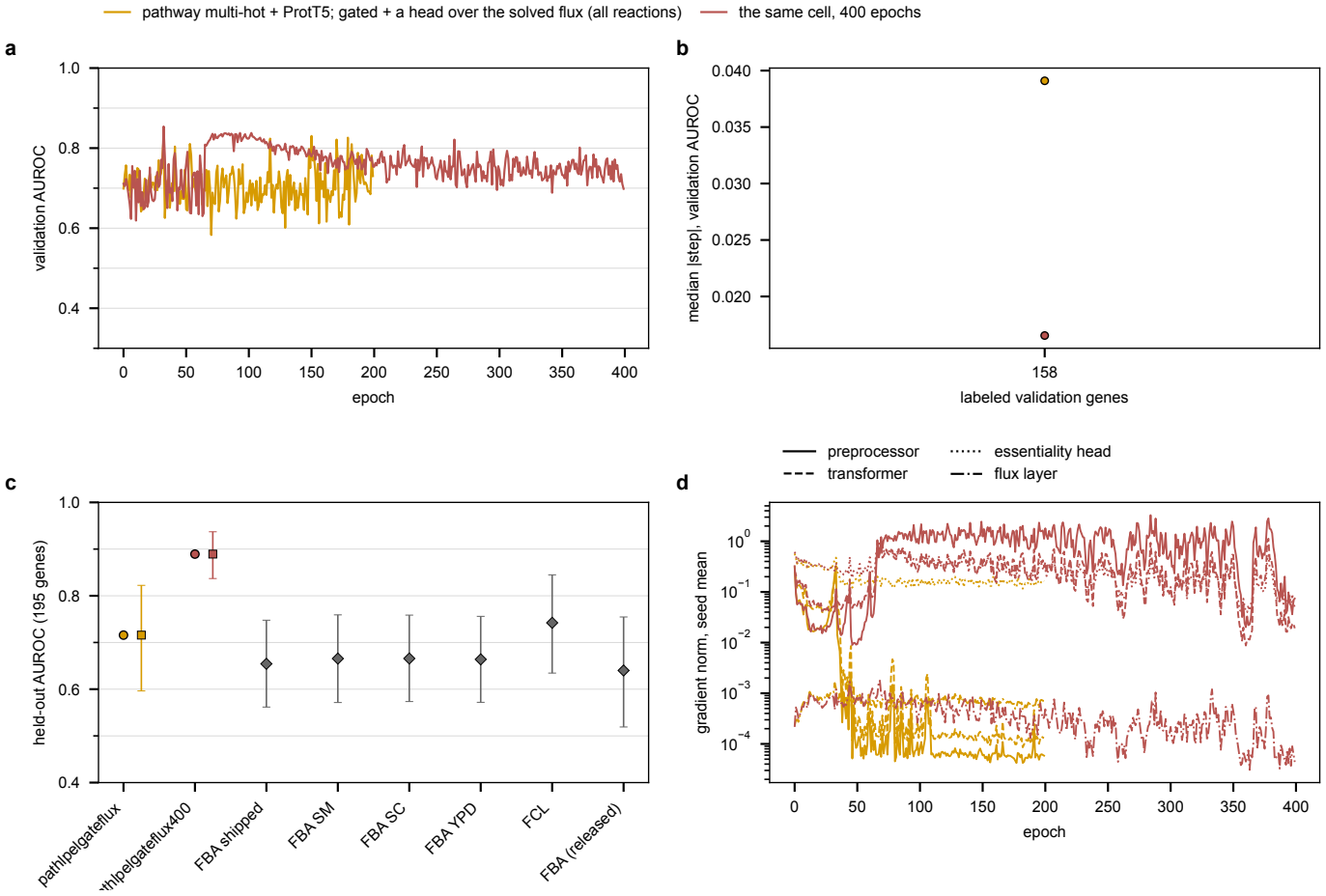



**Figure 51: Round 29, the unfit seed under a 400-epoch budget. a**, Validation AUROC per epoch for the two runs of seed 2701. **b**, The median absolute epoch-to-epoch step per cell against the validation size. **c**, Held-out AUROC per cell (points) and the gene bootstrap (square with interval), beside FBA on four media, Flux Cone Learning and the released FBA predictions on the same 195 genes. **d**, Gradient norm per module per epoch; the dash-dot line is the flux layer. Written by `experiments/028-gene-essentiality/scripts/wave12_readout.py -round w45`.

The transition is neither delayed nor absent on the seed; it is a chance event of the run. The 400-epoch run fits between epochs 60 and 80 (training loss 0.98 at epoch 60, 0.50 at 80, 0.07 at 399), selects epoch 82 and reads 0.889 [0.84, 0.94] on the held-out genes, against 0.716 for the 200-epoch run of the same seed, which never left a training loss of 0.95. The two runs are not the same trajectory: their training losses differ from the first epoch (1.128 against 1.125) and are unrelated by epoch 50 (0.987 against 0.951), so the seed fixes the initialization and not the path, and the solve inside the forward, the sampler and the card's kernels leave the path to chance. What the 200-epoch budget could not tell apart, a seed that fits late from a seed that does not fit, is therefore the wrong question. Across the seventeen elastic flux-head cells of rounds 21, 23 and 25 (six on ProtT5, six on the pathway input, five folds), the training loss first falls under 0.6 between epochs 39 and 137 in fourteen and never in three, and the two latest transitions (epochs 136 and 137) read 0.864 and 0.702 held out, so a late transition is no guarantee either. Past the transition the 400-epoch run overfits (validation AUROC 0.836 at epoch 82, 0.70 at 399, with the prediction spread growing twelvefold), so the longer budget buys nothing after the selected epoch. The scalable form of round 27 is not free of the chance: its rerun of the same seed fit in one attempt and not in the other (0.911 against 0.772), with the solve cached and nothing learned in the layer, so the layer's ceilings are not where the chance enters.

## 4.24 Round 32: the chance is the card's kernels ✗

Round 29 left three places for the chance to enter a seeded run with no loader workers: the solve inside the forward, the sampler, and the card's kernels. Round 32 pins the last of them and runs the round-27 cell on seed 2701, the seed that fit in one attempt and not in another, twice with cuDNN's autotuner off, its deterministic kernels on, torch's deterministic algorithms on (warning rather than failing on an operation without one) and cuBLAS's workspace fixed, and twice without, the four in one pack on one card (job 2405; Table 51, Figure 52). The read is per pair: whether the two pinned runs are one trajectory, and whether the two free runs are.

**Table 51:** Round 32, the unfit seed with the card's kernels pinned. The round-27 cell (the head over the cached GEM-box solve, pathway multi-hot and ProtT5) on seed 2701 run twice with cuDNN's autotuner off, its deterministic kernels on, torch's deterministic algorithms on and cuBLAS's workspace pinned, and twice without, in one pack on one card. Batch 32, class weight, 200 epochs; the read is per pair. Bold marks the highest value in a column. ↑ (↓) marks an arm whose held-out AUROC is above (below) the first arm's on every one of the same three seeds.

| arm | mean | sd | seeds | pooled [95% CI] | selected epochs | val step |
|---|---|---|---|---|---|---|
| pathway + ProtT5; gated + head over the GEM-box solve's flux (cached, nothing learned in the layer) | 0.882 | 0.055 | 0.772 to 0.917 (6) | 0.917 [0.87, 0.96] | 42 to 61 | 0.0176 |
| round-27 cell, seed 2701, kernels pinned, first run | 0.894 | 0.000 | 0.894 | 0.894 [0.84, 0.94] | 91 | 0.0138 |
| the same, pinned, second run | 0.894 | 0.000 | 0.894 | 0.894 [0.84, 0.94] | 91 | 0.0138 |
| the same, kernels free, first run | 0.782 | 0.000 | 0.782 | 0.782 [0.68, 0.87] | 146 | 0.0313 |
| the same, free, second run | **0.920** | 0.000 | 0.920 | **0.920** [0.88, 0.96] | 52 | 0.0129 |
| round-27 cell, kernels pinned, one run per seed (seeds 2700, 2702 to 2705) | 0.904 | 0.006 | 0.897 to 0.913 (5) | 0.911 [0.86, 0.95] | 38 to 60 | 0.0158 |

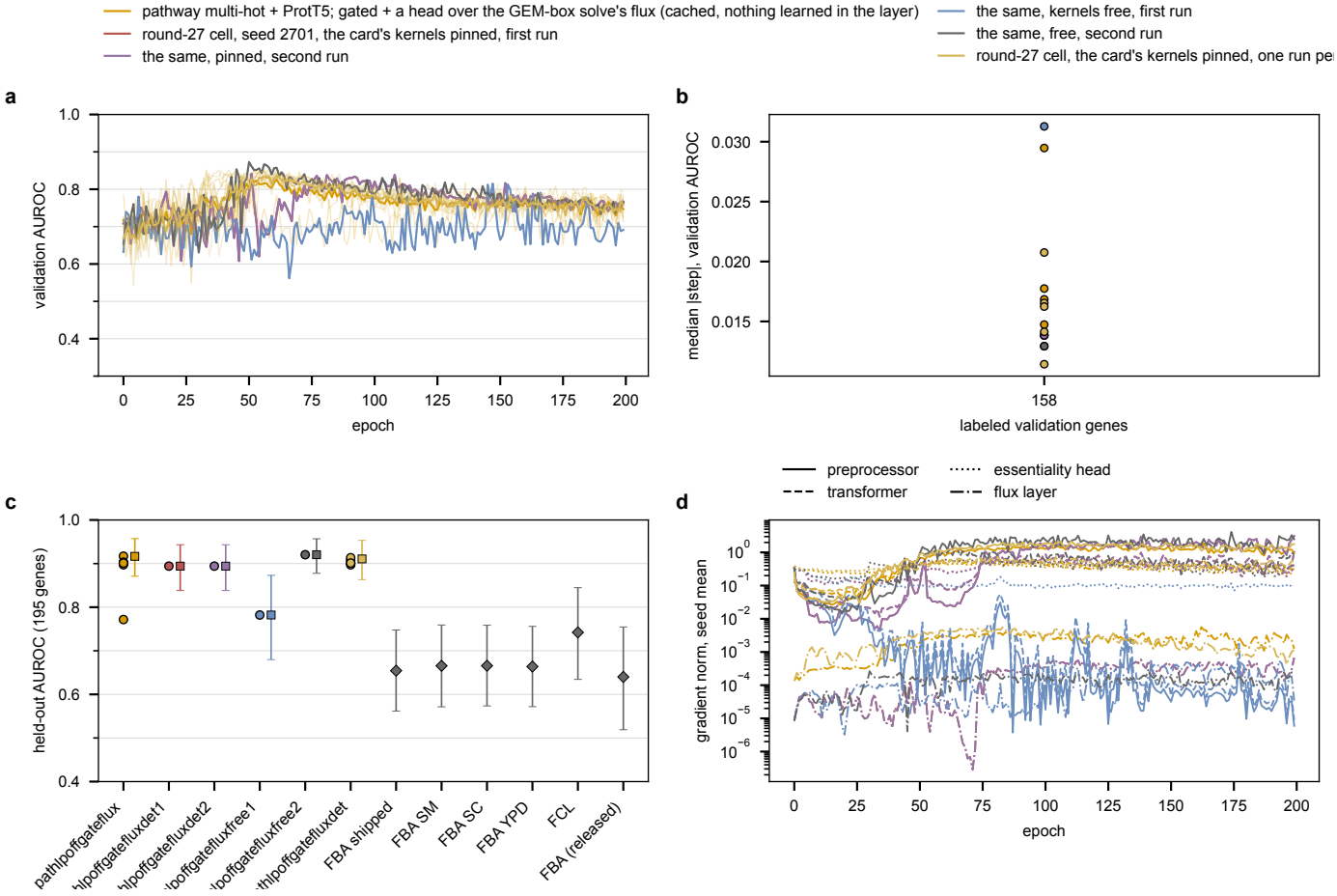



**Figure 52: Round 32, seed 2701 with the card's kernels pinned and free, and the other five seeds pinned. a**, Validation AUROC per epoch for the four runs of seed 2701, the pinned runs of seeds 2700 and 2702 to 2705, and the six free round-27 cells. **b**, The median absolute epoch-to-epoch step per cell against the validation size. **c**, Held-out AUROC per cell (points) and the gene bootstrap (square with interval), beside FBA on four media, Flux Cone Learning and the released FBA predictions on the same 195 genes. **d**, Gradient norm per module per epoch; the dash-dot line is the flux layer. Written by `experiments/028-gene-essentiality/scripts/wave12_readout.py -round w48`.

The chance is the card's. The two pinned runs are identical to four decimals in training loss and validation AUROC at every one of the 200 epochs, cross the fitting transition at epoch 86, select epoch 91 and read 0.894 [0.84, 0.94] held out, the same 195 logits; the two free runs of the same seed differ from their first epoch (training loss 1.2827 against 1.2768)

and end as a fit run and an unfit one, 0.920 (transition at epoch 58, selected 52) against 0.782 (no transition, training loss 1.08 at epoch 199). Nothing in the training path warned under the deterministic switch, so every operation the cell runs has a deterministic kernel; the one warning of the run is a median inside the post-training feasibility audit. Pinning cost about five percent of wall time (205 minutes per cell against 195 at four per card). With the run reproducible, whether a seed fits becomes a property of the seed, and the fit rate over seeds is round 32's second read: pinned once each, seeds 2700, 2702 and 2703 fit at epochs 49, 65 and 65 and read 0.897, 0.908 and 0.901 (job 2413; mean 0.902 over the three), and seeds 2704 and 2705 fit at epochs 40 and 57 and read 0.913 and 0.901 (job 2419), so the five pinned seeds beside 2701 read 0.897 to 0.913 (mean 0.904, sd 0.006; pooled 0.911 [0.86, 0.95]) and six pinned seeds fit six times where the six free runs of the same seeds fit five (0.882 mean with the miss, 0.917 pooled). Pinned, the form's number is the seed's, at 141 to 157 minutes per cell at two or three per card.

## 4.25 Round 45: the layer-free cell under the pinned kernels ✗

The same read on the cell the document ends with, the layer-free form of round 33 (the $\log(1 + |v|)$ vector in the affine, no layer): seed 2700 twice with the kernels pinned and twice free, in one pack on one card (job 2430; Table 52, Figure 53), and seeds 2701 to 2705 pinned once each (job 2431).

**Table 52:** Round 45, the layer-free cell with the card's kernels pinned. The round-33 cell (the $\log(1 + |v|)$ vector in the affine, pathway multi-hot and ProtT5, no layer) on seed 2700 run twice with cuDNN's autotuner off, its deterministic kernels on, torch's deterministic algorithms on and cuBLAS's workspace pinned, and twice without, in one pack on one card. Batch 32, class weight, 200 epochs; the read is per pair. Bold marks the highest value in a column. ↑ (↓) marks an arm whose held-out AUROC is above (below) the first arm's on every one of the same three seeds.

| arm | mean | sd | seeds | pooled [95% CI] | selected epochs | val step |
|---|---|---|---|---|---|---|
| the vector as log1p, default initialization | 0.908 | 0.008 | 0.896 to 0.919 (6) | 0.912 [0.87, 0.95] | 33 to 58 | 0.0043 |
| round-33 cell (log1p vector in the affine, no layer), seed 2700, kernels pinned, first run | 0.911 | 0.000 | 0.911 | 0.911 [0.87, 0.95] | 39 | 0.0033 |
| the same, pinned, second run | 0.911 | 0.000 | 0.911 | 0.911 [0.87, 0.95] | 51 | 0.0042 |
| the same, kernels free, first run | 0.905 | 0.000 | 0.905 | 0.905 [0.86, 0.95] | 43 | 0.0045 |
| the same, free, second run | **0.916** | 0.000 | 0.916 | **0.916** [0.87, 0.96] | 41 | 0.0051 |
| round-33 cell, kernels pinned, one run per seed (seeds 2701 to 2705) | 0.903 | 0.004 | 0.899 to 0.909 (5) | 0.911 [0.86, 0.95] | 39 to 67 | 0.0049 |

The pooled cell is not made reproducible by the switch that made the LP cell so. Its two pinned runs differ at every one of the 200 epochs, from the first training loss on (1.1625 against 1.1564), as the two free runs do (1.1552 against 1.1600), with the switch active in both processes (cuDNN's autotuner off, its deterministic kernels on, torch's deterministic algorithms on, cuBLAS's workspace fixed) and no operation warning that it lacks a deterministic implementation. All four runs fit (training loss under 0.6 at epochs 41 to 50) and read 0.911 and 0.911 pinned, 0.905 and 0.916 free, at 26 minutes per pinned cell against 22 free at four per card. So the divergence of two runs of this cell enters through something torch's deterministic mode does not cover, and what it is remains unmeasured (plausible, unverified: a path the LP form does not take, since that form's pinned pair was one trajectory); what the round settles is that the chance is harmless on this form: pinned once each, seeds 2701 to 2705 fit at epochs 38 to 65 and read 0.906, 0.899, 0.900, 0.909 and 0.902 (mean 0.903, sd 0.004; pooled 0.911 [0.86, 0.95]), the free six-seed run's 0.908 and 0.912, and every one of the twenty-seven $\log(1 + |v|)$ cells on the split file, pinned or free, fit within the budget.

Round 46 brackets the path (job 2433; Table 53, Figure 54): two more pinned pairs on seed 2700, the layer-free cell on ProtT5 alone (no pathway block, the vector in the affine) and the round-22 affine cell (pathway multi-hot and ProtT5, the four ratios, no solve columns). The ProtT5-alone pair is one trajectory, identical in every logged quantity at every one of the 200 epochs (training loss 1.2072 at epoch 0 in both, selected epoch 68 in both, 0.878 [0.81, 0.94] held out, the same logits), so the solve columns, the affine and the ProtT5 path are covered by the switch; the pathway pair differs from epoch 0 (1.2322 against 1.2530) at every epoch and ends at 0.884 and 0.874 with epochs 43 and 32 selected. The divergence enters with the pathway multi-hot in the pooled arm, and not with the multi-hot as such, since the LP cell of round 32 carries the same two embeddings under the `flux_off` arm and was one trajectory pinned. Which operation on the multi-hot in the pooled arm's path lacks a deterministic kernel that does not warn is the next bracket (round 49: the pooled arm on pathway and ProtT5 with no affine readout, on the pathway multi-hot alone with the affine, and on the GO multi-hot with ProtT5 and the affine, each pinned twice).

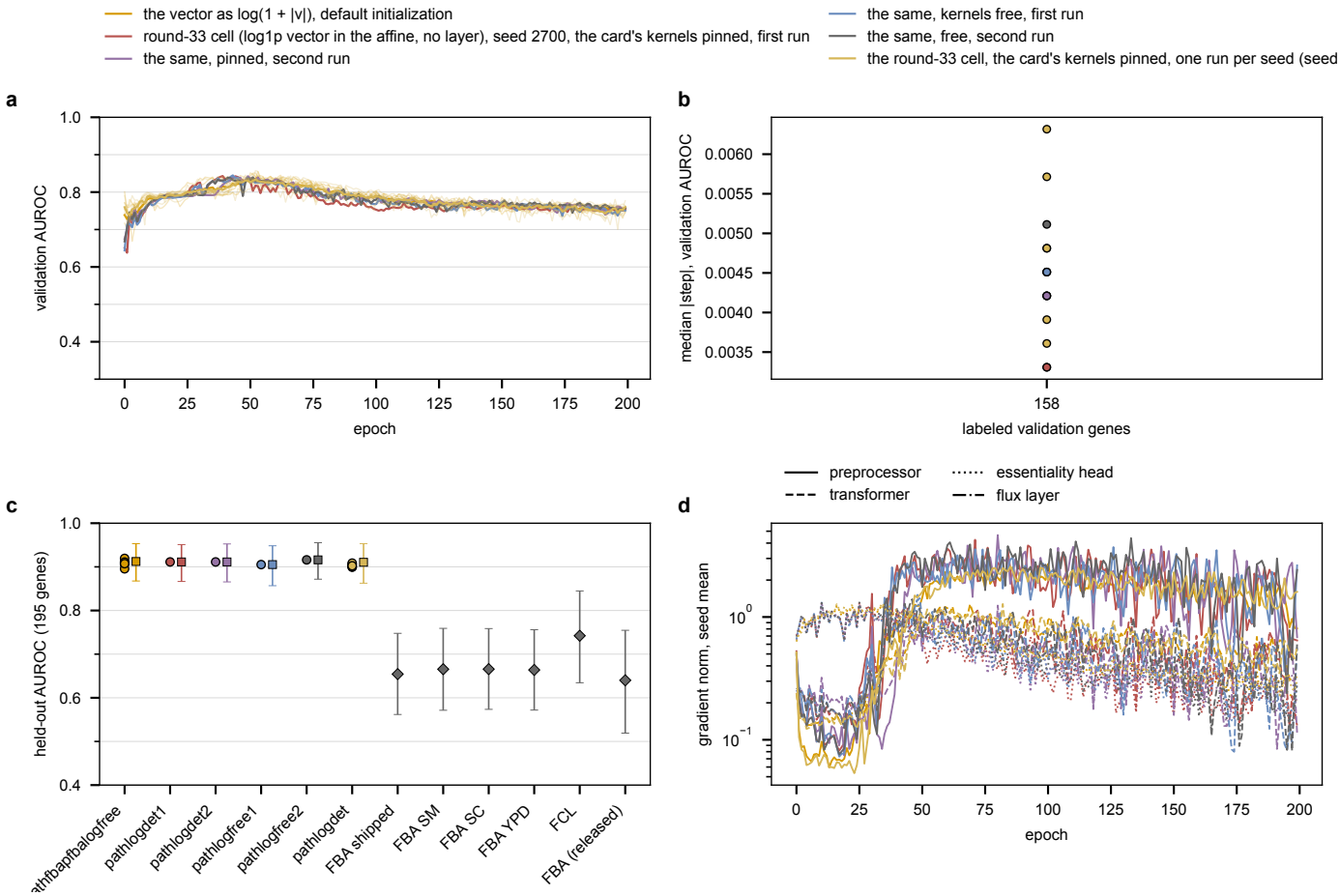



**Figure 53: Round 45, the layer-free cell with the card's kernels pinned and free.** **a**, Validation AUROC per epoch for the four runs of seed 2700, the pinned runs of seeds 2701 to 2705, and the six free round-33 cells. **b**, The median absolute epoch-to-epoch step per cell against the validation size. **c**, Held-out AUROC per cell (points) and the gene bootstrap (square with interval), beside FBA on four media, Flux Cone Learning and the released FBA predictions on the same 195 genes. **d**, Gradient norm per module per epoch. Written by `experiments/028-gene-essentiality/scripts/wave12_readout.py -round w61`.

**Table 53:** Round 46, bracketing the pooled cell's divergence. Two pinned pairs on seed 2700: the round-22 affine cell (pathway multi-hot and ProtT5, the four ratios only, no solve columns) and the layer-free cell on ProtT5 alone (no pathway block), each run twice with the card's kernels pinned as in round 32. Batch 32, class weight, 200 epochs; the read is per pair. Bold marks the highest value in a column. ↑ (↓) marks an arm whose held-out AUROC is above (below) the first arm's on every one of the same three seeds.

| arm | mean | sd | seeds | pooled [95% CI] | selected epochs | val step |
|---|---|---|---|---|---|---|
| round-22 affine cell (pathway + ProtT5, the four ratios, no solve columns), seed 2700, kernels pinned, first run | **0.884** | 0.000 | 0.884 | **0.884** [0.82, 0.94] | 43 | 0.0054 |
| the same, pinned, second run | 0.874 | 0.000 | 0.874 | 0.874 [0.81, 0.93] | 32 | 0.0045 |
| layer-free cell on ProtT5 alone (no pathway block), seed 2700, kernels pinned, first run | 0.878 | 0.000 | 0.878 | 0.878 [0.81, 0.94] | 68 | 0.0045 |
| the same, pinned, second run | 0.878 | 0.000 | 0.878 | 0.878 [0.81, 0.94] | 68 | 0.0045 |

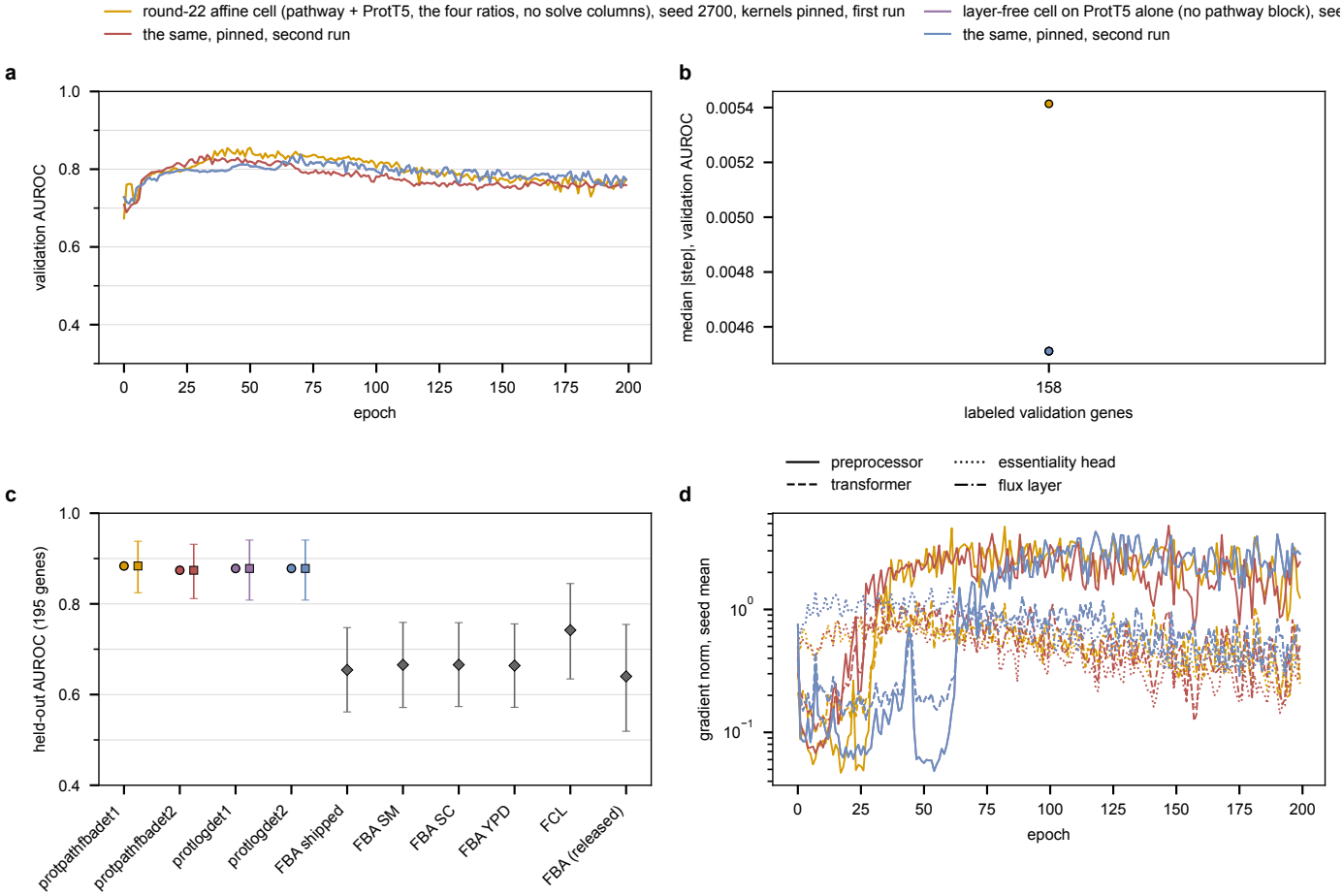



**Figure 54: Round 46, two pinned pairs that bracket the pooled cell's divergence. a**, Validation AUROC per epoch for the ProtT5-alone layer-free pair and the pathway-and-ProtT5 affine pair, each run twice with the card's kernels pinned. **b**, The median absolute epoch-to-epoch step per cell against the validation size. **c**, Held-out AUROC per cell (points) and the gene bootstrap (square with interval), beside FBA on four media, Flux Cone Learning and the released FBA predictions on the same 195 genes. **d**, Gradient norm per module per epoch. Written by `experiments/028-gene-essentiality/scripts/wave12_readout.py -round w62`.

Round 49 narrows it to the pathway embedding itself (job 2437; Table 58, Figure 59): three more pinned pairs on seed 2700 in the pooled arm. The round-9 cell (pathway and ProtT5, the plain pooled head, no affine) differs from epoch 0 (1.2452 against 1.2560) and reads 0.848 and 0.850; the affine on the pathway multi-hot alone, with no ProtT5, differs from epoch 0 (1.2303 against 1.2374) and reads 0.729 and 0.768 with epochs 186 and 57 selected; the affine on the GO multi-hot and ProtT5 is one trajectory (1.5831 in both at epoch 0, epoch 123 selected in both, 0.830 in both, the same logits). So the divergence is neither the readout nor multi-hot inputs as a class but the pathway multi-hot's own path: a cell diverges pinned exactly when the pathway block is on the node. The pathway dataset is rebuilt at every launch (its cell log prints the pathway count each run) and the rebuild enumerates the 173 pathway columns from a Python set, so the column order can differ between processes under Python's per-process hash randomization while the seeded weights do not; it is the reading the GO block supports, since that dataset sorts its columns. The rebuild at every launch is an unconditional call to the dataset's own process step after the base constructor has loaded the cached file, so each run served the column order the previous launch, or the previous cell of its own pack, wrote, and rewrote the file for the next; with the enumeration sorted and the call removed, two processes under different hash seeds build the same 6,607 by 173 matrix (identical bytes), and the cached file's columns differ from both.

Rounds 51 and 52 close it (jobs 2441 and 2442; Tables 56 and 57, Figures 57 and 58). On the old code with Python's hash seed fixed, the layer-free pair is one trajectory (identical at all 200 epochs, 1.1484 at epoch 0, epoch 39 selected, 0.913 in both) where without the hash seed it differed at every epoch (round 45), and the round-9 pooled pair still differs (1.2564 against 1.2516, 0.860 and 0.853): under the old code the first cell of the pack served the file the previous launch had written under another hash seed and rewrote it, so the pair's two cells saw two column orders, while the pack's third and fourth cells both served files written under the fixed seed. On the fixed code with no hash seed, both pairs are one trajectory: the round-9 cell identical at all 200 epochs (1.2488 at epoch 0, epoch 33 selected, 0.865 in both) and the layer-free cell identical (1.1478, epoch 39, 0.912 in both). The chance in the pooled cell was the pathway one-hot's column order, an input permuted between processes and not a kernel; with the dataset fixed and the card's kernels pinned, every cell in the document is one trajectory per seed. The fix does not move the number (round 61, job 2456; Table 54, Figure 55): the round-33 cell on the fixed code, six free seeds, reads 0.904 (sd 0.010; pooled 0.911 [0.87, 0.95]), fitting at epochs 39 to 65, against 0.908 and 0.912 on the six seeds that each served the pathway columns in the previous launch's order, so the permutation was a per-run relabeling of an input the cell reads column by column, and every number in the document stands.

**Table 54:** Round 61, the reference cell on the fixed one-hot code. The round-33 cell (six free seeds, each having served the pathway one-hot in the column order the previous launch wrote) against the same cell on the code that enumerates the columns sorted and processes only when the file is absent, six free seeds. Batch 32, class weight, 200 epochs, seeds 2700 to 2705. Bold marks the highest value in a column. ↑ (↓) marks an arm whose held-out AUROC is above (below) the first arm's on every one of the same three seeds.

| arm | mean | sd | seeds | pooled [95% CI] | selected epochs | val step |
|---|---|---|---|---|---|---|
| the vector as log1p, default initialization | **0.908** | 0.008 | 0.896 to 0.919 (6) | **0.912** [0.87, 0.95] | 33 to 58 | 0.0043 |
| round-33 cell on the fixed pathway one-hot code (sorted columns, no rebuild) | 0.904 | 0.010 | 0.887 to 0.914 (6) | 0.911 [0.87, 0.95] | 39 to 65 | 0.0042 |

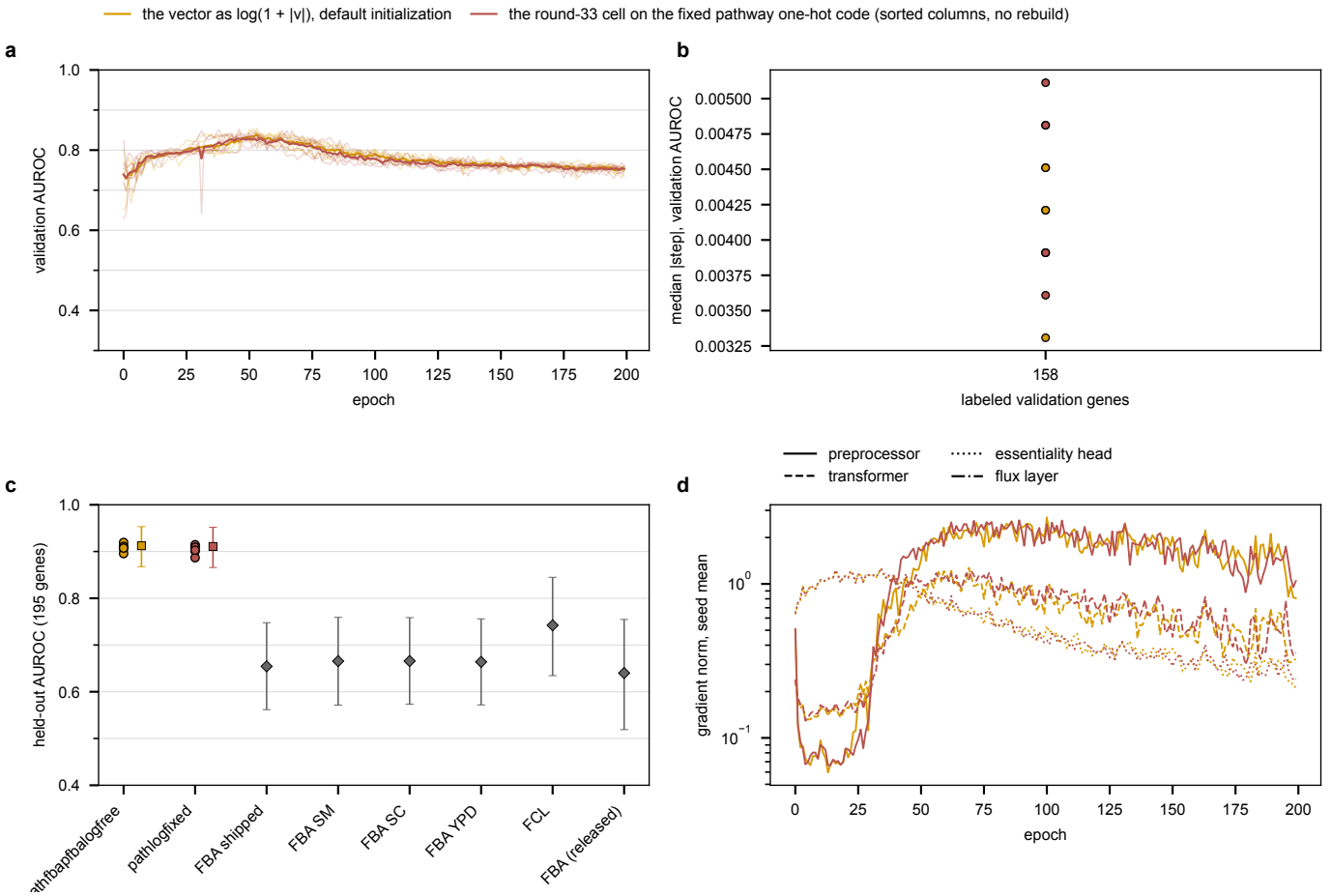



**Figure 55: Round 61, the reference cell on the fixed one-hot code.** **a**, Validation AUROC per epoch, one thin line per seed and the seed mean per arm. **b**, The median absolute epoch-to-epoch step per cell against the validation size. **c**, Held-out AUROC per cell (points), the seed mean and the gene bootstrap of the pooled per-gene logits (square with interval), beside FBA on four media, Flux Cone Learning and the released FBA predictions on the same 195 genes. **d**, Gradient norm per module per epoch, seed mean. Written by `experiments/028-gene-essentiality/scripts/wave12_readout.py -round w77`.

On the fixed input the free run is already one trajectory per seed (round 64, job 2463; Table 55, Figure 56). The same six seeds run once each with the card's kernels pinned reproduce the six free runs of round 61 exactly: every logged

quantity is identical at every one of the 200 epochs on every seed, the same selected epochs (39, 52, 52, 46, 65 and 42), the same held-out numbers (0.912, 0.914, 0.901, 0.887, 0.909 and 0.903) and the same logits, at 36 minutes a cell against 30 free. So the chance the pinned-kernel rounds were chasing on the pooled cell was the input permutation alone, the cell's own forward is deterministic on this card without the switch, and the switch is worth its 20 percent of wall time only on the LP form, whose free runs did diverge (round 32). Under the fold protocol on the fixed code (round 66, job 2465, one seed pinned) the cell reads 0.912, 0.911, 0.830, 0.886 and 0.914 on the folds, the third selecting at epoch 6 before its transition, and 0.917 [0.87, 0.96] as the ensemble, the permuted input's 0.919; with three seeds per fold (jobs 2473 and 2474) the folds read 0.911, 0.909, 0.913, 0.894 and 0.915 as seed-mean logits and the ensemble 0.920 [0.87, 0.96], the permuted input's 0.920 exactly. So the fold numbers stand as well, and the early-selecting fold is the one place the form's selection rule, the validation maximum, still picks a pre-transition epoch, which the seed mean absorbs.

**Table 55:** Round 64, the reported cell pinned on the fixed one-hot code. The round-33 cell on the fixed code, six free seeds (round 61) and the same six seeds with the card's kernels pinned once each, beside the six seeds pinned on the permuted input (round 45, seed 2700's first run and seeds 2701 to 2705). Batch 32, class weight, 200 epochs. Bold marks the highest value in a column. ↑ (↓) marks an arm whose held-out AUROC is above (below) the first arm's on every one of the same three seeds.

| arm | mean | sd | seeds | pooled [95% CI] | selected epochs | val step |
|---|---|---|---|---|---|---|
| round-33 cell on the fixed pathway one-hot code (sorted columns, no rebuild) | 0.904 | 0.010 | 0.887 to 0.914 (6) | 0.911 [0.87, 0.95] | 39 to 65 | 0.0042 |
| round-33 cell, kernels pinned, one run per seed (seeds 2701 to 2705) | 0.903 | 0.004 | 0.899 to 0.909 (5) | 0.911 [0.86, 0.95] | 39 to 67 | 0.0049 |
| round-33 cell (log1p vector in the affine, no layer), seed 2700, kernels pinned, first run | **0.911** | 0.000 | 0.911 | **0.911** [0.87, 0.95] | 39 | 0.0033 |
| round-33 cell on the fixed one-hot code, kernels pinned, one run per seed | 0.904 | 0.010 | 0.887 to 0.914 (6) | 0.911 [0.87, 0.95] | 39 to 65 | 0.0042 |

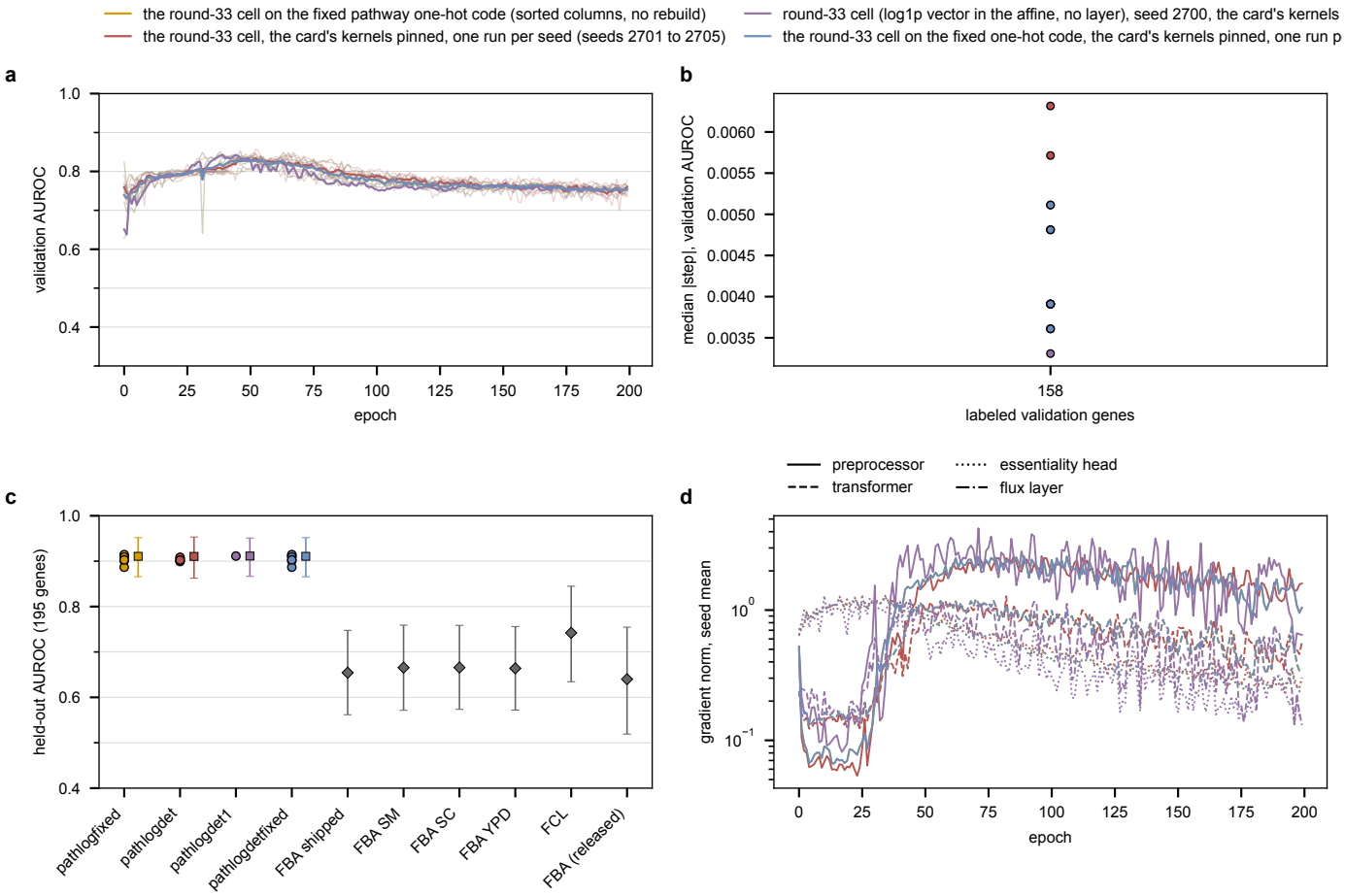



**Figure 56: Round 64, the reported cell pinned on the fixed one-hot code.** **a**, Validation AUROC per epoch, one thin line per seed and the seed mean per arm; the pinned and free arms on the fixed code coincide. **b**, The median absolute epoch-to-epoch step per cell against the validation size. **c**, Held-out AUROC per cell (points), the seed mean and the gene bootstrap of the pooled per-gene logits (square with interval), beside FBA on four media, Flux Cone Learning and the released FBA predictions on the same 195 genes. **d**, Gradient norm per module per epoch, seed mean. Written by `experiments/028-gene-essentiality/scripts/wave12_readout.py -round w80`.

**Table 56:** Round 51, the hash seed fixed on the old one-hot code. The round-9 pooled cell (pathway multi-hot and ProtT5, the plain pooled head) and the round-33 layer-free cell, each run twice on seed 2700 with the card's kernels pinned and Python's hash seed fixed, on the code that rebuilt the pathway one-hot from a set at every launch. Batch 32, class weight, 200 epochs; the read is per pair. Bold marks the highest value in a column. ↑ (↓) marks an arm whose held-out AUROC is above (below) the first arm's on every one of the same three seeds.

| arm | mean | sd | seeds | pooled [95% CI] | selected epochs | val step |
|---|---|---|---|---|---|---|
| round-9 pooled cell, kernels pinned, PYTHONHASHSEED=0, old one-hot code, first run | 0.860 | 0.000 | 0.860 | 0.860 [0.79, 0.93] | 31 | 0.0036 |
| the same, second run | 0.853 | 0.000 | 0.853 | 0.853 [0.78, 0.92] | 54 | 0.0039 |
| round-33 layer-free cell, kernels pinned, PYTHONHASHSEED=0, old one-hot code, first run | **0.913** | 0.000 | 0.913 | **0.913** [0.87, 0.95] | 39 | 0.0039 |
| the same, second run | **0.913** | 0.000 | 0.913 | **0.913** [0.87, 0.95] | 39 | 0.0039 |

**Table 57:** Round 52, the fixed one-hot code without a hash seed. The same two cells run twice each on seed 2700 with the card's kernels pinned and no hash seed, on the code that enumerates the one-hot's columns sorted and processes only when the file is absent. Batch 32, class weight, 200 epochs; the read is per pair. Bold marks the highest value in a column. ↑ (↓) marks an arm whose held-out AUROC is above (below) the first arm's on every one of the same three seeds.

| arm | mean | sd | seeds | pooled [95% CI] | selected epochs | val step |
|---|---|---|---|---|---|---|
| round-9 pooled cell, kernels pinned, no hash seed, fixed one-hot code, first run | 0.865 | 0.000 | 0.865 | 0.865 [0.80, 0.93] | 33 | 0.0057 |
| the same, second run | 0.865 | 0.000 | 0.865 | 0.865 [0.80, 0.93] | 33 | 0.0057 |
| round-33 layer-free cell, kernels pinned, no hash seed, fixed one-hot code, first run | **0.912** | 0.000 | 0.912 | **0.912** [0.86, 0.95] | 39 | 0.0039 |
| the same, second run | **0.912** | 0.000 | 0.912 | **0.912** [0.86, 0.95] | 39 | 0.0039 |

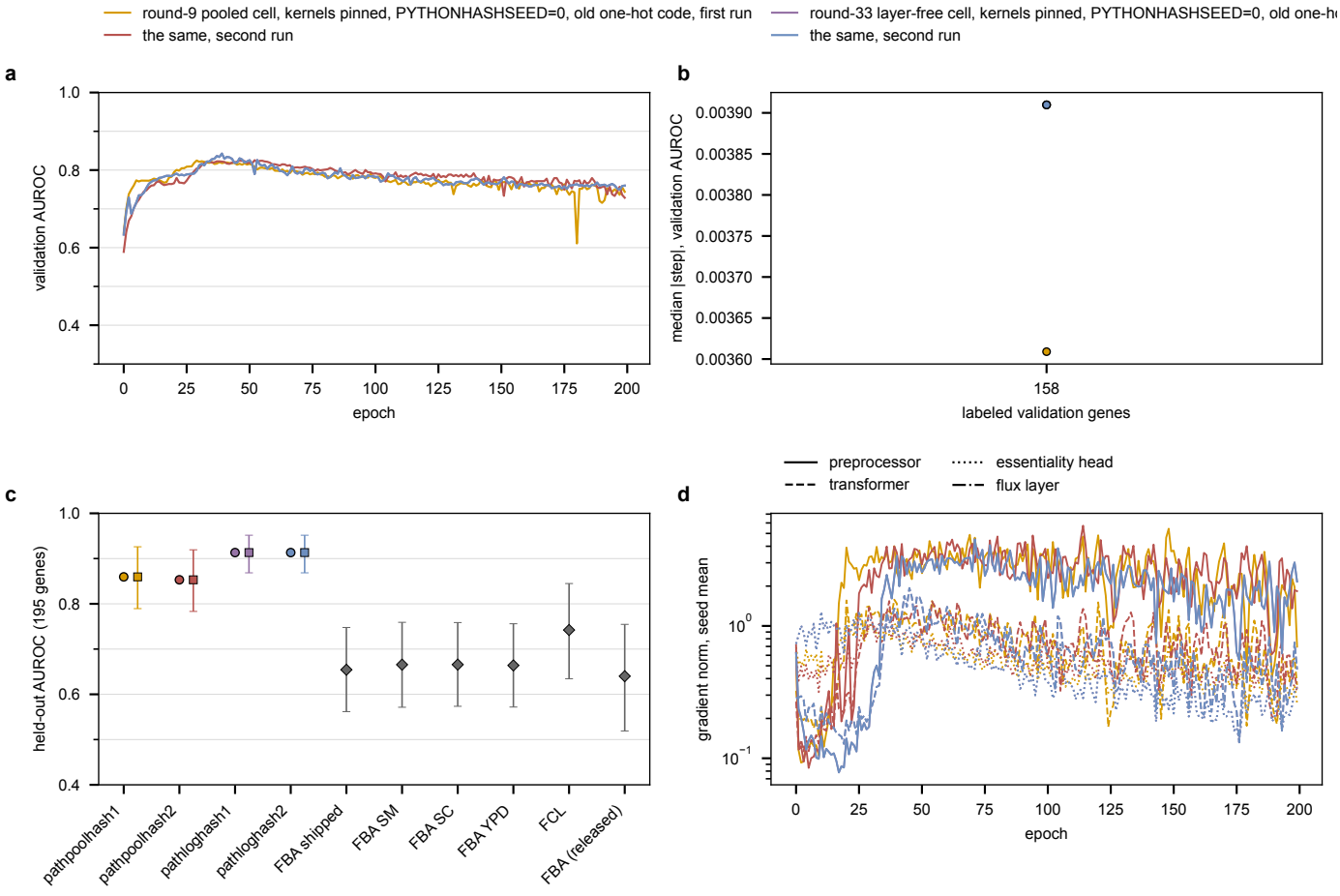



**Figure 57: Round 51, the hash seed fixed on the old one-hot code.** **a**, Validation AUROC per epoch for the round-9 pooled pair and the layer-free pair, each run twice with the card's kernels pinned and the hash seed fixed. **b**, The median absolute epoch-to-epoch step per cell against the validation size. **c**, Held-out AUROC per cell (points) and the gene bootstrap (square with interval), beside FBA on four media, Flux Cone Learning and the released FBA predictions on the same 195 genes. **d**, Gradient norm per module per epoch. Written by `experiments/028-gene-essentiality/scripts/wave12_readout.py -round w67`.

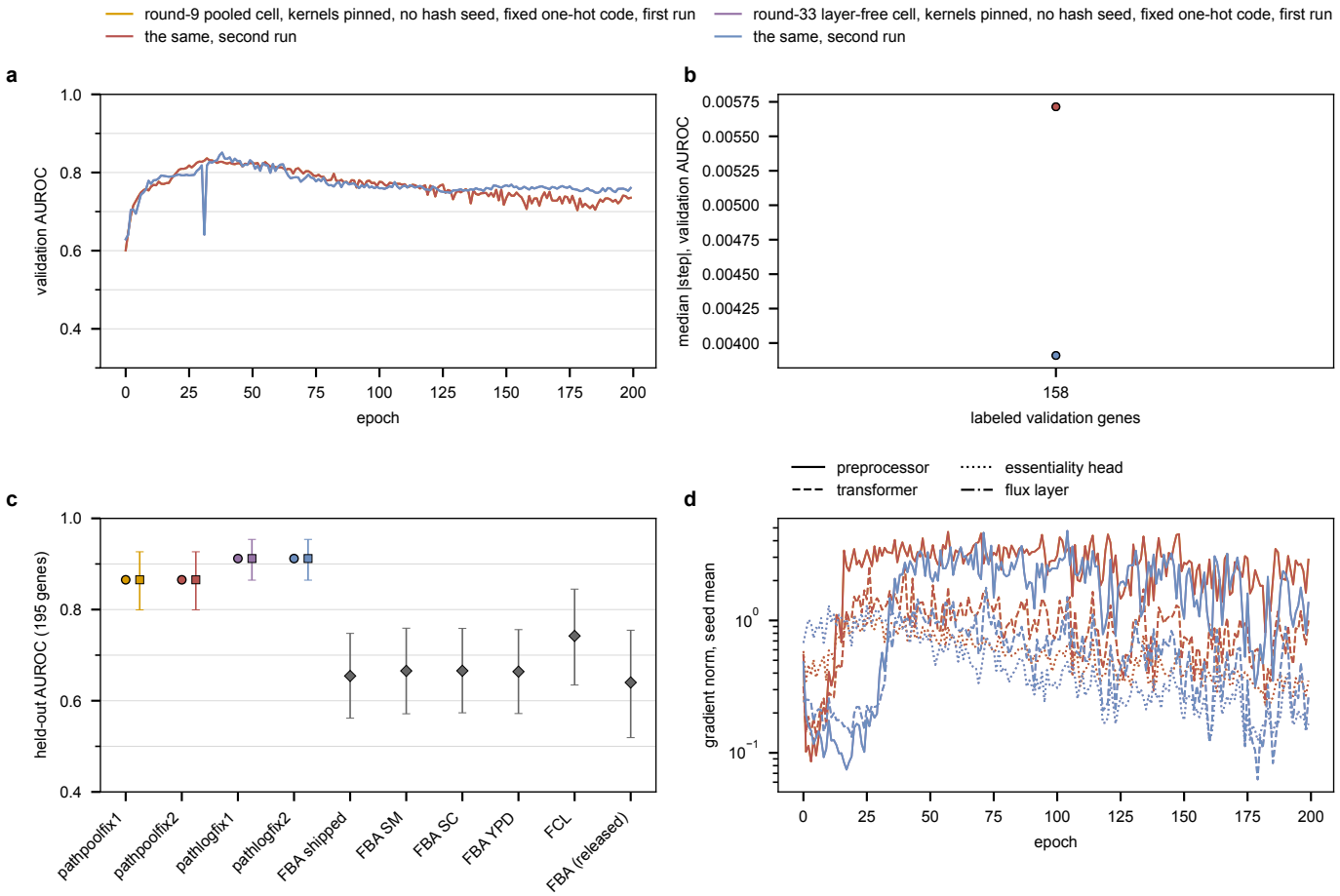



**Figure 58: Round 52, the fixed one-hot code without a hash seed.** **a**, Validation AUROC per epoch for the same two pairs on the code that enumerates the one-hot's columns sorted, each run twice with the card's kernels pinned. **b**, The median absolute epoch-to-epoch step per cell against the validation size. **c**, Held-out AUROC per cell (points) and the gene bootstrap (square with interval), beside FBA on four media, Flux Cone Learning and the released FBA predictions on the same 195 genes. **d**, Gradient norm per module per epoch. Written by `experiments/028-gene-essentiality/scripts/wave12_readout.py -round w68`.

**Table 58:** Round 49, the second bracket of the pooled cell's divergence. Three pinned pairs on seed 2700 in the pooled arm: the round-9 cell (pathway multi-hot and ProtT5, the plain pooled head, no affine readout), the affine cell on the pathway multi-hot alone, and the affine cell on the GO multi-hot and ProtT5, each run twice with the card's kernels pinned as in round 32. Batch 32, class weight, 200 epochs; the read is per pair. Bold marks the highest value in a column. ↑ (↓) marks an arm whose held-out AUROC is above (below) the first arm's on every one of the same three seeds.

| arm | mean | sd | seeds | pooled [95% CI] | selected epochs | val step |
|---|---|---|---|---|---|---|
| round-9 pooled cell (pathway + ProtT5, no affine readout), seed 2700, kernels pinned, first run | 0.848 | 0.000 | 0.848 | 0.848 [0.78, 0.92] | 62 | 0.0036 |
| the same, pinned, second run | **0.850** | 0.000 | 0.850 | **0.850** [0.78, 0.92] | 33 | 0.0051 |
| affine cell on the pathway multi-hot alone (no ProtT5), seed 2700, kernels pinned, first run | 0.729 | 0.000 | 0.729 | 0.729 [0.62, 0.82] | 186 | 0.0009 |
| the same, pinned, second run | 0.768 | 0.000 | 0.768 | 0.768 [0.67, 0.86] | 57 | 0.0012 |
| affine cell on GO + ProtT5 (no pathway block), seed 2700, kernels pinned, first run | 0.830 | 0.000 | 0.830 | 0.830 [0.75, 0.91] | 123 | 0.0051 |
| the same, pinned, second run | 0.830 | 0.000 | 0.830 | 0.830 [0.75, 0.91] | 123 | 0.0051 |

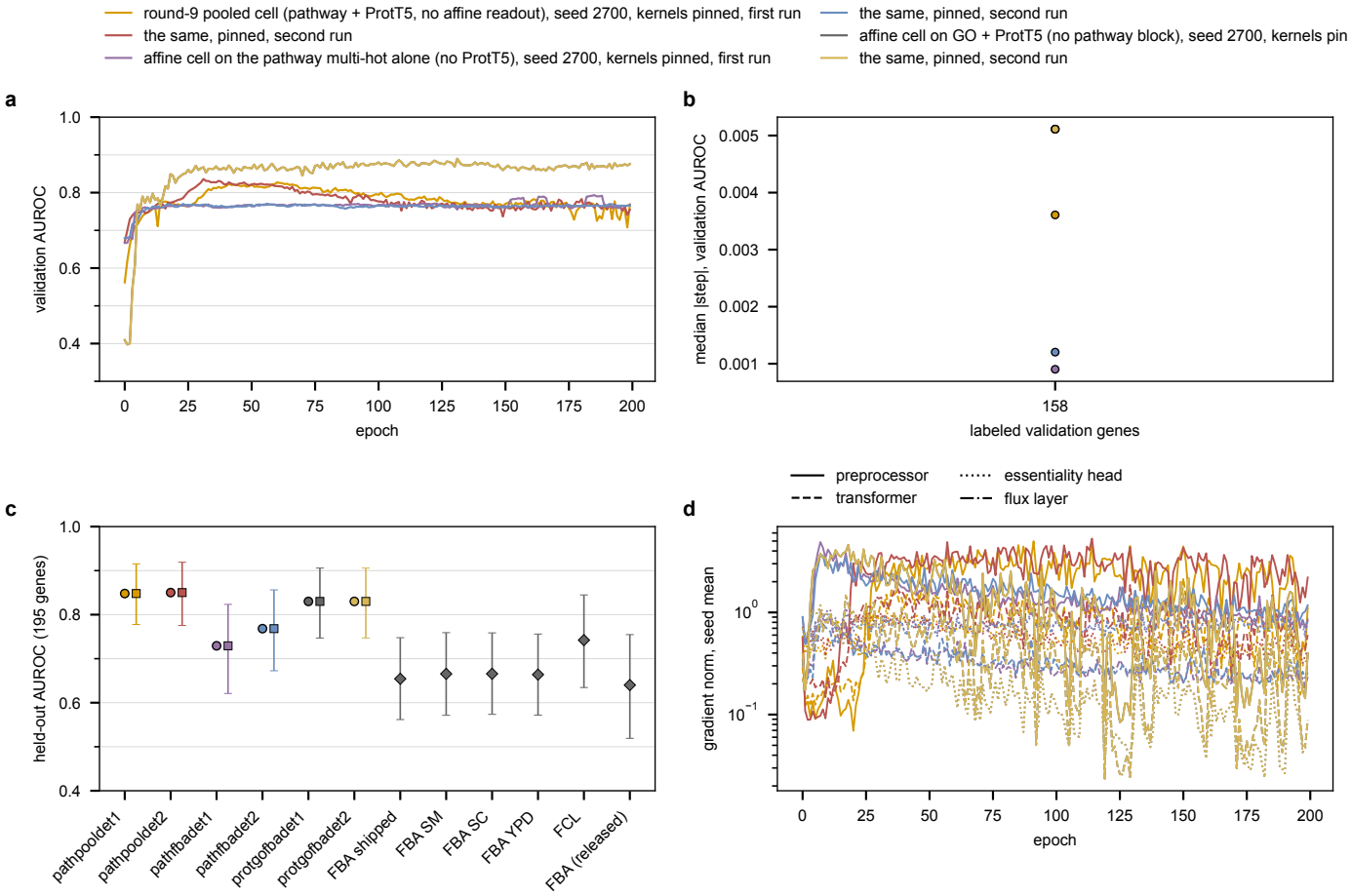



**Figure 59: Round 49, three pinned pairs in the pooled arm. a**, Validation AUROC per epoch for the round-9 pooled pair, the pathway-alone affine pair and the GO-and-ProtT5 affine pair, each run twice with the card's kernels pinned. **b**, The median absolute epoch-to-epoch step per cell against the validation size. **c**, Held-out AUROC per cell (points) and the gene bootstrap (square with interval), beside FBA on four media, Flux Cone Learning and the released FBA predictions on the same 195 genes. **d**, Gradient norm per module per epoch. Written by `experiments/028-gene-essentiality/scripts/wave12_readout.py -round w65`.

## 4.26 Round 31: the layer-free form ✗

Round 27's form still carries a flux layer in the graph: the program is solved once per genotype and cached, but the solve runs inside the forward pass, the flux enters through the gated head, and the cell takes 83 minutes at two per card. With nothing learned in the layer the vector is a function of the genotype and the medium alone, so it can be precomputed and appended to the affine's input as columns, with no layer in the graph, which is the form a model over the database would use. Round 31 runs that form on the pathway input: the deletion's own pFBA flux vector from the reference table (`ess_reference_flux_table.py`, YPD, 4,131 columns, each standardized over the labeled genes) appended to the four-ratio affine, alone and with the eight wild-type solve statistics beside it, six seeds each in two six-cell packs (jobs 2402 and 2404; Table 59, Figure 61). The comparators are the affine and the cached-solve head on the same input, and the wall time is the read beside the number.

Figure 60 draws the affine with the solve's columns, the form round 11 first ran on ProtT5.

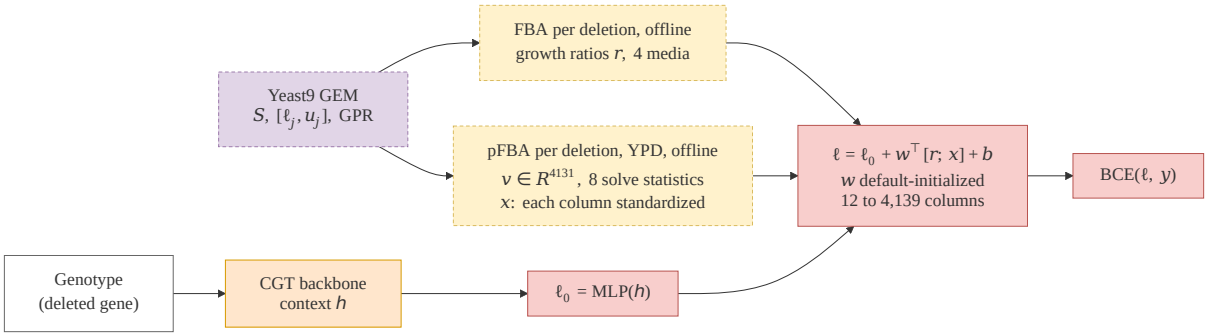



**Figure 60: The affine widened with solve features, rounds 11 and 31.** The FBA affine of Figure 23 with more columns from the same offline solves: the deletion's parsimonious FBA flux vector over 4,131 reactions and eight statistics of the wild-type solve on the gene's reactions, each column standardized over the labeled genes, under a default-initialized weight. Round 11 ran it on ProtT5 and round 31 on ProtT5 with the pathway multi-hot. Colors as in Figure 16.

**Table 59:** Round 31, the layer-free form. The round-22 affine on ProtT5 with the pathway multi-hot and the round-27 head over the cached GEM-box solve (the comparators), against the deletion's own pFBA flux vector precomputed and appended to the affine's input with no flux layer in the graph, and the same with the eight wild-type solve statistics beside it. Batch 32, class weight, 200 epochs, seeds 2700 to 2705. Bold marks the highest value in a column. ↑ (↓) marks an arm whose held-out AUROC is above (below) the first arm's on every one of the same three seeds.

| arm | mean | sd | seeds | pooled [95% CI] | selected epochs | val step |
|---|---|---|---|---|---|---|
| pathway + ProtT5 + FBA ratios | 0.890 | 0.005 | 0.889, 0.895, 0.885 | 0.890 [0.83, 0.94] | 31, 45, 55 | 0.0054 |
| pathway + ProtT5; gated + head over the GEM-box solve's flux (cached, nothing learned in the layer) | 0.882 | 0.055 | 0.772 to 0.917 (6) | **0.917** [0.87, 0.96] | 42 to 61 | 0.0176 |
| pathway + ProtT5; FBA affine + the deletion's pFBA vector as an input (no flux layer) | 0.832 | 0.071 | 0.741 to 0.923 (6) | 0.871 [0.79, 0.94] | 2 to 59 | 0.0039 |
| the same, with the eight wild-type solve statistics beside the pFBA vector | **0.892** | 0.023 | 0.849 to 0.911 (6) | 0.901 [0.85, 0.95] | 34 to 106 | 0.0031 |

The vector alone reads 0.804, 0.923, 0.741, 0.814, 0.911 and 0.797 on seeds 2700 to 2705 (mean 0.832, sd 0.071; pooled 0.871 [0.79, 0.94]), under the affine's 0.890 and the cached-solve head's 0.917; with the statistics beside it 0.908, 0.849, 0.892, 0.885, 0.907 and 0.911 (mean 0.892, sd 0.023; pooled 0.901 [0.85, 0.95]), one number with the affine. The vector-alone cells select early: four of the six select epochs 2 to 13, where the training loss is still 0.84 to 0.91 and the fitting transition (the training loss under 0.6) comes at epochs 28 to 46, or never on seed 2702, whose loss ends at 0.76. On those four the validation curve peaks at 0.81 to 0.85 before the backbone fits and falls as it does, so the selected model is the affine's initial projection of the vector over an unfit backbone; the two cells that select after the transition (seeds 2701 and 2704, epochs 59 and 42) read 0.923 and 0.911, the level of the cached-solve head. The affine here is a default-initialized linear layer over 4,135 columns, a weight norm of 0.59 before any training, where the cached-solve head is zero-initialized over $\log(1+|v^\star|)$ and starts as the backbone alone; whether that construction and not the solve inside the graph is the 0.05 is round 33's question, in flight at the time of writing (job 2406: the same vector with the solve columns zero-initialized, and as $\log(1+|v|)$ with zero initialization, six seeds each). The wall time is 31 minutes per cell at six per card, against 40 for the affine at three and 83 for the cached-solve head at two: with the vector precomputed the cell costs what the affine costs, and nothing on the card waits on a solver.

## 4.27 Round 33: the encoding is the layer-free form's number ✗

Round 31's affine differs from the cached-solve head in two ways, a default rather than zero initialization and standardized columns rather than $\log(1+|v|)$, and round 33 crosses them: the same pFBA vector with the solve columns zero-initialized, as $\log(1+|v|)$ under the default initialization, and as $\log(1+|v|)$ zero-initialized, six seeds each paired against round 31's cell (jobs 2406, 2407 and 2408; Table 60, Figure 63).

Figure 62 draws the form with both changes.

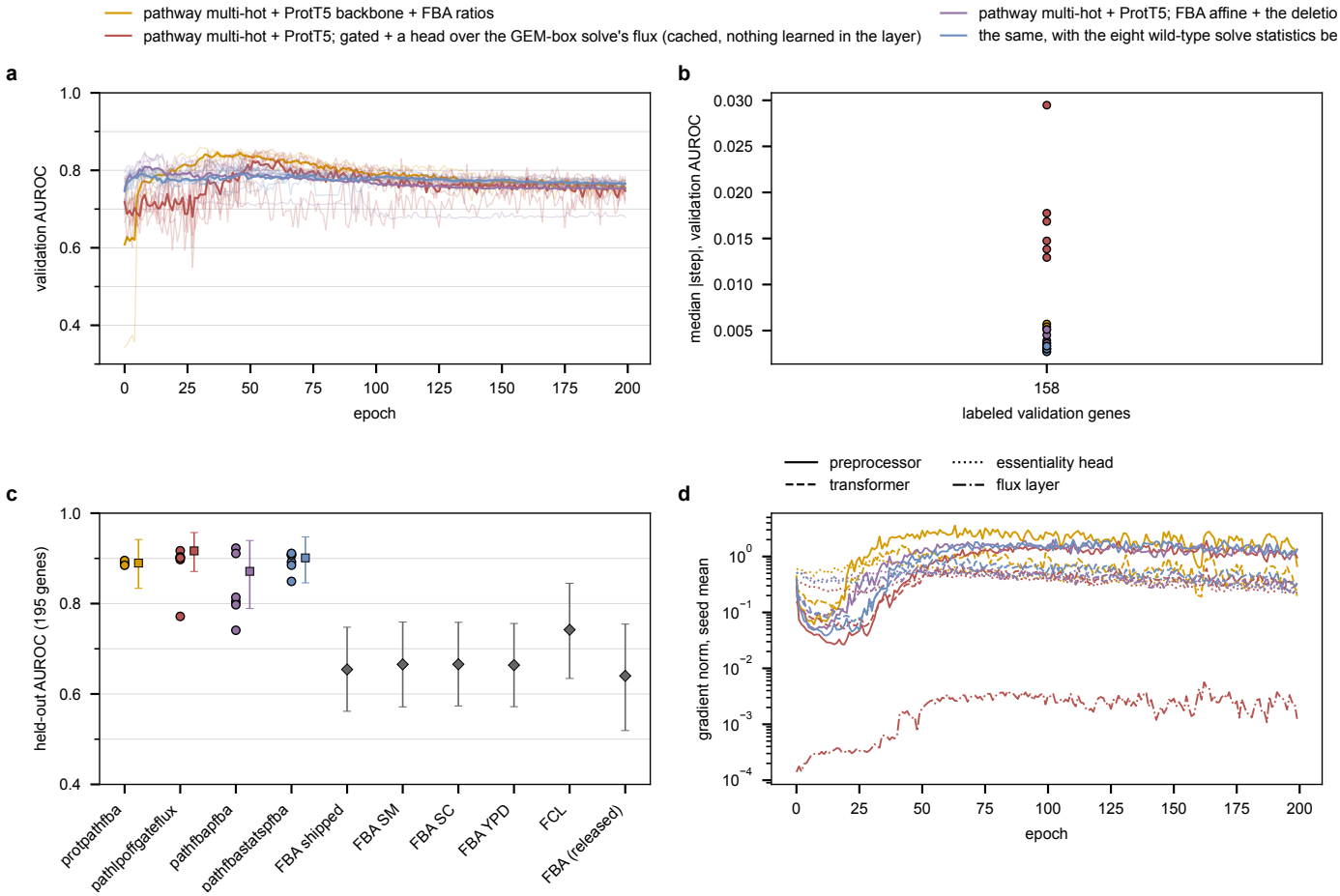



**Figure 61: Round 31, the layer-free form.** **a**, Validation AUROC per epoch, one thin line per seed and the seed mean per arm. **b**, The median absolute epoch-to-epoch step per cell against the validation size. **c**, Held-out AUROC per cell (points), the seed mean and the gene bootstrap of the pooled per-gene logits (square with interval), beside FBA on four media, Flux Cone Learning and the released FBA predictions on the same 195 genes. **d**, Gradient norm per module per epoch, seed mean. Written by `experiments/028-gene-essentiality/scripts/wave12_readout.py -round w47`.

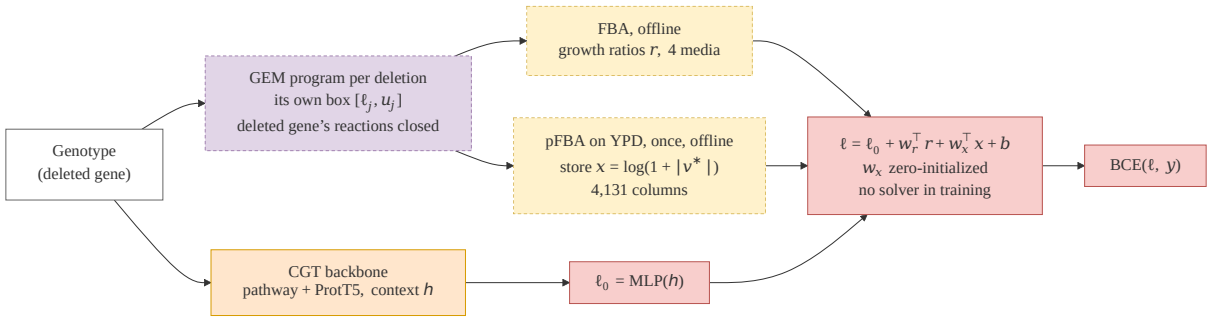



**Figure 62: The layer-free form, round 33.** No flux layer is in the graph. One parsimonious FBA solve per deletion on YPD is run offline, stored as $\log(1 + |v^*|)$ over the 4,131 reactions, and read beside the four growth ratios by an affine on the logit whose weights on the stored columns start at zero, so at initialization the cell is the FBA affine of Figure 23. Against Figure 60 the encoding and the initialization changed; against Figure 48 the solve left the forward pass. Colors as in Figure 16.

**Table 60:** Round 33, the layer-free form's encoding and initialization. The round-27 head over the cached solve (the comparator) and round 31's cell, the deletion's pFBA vector standardized under the affine's default initialization, against the same vector with the solve columns zero-initialized, the vector as $\log(1 + |v|)$ under the default initialization, and as $\log(1 + |v|)$ zero-initialized, the flux head's own construction with no layer in the graph. Batch 32, class weight, 200 epochs, seeds 2700 to 2705. Bold marks the highest value in a column. ↑ (↓) marks an arm whose held-out AUROC is above (below) the first arm's on every one of the same three seeds.

| arm | mean | sd | seeds | pooled [95% CI] | selected epochs | val step |
|---|---|---|---|---|---|---|
| pathway + ProtT5; gated + head over the GEM-box solve's flux (cached, nothing learned in the layer) | 0.882 | 0.055 | 0.772 to 0.917 (6) | **0.917** [0.87, 0.96] | 42 to 61 | 0.0176 |
| pathway + ProtT5; FBA affine + the deletion's pFBA vector as an input (no flux layer) | 0.832 | 0.071 | 0.741 to 0.923 (6) | 0.871 [0.79, 0.94] | 2 to 59 | 0.0039 |
| the same vector, the affine's solve columns zero-initialized | 0.836 | 0.054 | 0.792 to 0.916 (6) | 0.868 [0.78, 0.94] | 9 to 124 | 0.0038 |
| the vector as log1p, default initialization | 0.908 | 0.008 | 0.896 to 0.919 (6) | 0.912 [0.87, 0.95] | 33 to 58 | 0.0043 |
| the vector as log1p, solve columns zero-initialized (the flux head's construction, no layer) | **0.910** | 0.003 | 0.906 to 0.913 (6) | 0.910 [0.87, 0.95] | 36 to 66 | 0.0047 |

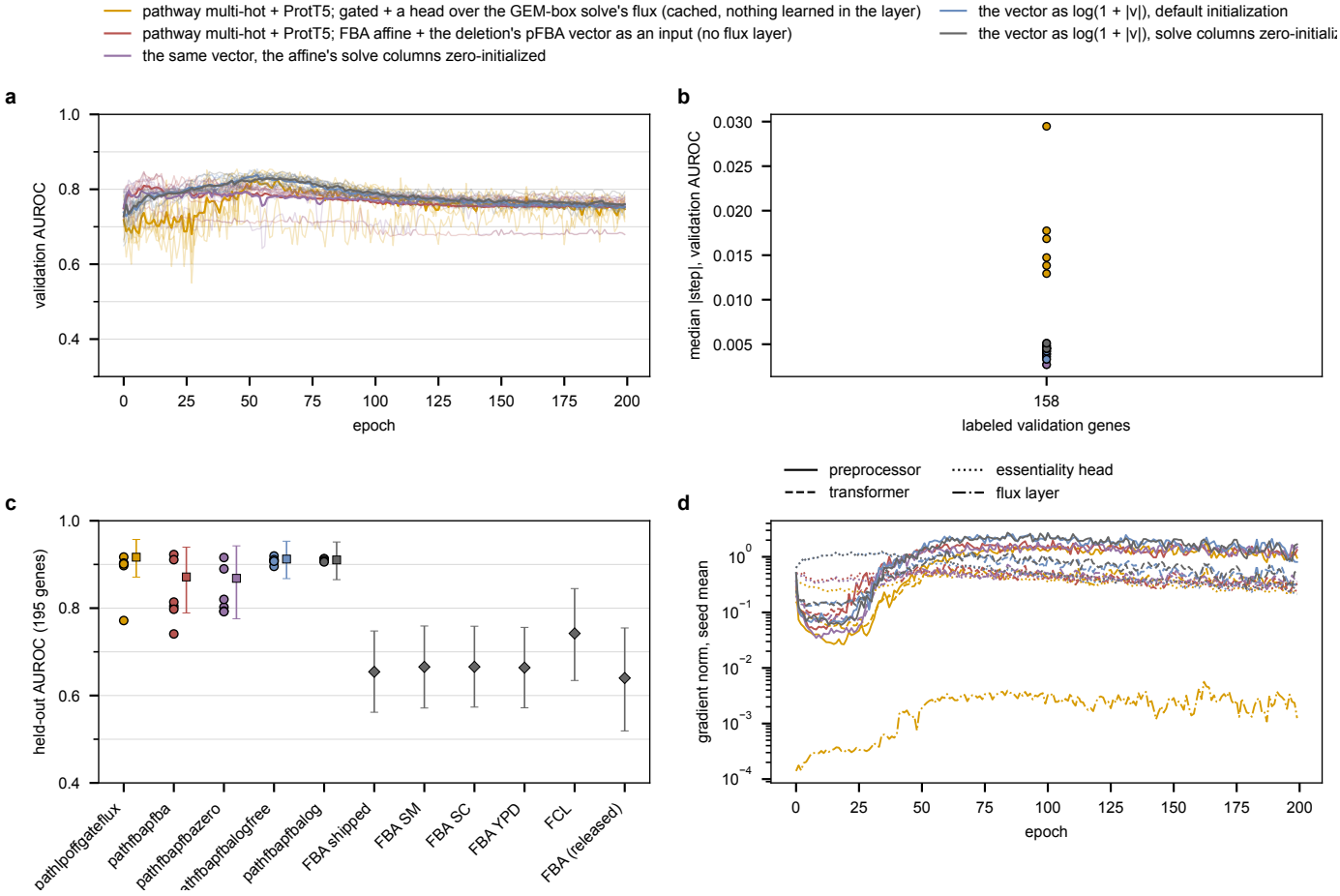



**Figure 63: Round 33, the layer-free form's encoding and initialization.** **a**, Validation AUROC per epoch, one thin line per seed and the seed mean per arm. **b**, The median absolute epoch-to-epoch step per cell against the validation size. **c**, Held-out AUROC per cell (points), the seed mean and the gene bootstrap of the pooled per-gene logits (square with interval), beside FBA on four media, Flux Cone Learning and the released FBA predictions on the same 195 genes. **d**, Gradient norm per module per epoch, seed mean. Written by `experiments/028-gene-essentiality/scripts/wave12_readout.py -round w49`.

The encoding is the whole effect and the initialization is none of it. The standardized vector reads 0.832 under the default initialization and 0.836 zero-initialized (0.802, 0.793, 0.820, 0.890, 0.916 and 0.792 on seeds 2700 to 2705; pooled 0.868 [0.78, 0.94]), and the zero-initialized cells peak early exactly as the default ones did: five of six reach their validation maximum between epochs 1 and 13, before the training loss falls, and select epochs 9 to 13 on three, so the early peak is not the affine's random projection but what the standardized columns do to the backbone's fitting. As $\log(1 + |v|)$ the same vector reads 0.910, 0.919, 0.911, 0.906, 0.896 and 0.907 under the default initialization (mean 0.908, sd 0.008; pooled 0.912 [0.87, 0.95]) and 0.913, 0.910, 0.909, 0.913, 0.909 and 0.906 zero-initialized (mean 0.910, sd 0.003; pooled 0.910 [0.87, 0.95]), every one of the twelve selecting after the transition (epochs 33 to 66), no miss, and the lowest seed-to-seed spread of any arm in the document. That is the cached-solve head's number (0.917 pooled from 0.882 mean with one miss in six) with no layer in the graph, no solver in training and no chance of the run to miss the transition, at 31 minutes per cell in a pack of six and 11 minutes alone (job 2408). Under the family protocol the form reads 0.910 alone and 0.948 [0.91, 0.98] averaged with the six-seed GO convolution, the maximum of the thirteen split-file averages tried (Section 5.3). Under the reported protocol (job 2409, 26 minutes per cell at five per card) the

form reads 0.912, 0.901, 0.838, 0.912 and 0.913 on the folds (mean 0.895) and 0.920 [0.87, 0.96] as the fold ensemble, the highest transformer ensemble in the document and one number with the GO convolution's 0.917 and the scalable form's 0.912 (Table 48); as a fold family it reads 0.920 alone and 0.953 [0.92, 0.98] with the GO fold models, level with the scalable form's 0.954 (Table 86). With three seeds per fold (job 2412, each fold model the seed-mean logit) the folds read 0.913, 0.905, 0.921, 0.904 and 0.910 (mean 0.911) and the ensemble 0.919 [0.87, 0.96], the single-seed number. The 0.838 fold selected epoch 6 with its training loss at 1.05, an early peak of the kind the standardized encoding produced on every cell; its seed 2702 did the same (epoch 22, loss 1.02, 0.841) and its seed 2701 did not (epoch 40, 0.908), and no other fold or seed did in the twenty-seven $\log(1 + |v|)$ cells run, so the early peak is rare under this encoding and, where it occurs, sits with one fold's validation fifth rather than with the seed.

The eight wild-type solve statistics, which lifted the standardized vector from 0.871 to 0.901 in round 31, cost the $\log(1 + |v|)$ vector 0.02 (round 40, job 2421; Table 61, Figure 64): beside it they read 0.882, 0.890, 0.897, 0.893, 0.892 and 0.879 (mean 0.889, sd 0.007; pooled 0.891 [0.83, 0.95]), under the vector alone on every seed, with every cell selecting after the transition (epochs 33 to 56). The statistics are standardized columns, and what they added to a standardized vector they take from an encoded one; the store the cell wants is the vector in the head's encoding and nothing beside it.

**Table 61:** Round 40, the solve statistics beside the log1p vector. Round 31's cell with the eight wild-type solve statistics beside the standardized vector and round 33's cell with the $\log(1 + |v|)$ vector alone (the comparators), against the statistics beside the $\log(1 + |v|)$ vector. Batch 32, class weight, 200 epochs, seeds 2700 to 2705. Bold marks the highest value in a column. ↑ (↓) marks an arm whose held-out AUROC is above (below) the first arm's on every one of the same three seeds.

| arm | mean | sd | seeds | pooled [95% CI] | selected epochs | val step |
|---|---|---|---|---|---|---|
| the same, with the eight wild-type solve statistics beside the pFBA vector | 0.892 | 0.023 | 0.849 to 0.911 (6) | 0.901 [0.85, 0.95] | 34 to 106 | 0.0031 |
| the vector as log1p, default initialization | **0.908** | 0.008 | 0.896 to 0.919 (6) | **0.912** [0.87, 0.95] | 33 to 58 | 0.0043 |
| round-33 cell with the eight wild-type solve statistics beside the log1p vector | 0.889 | 0.007 | 0.879 to 0.897 (6) | 0.891 [0.83, 0.95] | 33 to 56 | 0.0043 |

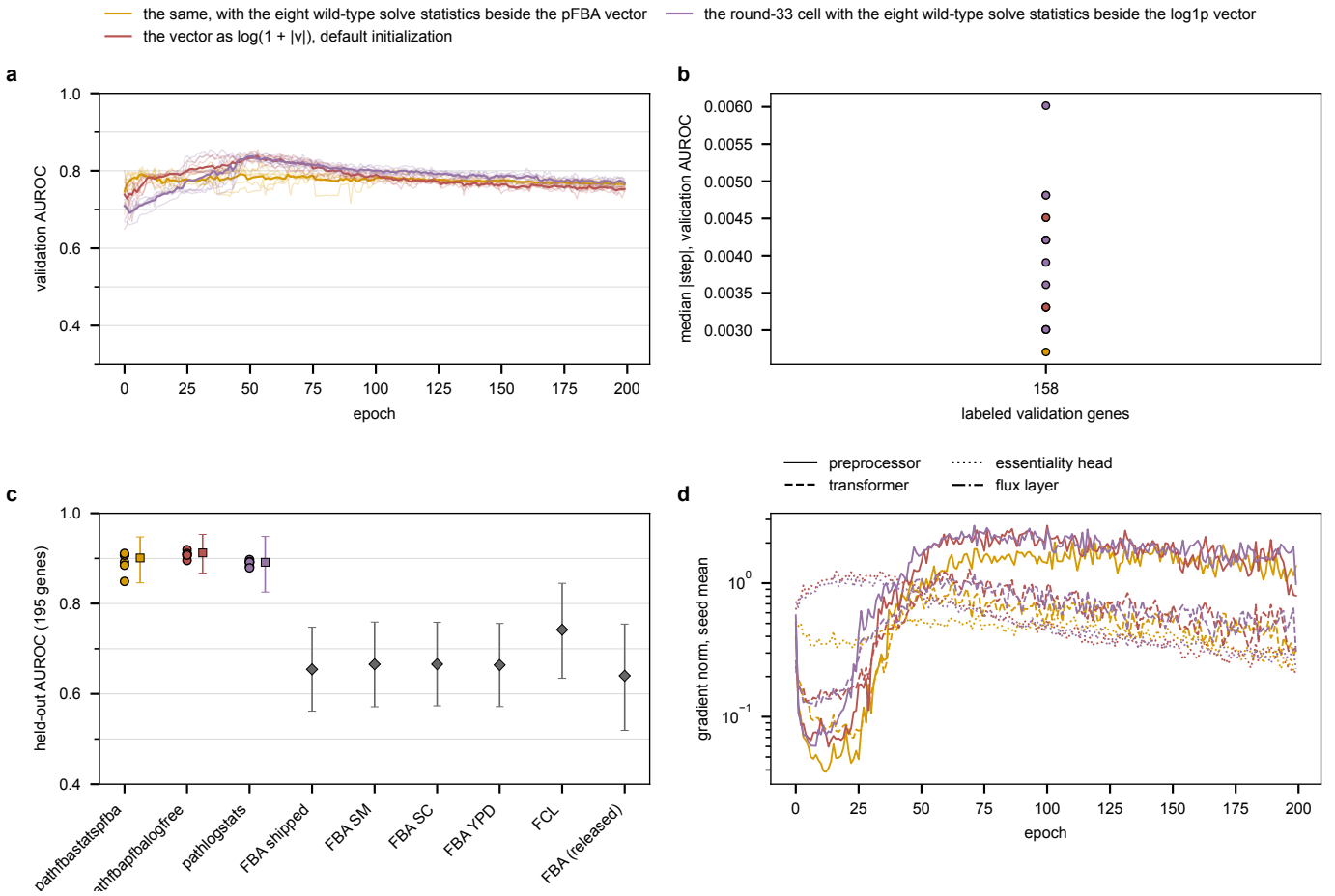



**Figure 64: Round 40, the solve statistics beside the log1p vector.** **a**, Validation AUROC per epoch, one thin line per seed and the seed mean per arm. **b**, The median absolute epoch-to-epoch step per cell against the validation size. **c**, Held-out AUROC per cell (points), the seed mean and the gene bootstrap of the pooled per-gene logits (square with interval), beside FBA on four media, Flux Cone Learning and the released FBA predictions on the same 195 genes. **d**, Gradient norm per module per epoch, seed mean. Written by `experiments/028-gene-essentiality/scripts/wave12_readout.py -round w56`.

A nonlinear read of the vector adds nothing to the linear one (round 41, jobs 2422 and 2423; Table 62, Figure 66). One GELU hidden layer between the affine's input and its logit reads 0.915, 0.911, 0.908, 0.907, 0.901 and 0.914 at width 64 (mean 0.910, sd 0.005; pooled 0.915 [0.87, 0.95]) and 0.911, 0.909, 0.899, 0.909, 0.911 and 0.895 at width 256 (mean 0.905; pooled 0.909 [0.86, 0.95]), against the linear affine's 0.908 and 0.912, every one of the twelve selecting after the transition (epochs 42 to 67). At the resolution of 195 genes the linear read of $\log(1 + |v|)$ is the vector's ceiling, and what is left of the head-form question is the vector entering the cell's representation rather than its logit.

Figure 65 draws the round-41 read.

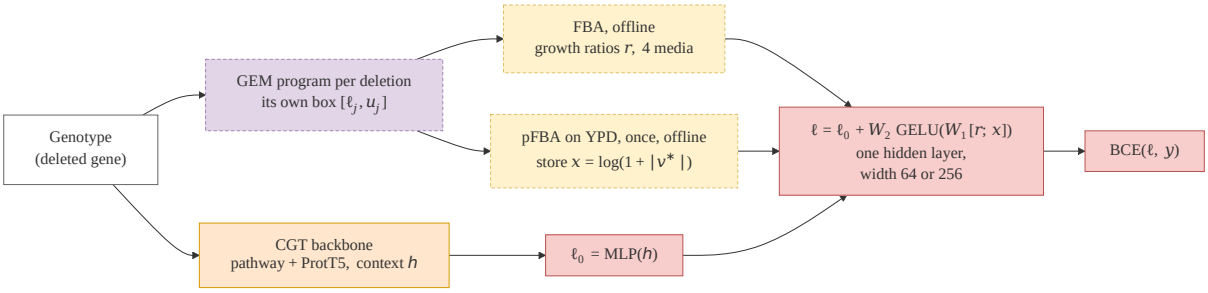



**Figure 65: A hidden layer in the affine, round 41.** The path of Figure 62 with the linear read of the ratios and the stored vector replaced by one GELU hidden layer of width 64 or 256. Colors as in Figure 16.

**Table 62:** Round 41, a hidden layer in the affine over the log1p vector. Round 33's cell, the linear affine over the media ratios and the $\log(1 + |v|)$ vector (the comparator), against the same cell with one GELU hidden layer of width 64 or 256 between that input and the logit. Batch 32, class weight, 200 epochs, seeds 2700 to 2705. Bold marks the highest value in a column. ↑ (↓) marks an arm whose held-out AUROC is above (below) the first arm's on every one of the same three seeds.

| arm | mean | sd | seeds | pooled [95% CI] | selected epochs | val step |
|---|---|---|---|---|---|---|
| the vector as log(1 + \|v\|), default initialization | 0.908 | 0.008 | 0.896 to 0.919 (6) | 0.912 [0.87, 0.95] | 33 to 58 | 0.0043 |
| round-33 cell with a GELU hidden layer of width 64 in the affine | **0.909** | 0.005 | 0.901 to 0.915 (6) | **0.915** [0.87, 0.95] | 49 to 67 | 0.0053 |
| round-33 cell with a GELU hidden layer of width 256 in the affine | 0.905 | 0.007 | 0.895 to 0.911 (6) | 0.909 [0.86, 0.95] | 42 to 58 | 0.0049 |

The cell's capacity does not move the number either, until it breaks the fit (round 48, jobs 2439, 2443 and 2444; Table 63, Figure 67). The round-33 cell has 25 thousand parameters. At width 24 (49 thousand) it reads 0.911 on six seeds (sd 0.005; pooled 0.912 [0.87, 0.95]); at width 48 (125 thousand) 0.910, 0.917, 0.920, 0.914, 0.909 and 0.913 (mean 0.914, sd 0.004; pooled 0.915 [0.87, 0.95]), every seed fitting at epochs 29 to 101, the highest seed mean of any single arm on the split file and inside the width-12 cell's interval; with two transformer blocks (graph regularization off, as every two-block cell runs) 0.909 on three (pooled 0.913); at width 24 with the two-layer preprocessor (767 thousand) 0.910 on six (pooled 0.912). At width 96 (371 thousand) the cell misses the fitting transition on two seeds of three (training loss 0.90 and 0.92 at epoch 199, reading 0.874 and 0.804) and reads 0.892 on the one that fits, the first layer-free cell on the split file to miss. So from 25 to 125 thousand parameters the form reads one number, and the transition, not the capacity, is what a wider cell puts at risk. Under the fold protocol (round 54, job 2447) the width-48 cell reads 0.881, 0.912, 0.927, 0.880 and 0.908 on the folds, every fold fit at epochs 34 to 55, and 0.917 [0.87, 0.96] as the ensemble on one seed and 0.920 [0.87, 0.96] on three (jobs 2450 and 2451; folds 0.917, 0.895, 0.919, 0.894 and 0.916 as seed-mean logits), the width-12 cell's 0.919 on one seed and 0.920 on three (Table 48).

**Table 63:** Round 48, the capacity of the layer-free cell. The round-33 cell (width 12, one block, one-layer preprocessor, the $\log(1 + |v|)$ vector in the affine) against the same cell at widths 24, 48 and 96, with two transformer blocks (graph regularization off, as every two-block cell runs), and at width 24 with the two-layer preprocessor (the round-28 wide recipe). Batch 32, class weight, 200 epochs, seeds 2700 to 2705 (2700 to 2702 at width 96 and with two blocks). Bold marks the highest value in a column. ↑ (↓) marks an arm whose held-out AUROC is above (below) the first arm's on every one of the same three seeds.

| arm | mean | sd | seeds | pooled [95% CI] | selected epochs | val step |
|---|---|---|---|---|---|---|
| the vector as log(1 + \|v\|), default initialization | 0.908 | 0.008 | 0.896 to 0.919 (6) | 0.912 [0.87, 0.95] | 33 to 58 | 0.0043 |
| round-33 cell at width 24 | 0.911 | 0.005 | 0.902 to 0.917 (6) | 0.912 [0.87, 0.95] | 38 to 56 | 0.0042 |
| round-33 cell at width 48 | **0.914** | 0.004 | 0.909 to 0.920 (6) | **0.915** [0.87, 0.95] | 27 to 104 | 0.0041 |
| round-33 cell at width 96 | 0.856 | 0.046 | 0.874, 0.804, 0.892 | 0.893 [0.82, 0.95] | 83, 137, 113 | 0.0022 |
| round-33 cell with two transformer blocks (graph regularization off) | 0.909 | 0.005 | 0.913, 0.911, 0.904 | 0.912 [0.87, 0.95] | 39, 53, 45 | 0.0048 |
| round-33 cell at width 24 with the two-layer preprocessor | 0.909 | 0.007 | 0.901 to 0.918 (6) | 0.912 [0.87, 0.95] | 50 to 79 | 0.0035 |

The update budget is not what the form needs either (round 56, job 2449; Table 64, Figure 68). At batch 64, half the updates of batch 32 at the same learning rate and epoch count, the cell reads 0.907, 0.892 and 0.907 (mean 0.902; pooled 0.909 [0.86, 0.95]), fitting at epochs 45 to 52; at batch 128, a quarter of the updates, 0.903, 0.913 and 0.904 (mean 0.907; pooled 0.908 [0.86, 0.95]), fitting at epochs 60 to 72, with the training loss still at 0.13 to 0.27 at epoch 199 where batch 32 reaches 0.08 to 0.13. Six cells of six fit and read the batch-32 number with a quarter of the optimizer

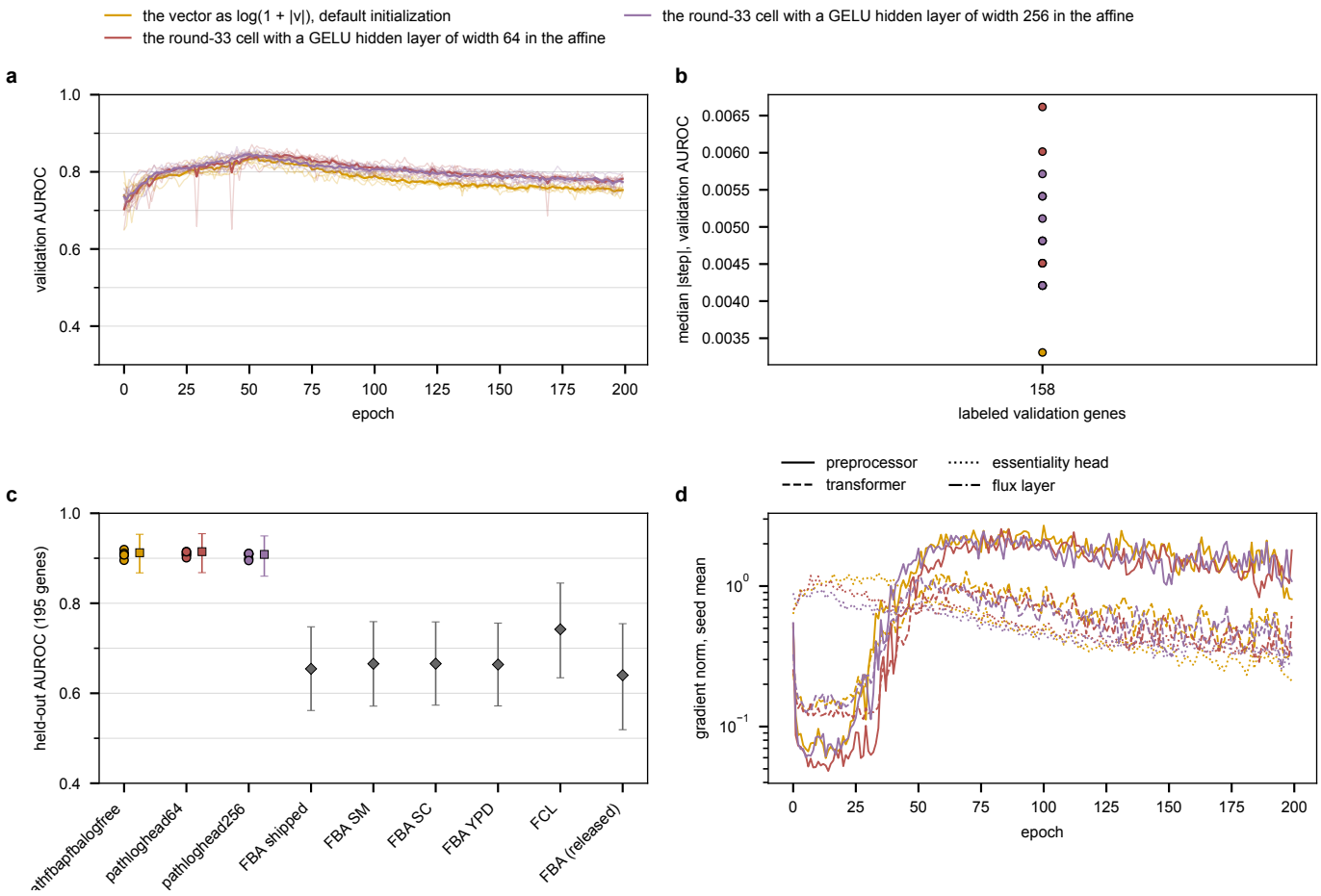



**Figure 66: Round 41, a hidden layer in the affine over the log1p vector.** **a**, Validation AUROC per epoch, one thin line per seed and the seed mean per arm. **b**, The median absolute epoch-to-epoch step per cell against the validation size. **c**, Held-out AUROC per cell (points), the seed mean and the gene bootstrap of the pooled per-gene logits (square with interval), beside FBA on four media, Flux Cone Learning and the released FBA predictions on the same 195 genes. **d**, Gradient norm per module per epoch, seed mean. Written by `experiments/028-gene-essentiality/scripts/wave12_readout.py -round w57`.

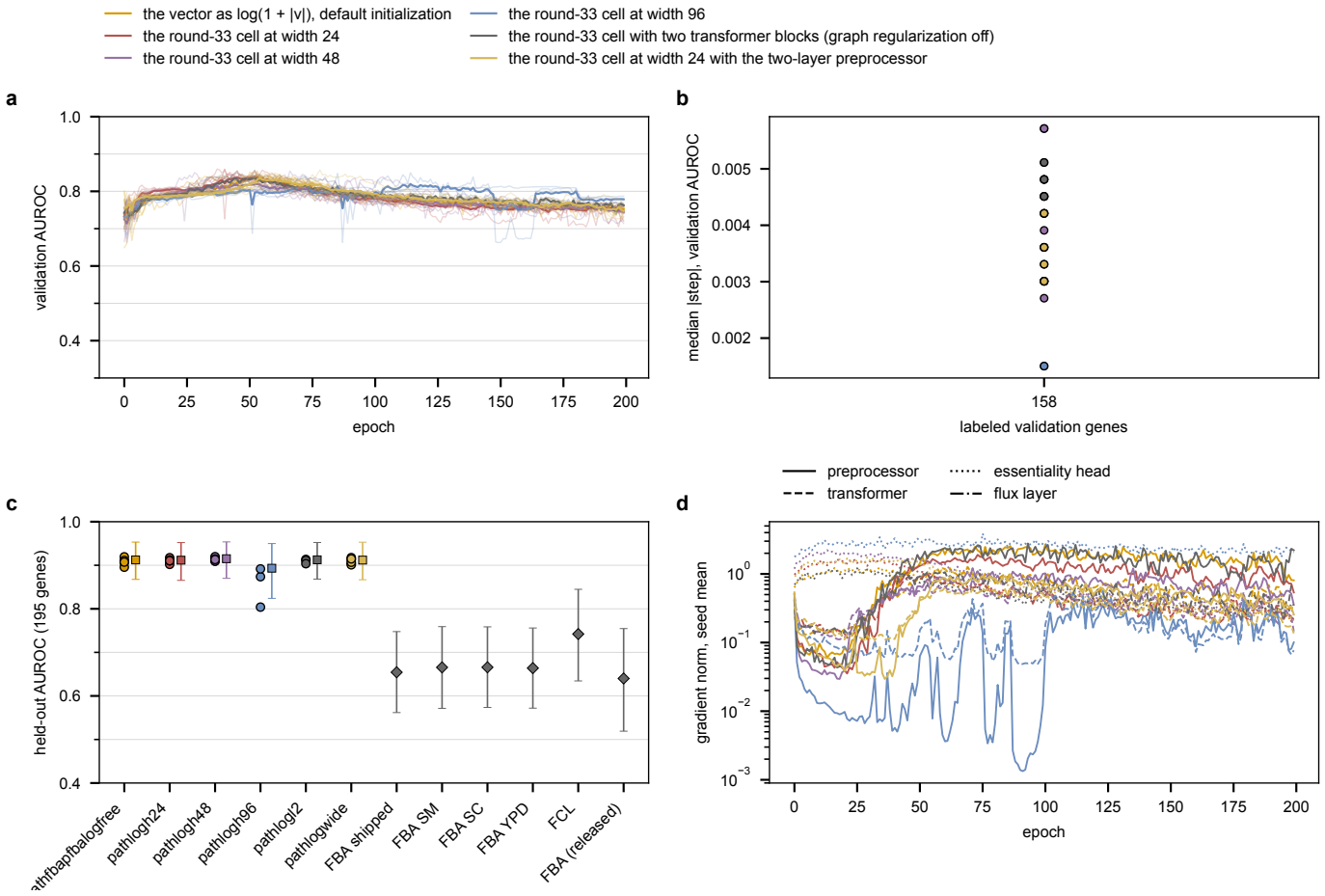



**Figure 67: Round 48, the capacity of the layer-free cell.** **a**, Validation AUROC per epoch, one thin line per seed and the seed mean per arm. **b**, The median absolute epoch-to-epoch step per cell against the validation size. **c**, Held-out AUROC per cell (points), the seed mean and the gene bootstrap of the pooled per-gene logits (square with interval), beside FBA on four media, Flux Cone Learning and the released FBA predictions on the same 195 genes. **d**, Gradient norm per module per epoch, seed mean. Written by `experiments/028-gene-essentiality/scripts/wave12_readout.py -round w64`.

steps, which is the property a store-backed model over millions of records needs: the store's read is one gather per sample, and the cell's fit does not depend on the step count within a factor of four.

**Table 64:** Round 56, the batch axis of the layer-free cell. The round-33 cell at batch 32 (seeds 2700 to 2705) against the same cell at batch 64 and at batch 128 (seeds 2700 to 2702), all at 200 epochs, so the larger batches see a half and a quarter of the updates. Class weight, the same learning rate. Bold marks the highest value in a column. ↑ (↓) marks an arm whose held-out AUROC is above (below) the first arm's on every one of the same three seeds.

| arm | mean | sd | seeds | pooled [95% CI] | selected epochs | val step |
|---|---|---|---|---|---|---|
| the vector as log(1 + \|v\|), default initialization | **0.908** | 0.008 | 0.896 to 0.919 (6) | **0.912** [0.87, 0.95] | 33 to 58 | 0.0043 |
| the round-33 cell at batch 64 | 0.902 | 0.008 | 0.907, 0.892, 0.907 | 0.909 [0.86, 0.95] | 47, 53, 43 | 0.0047 |
| the round-33 cell at batch 128 | 0.907 | 0.006 | 0.903, 0.913, 0.904 | 0.908 [0.86, 0.95] | 71, 71, 63 | 0.0053 |

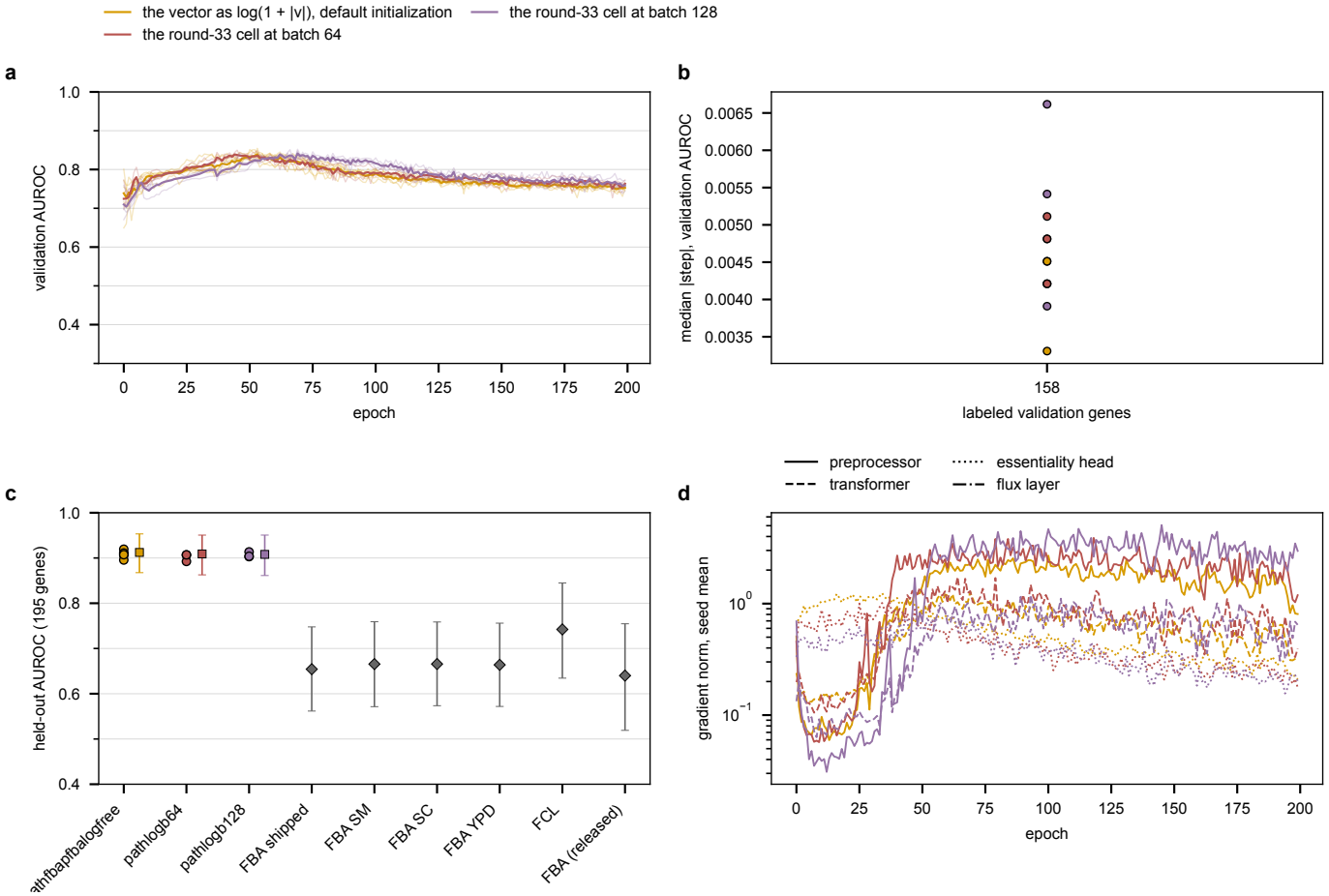



**Figure 68: Round 56, the batch axis of the layer-free cell.** **a**, Validation AUROC per epoch, one thin line per seed and the seed mean per arm. **b**, The median absolute epoch-to-epoch step per cell against the validation size. **c**, Held-out AUROC per cell (points), the seed mean and the gene bootstrap of the pooled per-gene logits (square with interval), beside FBA on four media, Flux Cone Learning and the released FBA predictions on the same 195 genes. **d**, Gradient norm per module per epoch, seed mean. Written by `experiments/028-gene-essentiality/scripts/wave12_readout.py -round w72`.

## 4.28 Round 35: the solve does not substitute for the annotation ✗

Since round 14 the open question on the pathway multi-hot has been whether a solve-derived quantity would substitute for the annotation. The layer-free form is the direct test: the same $\log(1+|v|)$ vector appended to the affine on ProtT5 alone, with no pathway block, six seeds (job 2410; Table 65, Figure 69), against the affine on ProtT5 (round 5), the cached-solve head on ProtT5 (round 24) and the same form with the pathway block (round 33).

**Table 65:** Round 35, the layer-free form on ProtT5 alone. The round-5 affine on ProtT5, the round-24 head over the cached GEM-box solve on ProtT5 and the round-33 layer-free cell with the pathway multi-hot (the comparators), against the deletion's pFBA vector as $\log(1 + |v|)$ appended to the affine on ProtT5 alone. Batch 32, class weight, 200 epochs, seeds 2700 to 2705 where run. Bold marks the highest value in a column. ↑ (↓) marks an arm whose held-out AUROC is above (below) the first arm's on every one of the same three seeds.

| arm | mean | sd | seeds | pooled [95% CI] | selected epochs | val step |
|---|---|---|---|---|---|---|
| context head + FBA growth ratios (round-3 levers) | 0.821 | 0.020 | 0.814, 0.805, 0.844 | 0.838 [0.76, 0.91] | 28, 70, 61 | 0.0036 |
| gated + head over the GEM-box solve's flux (all reactions) | 0.852 | 0.019 | 0.873, 0.836, 0.847 ↑ | 0.860 [0.78, 0.93] | 68, 164, 49 | 0.0184 |
| the vector as log1p, default initialization | **0.908** | 0.008 | 0.896 to 0.919 (6) | **0.912** [0.87, 0.95] | 33 to 58 | 0.0043 |
| ProtT5 alone; FBA affine + the deletion's pFBA vector as log1p (no pathway block, no layer) | 0.863 | 0.018 | 0.829 to 0.878 (6) | 0.868 [0.79, 0.94] | 66 to 160 | 0.0036 |

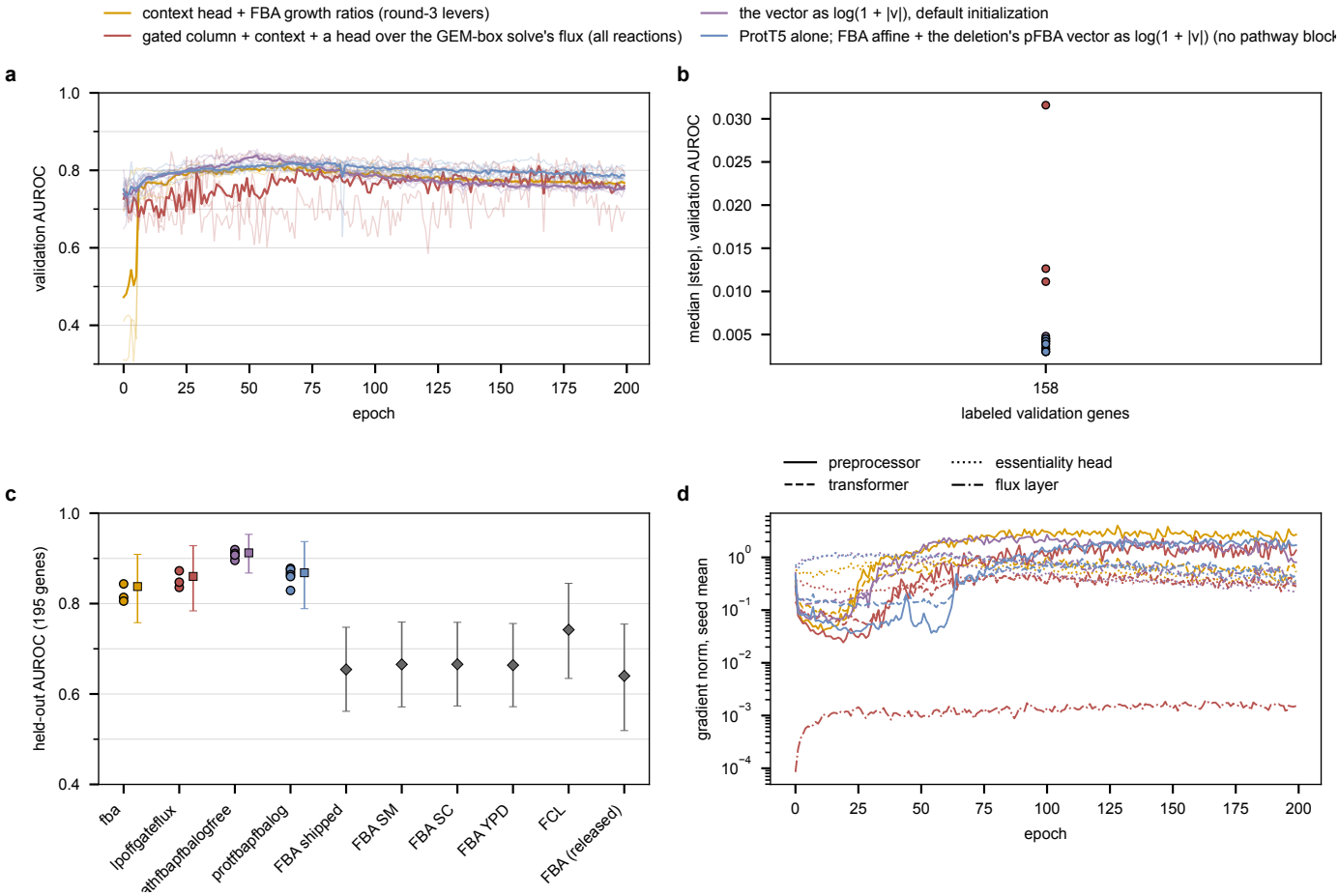



**Figure 69: Round 35, the layer-free form on ProtT5 alone.** **a**, Validation AUROC per epoch, one thin line per seed and the seed mean per arm. **b**, The median absolute epoch-to-epoch step per cell against the validation size. **c**, Held-out AUROC per cell (points), the seed mean and the gene bootstrap of the pooled per-gene logits (square with interval), beside FBA on four media, Flux Cone Learning and the released FBA predictions on the same 195 genes. **d**, Gradient norm per module per epoch, seed mean; the dash-dot line is the flux layer where there is one. Written by `experiments/028-gene-essentiality/scripts/wave12_readout.py -round w51`.

On ProtT5 alone the vector reads 0.878, 0.869, 0.875, 0.829, 0.864 and 0.860 (mean 0.863, sd 0.018; pooled 0.868 [0.79, 0.94]), above the affine's 0.821 (pooled 0.838) and level with the cached-solve head's 0.852 (pooled 0.860) on the same backbone, and 0.045 under the same form with the pathway block (0.908, pooled 0.912). The six cells fit later than their pathway counterparts (training loss under 0.6 at epochs 79 to 160 against 33 to 66) and select epochs 66 to 160. So the solve adds what the layer's head added on this backbone, 0.03 to 0.04 over the affine, and the annotation adds 0.045 on top of it: the two are not one signal, and the vector does not substitute for the pathway block any more than the block substitutes for the vector (0.890 with the block and the ratios alone, round 22).

## 4.29 Round 43: the solve on the node ✗

Every layer-free cell so far reads the vector at the logit, through the affine. The multi-phenotype form wants it in the representation, where every head can read it, so round 43 puts the same $\log(1 + |v|)$ vector on the deleted gene's node as a gene embedding (`torchcell/datasets/solve_vector.py`: 4,131 columns per gene from the pFBA store, the zero

row for a gene the store does not cover) beside the pathway multi-hot and ProtT5, with the four-ratio affine only, and beside it the same with the vector in the affine as well, six seeds each (jobs 2425 and 2426; Table 66, Figure 71).

Figure 70 draws the node path of this round and of round 44.

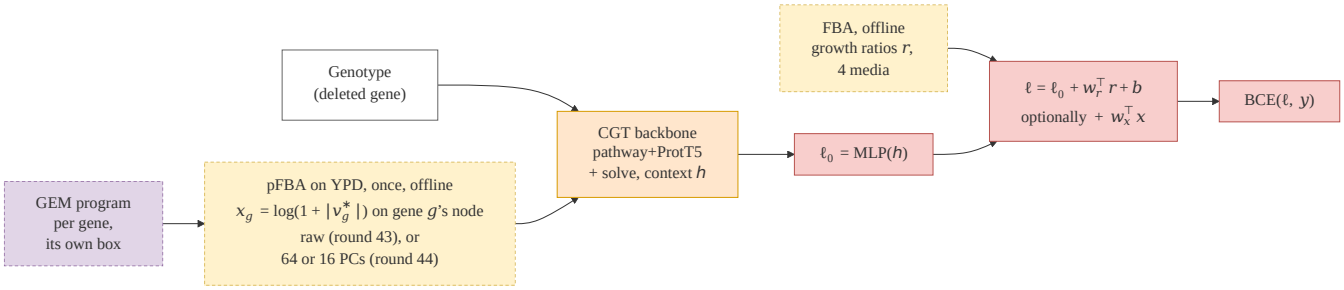



**Figure 70: The solve on the node, rounds 43 and 44.** The stored vector enters the representation instead of the logit: each gene's node carries its own deletion vector as an embedding beside the pathway multi-hot and ProtT5, raw in round 43 and reduced to 64 or 16 principal components in round 44. The affine on the logit keeps the four ratios, and one arm of each round keeps the vector there as well. Colors as in Figure 16.

**Table 66:** Round 43, the solved flux on the node. The round-22 affine on ProtT5 with the pathway multi-hot and the round-33 cell with the $\log(1+|v|)$ vector appended to the affine (the comparators), against the same vector entering as a node embedding beside the pathway multi-hot and ProtT5, with the four-ratio affine only. Batch 32, class weight, 200 epochs, seeds 2700 to 2705. Bold marks the highest value in a column. ↑ (↓) marks an arm whose held-out AUROC is above (below) the first arm's on every one of the same three seeds.

| arm | mean | sd | seeds | pooled [95% CI] | selected epochs | val step |
|---|---|---|---|---|---|---|
| pathway + ProtT5 + FBA ratios | 0.890 | 0.005 | 0.889, 0.895, 0.885 | 0.890 [0.83, 0.94] | 31, 45, 55 | 0.0054 |
| the vector as log1p, default initialization | **0.908** | 0.008 | 0.896 to 0.919 (6) | **0.912** [0.87, 0.95] | 33 to 58 | 0.0043 |
| pathway + ProtT5 + the deletion's log1p flux vector as a NODE embedding; the four-ratio affine only | 0.875 | 0.010 | 0.863 to 0.886 (6) | 0.894 [0.82, 0.95] | 84 to 118 | 0.0041 |
| the same, with the log1p vector appended to the affine as well | 0.889 | 0.009 | 0.875 to 0.901 (6) | 0.893 [0.82, 0.95] | 69 to 101 | 0.0022 |

On the node the vector costs what it adds at the logit. With the ratios alone in the affine it reads 0.864, 0.886, 0.873, 0.863, 0.885 and 0.880 (mean 0.875, sd 0.010; pooled 0.894 [0.82, 0.95]), under the affine without any vector (0.890) and 0.03 under the vector in the affine (0.908, pooled 0.912); with the vector in the affine as well it reads 0.875, 0.895, 0.892, 0.890, 0.883 and 0.901 (mean 0.889; pooled 0.893), the affine-alone number and still under the logit form. Both node arms fit late (training loss under 0.6 at epochs 71 to 120 against 33 to 66 for the logit form) and select epochs 69 to 118. The 4,131 columns enter the width-12 preprocessor as raw input, so the vector's read is a projection with 12 weights per reaction learned from 629 genes, most of the cell's parameters, where the affine reads it with one weight per reaction on the logit; what the representation path needs is a compact vector, and round 44 reduces it before the node.

A compact vector on the node fits on time and still reads nothing the backbone did not have (round 44, jobs 2427 to 2429; Table 67, Figure 72). The 1,121 $\log(1+|v|)$ rows are close to low rank: principal components fitted on them hold an explained variance of 1.000 at 64 and 0.999 at 16. As the node embedding, the 64-component vector reads 0.879, 0.872, 0.888, 0.908, 0.905 and 0.897 (mean 0.892, sd 0.014; pooled 0.894 [0.84, 0.94]) and the 16-component vector 0.906, 0.891, 0.906, 0.892, 0.895 and 0.891 (mean 0.897; pooled 0.901 [0.85, 0.95]), the affine-alone level (0.890), with transitions at epochs 27 to 57 as the logit form's; with the full vector in the affine as well, the 64-component node embedding reads 0.908, 0.914, 0.907, 0.903, 0.906 and 0.909 (mean 0.908, sd 0.004; pooled 0.909), the logit form's number and no more. So the representation path carries the vector without cost once it is compact, and carries nothing the logit path does not: at this resolution the solve's information about the deleted gene is read at the logit, and a representation-borne vector would need a head that reads it there, which the context head over the pooled tokens does not.

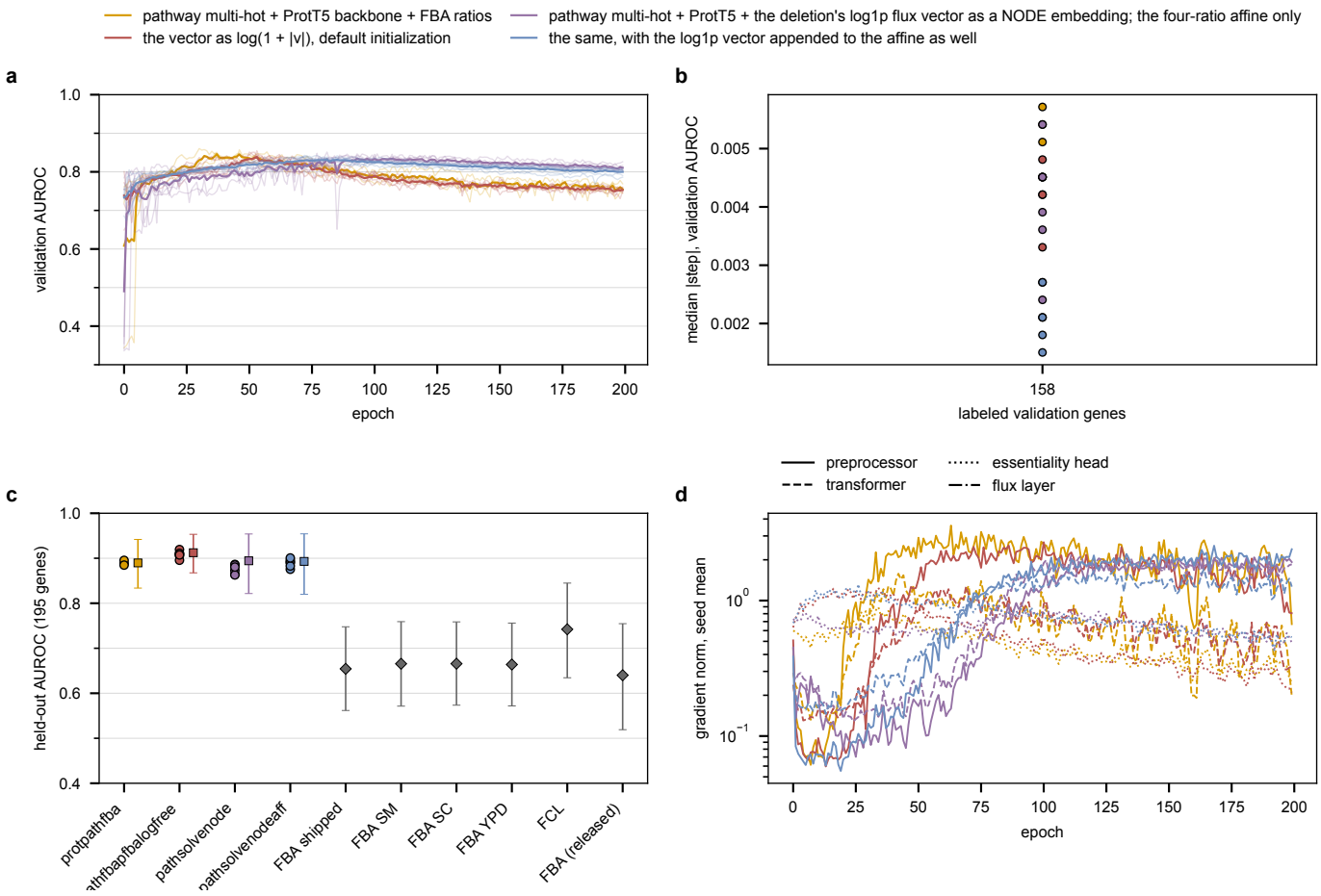



**Figure 71: Round 43, the solve on the node.** **a**, Validation AUROC per epoch, one thin line per seed and the seed mean per arm. **b**, The median absolute epoch-to-epoch step per cell against the validation size. **c**, Held-out AUROC per cell (points), the seed mean and the gene bootstrap of the pooled per-gene logits (square with interval), beside FBA on four media, Flux Cone Learning and the released FBA predictions on the same 195 genes. **d**, Gradient norm per module per epoch, seed mean. Written by `experiments/028-gene-essentiality/scripts/wave12_readout.py -round w59`.

**Table 67:** Round 44, the reduced vector on the node. The round-33 cell (the vector in the affine) and the round-43 cell (the raw 4,131-column vector as the node embedding) as the comparators, against the vector reduced by principal components fitted on the covered genes' $\log(1 + |v|)$ rows, 64 and 16 components, as the node embedding with the four-ratio affine, and the 64-component embedding with the full vector in the affine as well. Batch 32, class weight, 200 epochs, seeds 2700 to 2705. Bold marks the highest value in a column. ↑ (↓) marks an arm whose held-out AUROC is above (below) the first arm's on every one of the same three seeds.

| arm | mean | sd | seeds | pooled [95% CI] | selected epochs | val step |
|---|---|---|---|---|---|---|
| the vector as log1p, default initialization | **0.908** | 0.008 | 0.896 to 0.919 (6) | **0.912** [0.87, 0.95] | 33 to 58 | 0.0043 |
| pathway + ProtT5 + the deletion's log1p flux vector as a NODE embedding; the four-ratio affine only | 0.875 | 0.010 | 0.863 to 0.886 (6) | 0.894 [0.82, 0.95] | 84 to 118 | 0.0041 |
| the vector reduced by PCA to 64 components as the node embedding; the four-ratio affine only | 0.892 | 0.014 | 0.872 to 0.908 (6) | 0.894 [0.84, 0.94] | 32 to 68 | 0.0035 |
| the vector reduced by PCA to 16 components as the node embedding; the four-ratio affine only | 0.897 | 0.007 | 0.891 to 0.906 (6) | 0.901 [0.85, 0.95] | 26 to 55 | 0.0038 |
| the 64-component node embedding, with the full log1p vector in the affine as well | 0.908 | 0.004 | 0.903 to 0.914 (6) | 0.909 [0.86, 0.95] | 36 to 75 | 0.0042 |

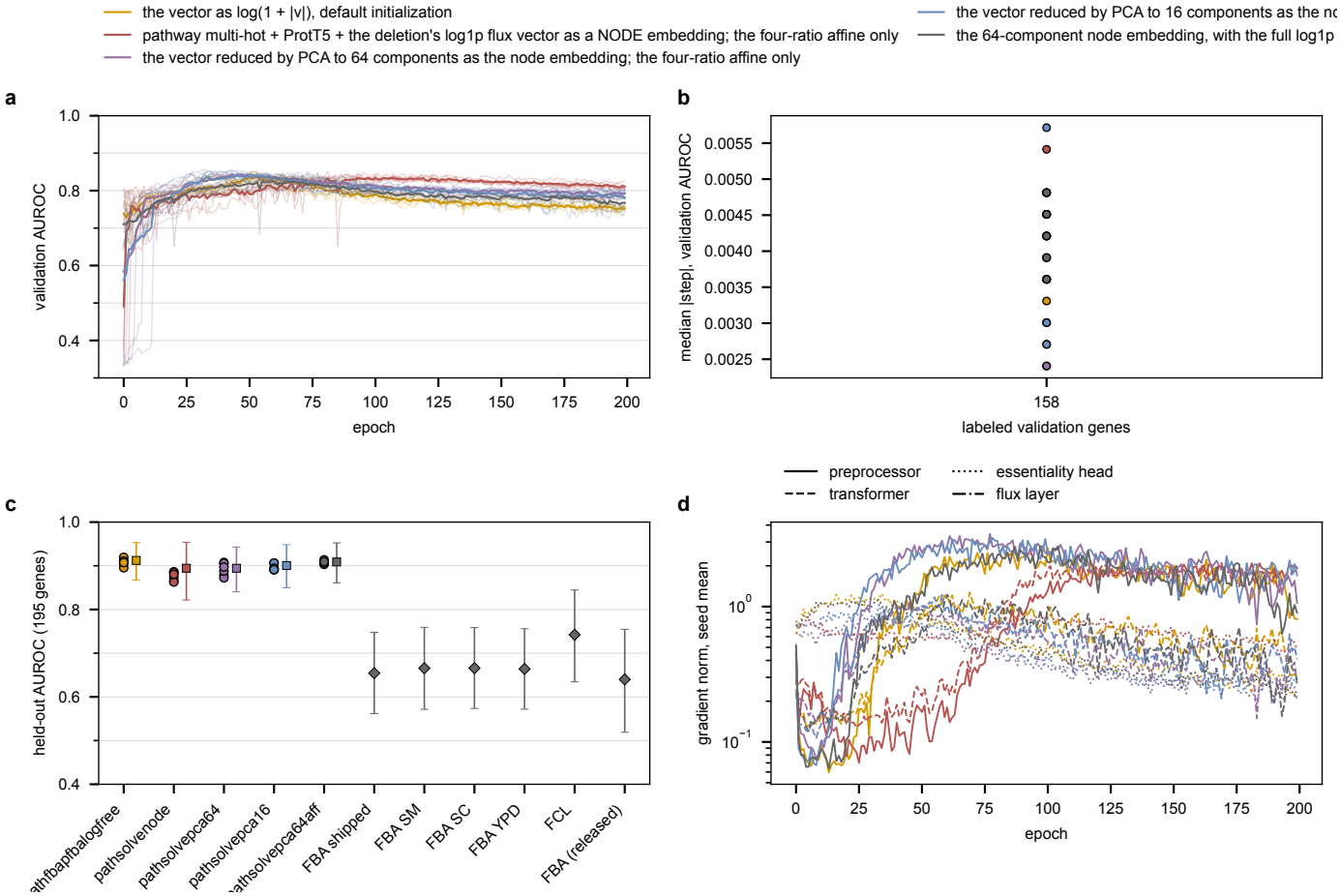



**Figure 72: Round 44, the reduced vector on the node.** **a**, Validation AUROC per epoch, one thin line per seed and the seed mean per arm. **b**, The median absolute epoch-to-epoch step per cell against the validation size. **c**, Held-out AUROC per cell (points), the seed mean and the gene bootstrap of the pooled per-gene logits (square with interval), beside FBA on four media, Flux Cone Learning and the released FBA predictions on the same 195 genes. **d**, Gradient norm per module per epoch, seed mean. Written by `experiments/028-gene-essentiality/scripts/wave12_readout.py -round w60`.

## 4.30 Round 36: a store from another solver's point ✗

The solve read of Section 4.31 is a probe; round 36 is the training test. The growth-maximizing program of round 27 was solved once per labeled gene by HiGHS 1.15's dual simplex, a vertex, and by its interior point without crossover at KKT $10^{-4}$, a point inside the optimal face and the kind of point a first-order solve on the card returns (`ess_reference_flux_table.py -solver`, 122 s a store; six of the 1,121 interior-point solves stopped short and keep the zero flux), and the round-33 cell was trained over each store, six seeds, paired against the pFBA store (job 2411; Table 68, Figure 74).

Figure 73 draws the form with the store swapped.

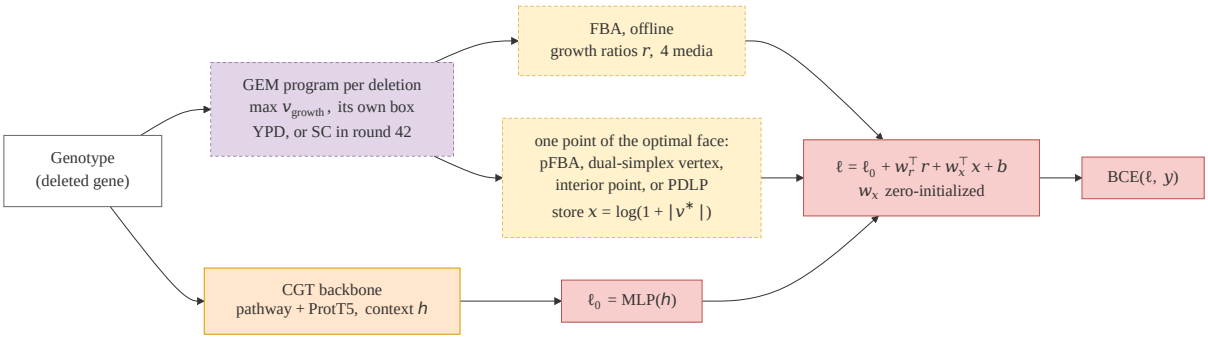



**Figure 73: The layer-free form over another solver's point, rounds 36, 38 and 42.** The path of Figure 62 with only the store changed: the growth-maximizing program solved once per deletion by the dual simplex (a vertex), by the interior point without crossover, or by PDLP, each stored in the same encoding, and in round 42 the parsimonious solve on synthetic complete medium. Colors as in Figure 16.

**Table 68:** Round 36, the layer-free cell over a store from another solver. The round-33 cell (the pFBA vector as $\log(1 + |v|)$ appended to the affine, pathway multi-hot and ProtT5, no layer; the comparator) against the same cell over the growth program solved per gene by HiGHS 1.15's dual simplex (a vertex) and by its interior point without crossover at KKT $10^{-4}$ (a point inside the optimal face). Batch 32, class weight, 200 epochs, seeds 2700 to 2705. Bold marks the highest value in a column. ↑ (↓) marks an arm whose held-out AUROC is above (below) the first arm's on every one of the same three seeds.

| arm | mean | sd | seeds | pooled [95% CI] | selected epochs | val step |
|---|---|---|---|---|---|---|
| the vector as log1p, default initialization | 0.908 | 0.008 | 0.896 to 0.919 (6) | **0.912** [0.87, 0.95] | 33 to 58 | 0.0043 |
| round-33 cell over the growth program's dual-simplex vertex (store by HiGHS simplex) | 0.840 | 0.051 | 0.767 to 0.887 (6) | 0.870 [0.79, 0.94] | 12 to 74 | 0.0071 |
| round-33 cell over the growth program's interior point without crossover (store by HiGHS ipm, KKT 1e-4) | **0.911** | 0.004 | 0.905 to 0.916 (6) | 0.911 [0.86, 0.95] | 52 to 83 | 0.0037 |

The interior point is as good a store as the parsimonious vertex and the growth program's own vertex is not. Over the interior-point store the cell reads 0.910, 0.909, 0.911, 0.905, 0.916 and 0.913 (mean 0.911, sd 0.004; pooled 0.911 [0.86, 0.95]), every seed selecting after the transition (epochs 52 to 83), against 0.908 and 0.912 pooled over the pFBA store. Over the dual-simplex vertex it reads 0.767, 0.887, 0.865, 0.856, 0.786 and 0.878 (mean 0.840, sd 0.051; pooled 0.870 [0.79, 0.94]): two seeds peak early (epochs 22 and 12, training loss 0.83 and 0.92) and the four that fit read 0.856 to 0.887, under every interior-point and pFBA cell. The three stores are three points of the same program's optimal face, one growth value (Section 4.31); what differs is which flux the point carries on the reactions the objective does not pin, and the vertex the dual simplex lands on is the one of the three the head reads worst. Plausible mechanism, unverified: a vertex of the growth face carries extremal flux through the free directions, the loops and the parallel routes, which pFBA's minimum-flux stage removes and the interior point spreads, and that extremal flux is what the head cannot use. The vectors themselves say what the vertex is made of (Table 69, `store_stats.py`): the dual-simplex vertex carries a median total absolute flux of 39,129 mmol/gDW/h per genotype with 339 reactions at the layer's ±100 clamp on every one of the 1,121 genotypes and on the wild type, where the pFBA vector carries 495 with none at the clamp, and the interior point and PDLP 2,463 and 1,973 with nine; the vertex's flux is loops spinning at the bound. The description is measured; that this is why the head reads it worst is not. What the round settles for the store is that the point a first-order solve on the card returns trains the cell to the number the CPU's parsimonious solve does, so the store can be built with the batched solver and nothing on the CPU.

**Table 69:** The four solve stores over the 1,121 labeled genes, one flux vector per single-gene deletion of the same growth-maximizing program (YPD, the GEM box clamped to ±100 mmol/gDW/h): rows the solver reported non-optimal and stored as the zero flux; the median total absolute flux per genotype; the median number of reactions carrying flux above $10^{-6}$; the median number of reactions at the ±100 clamp, and the share of genotypes with any; the median growth; and the vector's distance from the pFBA store's in the head's encoding (median relative $L_2$ of $\log(1 + |v|)$, and the mean share of reactions moved by more than 0.01). Written by `store_stats.py`.

| store | zero rows | $\sum \lvert v \rvert$ | carrying | at clamp | any clamp (%) | growth | rel. $L_2$ | moved (%) |
|---|---|---|---|---|---|---|---|---|
| pFBA (two-stage, scipy HiGHS) | 0 | 495 | 505 | 0 | 2 | 0.5352 | 0.00 | 0.0 |
| dual simplex (vertex) | 0 | 39,129 | 913 | 339 | 100 | 0.5352 | 6.12 | 12.0 |
| interior point, no crossover | 6 | 2,463 | 1,231 | 9 | 99 | 0.5352 | 1.65 | 16.1 |
| PDLP, no crossover | 12 | 1,973 | 1,136 | 9 | 99 | 0.5352 | 1.40 | 5.6 |

Under the reported protocol (round 37, jobs 2414, 2416 and 2418: five folds, three seeds per fold, each fold model the seed-mean logit) the interior-point store reads 0.915, 0.852, 0.887, 0.884 and 0.926 on the folds (mean 0.893) and 0.912 [0.86, 0.95] as the ensemble, 0.909 on one seed, against the pFBA store's 0.919 (Table 48); folds 1 and 2 select late on every seed (epochs 95 to 196, training loss 0.05 to 0.30 at selection) and read 0.84 to 0.89, where the pFBA store's cells

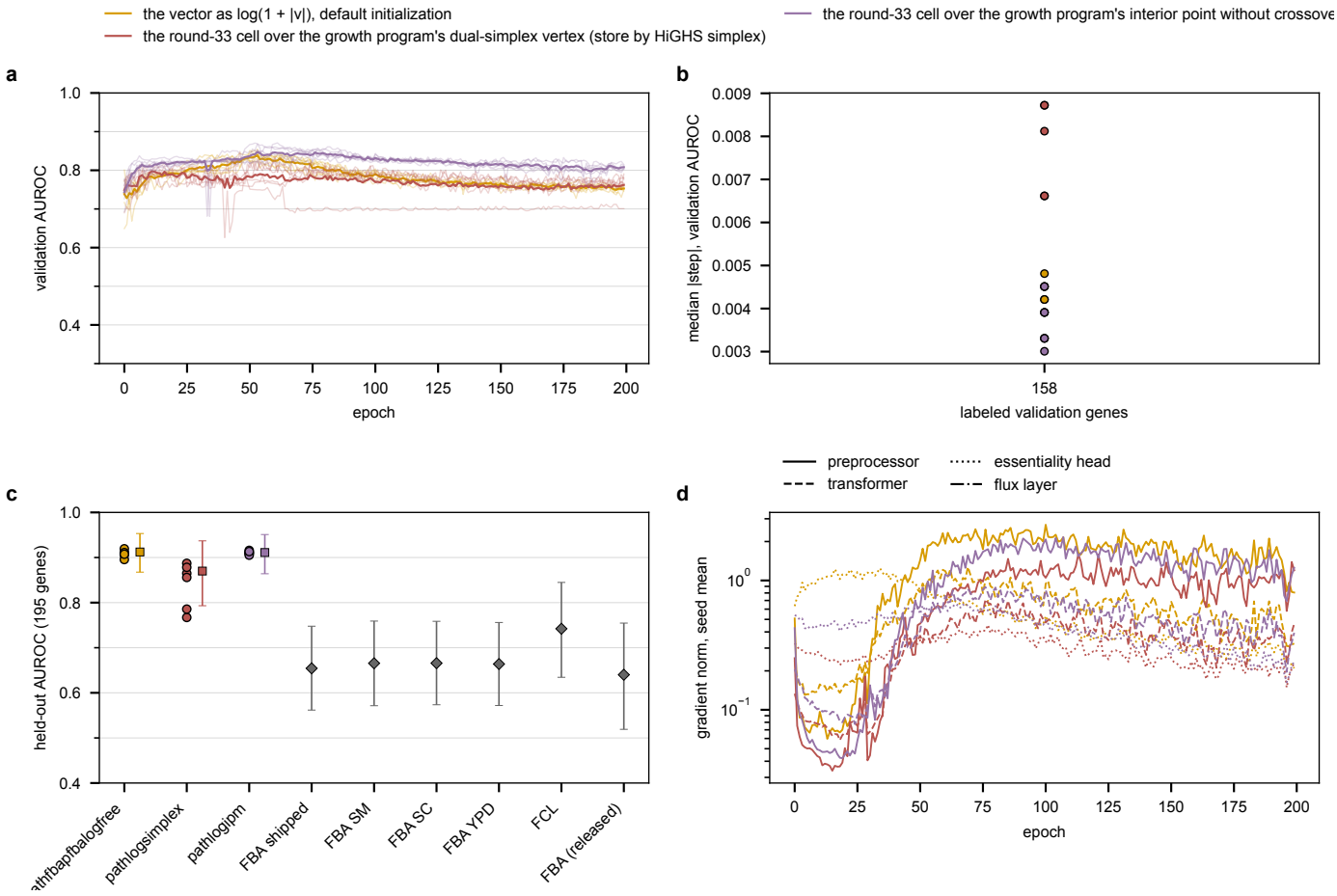



**Figure 74: Round 36, the layer-free cell over a store from another solver.** **a**, Validation AUROC per epoch, one thin line per seed and the seed mean per arm. **b**, The median absolute epoch-to-epoch step per cell against the validation size. **c**, Held-out AUROC per cell (points), the seed mean and the gene bootstrap of the pooled per-gene logits (square with interval), beside FBA on four media, Flux Cone Learning and the released FBA predictions on the same 195 genes. **d**, Gradient norm per module per epoch, seed mean. Written by `experiments/028-gene-essentiality/scripts/wave12_readout.py -round w52`.

on the same folds selected epochs 6 to 60, so the two stores' fold models are one number as ensembles and differ in how the cell trains on two of the five folds. As a fold family it reads 0.913 alone and 0.951 [0.92, 0.98] with the GO fold models (Table 86).

The PDLP store itself closes the question (round 38, job 2417; Table 72, Figure 77). The same program solved per gene by HiGHS's PDLP without crossover at KKT $10^{-4}$, the algorithm cuOpt runs on the card, written by the solve read of Section 4.31 (twelve of 1,121 programs stopped short and keep the zero flux), trains the round-33 cell to 0.906, 0.901, 0.910, 0.899, 0.897 and 0.908 (mean 0.903, sd 0.005; pooled 0.907 [0.86, 0.95]), every seed selecting after the transition (epochs 41 to 85), one number with the pFBA store's 0.912 and the interior point's 0.911. Under the reported protocol (round 39, job 2420, one seed) it reads 0.921, 0.859, 0.887, 0.878 and 0.907 on the folds and 0.909 [0.86, 0.95] as the ensemble, the interior-point store's single-seed 0.909 against the pFBA store's 0.920, with the same fold, the second, selecting late (epoch 187) and low on both first-order stores; with three seeds per fold (job 2432, each fold model the seed-mean logit) the folds read 0.915, 0.881, 0.884, 0.880 and 0.908 and the ensemble 0.911 [0.86, 0.95], the interior-point store's three-seed 0.912 (Table 48); as a fold family 0.915 alone and 0.949 [0.91, 0.98] with the GO fold models. Three stores from three solvers that stop inside or at the parsimonious edge of the optimal face train the cell to one number on the split file and to numbers within each other's intervals under the fold protocol; the growth vertex alone does not.

The tolerance of the first-order store does not move the number, and its uncertified fraction does (rounds 53 and 55, jobs 2445 and 2448; Table 70, Figure 75). The transfer read of Section 4.31 wrote PDLP's stores at KKT $10^{-6}$ and $10^{-8}$, with 70 and 320 of the 1,121 programs uncertified within 120 s and kept as zero rows against 12 at $10^{-4}$. Over the $10^{-6}$ store the cell reads 0.905 (sd 0.007; pooled 0.908 [0.86, 0.95]), the $10^{-4}$ store's 0.903 and 0.907, every seed fitting at epochs 47 to 66; over the $10^{-8}$ store 0.884, 0.896, 0.875, 0.899, 0.890 and 0.878 (mean 0.887, sd 0.010; pooled 0.892 [0.84, 0.94]), every seed fitting at epochs 35 to 50, 0.02 under the other two. Twelve and seventy zero rows cost nothing and 320 cost 0.02, and the cost is the rows, not the tighter point (round 57, job 2452): the $10^{-8}$ store with its uncertified rows filled from the $10^{-6}$ store (250 rows) and the $10^{-4}$ store (60), ten rows zero in every store, the second pass a batched solver on the card would run, reads 0.906 (sd 0.012; pooled 0.909 [0.86, 0.95]), the $10^{-4}$ and $10^{-6}$ stores' number, every seed fitting at epochs 49 to 68. So a first-order solver on the card can run at whatever tolerance it certifies most programs at, and what it fails to certify it re-solves looser; the store's number does not depend on which pass a row came from.

**Table 70:** Rounds 53, 55 and 57, the PDLP stores at KKT $10^{-6}$ and $10^{-8}$. The round-33 cell over the pFBA store and over the PDLP store at KKT $10^{-4}$ (round 38, 12 programs uncertified) as the comparators, against the same cell over the PDLP stores at KKT $10^{-6}$ (70 of 1,121 programs uncertified within 120 s and kept as zero flux) and $10^{-8}$ (320 uncertified) written by the transfer read, and over the $10^{-8}$ store with its uncertified rows filled from the $10^{-6}$ and $10^{-4}$ stores. Batch 32, class weight, 200 epochs, seeds 2700 to 2705. Bold marks the highest value in a column. ↑ (↓) marks an arm whose held-out AUROC is above (below) the first arm's on every one of the same three seeds.

| arm | mean | sd | seeds | pooled [95% CI] | selected epochs | val step |
|---|---|---|---|---|---|---|
| the vector as log1p, default initialization | **0.908** | 0.008 | 0.896 to 0.919 (6) | **0.912** [0.87, 0.95] | 33 to 58 | 0.0043 |
| round-33 cell over the growth program's PDLP point (store by HiGHS PDLP, KKT 1e-4; the cuOpt algorithm) | 0.903 | 0.005 | 0.897 to 0.910 (6) | 0.907 [0.86, 0.95] | 41 to 85 | 0.0048 |
| round-33 cell over the PDLP store at KKT 1e-6 (70 uncertified rows kept as zero flux) | 0.905 | 0.007 | 0.897 to 0.913 (6) | 0.908 [0.86, 0.95] | 47 to 66 | 0.0050 |
| round-33 cell over the PDLP store at KKT 1e-8 (320 uncertified rows kept as zero flux) | 0.887 | 0.010 | 0.875 to 0.899 (6) | 0.892 [0.84, 0.94] | 38 to 52 | 0.0048 |
| round-33 cell over the PDLP 1e-8 store with its uncertified rows filled from the 1e-6 and 1e-4 stores | 0.906 | 0.012 | 0.890 to 0.923 (6) | 0.909 [0.86, 0.95] | 49 to 68 | 0.0052 |

The store's medium does not move the number either (round 42, job 2424; Table 71, Figure 76). Every store above is solved on YPD, the medium the essentiality calls were scored on; the same two-stage solve with the exchanges set to synthetic complete medium (wild-type growth 0.5428 against 0.5352) trains the cell to 0.918, 0.914, 0.908, 0.905, 0.915 and 0.901 (mean 0.910, sd 0.007; pooled 0.912 [0.87, 0.95]), the YPD store's 0.908 and 0.912, with every seed selecting near the transition (epochs 28 to 52). On this endpoint the vector's information is the deletion's, not the medium's; whether a store per medium carries anything for a phenotype scored across media is the question a database of media conditions asks, and the 195 genes scored on one medium cannot.

**Table 71:** Round 42, the store on synthetic complete medium. Round 33's cell over the pFBA store built on YPD (the comparator) against the same cell over the pFBA store built on synthetic complete medium, the solve's exchanges set to SC and everything else the same. Batch 32, class weight, 200 epochs, seeds 2700 to 2705. Bold marks the highest value in a column. ↑ (↓) marks an arm whose held-out AUROC is above (below) the first arm's on every one of the same three seeds.

| arm | mean | sd | seeds | pooled [95% CI] | selected epochs | val step |
|---|---|---|---|---|---|---|
| the vector as log1p, default initialization | 0.908 | 0.008 | 0.896 to 0.919 (6) | 0.912 [0.87, 0.95] | 33 to 58 | 0.0043 |
| round-33 cell over the pFBA store built on synthetic complete medium (SC) instead of YPD | **0.910** | 0.007 | 0.901 to 0.918 (6) | **0.912** [0.87, 0.95] | 28 to 52 | 0.0037 |

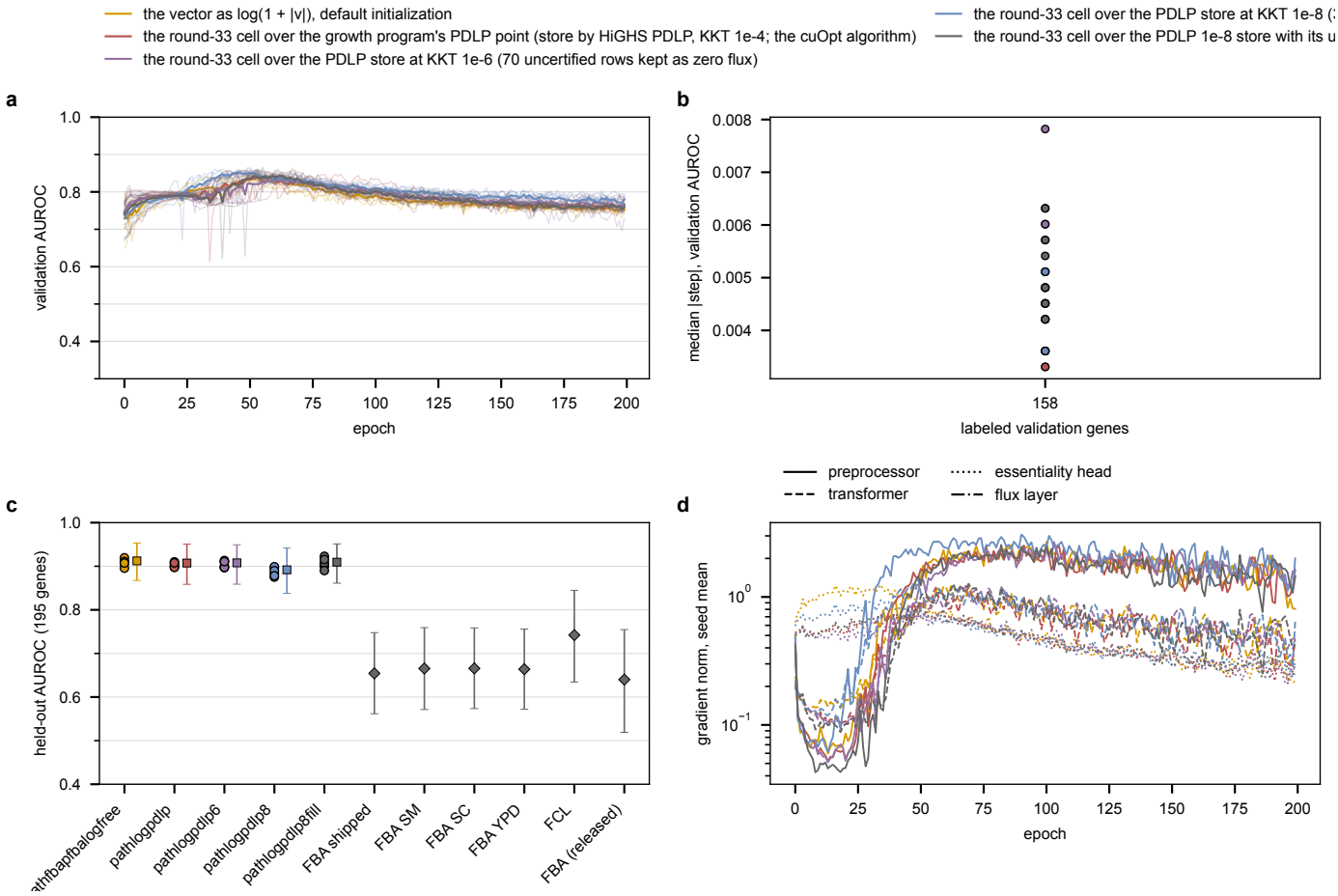



**Figure 75: Rounds 53, 55 and 57, the PDLP stores at KKT $10^{-6}$ and $10^{-8}$, and the $10^{-8}$ store with its uncertified rows filled.** **a**, Validation AUROC per epoch, one thin line per seed and the seed mean per arm. **b**, The median absolute epoch-to-epoch step per cell against the validation size. **c**, Held-out AUROC per cell (points), the seed mean and the gene bootstrap of the pooled per-gene logits (square with interval), beside FBA on four media, Flux Cone Learning and the released FBA predictions on the same 195 genes. **d**, Gradient norm per module per epoch, seed mean. Written by `experiments/028-gene-essentiality/scripts/wave12_readout.py -round w69`.

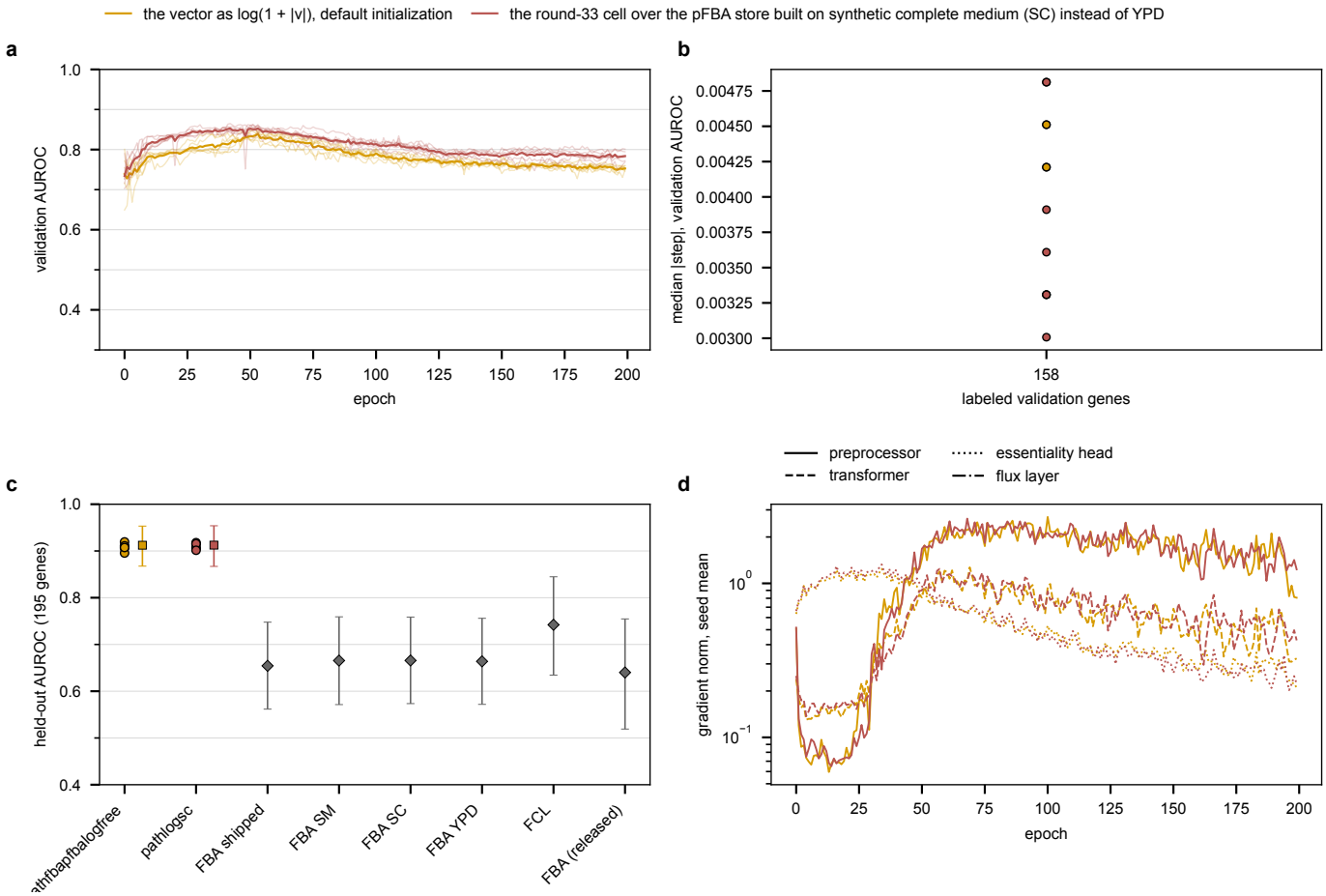



**Figure 76: Round 42, the store on synthetic complete medium.** **a**, Validation AUROC per epoch, one thin line per seed and the seed mean per arm. **b**, The median absolute epoch-to-epoch step per cell against the validation size. **c**, Held-out AUROC per cell (points), the seed mean and the gene bootstrap of the pooled per-gene logits (square with interval), beside FBA on four media, Flux Cone Learning and the released FBA predictions on the same 195 genes. **d**, Gradient norm per module per epoch, seed mean. Written by `experiments/028-gene-essentiality/scripts/wave12_readout.py -round w58`.

**Table 72:** Round 38, the layer-free cell over the PDLP store. The round-33 cell over the pFBA store and over the interior-point store (the comparators), against the same cell over the growth program solved per gene by HiGHS 1.15's PDLP without crossover at KKT $10^{-4}$, the algorithm a batched solve on the card runs, written by `solve_tolerance_read.py`. Batch 32, class weight, 200 epochs, seeds 2700 to 2705. Bold marks the highest value in a column. ↑ (↓) marks an arm whose held-out AUROC is above (below) the first arm's on every one of the same three seeds.

| arm | mean | sd | seeds | pooled [95% CI] | selected epochs | val step |
|---|---|---|---|---|---|---|
| the vector as log1p, default initialization | 0.908 | 0.008 | 0.896 to 0.919 (6) | **0.912** [0.87, 0.95] | 33 to 58 | 0.0043 |
| round-33 cell over the growth program's interior point without crossover (store by HiGHS ipm, KKT 1e-4) | **0.911** | 0.004 | 0.905 to 0.916 (6) | 0.911 [0.86, 0.95] | 52 to 83 | 0.0037 |
| round-33 cell over the growth program's PDLP point (store by HiGHS PDLP, KKT 1e-4; the cuOpt algorithm) | 0.903 | 0.005 | 0.897 to 0.910 (6) | 0.907 [0.86, 0.95] | 41 to 85 | 0.0048 |

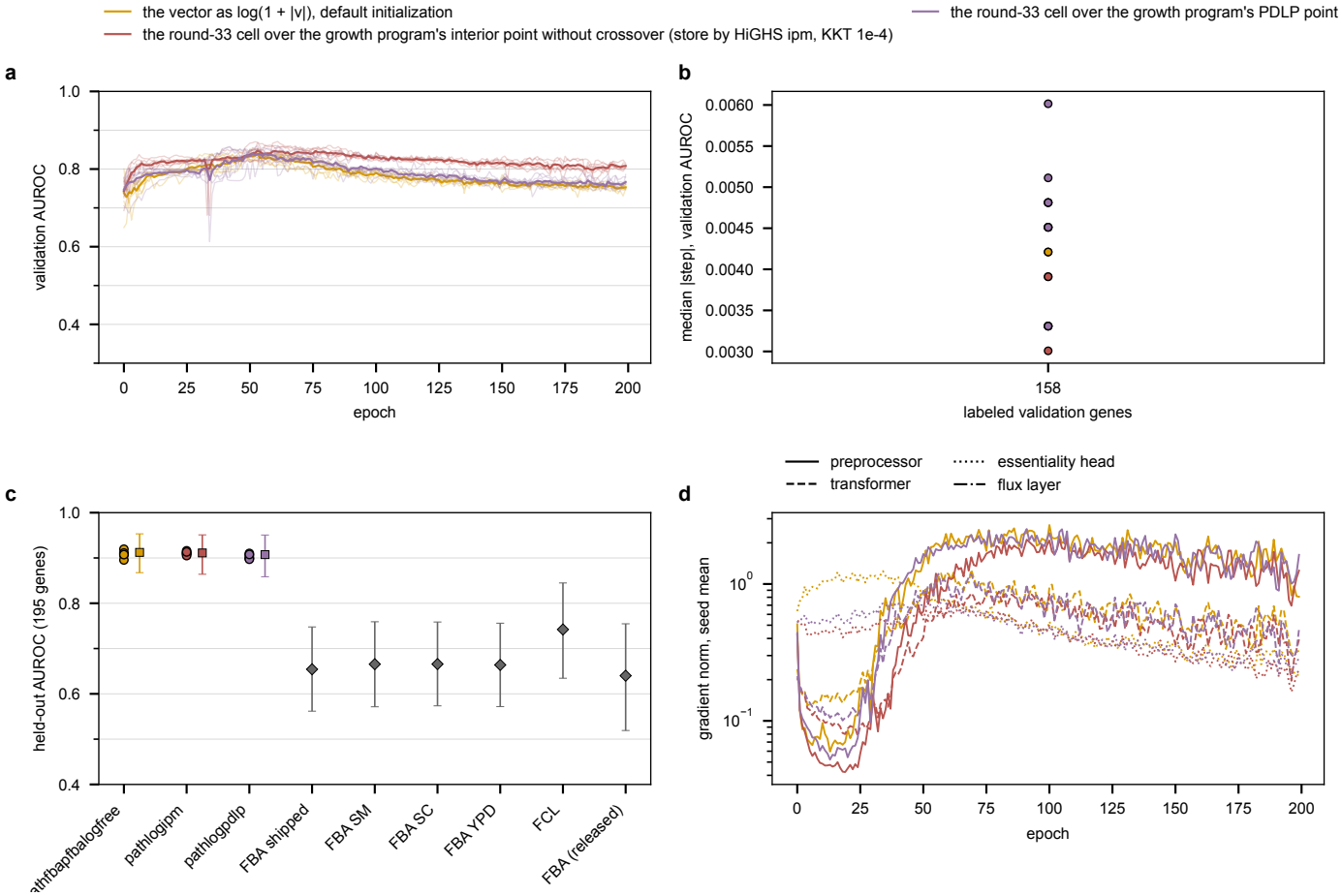



**Figure 77: Round 38, the layer-free cell over the PDLP store.** **a**, Validation AUROC per epoch, one thin line per seed and the seed mean per arm. **b**, The median absolute epoch-to-epoch step per cell against the validation size. **c**, Held-out AUROC per cell (points), the seed mean and the gene bootstrap of the pooled per-gene logits (square with interval), beside FBA on four media, Flux Cone Learning and the released FBA predictions on the same 195 genes. **d**, Gradient norm per module per epoch, seed mean. Written by `experiments/028-gene-essentiality/scripts/wave12_readout.py -round w54`.

## 4.31 The solve across solvers: the argmax is not unique ✗

The head over the solved flux reads one solver's answer. A batched solve on the card (NVIDIA's cuOpt) is a first-order method, PDLP, run to a $10^{-4}$ relative tolerance in its own benchmark, and it stops at a point of the optimal face rather than at a vertex, so before any card is involved the question is whether the head's columns survive the change of solver. HiGHS 1.15 carries the PDLP algorithm on the CPU, an interior-point solver that can stop before crossover, and the simplex at any tolerance, so the round-27 program was solved for every labeled gene five ways (Table 73). The read on each is the growth value, the flux vector's distance from the reference in the head's own encoding, and a linear probe over that encoding, fitted on the training genes' reference vectors and scored on each solver's held-out vectors, which is what a head trained on exact solves sees when it is served another solver's, beside the same probe fitted within each solver.

**Table 73:** The round-27 program (the GEM box clamped to ±100 mmol/gDW/h, YPD on the exchanges, the gene's reactions closed through the GPR, maximize growth) solved for every one of the 1,121 labeled genes by five solvers of HiGHS 1.15.1, one thread each. "Optimal" counts the programs the solver reported optimal within 120 s; "s" is the median wall clock per solve; "growth AUROC" scores 1− the knockout-to-wild-type growth ratio on the 195 held-out genes; "moved" is the mean fraction of the 4,131 reactions whose $\log(1+|v|)$ differs from the reference's by more than 0.01, and "rel. $L_2$" the median relative distance of that vector. The probe is a logistic regression over $\log(1+|v|)$ ($C = 0.01$, balanced) fitted on the 629 training genes: "fit on reference" is fitted on the reference solver's vectors and scored on this solver's held-out vectors, what a head trained on exact solves sees when served another solver's; "fit within" is fitted and scored on this solver's own. Gene bootstrap, 2,000 draws. Written by `solve_tolerance_read.py`. Bold marks the highest within-solver probe.

| solver | optimal | s | growth AUROC | moved (%) | rel. $L_2$ | probe, fit on reference | probe, fit within |
|---|---|---|---|---|---|---|---|
| dual simplex, default tolerances (reference) | 1121/1121 | 0.11 | 0.632 | 0.0 | 0.00 | 0.756 [0.63, 0.86] | 0.756 [0.63, 0.86] |
| primal simplex, default tolerances (another exact vertex) | 1121/1121 | 0.19 | 0.635 | 4.0 | 0.37 | 0.543 [0.42, 0.67] | 0.673 [0.55, 0.79] |
| dual simplex, feasibility tolerances $10^{-4}$ | 1121/1121 | 0.16 | 0.648 | 3.4 | 0.41 | 0.503 [0.37, 0.64] | 0.653 [0.51, 0.77] |
| interior point, no crossover, KKT $10^{-4}$ | 1115/1121 | 0.10 | 0.602 | 16.4 | 0.88 | 0.577 [0.45, 0.70] | **0.776** [0.67, 0.86] |
| PDLP (the cuOpt algorithm, on the CPU), KKT $10^{-4}$ | 1109/1121 | 13.57 | 0.675 | 13.4 | 0.96 | 0.685 [0.57, 0.79] | 0.720 [0.62, 0.82] |

The growth value is the same under every solver, to $3 \times 10^{-7}$ for the simplex at $10^{-4}$ and $7 \times 10^{-6}$ for the interior point, and its ranking of the 195 genes is FBA's 0.60 to 0.68 whichever solver produced it. The flux vector is not the same: the primal simplex, an exact solve at the same tolerances, lands on another vertex of the optimal face and moves 4 percent of the reactions by more than 0.01 in $\log(1+|v|)$ (median relative distance 0.37); the dual simplex at $10^{-4}$ moves 3 percent; the interior point without crossover and PDLP, which stop inside the face, move 16 and 13 percent (0.88 and 0.96). The probe fitted on the reference vertices reads 0.756 [0.63, 0.86] on the reference's own held-out vectors and 0.54, 0.50, 0.58 and 0.69 on the other four solvers', at chance on the other exact vertex, so a head trained on one solver's vertex does not read another's; and fitted within each solver it reads 0.67, 0.65, 0.78 and 0.72, the interior point's 0.776 [0.67, 0.86] one number with the reference vertex's 0.756. The argmax of the program is degenerate and the ranking the head extracts is a property of the point the solver returns, not of the program; that point is as rankable from inside the face as from a vertex, and a head must be trained on the solver it will be served with. On the CPU the PDLP solve took 11.7 s a program against 0.1 for the simplex and 0.09 for the interior point, and reported 12 of 1,121 programs non-optimal within 120 s, so the card is where PDLP's batch would have to pay for itself; the interior-point solve without crossover is the CPU form that returns the same kind of point at the simplex's cost.

The read above left the change of tolerance together with the change of point, and the transfer read separates them (`solve_tolerance_transfer.py`, job 2436, 93 minutes at twenty workers; Tables 74, 75 and 76): the same programs under the interior point without crossover and under PDLP at KKT $10^{-4}$, $10^{-6}$ and $10^{-8}$, with a probe fitted on every solver's training vectors and scored on every solver's held-out vectors. Two of the six settings do not certify: the interior point without crossover reports 1,115 programs optimal at $10^{-4}$ and 112 at $10^{-6}$ and $10^{-8}$, and PDLP certifies 1,109, 1,051 and 801 at 19, 21 and 25 s a program, the $10^{-8}$ shortfall the 120 s cap. Across tolerances of one first-order solver the point is the same point: PDLP's $10^{-4}$ and $10^{-6}$ vectors move 0.6 percent of the reactions against each other (3 percent against $10^{-8}$), and the probe fitted at $10^{-4}$ scores the $10^{-6}$ vectors at 0.709 against its own 0.716, the probe fitted at $10^{-6}$ the $10^{-4}$ vectors at 0.717 against its own 0.702. Across solvers the same probes lose 0.05 to 0.35 (the reference vertex's probe 0.68 on PDLP's points, PDLP's 0.55 on the vertex, the interior point's 0.39). So a head trained on a first-order store at one tolerance is served at another without loss, and the solver, not the tolerance, is what the store and the head must share; the $10^{-6}$ store is written (70 programs uncertified and kept as zero flux) for the training read.

**Table 74:** The round-27 program for the 1,121 labeled genes under HiGHS 1.15.1's dual simplex, its interior point without crossover and its PDLP, the last two at KKT tolerances $10^{-4}$, $10^{-6}$ and $10^{-8}$, one thread each, 120 s per solve. "Optimal" counts the programs the solver certified; a program it did not certify keeps the zero flux. "s" and "iterations" are medians per solve; "growth diff." is the largest absolute difference of the knockout-to-wild-type growth ratio from the simplex's, and "growth AUROC" scores 1− that ratio on the 195 held-out genes. Written by `solve_tolerance_transfer.py`.

| solver | optimal | s | iterations | growth diff. | growth AUROC |
|---|---|---|---|---|---|
| dual simplex, default (anchor) | 1121/1121 | 0.16 | 1821 | 0.0e+00 | 0.632 |
| interior point $10^{-4}$ | 1115/1121 | 0.15 | 32 | 6.5e-06 | 0.602 |
| interior point $10^{-6}$ | 112/1121 | 0.16 | 35 | 1.0e+00 | 0.455 |
| interior point $10^{-8}$ | 112/1121 | 0.16 | 36 | 1.0e+00 | 0.503 |
| PDLP $10^{-4}$ | 1109/1121 | 19.13 | 422720 | 1.0e+00 | 0.675 |
| PDLP $10^{-6}$ | 1051/1121 | 20.88 | 461280 | 1.0e+00 | 0.676 |
| PDLP $10^{-8}$ | 801/1121 | 25.06 | 552480 | 1.0e+00 | 0.599 |

**Table 75:** A logistic regression over $\log(1+|v|)$ ($C = 0.01$, balanced, standardized on the fitting solver's training genes) fitted on the 629 training genes' vectors from the row's solver and scored on the 195 held-out genes' vectors from the column's, AUROC; the diagonal is the within-solver read, an off-diagonal cell within a family is the transfer across tolerance and across families the transfer across point. Settings that certified fewer than half the programs are omitted. Bold marks the highest within-solver probe. Written by `solve_tolerance_transfer.py`.

| fitted on / scored on | simplex | ipm $10^{-4}$ | PDLP $10^{-4}$ | PDLP $10^{-6}$ | PDLP $10^{-8}$ |
|---|---|---|---|---|---|
| simplex | 0.756 | 0.577 | 0.680 | 0.678 | 0.649 |
| ipm $10^{-4}$ | 0.390 | **0.769** | 0.645 | 0.629 | 0.632 |
| PDLP $10^{-4}$ | 0.547 | 0.620 | 0.716 | 0.709 | 0.662 |
| PDLP $10^{-6}$ | 0.488 | 0.615 | 0.717 | 0.702 | 0.671 |
| PDLP $10^{-8}$ | 0.472 | 0.532 | 0.645 | 0.632 | 0.683 |

**Table 76:** The mean percentage of the 4,131 reactions whose $\log(1+|v|)$ differs by more than 0.01 between the row's and the column's solver, over the 1,121 programs. Written by `solve_tolerance_transfer.py`.

| solver | simplex | ipm $10^{-4}$ | PDLP $10^{-4}$ | PDLP $10^{-6}$ | PDLP $10^{-8}$ |
|---|---|---|---|---|---|
| simplex | 0.0 | 16.4 | 13.4 | 13.6 | 13.9 |
| ipm $10^{-4}$ | 16.4 | 0.0 | 15.2 | 15.5 | 16.9 |
| PDLP $10^{-4}$ | 13.4 | 15.2 | 0.0 | 0.6 | 3.4 |
| PDLP $10^{-6}$ | 13.6 | 15.5 | 0.6 | 0.0 | 2.9 |
| PDLP $10^{-8}$ | 13.9 | 16.9 | 3.4 | 2.9 | 0.0 |

## 4.32 Where the gradient goes ✗

Every cell logs a gradient norm per module each epoch, and read across the module arms (Figure 78) they say which of the three failure modes each arm is in. On the forced arms the whole gradient passes through the layer and the layer passes it on: the flux layer carries 10 to 40 percent of the summed module norm and the preprocessor's norm climbs to 1 to 3, the same level as under the pooled head, so the backbone is trained through the layer and the growth column is a trained, free coordinate that ranks no better than the head. On the gated arms the layer carries 0.3 to 3 percent, the head carries the rest, and the gate ends between $-0.2$ and $+0.2$ across seeds under constraint scale 0 (seed mean $+0.05$) and at $+0.2$ on every seed with the FBA affine beside it, so the backbone is trained through the MLP and the flux path is a small correction. On the warm-started forced arm with $\alpha$ frozen the backbone receives no gradient at all: the preprocessor's norm sits at 0.02 to 0.1 for the whole run, two orders below every other arm, because the flux is 99 percent a fixed reference and only the two parameters of the growth affine train; the cell is the affine on the projected reference, and the projection reversed its sign. The warm-started gated arms leave the flux path at 0.3 percent of the gradient and push the gate to $-0.8$, which is the head learning to subtract a sign-reversed growth column. None of the arms is gradient-starved except the one that was meant to be a fixed function, so the alternating module's null on this endpoint is not a gradient-flow problem; the gradient reaches the layer, and what that layer computes is not growth. The layer that solves has the opposite gradient problem, which rounds 15b and 15c measure: its gradient is exactly the dual of the binding bounds, which is zero when no gene-owned bound binds and absent when the program is infeasible.

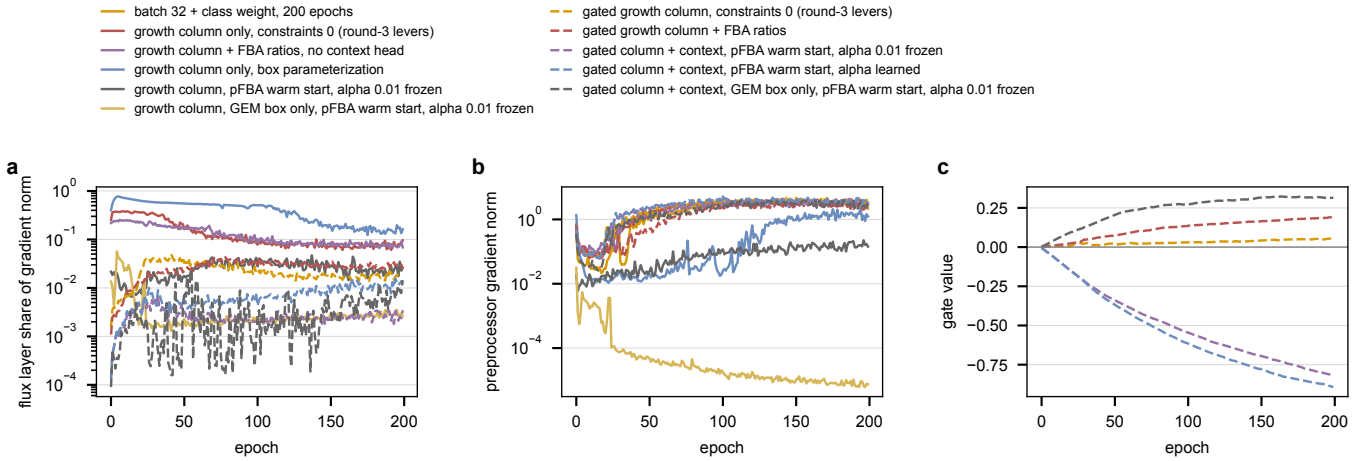



**Figure 78: Where the gradient goes on the module arms.** **a**, The flux layer's share of the summed per-module gradient norm per epoch, seed mean, on every module arm of rounds 4 to 13; solid lines are the forced arms and the pooled reference, dashed the gated arms. **b**, The preprocessor's gradient norm per epoch, seed mean, on the same arms. **c**, The gate's value per epoch on the gated arms. Written by `experiments/028-gene-essentiality/scripts/grad_flow_readout.py`.

## 4.33 Why the solve helps and the layer does not ✗

What FBA contributes to the prediction is one number per deletion: the maximum of growth over the polytope $\{Sv = 0,\ \ell \le v \le u\}$ with the deleted gene's reactions closed. That number ranks the held-out genes at 0.65 alone, its four-medium version lifts a logistic regression, a random forest and the transformer by 0.05 each, and the solve has more to give than the number: eight statistics of the gene's reactions under the wild-type solve reach 0.80 with no sequence, and the whole deletion flux vector 0.77. The value is in the optimum over the constraint set and in the flux that attains it.

The module holds the same constraint set and computes something else. Its flux is $K$ alternating projections of a proposal the backbone writes, and the rounds measure, in order, that the projections stop short of the polytope (58 percent of reactions out of bounds when the balance is exact, a balance residual of 0.85 to 2.0 when the bounds are), that the growth column of a flux off the polytope is a free coordinate the readout learns to use with either sign (the probe: 0.23 on the held-out genes under trained weights, Spearman $-0.25$ with the solve), that the learned parts of the layer are not what the column carries (the shuffled-$k_{cat}$ control matches the real table), and that the layer's constraint set rejects the solve when the solve is supplied as the proposal (round 10: the reference violates the anchored box on a third of its reactions and the returned flux sits half the reference's norm away, sign reversed). The gradient is not the cause; it reaches the layer on every arm that lets it.

The comparison also bounds what a feasible layer could add. A layer that reached the polytope would return a feasible flux, not the optimal one, unless the backbone learned to write proposals whose projection is the optimum; the solve finds the optimum by construction. And a growth column is one number, where the solve's flux vector scores 0.11 to 0.13 more AUROC on the ladder than its growth ratio does. A module that helped on this endpoint would therefore have to solve rather than project, with the backbone setting the bounds it solves under (the capacity ceilings from the abundance head are that interface already), and read more of the resulting flux than its growth column. The first of those is a linear program per genotype per forward pass with the gradient taken through its dual, about 50 milliseconds each. Rounds 13, 15 and 15b test both routes on both boxes. On the GEM's own box the solve survives the projections (round 13) and the program that solves is feasible and ranks at 0.73 before training (round 15b), and in both cases the gated module cell reaches the affine's level, 0.816 and 0.824, three of three seeds above the round-4 gated cell, with the growth column entering through a positive gate. On the anchored box the projections reverse the solve (round 10) and the program has no feasible point (round 15), because the enzyme-capacity ceilings as parameterized exclude growth; with those ceilings scaled to admit growth the program solves, the ceilings bind, and on one of three seeds the growth column alone ranks the held-out genes at 0.84 with no head and no appended ratio (round 15c), the module's upper bound so far. The module helps, then, exactly when its flux is on the polytope the solve optimizes over, and the constraint set that was meant to make it more than the solve is what kept it off until it was calibrated.

## 4.34 Why the vertex is the worst store ✗

Four stores of the same optimal face trained the same cell to 0.912, 0.911, 0.907 and, for the growth program's own dual-simplex vertex, 0.840 (Section 4.30). Two candidate readings fail on the numbers, and one survives (Table 77).

**Table 77:** The four stores of the same growth program's optimal face over the 1,121 labeled genes, in the affine's encoding $\log(1 + |v|)$, beside the AUROC the layer-free cell reached over each (rounds 36 and 38, six seeds pooled). Columns: the Spearman correlation between a deletion's distance from the wild-type row and its drop in growth; the share of viable deletions whose row IS the wild-type row; the median distance of a lethal deletion's row; the number of reactions whose encoded column varies at all; the share of the total column variance in the ten widest columns and how many of those touch the ±100 clamp, where a loop can spin; and what one column can do, the best single-column AUROC and the number of columns above 0.7. Written by `store_discriminability.py`.

| store | trained | $\rho$ | wt rows (%) | lethal dist. | varying | top 10 (%) | clamp | best col. | $> 0.7$ |
|---|---|---|---|---|---|---|---|---|---|
| pFBA | **0.912** | +0.70 | 59 | 14.2 | 860 | 43 | 0 | 0.738 | 463 |
| dual simplex | 0.840 | +0.57 | 59 | 36.2 | 1,548 | 13 | 10 | 0.735 | 298 |
| interior point | 0.911 | +0.56 | 60 | 27.7 | 2,743 | 9 | 0 | 0.755 | 836 |
| PDLP | 0.907 | +0.31 | 0 | 14.0 | 2,744 | 16 | 1 | 0.751 | 370 |

The first candidate is smoothness: a store whose rows move with the deletion should be the one a head can read. It does not order the stores. The correlation between a deletion's distance from the wild-type row and its drop in growth is 0.70 for pFBA, 0.57 for the vertex, 0.56 for the interior point and 0.31 for PDLP, while the trained reads put the vertex alone at the bottom and the other three within 0.005 of each other. The second candidate is motion at all on the genes that survive: three of the four stores return the wild-type row exactly for 59 to 60 percent of the viable deletions, and PDLP never does, yet PDLP reads 0.907, level with the stores whose rows for those genes carry nothing.

What does separate the vertex is where its variance sits. Its ten widest columns are all at the ±100 clamp, the bound at which a loop can spin, against none for pFBA, none for the interior point and one for PDLP; its lethal rows sit

36.2 from the wild-type row against 14.2, 27.7 and 14.0; and of its 1,548 varying columns only 298 rank the labeled genes above 0.7 on their own, the fewest of the four, where the interior point has 836 of 2,743. So the vertex moves further and in more places than the interior point and gets less for it: an extreme point puts its flux on the loops the bounds allow, and those columns vary between genotypes without tracking essentiality. Stated as a hypothesis, since four stores are an association rather than a demonstration: what costs the vertex store 0.07 is loop-carrying columns at the clamp, not any failure of its rows to move. A store built from a first-order or interior point has no such columns by construction, which is the property to check on the next store rather than assume.

No store has a column that does the job alone. The best single column reads 0.735 to 0.755 across all four, against the 0.84 to 0.91 the trained cells reach, so the affine is combining columns in every case and the question is only how many of them carry anything.

Round 68 runs the intervention that hypothesis implies. Every reaction that reaches the clamp on any genotype is zeroed in a copy of the store, which removes 429 of the vertex store's 4,131 columns and exactly one of the pFBA store's, and the same cell is trained over each on the six seeds the stores were read with (Table 78, Figure 80).

Figure 79 draws the cut.

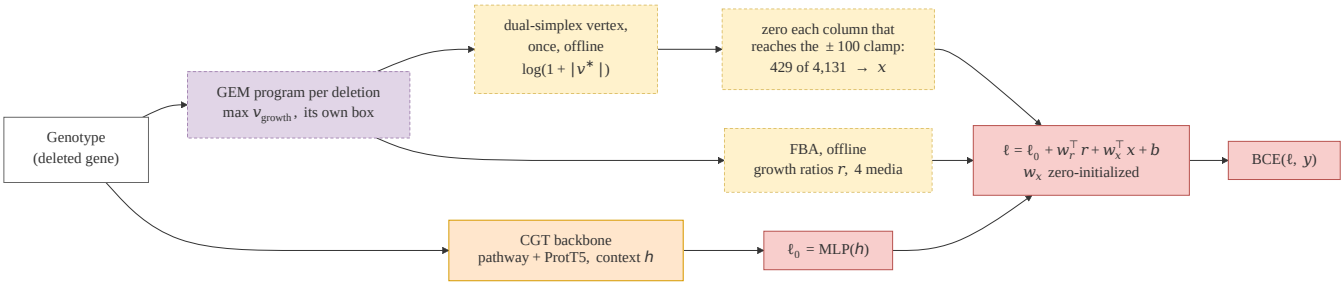



**Figure 79: The vertex store with its clamp columns dropped, rounds 68 to 70.** The path of Figure 73 over the dual-simplex vertex store, with every column that reaches the ±100 flux clamp on any genotype set to zero before the affine reads it, 429 of 4,131 columns. Round 69 drops instead the 429 widest columns that never reach the clamp, and round 70 the 25 or the 100 widest clamp columns. Colors as in Figure 16.

**Table 78:** Round 68, the clamp-carrying columns dropped from two stores. The layer-free $\log(1 + |v|)$ cell over the pFBA store and over the growth program's dual-simplex vertex store, each beside the same store with every reaction that reaches the ±100 clamp on any genotype zeroed (`declamp_store.py`: 1 of 4,131 columns from the pFBA store, 429 from the vertex store), and beside the specificity control, the vertex store with the same 429 columns taken instead as the widest that never reach the clamp (round 69, `-mode topvar`), and beside the dose, the 25 and the 100 widest of those 429 clamp columns dropped instead of all of them (round 70, `-mode partial`). Batch 32, class weight, 200 epochs, seeds 2700 to 2705. Bold marks the highest value in a column. ↑ (↓) marks an arm whose held-out AUROC is above (below) the first arm's on every one of the same three seeds.

| arm | mean | sd | seeds | pooled [95% CI] | selected epochs | val step |
|---|---|---|---|---|---|---|
| the vector as log1p, default initialization | **0.908** | 0.008 | 0.896 to 0.919 (6) | **0.912** [0.87, 0.95] | 33 to 58 | 0.0043 |
| layer-free log1p, pFBA store, clamp columns dropped | 0.906 | 0.008 | 0.891 to 0.914 (6) | 0.911 [0.86, 0.95] | 47 to 64 | 0.0046 |
| round-33 cell over the growth program's dual-simplex vertex (store by HiGHS simplex) | 0.840 | 0.051 | 0.767 to 0.887 (6) | 0.870 [0.79, 0.94] | 12 to 74 | 0.0071 |
| layer-free log1p, vertex store, clamp columns dropped | 0.888 | 0.011 | 0.876 to 0.906 (6) | 0.893 [0.84, 0.94] | 44 to 72 | 0.0043 |
| layer-free log1p, vertex store, 25 widest clamp columns dropped | 0.874 | 0.013 | 0.851 to 0.887 (6) | 0.882 [0.82, 0.94] | 44 to 137 | 0.0047 |
| layer-free log1p, vertex store, 100 widest clamp columns dropped | 0.896 | 0.022 | 0.854 to 0.914 (6) | 0.896 [0.83, 0.95] | 47 to 68 | 0.0069 |
| layer-free log1p, vertex store, as many non-clamp columns dropped | 0.846 | 0.009 | 0.831 to 0.856 (6) | 0.847 [0.76, 0.92] | 39 to 86 | 0.0060 |

The cut recovers most of the vertex store's deficit and does nothing to the control. The vertex store goes from 0.840 to 0.888 as a seed mean, up on six of six seeds with a paired mean of +0.049. The pFBA store, where the same rule removes one column, goes from 0.908 to 0.906, up on two seeds of six and down on four, a paired mean of −0.003 inside its own seed spread. So dropping clamp columns is not itself a help, and what it removes from the vertex store is most of what that store was costing: the remaining gap to the pFBA store is 0.020 against 0.068 before the cut.

Round 69 is the specificity control, and it separates two things the cut does at once. Dropping 429 columns from the same store, taken instead as the widest that never reach the clamp, leaves 1,119 varying columns against the clamp cut's 1,123, so the two interventions differ in which columns go and in little else. The score does not follow: that cut reads 0.846, a paired mean of +0.007 on two seeds of six, against the clamp cut's +0.049 on six of six. The seed spread does follow. Both cuts collapse it, from 0.051 to 0.011 for the clamp columns and to 0.009 for the others, and both remove the early peaks, selecting epochs 39 to 86 where the store they were cut from selected 12 and 22 on its two worst seeds. So the vertex store has two problems and they are not the same problem: its widest columns, whatever they are, make the validation curve peak early and the seeds disagree, and its clamp columns, specifically, cost it 0.05 of AUROC. Only the second is about loops.

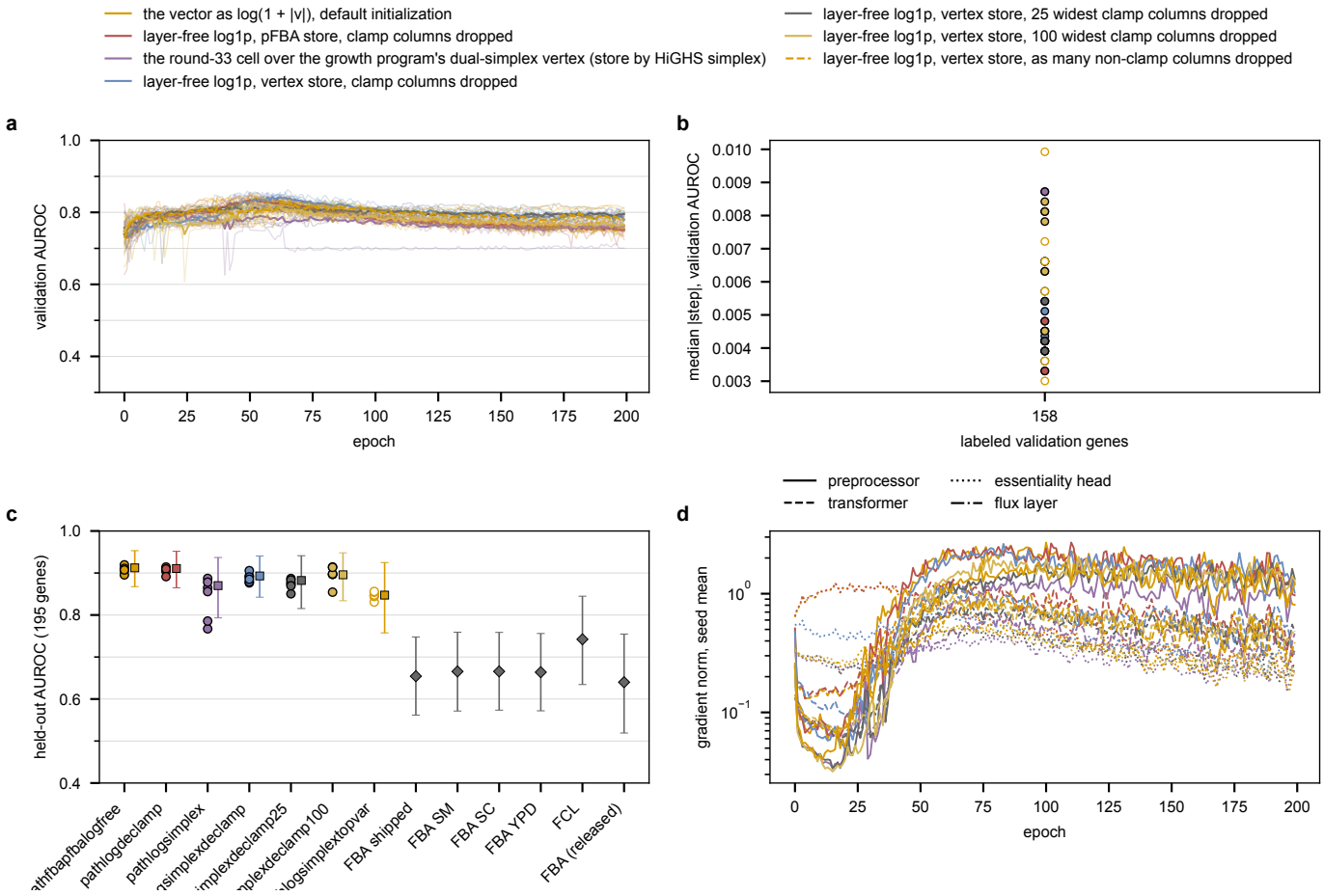



**Figure 80: Round 68, the clamp columns dropped.** **a**, Validation AUROC per epoch, one thin line per seed and the seed mean per arm. **b**, The median absolute epoch-to-epoch step per cell against the validation size. **c**, Held-out AUROC per cell (points), the seed mean and the gene bootstrap of the pooled per-gene logits (square with interval), beside FBA on four media, Flux Cone Learning and the released FBA predictions on the same 195 genes. **d**, Gradient norm per module per epoch, seed mean. Written by `experiments/028-gene-essentiality/scripts/wave12_readout.py -round w84`.

Round 73 puts the cut store on the protocol the other stores are read with. Its five fold models, standardized logits averaged, read 0.901 [0.85, 0.95] with a fold mean of 0.872, against the pFBA store's 0.920, the interior point's 0.909 and PDLP's 0.909 (Table 48). The uncut vertex store has no fold number, so this is not a paired read, but the cut store lands with the three that never had the deficit rather than 0.07 below them, and its residual 0.02 to the pFBA store is what the pooled seeds also showed. One of its five folds still selects epoch 12 and reads 0.808, so the early peak is reduced rather than gone.

Round 70 places that 0.05 on a few columns rather than on the set. Dropping the 25 widest clamp columns, 0.6 percent of the store's reactions, reads 0.874, two thirds of the way from the uncut store to the full cut; dropping the 100 widest reads 0.896, which is the full cut's number and, within a seed spread of 0.022, its ceiling. Cutting the remaining 329 buys nothing: all 429 read 0.888, below the 100 and level with it inside the spread, though on six of six seeds against the 100's five. So the vertex store's deficit is not spread over its loops. It is carried by a hundred columns, and a quarter of it by twenty-five of them, out of 4,131. The vertex is a worse store because an extreme point spends its flux on the loops its bounds allow, and those columns vary between genotypes without tracking essentiality. What the cut does not recover, and what is still open, is the last 0.02.

## 4.35 The capacity box under the two $k_{cat}$ tiers ✗

The module's own parameter sources were fixed for every round: the turnover tier and the thermodynamic route were set once and never moved. Rounds 15 and 15c ran the predicted tier, 97 assayed catalytic units, 782 from homologs and 2,831 from the sequence predictors, 18 on the organism default. What the measured tier does instead is stop before the predictors: the same 97 assays, the same 782 homologs, and the remaining 2,849 units on the default. Since the capacity ceiling of a reaction is its units' turnover times their abundance, the tier sets the ceiling vector, and round 15 found the anchored program infeasible under whichever ceilings it had. Running the round-15 calibration on both tiers (Table 79) measures which of the two boxes that was.

**Table 79:** The anchored layer's growth program at initialization under each $k_{cat}$ tier, over one batch of 32 held-out genotypes and a geometric grid of capacity scales: the share of genotypes whose program is feasible and the median growth, mmol/gDW/h. The upper block is the parameter table each tier resolves, with the nominal ceiling the box carries at scale 1 over the 2,709 reactions that have a gene-protein-reaction rule; the coarse counts place the homolog tier with the predictions, so the measured tier's 782 are homolog values and its 97 are assays. The measured tier leaves 2,849 of 3,728 catalytic units on the organism default, which is larger than most predicted turnovers, so its box is the looser of the two. Written by `capacity_calibration.py -kcat-tier {measured,predicted}` from the round-15 cell.

| capacity scale | 1 | 3 | 10 | 30 | 100 | 300 | 1,000 | 3,000 | 10,000 |
|---|---|---|---|---|---|---|---|---|---|
| measured, units measured / predicted / default | 97 / 782 / 2,849 of 3,728 | | | | | | | | |
| measured, $k_{cat}$ per hour (median, 5th to 95th) | 34,560 (4,045 to 401,940) | | | | | | | | |
| measured, nominal ceiling (median, 5th to 95th) | 34.6 (5.3 to 407.5) | | | | | | | | |
| predicted, units measured / predicted / default | 97 / 3,613 / 18 of 3,728 | | | | | | | | |
| predicted, $k_{cat}$ per hour (median, 5th to 95th) | 15,356 (1,896 to 435,600) | | | | | | | | |
| predicted, nominal ceiling (median, 5th to 95th) | 16.5 (2.0 to 422.1) | | | | | | | | |
| measured, genotypes feasible | 0.00 | 0.56 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 |
| measured, median growth | 0.000 | 0.000 | 0.019 | 0.057 | 0.156 | 0.280 | 0.532 | 0.535 | 0.535 |
| predicted, genotypes feasible | 0.00 | 0.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 |
| predicted, median growth | 0.000 | 0.000 | 0.010 | 0.042 | 0.103 | 0.126 | 0.203 | 0.386 | 0.535 |

The measured tier is the looser box, and by more than the count of assays would suggest. Its median turnover is the organism default, 34,560 per hour against the predictors' median of 15,356, so the median nominal ceiling is 34.6 against 16.5 mmol/gDW/h, and the program admits growth for 56 percent of the batch at a capacity scale of 3 where the predicted tier admits none. The two tiers separate further as the scale rises: at 1,000, the upper scale round 15c ran, the measured tier's median growth is 0.532 against the predicted tier's 0.203, where 0.535 is what both tiers reach at 10,000 and the ceilings no longer bind. So on the measured tier the capacity constraint is within 0.003 mmol/gDW/h of absent at the scale the module was run at, and the tier and the scale are one knob rather than two: a tier with fewer predictions is a looser box, which the calibration then scales less. What the calibration does not say is how a trained cell reads under the measured tier, since round 15c's result is a growth column that ranked the held-out genes at 0.84 on one seed of three with the ceilings binding at 3 to 8 percent of reactions, and a box that does not bind has no ceilings to bind.

Round 67 trains it. Round 15c's two cells run again at capacity scale 100, where both tiers still bind, with the measured tier in place of the predicted one and everything else held, six seeds each (Table 80, Figure 81).

**Table 80:** Round 67, the capacity ceilings from the measured turnover tier. Round 15c's two cells at capacity scale 100, where both turnover tiers still bind, on the predicted tier they were run with and on the measured tier, which leaves 2,849 of 3,728 catalytic units on the organism default and is the looser box (Section 4.35). Forced and gated, constraint scale 0, batch 32, class weight, 200 epochs, seeds 2700 to 2705. Bold marks the highest value in a column. ↑ (↓) marks an arm whose held-out AUROC is above (below) the first arm's on every one of the same three seeds.

| arm | mean | sd | seeds | pooled [95% CI] | selected epochs | val step |
|---|---|---|---|---|---|---|
| growth column, the layer that solves, capacity x100 | 0.683 | 0.151 | 0.500 to 0.843 (6) | 0.803 [0.71, 0.89] | 2 to 58 | 0.0005 |
| gated column + context, the layer that solves, capacity x100 | 0.800 | 0.015 | 0.786 to 0.820 (6) | 0.801 [0.72, 0.88] | 6 to 98 | 0.0035 |
| growth column, capacity x100, measured $k_{cat}$ tier | 0.707 | 0.044 | 0.669 to 0.759 (6) | 0.784 [0.69, 0.87] | 10 to 174 | 0.0054 |
| gated column + context, capacity x100, measured $k_{cat}$ tier | **0.812** | 0.014 | 0.787 to 0.827 (6) | **0.811** [0.73, 0.89] | 6 to 96 | 0.0038 |

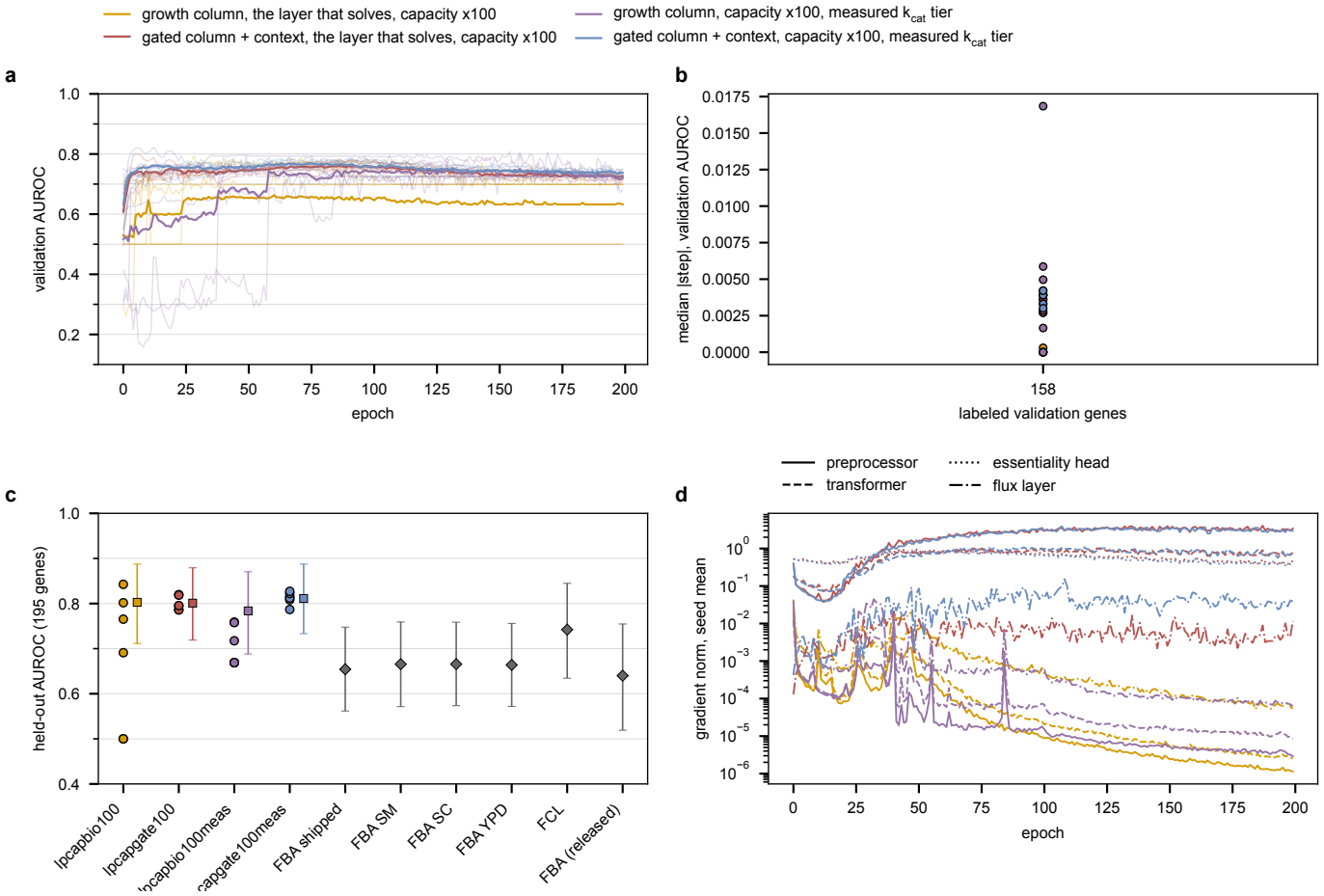



**Figure 81: Round 67, the capacity ceilings from the measured turnover tier. a**, Validation AUROC per epoch, one thin line per seed and the seed mean per arm. **b**, The median absolute epoch-to-epoch step per cell against the validation size. **c**, Held-out AUROC per cell (points), the seed mean and the gene bootstrap of the pooled per-gene logits (square with interval), beside FBA on four media, Flux Cone Learning and the released FBA predictions on the same 195 genes. **d**, Gradient norm per module per epoch, seed mean; the dash-dot line is the flux layer. Written by `experiments/028-gene-essentiality/scripts/wave12_readout.py -round w83`.

The looser box trades a failure for a ceiling. On the gated cell the tier is worth little: 0.800 against 0.812 as seed means, four seeds of six up, a paired mean of +0.012 against a seed spread of 0.015. On the forced growth column the trade is the finding. Under the predicted tier the arm is bimodal: two seeds of six enter the absorbing infeasible state and select epoch 2 at 0.500, while two others read above 0.80, at 0.8425 and 0.8017, for a seed mean of 0.683 with a spread of 0.151. Under the measured tier no seed collapses and none exceeds 0.76, the six reading 0.669 to 0.759 for a mean of 0.707 with a spread of 0.044. So the looser box removes the absorbing state, which is what the calibration predicted of a box that admits growth at a third of the scale, and it removes the high end with it. Round 15c's 0.84 is not a unique event, since a second seed of six reaches 0.80, but it belongs to the tighter box, which either collapses or ranks well.

Round 72 asks whether the bimodality can be removed without the high end going with it, and the answer is that it cannot, at least by the two constructions that remove it (Table 81, Figure 82). The hard program on the tight box, now on twelve seeds, collapses on two and clears 0.80 on four, with a mean of 0.699 and a spread of 0.115. The two ways of stopping the collapse are the looser turnover tier and the elastic program, which prices unmet forced flux instead of refusing the genotype. Both work, and both land in the same place: no seed of six at chance, no seed of six above 0.80, means of 0.707 and 0.715, spreads of 0.044 and 0.049. Three constructions, one mean to within 0.016 and a spread that differs by a factor of two and a half. What the collapse buys is not accuracy on average but a tail, and the 0.84 lives

in it.

**Table 81:** Round 72, the forced growth column under every construction tried. The growth column of the layer that solves, read on the capacity box at scale 100 under the hard program (twelve seeds) and under the two constructions that remove its absorbing infeasible state: the looser measured turnover tier and the elastic program with unmet forced flux priced. The gated cells of the same two programs are shown for reference. Constraint scale 0, batch 32, class weight, 200 epochs. Bold marks the highest value in a column. ↑ (↓) marks an arm whose held-out AUROC is above (below) the first arm's on every one of the same three seeds.

| arm | mean | sd | seeds | pooled [95% CI] | selected epochs | val step |
|---|---|---|---|---|---|---|
| growth column, the layer that solves, capacity x100 | 0.699 | 0.115 | 0.500 to 0.843 (12) | 0.807 [0.71, 0.89] | 2 to 97 | 0.0008 |
| growth column, capacity x100, measured $k_{cat}$ tier | 0.707 | 0.044 | 0.669 to 0.759 (6) | 0.784 [0.69, 0.87] | 10 to 174 | 0.0054 |
| growth column, elastic solve, x100 | 0.714 | 0.049 | 0.631 to 0.763 (6) | 0.637 [0.52, 0.75] | 10 to 180 | 0.0066 |
| gated column + context, the layer that solves, capacity x100 | 0.800 | 0.015 | 0.786 to 0.820 (6) | 0.801 [0.72, 0.88] | 6 to 98 | 0.0035 |
| gated column + context, elastic solve, x100 | **0.812** | 0.017 | 0.783 to 0.836 (6) | **0.807** [0.73, 0.88] | 6 to 97 | 0.0038 |

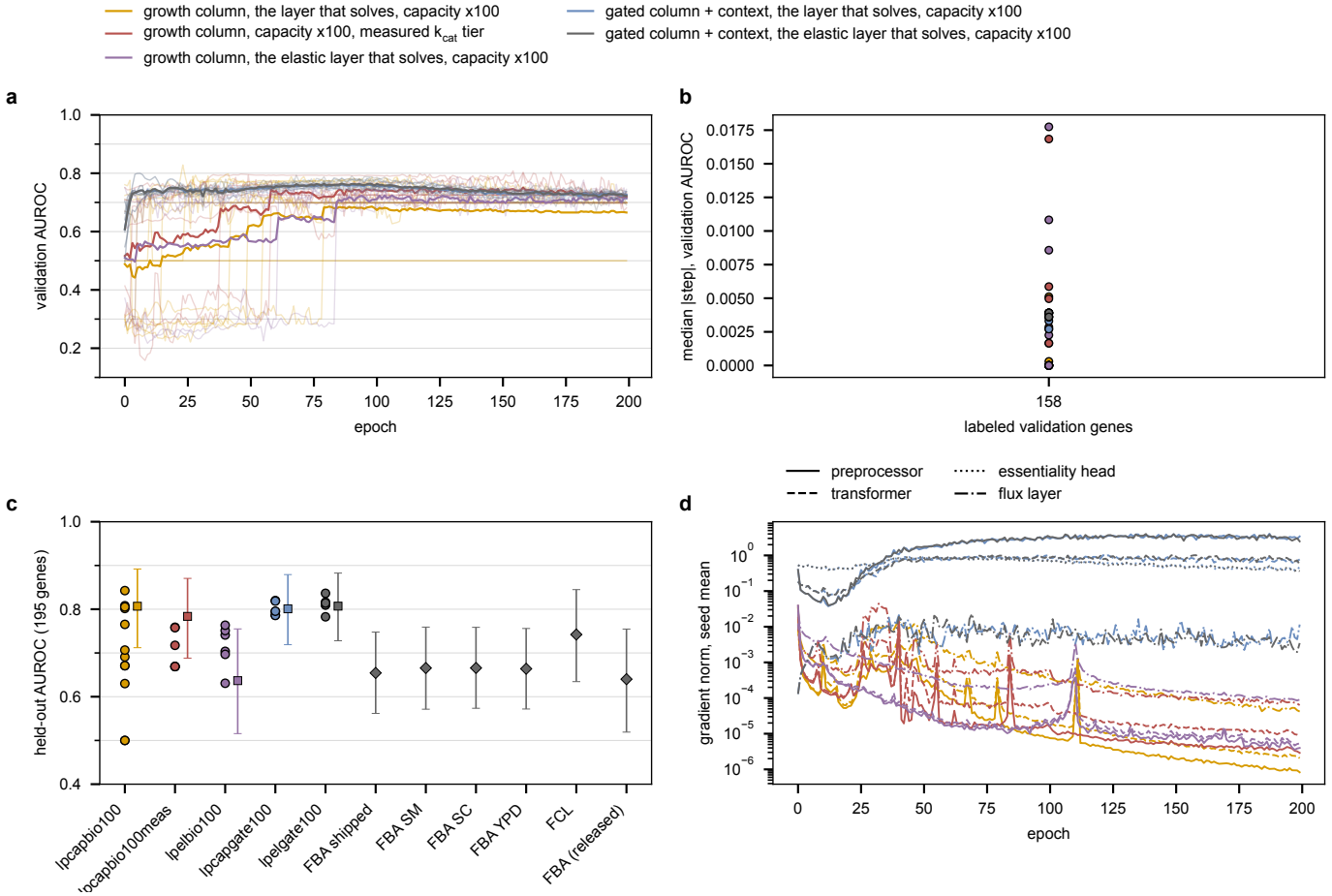



**Figure 82: Round 72, the forced growth column under every construction tried. a**, Validation AUROC per epoch, one thin line per seed and the seed mean per arm. **b**, The median absolute epoch-to-epoch step per cell against the validation size. **c**, Held-out AUROC per cell (points), the seed mean and the gene bootstrap of the pooled per-gene logits (square with interval), beside FBA on four media, Flux Cone Learning and the released FBA predictions on the same 195 genes. **d**, Gradient norm per module per epoch, seed mean; the dash-dot line is the flux layer. Written by `experiments/028-gene-essentiality/scripts/wave12_readout.py -round w89`.

The gated cells do not show any of this. Across the same three programs they read 0.800 and 0.812, with spreads near 0.015 and no seed at chance, so the gate is what makes the column safe to read and what caps it: the arm that never collapses never reaches 0.84 either. On this endpoint the module's best single number and its worst failure are the same construction seen twice.

Two details of the forced readout are worth keeping. Its seeds select epochs 2 to 174, so the collapse is visible as an epoch-2 selection rather than as a late failure. And three of the six measured-tier seeds read exactly 0.669, which is not a coincidence of rounding: the readout is the layer's growth column and nothing else, and it takes only 19 distinct values over the 195 held-out genes, so the ranking is a partition into 19 blocks and runs that place the genes in the same blocks score identically.

The other parameter source cannot matter here, and the same calibration is what says so. Thermodynamics enters the anchored layer as a loss term, every constraint weight is multiplied by a constraint scale the module rounds set

to zero, and the sign constraint that would put a reaction's direction in the box was never switched on. Rerunning the calibration with the route swapped from the merged external table to the model's own shipped column leaves the program identical at every capacity scale, feasibility and median growth agreeing to the last bit. So the thermodynamic route is recorded provenance on these rounds rather than a hyperparameter, and moving it is a question for a cell that prices the constraint.

## 4.36 The store at the database's scale ✗

The layer-free form (Section 4.27) makes the module a data transform: the growth program solved once per genotype and medium, its flux stored as $\log(1+|v^*|)$ and read through an affine at training time, with no solver in the loop. What sizes that transform for a model over the whole database is not the record count but the number of distinct programs, since the program depends on the genotype only through the GEM genes it deletes and on the environment only through the medium: a record whose deletions include no GEM gene reads the wild-type row, and two records deleting the same GEM genes on the same medium read one row. Table 82 measures that count over the 025 solid-growth build, the largest phenotype table in the database (13.5 million records, single to triple deletions on three media), by a cursor scan of its LMDB for each record's deleted genes and medium name (job 2469, 8 minutes on 20 processes), and then builds the store.

Figure 83 draws the read at that scale.

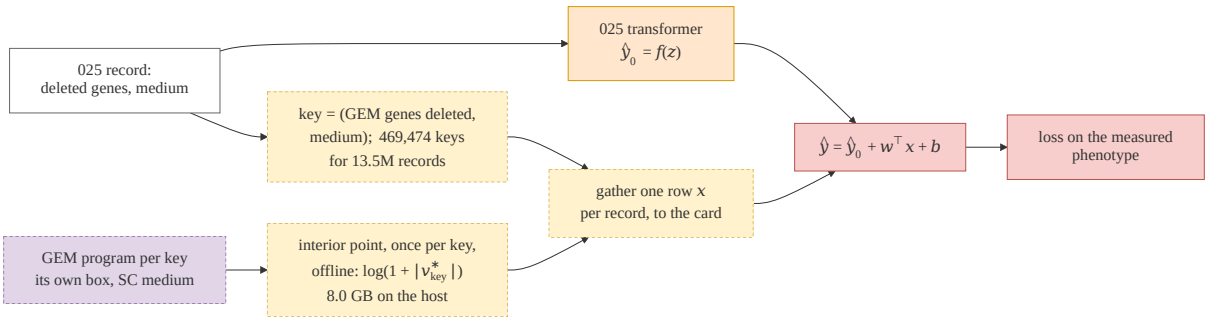



**Figure 83: The store at the database's scale.** The layer-free path on the 025 solid-growth transformer: a record's deleted GEM genes and its medium form a key, the program is solved once per key offline by the interior point, and at training time each record gathers its row from a table on the host and an affine of the row is added to the transformer's prediction. Colors as in Figure 16.

**Table 82:** The solve store keyed by (GEM gene subset, medium) over the 025 solid-growth build. The count (upper block) is a cursor scan of the build's LMDB for each record's deleted genes and medium name, written by `store_key_count.py`; the build (lower block) solves the round-27 growth program once per key with every GEM gene of the key deleted, written by `solid_growth_store.py` (job 2471). The build's selection media have no FBA recipe, so every key is solved on synthetic complete medium and keeps its own medium name; the store's medium axis is therefore one medium in this build. The scan counts a record under every medium name it carries, so the records served exceed the records in the build by the number of records that name two media.

| quantity | value |
|---|---|
| records in the build | 13,525,071 |
| records by deletions (1 / 2 / 3) | 5,694 / 13,142,645 / 376,732 |
| distinct media in the build | 3 |
| records deleting no GEM gene (read the wild-type row) | 8,986,762 |
| GEM genes among the deleted | 1,161 |
| distinct GEM subsets by size (1 / 2 / 3) | 1,110 / 449,830 / 8,157 |
| distinct (GEM subset, medium) keys | 469,474 |
| store as float32 (GB, estimated) | 7.8 |
| CPU hours per core, estimated (interior point / pFBA) | 20 / 130 |
| scan of the build (s) | 504 |
| *build* | |
| solver, medium | HiGHS ipm at $10^{-4}$, SC |
| keys solved | 469,474 |
| records the keys serve | 13,767,895 |
| wall (min) at 20 workers | 71 |
| lethal keys (growth 0 under the solve) | 62,905 |
| uncertified keys stored as the zero flux | 2,234 |
| wild-type growth | 0.5428 |

Two thirds of the records (8,986,762 of 13,525,071) delete no GEM gene and read the wild-type row; the rest map onto 459,098 distinct GEM subsets, almost all pairs, and 469,474 (subset, medium) keys, 3.5 percent of the record count. Solved by the interior-point method at $10^{-4}$ with every GEM gene of the key deleted (the round-36 store's solver, whose point trains to the pFBA store's number), the whole store took 71 minutes on 20 CPU cores (job 2471; 23.4 CPU hours, 0.18 s per program) and is 8.0 GB as float32 over 4,131 reactions, with 62,905 keys lethal under the solve and 2,234 uncertified, stored as the zero flux as in round 55, where the 320 uncertified rows of the $10^{-8}$ store cost 0.02 until they were filled from a looser solve. The build's SGA selection media have no FBA recipe (a synthetic dropout with drugs), so every key was solved on synthetic complete medium; on this endpoint the SC store reads the YPD store's number (round 42), and the store's medium axis is one medium in this build.

At training time the store is a table on the host and the read is a gather: the 025 transformer wrapped by `store_affine.py` in that experiment (one card per arm, jobs 2476 to 2479, launched 2026-09-18) looks up each record's row from its deleted genes, moves the batch's rows to the card and adds a linear map of them to the logit. The arm with the store trains at the control's step rate, 0.37 iterations per second on both, so the transform costs nothing measurable per step. Its number is not read: the two arms on two seeds have reached epoch 15 of 30, and through that epoch the validation Pearson on gene interaction of the store arm is within 0.022 of the control's at every epoch, with the sign of the paired difference varying by epoch and by seed; the fixed-epoch and best-epoch reads, paired by seed, are written by `store_affine_readout.py` when the runs end.

The solvers on the card were not run. Section 4.31 settled that a first-order point is a store (PDLP, the algorithm NVIDIA's cuOpt batches on the card, trains to 0.907 against the interior point's 0.911 and pFBA's 0.912), so the store could be built on the card; but with the program count set by the keys and not the records, the store over the whole build is an hour of CPU, under two epochs of the transformer that reads it (58 minutes per epoch on one card), and a faster solve shortens a step that is not the cost. Where a solver on the card would matter is the case this endpoint did not need and did not measure: a program re-solved under bounds the model learns (the fine-tuning form with the capacity ceilings of round 15c), or a medium axis with many recipes, each of which multiplies the programs by the number of parameter settings or of media, and a program solved inside the loop rather than before it.

## 4.37 The forms tested after the layer-free result ✗

After the layer-free result the module changed only in what the affine reads and, once, in how the backbone forms its context: six gene-rule counts beside the stored vector (Section 8.3, Figure 84), the class token sent through the perturbation operator (Section 8.4, Figure 85), the counts with no flux vector (Sections 8.6 and 8.10, Figure 86), and every column restricted to one medium (Section 8.7, Figure 87). Each is a small change to the path of Figure 62, and the cited sections hold the numbers.

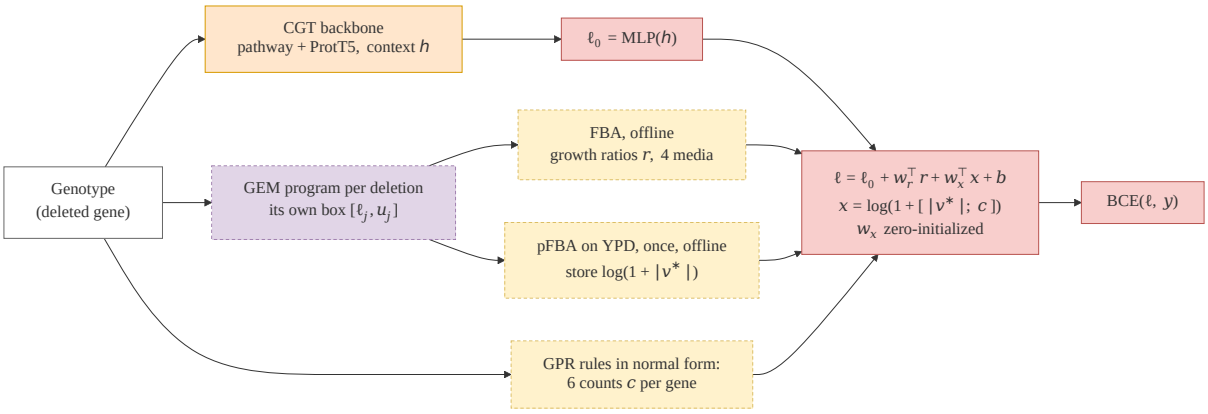



**Figure 84: The layer-free form with the gene-rule counts, round 77.** The path of Figure 62 with six columns appended to the store: counts read off each gene's gene-protein-reaction rules in disjunctive normal form, which need no solve and carry no medium. The affine reads them in the store's $\log(1 + \cdot)$ encoding under the same zero initialization. Colors as in Figure 16.

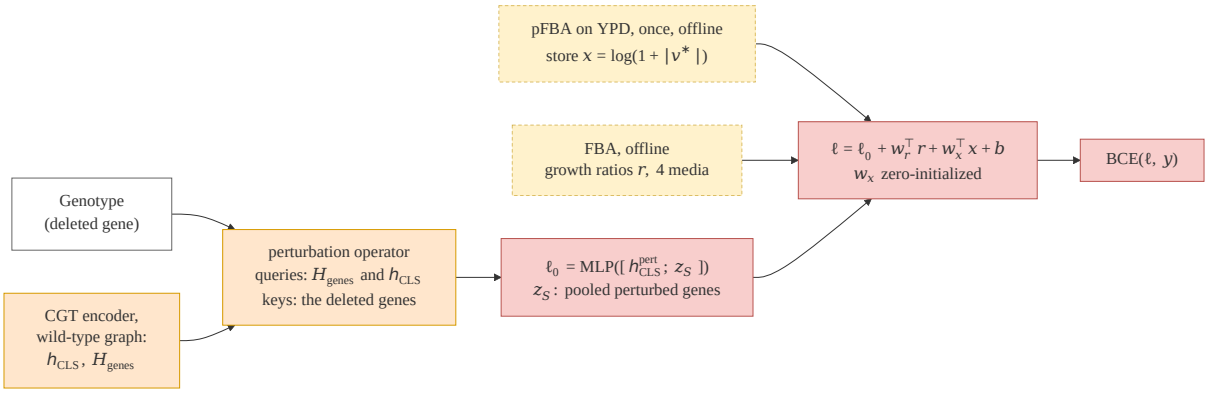



**Figure 85: The class token through the perturbation operator, round 79.** The metabolic path is that of Figure 62 and the change is in the backbone: the class token, encoded once on the wild-type graph, joins the gene tokens as one more query of the perturbation operator over the deleted-gene keys, so the head reads a strain-specific token beside the pooled perturbed genes. No parameter is added. Colors as in Figure 16.

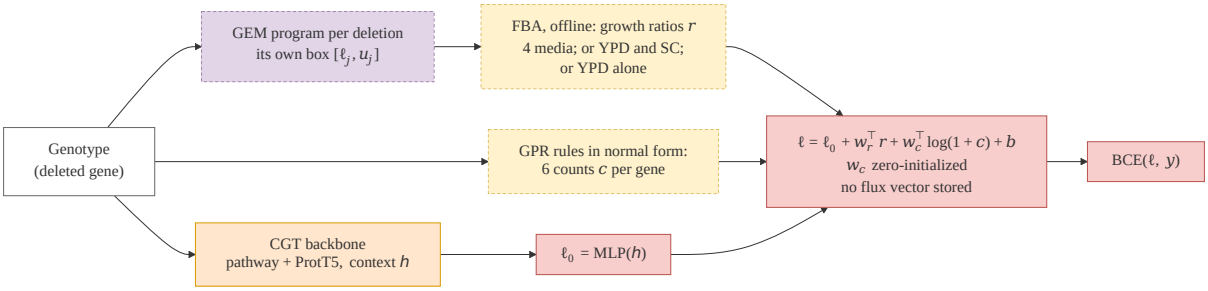



**Figure 86: The counts without the flux vector, rounds 82 to 98.** The path of Figure 84 with the stored flux vector removed, so the affine reads the growth ratios and the six rule counts only. Round 82 keeps all four media, and the later recipes of Section 8.10 read the YPD and SC ratios or the YPD ratio alone. Colors as in Figure 16.

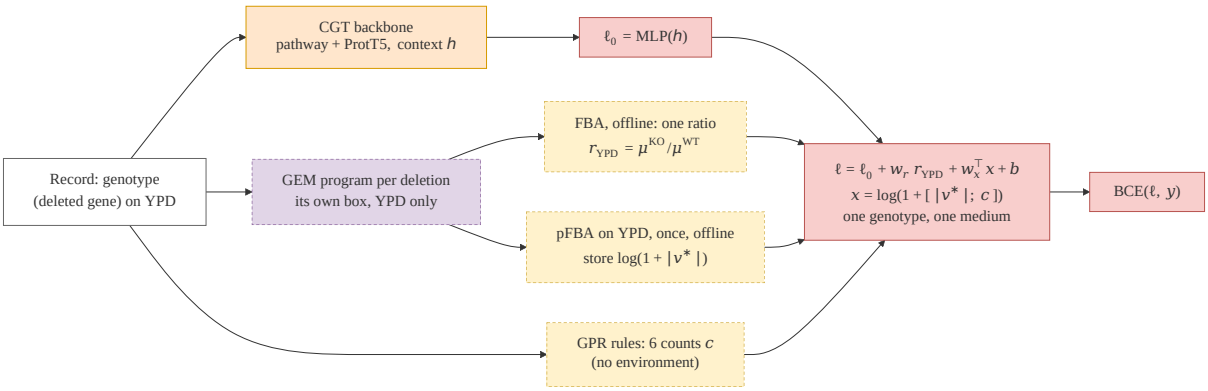



**Figure 87: Every column inside one experiment, round 83.** The path of Figure 84 restricted to one medium: the YPD growth ratio, the flux vector of the YPD solve and the rule counts, which carry no environment, so a record is one genotype in one medium. Colors as in Figure 16.

# 5 Which genes are missed, the two model families, and the genome-wide set ✗

The arms read here are the backbone cells of Section 3 and the module forms of Section 4, so every arm is defined before it is compared.

## 5.1 Which genes the best cells misrank ✗

At 195 genes the AUROC is a sum over a few dozen misrankings, so the genes behind it can be named. For every cell of nine arms, `hard_genes.py` reads the per-gene logits on the held-out genes and scores each gene's *misrank*, the fraction of the other class on the wrong side of it: for an essential gene the nonessential genes scoring above it, for a nonessential gene the essential genes scoring below it. One minus the class mean of the misrank is that class's share of the AUROC, and averaging over an arm's cells separates the genes an arm gets wrong from the ones a seed gets wrong (Figure 88, Table 83).

**Table 83:** The fifteen held-out genes of each class the best cells misrank most, sorted by the mean misrank over the arms. A gene's misrank in one cell is the fraction of the other class on the wrong side of it (for an essential gene, the nonessential genes scoring above it; for a nonessential gene, the essential genes scoring below it), averaged over the arm's cells; one minus the class mean of the misrank is that class's share of the AUROC. FBA is the knockout-to-wild-type growth ratio on YPD; pathways and GO the gene's counts in the pathway multi-hot and the propagated GO vector. Arms: A1: pathway + ProtT5 + affine (round 22); A2: indicator + ProtT5 + affine, folds (round 12); A3: flux head, ProtT5 (rounds 21, 24); A4: flux head, pathway + ProtT5 (rounds 23, 25); A5: cached solve + flux head, pathway + ProtT5 (round 27); A6: layer-free log1p vector + affine, pathway + ProtT5 (round 33); A7: ProtT5 + affine (round 5). Written by `hard_genes.py` from the per-gene prediction dumps.

| gene | A1 | A2 | A3 | A4 | A5 | A6 | A7 | FBA | pathways | GO |
|---|---|---|---|---|---|---|---|---|---|---|
| *essential genes* | | | | | | | | | | |
| YDR373W | 0.44 | 0.28 | 0.51 | 0.40 | 0.34 | 0.39 | 0.59 | 1.00 | 0 | 25 |
| YKL004W | 0.48 | 0.25 | 0.50 | 0.37 | 0.28 | 0.29 | 0.66 | 1.00 | 1 | 39 |
| YBL030C | 0.30 | 0.22 | 0.46 | 0.31 | 0.32 | 0.27 | 0.57 | 1.00 | 0 | 68 |
| YIL048W | 0.17 | 0.41 | 0.19 | 0.31 | 0.18 | 0.18 | 0.35 | 1.00 | 0 | 52 |
| YML125C | 0.37 | 0.46 | 0.40 | 0.35 | 0.30 | 0.27 | 0.41 | 1.00 | 0 | 27 |
| YAL038W | 0.13 | 0.42 | 0.62 | 0.21 | 0.14 | 0.11 | 0.49 | 1.00 | 2 | 81 |
| YMR296C | 0.14 | 0.31 | 0.36 | 0.35 | 0.29 | 0.12 | 0.35 | 1.00 | 1 | 36 |
| YDR062W | 0.15 | 0.15 | 0.40 | 0.36 | 0.28 | 0.11 | 0.27 | 1.00 | 1 | 36 |
| YBR011C | 0.10 | 0.10 | 0.16 | 0.37 | 0.24 | 0.10 | 0.14 | 1.00 | 0 | 18 |
| YDR050C | 0.11 | 0.14 | 0.19 | 0.07 | 0.03 | 0.01 | 0.40 | 1.00 | 2 | 85 |
| YDL120W | 0.12 | 0.10 | 0.07 | 0.14 | 0.12 | 0.09 | 0.21 | 1.00 | 0 | 50 |
| YGR277C | 0.05 | 0.04 | 0.03 | 0.34 | 0.17 | 0.16 | 0.05 | 1.00 | 1 | 64 |
| YDR454C | 0.27 | 0.15 | 0.03 | 0.11 | 0.05 | 0.21 | 0.08 | 0.00 | 5 | 65 |
| YGL055W | 0.22 | 0.04 | 0.02 | 0.08 | 0.05 | 0.06 | 0.08 | 0.00 | 3 | 51 |
| YMR281W | 0.07 | 0.23 | 0.06 | 0.13 | 0.12 | 0.10 | 0.12 | 1.00 | 0 | 64 |
| *nonessential genes* | | | | | | | | | | |
| YGL012W | 0.26 | 0.83 | 0.79 | 0.68 | 0.75 | 0.71 | 0.72 | 0.00 | 2 | 48 |
| YLR450W | 0.67 | 0.67 | 0.32 | 0.42 | 0.40 | 0.58 | 0.37 | 1.00 | 3 | 63 |
| YBR121C | 0.65 | 0.52 | 0.32 | 0.37 | 0.28 | 0.44 | 0.62 | 1.00 | 0 | 66 |
| YOL049W | 0.20 | 0.34 | 0.74 | 0.68 | 0.83 | 0.53 | 0.77 | 0.00 | 2 | 27 |
| YGR192C | 0.63 | 0.67 | 0.24 | 0.54 | 0.55 | 0.59 | 0.16 | 1.00 | 3 | 125 |
| YJR121W | 0.73 | 0.67 | 0.35 | 0.40 | 0.38 | 0.61 | 0.68 | 0.34 | 0 | 121 |
| YGR240C | 0.30 | 0.37 | 0.39 | 0.62 | 0.60 | 0.67 | 0.14 | 1.00 | 2 | 105 |
| YJR009C | 0.65 | 0.52 | 0.25 | 0.56 | 0.52 | 0.59 | 0.17 | 1.00 | 3 | 113 |
| YHR106W | 0.61 | 0.41 | 0.57 | 0.52 | 0.56 | 0.54 | 0.45 | 1.00 | 1 | 30 |
| YMR205C | 0.31 | 0.15 | 0.38 | 0.66 | 0.66 | 0.67 | 0.15 | 1.00 | 2 | 119 |
| YDR315C | 0.45 | 0.41 | 0.58 | 0.34 | 0.42 | 0.19 | 0.54 | 1.00 | 0 | 73 |
| YLR377C | 0.65 | 0.43 | 0.31 | 0.53 | 0.56 | 0.60 | 0.39 | 1.00 | 3 | 26 |
| YJL052W | 0.62 | 0.47 | 0.24 | 0.56 | 0.54 | 0.59 | 0.16 | 1.00 | 3 | 107 |
| YGR106C | 0.54 | 0.48 | 0.59 | 0.34 | 0.37 | 0.44 | 0.52 | 1.00 | 0 | 30 |
| YMR202W | 0.16 | 0.25 | 0.26 | 0.25 | 0.13 | 0.21 | 0.35 | 1.00 | 2 | 41 |

The solve's own call splits the essential genes. Nineteen of the 31 essential held-out genes have a knockout growth ratio of 1.0 on YPD, the solve calling them dispensable, and thirteen of the fifteen hardest are among those nineteen, while nine of the fifteen easiest have a ratio of 0. The hardest essential genes are the ones FBA misses, and every model misses them with it: AUR1, FRQ1, PET9 (the ADP/ATP carrier, covered in the GEM by AAC1 and AAC3), NEO1, PGA3, CDC19 (pyruvate kinase, covered by its paralog PYK2), LCB1 and LCB2 (serine palmitoyltransferase), IPP1, TPI1. The hardest nonessential genes are metabolic enzymes the models call essential: ERG4, HMG2, GSH2, the three glyceraldehyde-3-phosphate dehydrogenase isozymes TDH3, TDH2 and TDH1, ATP2, the two phosphofructokinase subunits PFK1 and PFK2, TRR2, ERG2 and FBP1, twelve of the fifteen with a ratio of 1.0 and most with 26 to 125 GO terms. Isozyme cover explains a minority of the missed essential genes. Of the 19 essential held-out genes the solve calls viable, the gene-protein-reaction rules (`isozyme_affine_store.py`, rules in disjunctive normal form) give an alternative gene set for every reaction of only four, PET9, CDC19, PGA3 and YDR302W; the other fifteen, AUR1, NEO1, LCB1 and LCB2 among them, each have at least one reaction that only they catalyze, which their deletion

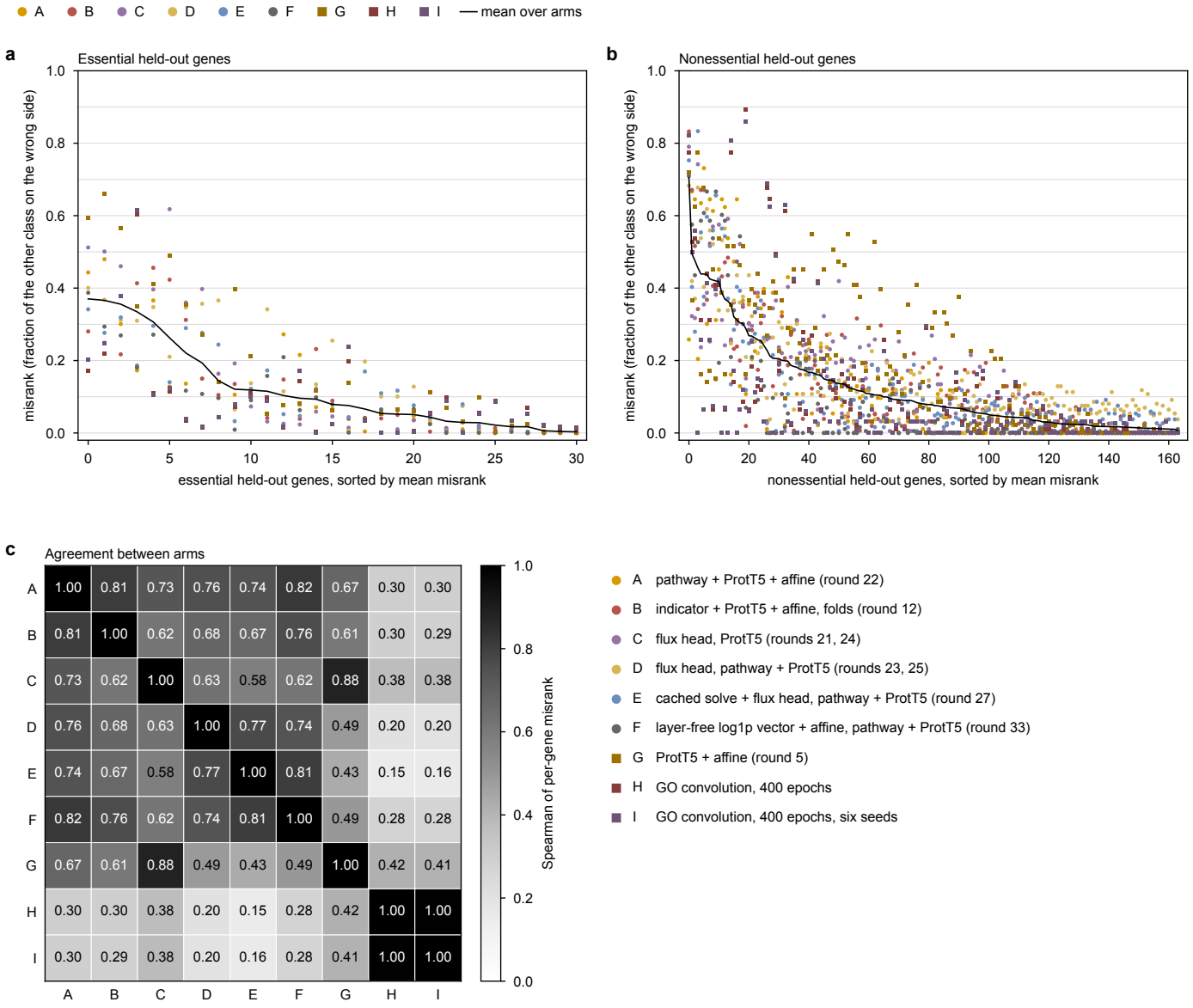



**Figure 88: Which held-out genes the best cells misrank.** Each arm has a letter, a color and a marker shape, shared by all three panels and listed in the key beside panel **c**; an arm's value for a gene is the mean over that arm's cells. **a**, The 31 essential held-out genes sorted by their mean misrank over arms, one marker per arm, the black line the mean over arms. **b**, The same for the 164 nonessential held-out genes. **c**, The Spearman correlation of the per-gene misrank between arms, with the value written in each cell. Arms: **A**, the pathway multi-hot with ProtT5 and the affine (round 22, three seeds); **B**, the round-12 fold models (five); **C**, the head over the solved flux on ProtT5 (rounds 21 and 24, six seeds); **D**, the same on the pathway input (rounds 23 and 25, six); **E**, that head over the cached GEM-box solve on the pathway input (round 27, six); **F**, the layer-free cell with the $\log(1 + |v|)$ vector in the affine (round 33, seeds 2700 to 2705); **G**, ProtT5 with the affine (round 5, three); **H** and **I**, graph convolution on GO at 400 epochs, three seeds and six. Written by `experiments/028-gene-essentiality/scripts/hard_genes.py`.

closes and which the growth program does not need. So the larger failure is a reaction the biomass objective does not require, and the smaller one a paralog that covers the gene in the solve; among the nonessential genes it is the reverse case, a paralog of an essential metabolic function that the annotation vectors make look essential. The arms agree on the genes they fail: the two affine cells at a Spearman of 0.81, the flux head on ProtT5 with the affine on ProtT5 at 0.88, the two flux heads on the pathway input with each other at 0.77, and those two least aligned with the rest (0.43 to 0.76), the arms whose means carry an unfit seed; the layer-free cell of round 33 sits with the transformer family, at 0.82 with the pathway affine and 0.81 with the cached-solve head, and at 0.28 with the GO convolution. Hypothesis, untested: a paralog-aware input, the GPR disjunction with an expression weight or the isozyme count of a gene's reactions, is what would move the fourteen; the solve's own vector cannot, since the solve reroutes through the paralog.

Graph convolution on GO fails on different genes. Its per-gene misrank correlates with the transformer arms' at 0.15 to 0.42 where those arms correlate with one another at 0.4 to 0.9, the same on three seeds and on six, and its class mean misrank on the essential genes is 0.091 against the pathway cell's 0.110. On the hardest essential genes it halves the misrank the solve leaves: FRQ1 from 0.44 to 0.17, AUR1 from 0.48 to 0.22, PGA3 from 0.37 to 0.11, LCB1 from 0.14 to 0.03, LCB2 from 0.15 to 0.01, while it does worse on NEO1 (0.17 to 0.60) and PET9 (0.30 to 0.38). Among the nonessential genes it repairs twelve the pathway cell misranks by more than 0.3 (TDH1, ACS1, ATP2, FBP1, TRR2 among them) and breaks six the pathway cell had right. Two models that read the same genes wrong for different reasons are the case for averaging them.

## 5.2 The essential genes every predictor ranks low ✗

If every predictor failed on the same essential genes, the shared failures would mark a branch of function that no input describes. The released split holds out 31 essential genes, too few to test that, so the intersection is taken on the ten repeated outer splits of Section 8.1, which hold out 874 of the 1,121 labeled genes at least once, 140 of them essential. Five predictors are read there: the transformer with the single-experiment module, the same transformer with no metabolic column, graph convolution on GO, the DCell hierarchy with a shared input (Section 8.8), and FBA alone as one minus the YPD growth ratio. A gene's misrank under one predictor is averaged over its seeds and over the splits that hold it out, and its consensus misrank is the lowest of the four learned predictors.

The intersection is nearly empty. The mean misrank of the essential genes is 0.133 for the module transformer, 0.193 without metabolism, 0.132 for graph convolution, 0.190 for the DCell hierarchy and 0.290 for FBA, and each learned predictor ranks 7 to 15 essential genes below the median nonessential gene of their block. No essential gene is below that median under all four: 111 of the 140 are below it under none, 16 under one, 9 under two and 4 under three. The consensus misrank averages 0.055, under half of the best single predictor's mean, and only one gene reaches 0.5, the biotin transporter *VHT1* (Table 84). The predictors disagree on which genes are hard: the per-gene misrank of the essential genes correlates at a Spearman of 0.78 between the two transformers, 0.40 between the module transformer and graph convolution, 0.43 between graph convolution and the DCell hierarchy although both read the same GO multi-hot, and 0.08 between graph convolution and FBA.

What the residual shares is a function, and the depth of a gene's annotation does not explain it. Across the 140 essential genes the consensus misrank has a Spearman correlation of $-0.002$ with the count of propagated GO biological-process terms and $-0.022$ with physical degree, and a term count under the class median is not enriched. Of 179 terms and features tested by the Mann-Whitney rule of Section 5.1, 11 pass $q < 0.1$. The largest group is the 70 essential genes whose FBA growth ratio is 1.0, at a consensus misrank of 0.082 against 0.028 for the rest (effect 0.78). Among GO terms the strongest are organic substance transport (10 genes, 0.190 against 0.045, effect 0.83), cellular component organization (14 genes, 0.120 against 0.048) and monoatomic ion transport (5 genes, 0.184 against 0.050). Test blocks overlap across splits, so these $q$-values rank terms and are not calibrated error rates. The top of the table is transporters and cofactor supply: *VHT1*, *HIP1*, *RIM2*, *THI80*, *NCE103*. One reading is that these are essential only given the strain the deletion collection was made in, BY4741, which carries *his3*, *leu2*, *met15* and *ura3* deletions and lives on imported histidine, leucine, methionine and uracil, while every FBA column here deletes one gene from the wild-type model. A probe tested that (`fba_background_probe.py`): delete the four background genes, take that strain as the reference, and recompute the knockout growth ratio of the 1,121 labeled genes. The background does not explain the residual at the resolution of the model. On YPD 63 ratios move by more than 0.01 and none of them is an essential gene moving toward lethal; the one gene that becomes lethal is the uracil permease *FUR4*, labeled nonessential, and the count of essential genes at a ratio of 1.0 goes from 77 to 83. FBA's AUROC over the labeled genes moves from 0.717 to 0.713 on YPD and from 0.718 to 0.714 on SC, with ratios rounded to six decimals because solver noise near a ratio of 1.0 reorders ties between identical runs. *HIP1* and *VHT1* stay at 1.0 because the model's rules let other permeases carry their substrates. Hypothesis, untested: those alternatives are repressed or too weak in rich medium, which is regulation and transporter affinity that neither the model nor GO encodes. The probe also found a defect in the modeled medium: YPD as defined here opens no uracil uptake, so a *ura3* deletion alone takes the model's YPD growth from 0.535 to zero (SC, which lists uracil, is unaffected), while BY4741 grows on YPD. The probe opens uracil uptake in both of its arms; the stored solves of the module were computed on the wild-type model, where the missing uptake does not bind. *ILV3*

and *ILV5*, which FBA ranks correctly and the learned predictors do not, are the converse case of a label the network explains and the annotation vectors do not.

**Table 84:** The twelve essential genes with the highest consensus misrank over the ten repeated outer splits. A misrank is the share of the nonessential genes of the same test block that score above the gene, averaged over seeds and over the splits that hold the gene out (held), so 0 is a perfect rank and 0.5 the median nonessential gene. The consensus is the lowest of the four learned predictors, in bold: the transformer with the single-experiment module (nine seeds per split), the same transformer with no metabolic column (nine), graph convolution on GO (six) and the DCell hierarchy with a shared input (three). FBA scores one minus the YPD growth ratio and is not part of the consensus. Ratio is the FBA knockout growth ratio on YPD, GO the count of propagated biological-process terms, degree the physical interaction degree. Written by `notes_tex_tables.py` from `consensus_missed.py`.

| gene | systematic | held | module | no metab. | GO conv. | DCell | FBA | ratio | GO | degree |
|---|---|---|---|---|---|---|---|---|---|---|
| *VHT1* | YGR065C | 2 | **0.50** | 0.58 | 0.75 | 0.68 | 0.57 | 1.00 | 14 | 13 |
| *NCE103* | YNL036W | 2 | 0.78 | 0.84 | 0.78 | **0.45** | 0.56 | 1.00 | 18 | 12 |
| *ACS2* | YLR153C | 3 | 0.38 | 0.38 | **0.32** | 0.51 | 0.55 | 1.00 | 60 | 80 |
| *HIP1* | YGR191W | 1 | 0.82 | 0.83 | 0.66 | **0.32** | 0.56 | 1.00 | 17 | 17 |
| *SDH3* | YKL141W | 4 | 0.40 | 0.99 | 0.59 | **0.31** | 0.06 | 0.94 | 38 | 19 |
| *ILV5* | YLR355C | 3 | 0.41 | 0.64 | **0.29** | 0.82 | 0.06 | 0.96 | 27 | 76 |
| *ARH1* | YDR376W | 3 | 0.70 | 0.51 | **0.28** | 0.57 | 0.56 | 1.00 | 26 | 10 |
| *RIM2* | YBR192W | 2 | 0.32 | 0.58 | 0.57 | **0.27** | 0.56 | 1.00 | 20 | 17 |
| *THI80* | YOR143C | 1 | 0.30 | 0.47 | 0.29 | **0.27** | 0.56 | 1.00 | 37 | 13 |
| *ILV3* | YJR016C | 6 | **0.24** | 0.42 | 0.30 | 0.32 | 0.06 | 0.96 | 22 | 35 |
| *CYR1* | YJL005W | 2 | 0.28 | **0.21** | 0.28 | 0.24 | 0.56 | 1.00 | 12 | 54 |
| *PET9* | YBL030C | 1 | **0.21** | 0.26 | 0.36 | 0.73 | 0.56 | 1.00 | 36 | 146 |

## 5.3 Averaging the two families ✗

The average is a post hoc read of predictions already made, with nothing fitted: each family's per-cell logits on the held-out genes are standardized and averaged over its seeds, the families are averaged with equal weight, and the mean is scored once (Table 85). The GO convolution and the pathway cell with the affine average to 0.933 [0.89, 0.97] from 0.909 and 0.891 alone; the GO convolution and the flux head on the pathway input, six seeds with two unfit among them, average to 0.941 [0.90, 0.98] from 0.909 and 0.911; all three to 0.935. With the six-seed families of the later rounds, the GO convolution on six seeds (0.905 as a family) and the scalable module form of round 27 (0.918 as a family, the highest single family) average to 0.946 [0.91, 0.98], and with the pathway affine beside them to 0.939; the layer-free form of round 33 (0.910 as a family) with the six-seed GO convolution averages to 0.948 [0.91, 0.98], and the same cell over the interior-point store (round 36, 0.912 as a family) to 0.944. Every average tried is in the table, so 0.948 is the maximum of fifteen post hoc choices and carries that bias, and the held-out genes were used to choose nothing else. It says what the per-gene analysis said: the two families carry different information about the same genes, and the sum of the annotation read through the graph and the solve read through the transformer is more than either. Under the reported protocol the same average holds (Table 86): the five GO fold models and the five pathway flux-head fold models, each family 0.917 and 0.897 as its own ensemble, average to 0.949 [0.91, 0.98]; the GO fold models with the reported cell's fold models (0.893 alone) to 0.944; all three to 0.945. With the scalable form's five fold models (round 27; 0.912 as its own ensemble and 0.922 as a standardized family, the highest single family under this protocol) the GO fold models average to 0.954 [0.92, 0.98], and with the reported cell's fold models beside them to 0.947; the layer-free form's five fold models (round 34; 0.920 as their own ensemble and as a family) with the GO fold models average to 0.953 [0.92, 0.98]. Eleven averages tried, all listed, the best the maximum of eleven. The interval of 0.954 excludes the reported 0.893 point, and the two families' own intervals overlap it, which is what two models that fail on different genes produce when averaged and what neither produces alone.

**Table 85:** Post hoc averages of existing predictions on the 195 held-out genes: each family's per-cell logits are standardized and averaged over its seeds, families are averaged with equal weight, and the mean is scored once with a 2,000-draw gene bootstrap. Nothing is fitted and every average tried is listed, so the best row is the maximum of seven post hoc choices, not a selected model. Written by `family_ensembles.py`. Bold marks the highest value in a column.

| average of | cells | AUROC [95% CI] |
|---|---|---|
| GO convolution (400 epochs, 3 seeds) | 3 | 0.909 [0.85, 0.95] |
| GO convolution (400 epochs, 6 seeds) | 6 | 0.905 [0.85, 0.95] |
| pathway + ProtT5 + affine (round 22, 3 seeds) | 3 | 0.891 [0.83, 0.94] |
| flux head, pathway + ProtT5 (rounds 23 and 25, 6 seeds) | 6 | 0.911 [0.86, 0.95] |
| cached solve + flux head, pathway + ProtT5 (round 27, 6 seeds) | 6 | 0.918 [0.87, 0.96] |
| layer-free log1p vector + affine, pathway + ProtT5 (round 33, 6 seeds) | 6 | 0.910 [0.86, 0.95] |
| GO convolution (400 epochs, 6 seeds) + layer-free log1p vector + affine, pathway + ProtT5 (round 33, 6 seeds) | 12 | **0.947** [0.91, 0.98] |
| layer-free log1p, interior-point store (round 36, 6 seeds) | 6 | 0.912 [0.86, 0.95] |
| GO convolution (400 epochs, 6 seeds) + layer-free log1p, interior-point store (round 36, 6 seeds) | 12 | 0.944 [0.90, 0.98] |
| GO convolution (400 epochs, 6 seeds) + cached solve + flux head, pathway + ProtT5 (round 27, 6 seeds) | 12 | 0.946 [0.91, 0.98] |
| GO convolution (400 epochs, 6 seeds) + pathway + ProtT5 + affine (round 22, 3 seeds) + cached solve + flux head, pathway + ProtT5 (round 27, 6 seeds) | 15 | 0.939 [0.90, 0.97] |
| GO convolution (400 epochs, 3 seeds) + pathway + ProtT5 + affine (round 22, 3 seeds) | 6 | 0.933 [0.89, 0.97] |
| GO convolution (400 epochs, 3 seeds) + flux head, pathway + ProtT5 (rounds 23 and 25, 6 seeds) | 9 | 0.941 [0.90, 0.98] |
| GO convolution (400 epochs, 3 seeds) + flux head, ProtT5 (rounds 21 and 24, 6 seeds) | 9 | 0.923 [0.87, 0.97] |
| GO convolution (400 epochs, 3 seeds) + pathway + ProtT5 + affine (round 22, 3 seeds) + flux head, pathway + ProtT5 (rounds 23 and 25, 6 seeds) | 12 | 0.935 [0.89, 0.97] |

**Table 86:** The same post hoc averages under the reported protocol: each family is its five fold models (one seed per fold, the cross-validation files of Section 3), standardized per fold and averaged, families averaged with equal weight, scored once on the 195 held-out genes with a 2,000-draw gene bootstrap. Six averages tried, all listed. Written by `family_ensembles.py -folds`. Bold marks the highest value in a column.

| average of | cells | AUROC [95% CI] |
|---|---|---|
| GO convolution, five folds (400 epochs) | 5 | 0.917 [0.87, 0.95] |
| indicator + ProtT5 + affine, five folds (round 12) | 5 | 0.897 [0.84, 0.94] |
| flux head, pathway + ProtT5, five folds (round 25) | 5 | 0.897 [0.84, 0.94] |
| cached solve + flux head, pathway + ProtT5, five folds (round 27) | 5 | 0.922 [0.88, 0.96] |
| layer-free log1p vector + affine, pathway + ProtT5, five folds (round 34) | 5 | 0.920 [0.87, 0.96] |
| GO convolution, five folds (400 epochs) + layer-free log1p vector + affine, pathway + ProtT5, five folds (round 34) | 10 | 0.953 [0.92, 0.98] |
| + GPR redundancy counts, pathway + ProtT5, five folds (round 80) | 5 | 0.930 [0.89, 0.97] |
| GO convolution, five folds (400 epochs) + + GPR redundancy counts, pathway + ProtT5, five folds (round 80) | 10 | **0.955** [0.92, 0.98] |
| YPD ratio + vector + counts, pathway + ProtT5, five folds (rounds 84 to 98) | 5 | 0.929 [0.88, 0.97] |
| GO convolution, five folds (400 epochs) + YPD ratio + vector + counts, pathway + ProtT5, five folds (rounds 84 to 98) | 10 | 0.953 [0.92, 0.98] |
| gene-rule counts, no solve, five folds (round 82) | 5 | 0.909 [0.86, 0.95] |
| GO convolution, five folds (400 epochs) + gene-rule counts, no solve, five folds (round 82) | 10 | 0.948 [0.91, 0.98] |
| layer-free log1p, interior-point store, five folds (round 37) | 5 | 0.913 [0.87, 0.95] |
| GO convolution, five folds (400 epochs) + layer-free log1p, interior-point store, five folds (round 37) | 10 | 0.951 [0.92, 0.98] |
| layer-free log1p, PDLP store, five folds (round 39) | 5 | 0.915 [0.87, 0.96] |
| GO convolution, five folds (400 epochs) + layer-free log1p, PDLP store, five folds (round 39) | 10 | 0.949 [0.91, 0.98] |
| GO convolution, five folds (400 epochs) + indicator + ProtT5 + affine, five folds (round 12) | 10 | 0.944 [0.90, 0.98] |
| GO convolution, five folds (400 epochs) + flux head, pathway + ProtT5, five folds (round 25) | 10 | 0.949 [0.91, 0.98] |
| GO convolution, five folds (400 epochs) + cached solve + flux head, pathway + ProtT5, five folds (round 27) | 10 | 0.954 [0.92, 0.98] |
| GO convolution, five folds (400 epochs) + indicator + ProtT5 + affine, five folds (round 12) + flux head, pathway + ProtT5, five folds (round 25) | 15 | 0.945 [0.91, 0.97] |
| GO convolution, five folds (400 epochs) + indicator + ProtT5 + affine, five folds (round 12) + cached solve + flux head, pathway + ProtT5, five folds (round 27) | 15 | 0.947 [0.91, 0.97] |

One model over both inputs does not recover the average. The deletion's solved flux as the round-33 cell encodes it, $\log(1 + |v|)$ over 4,131 reactions, put on the node beside the GO multi-hot and convolved over the union graph at 400 epochs reads 0.900, 0.912 and 0.914 (mean 0.909), the GO-alone rows' 0.909, and 0.910 with the ratios beside them; alone through the graph it reads 0.773 and on the MLP 0.709, and GO with it on the MLP 0.851 (Table 87). The convolution takes nothing from the solve columns that the average of the two families took (0.948), so what the average adds is not a matter of the graph model seeing the solve; the solve's ranking, as the transformer's affine reads it, is not what a convolution over these graphs extracts from the same columns.

The ten repeated splits agree (job 2684, three seeds per split). Against graph convolution on GO alone, the same convolution with the solve columns on the node reads −0.009 (higher on one of ten splits), with the four ratios beside them −0.010 (one of ten), the MLP on GO and the solve columns −0.013 (four of ten), and the solve columns alone through the graph −0.189 (none of ten). As node features the module does not carry to the graph family.

As an affine on the logit it does (job 2691, three seeds per split). The columns stay out of the message passing and a

zero-initialized linear map over them is added to each node's logit, the form the transformer uses. Against the same convolution on GO alone, rerun in the same job, the YPD ratio alone reads +0.011 over the ten splits (higher on ten of ten), the single-experiment recipe of ratio, vector and counts +0.018 (nine of ten) and the four-media recipe +0.021 (nine of ten); on the released split the three read 0.914, 0.911 and 0.910 against 0.908. At six seeds per split (job 2693) the three read +0.009 (ten of ten), +0.017 (seven of ten) and +0.019 (seven of ten), and 0.914, 0.911 and 0.911 against 0.905 on the released split. The interface that carries the module across model families is therefore the affine on the logit, with the solve computed offline, and it needs nothing from the architecture it is added to.

Giving both families the module does not improve their average. With the single-experiment transformer of Section 8.1 and the graph family carrying the same recipe as an affine, the fixed-in-advance average reads 0.897 over the ten splits against 0.907 with the plain GO convolution as the partner, although the graph family alone rises from 0.871 to 0.887. The average gains from the two families failing on different genes, and handing them the same columns makes them fail more alike. The fold protocol on the released genes says the same (job 2695, three seeds per fold): the graph family with the affine reads 0.930 [0.88, 0.97] as a five-fold ensemble against 0.917 without, and averaged with the transformer's single-experiment fold models it reads 0.941 against 0.953 for the plain graph family with the same partner.

The worst case for a module carried to the whole database is a record whose deletion touches no reaction, which reads the constant wild-type row. The genome-wide held-out set of Section 5.4 is entirely that case: none of its 4,674 genes is in the solve store and every one has a growth ratio of 1.0, so the affine adds the same constant to every held-out logit and cannot reorder them by itself. It still does not hurt (job 2696, three seeds, the same training and validation genes as the released split): graph convolution on GO reads 0.702 alone and 0.767 with the single-experiment recipe on its logit, higher on three of three seeds, and 0.704 with the YPD ratio alone; at six seeds (job 2702) the three read 0.698, 0.768 (six of six) and 0.696. Since the held-out logits differ only by the convolution's weights, the gain is in what the convolution learned. Hypothesis, untested: the affine explains the metabolic genes' labels during training, so the convolution's weights and its validation-selected epoch (87 to 93 against 382 to 397 without the affine) fit the remaining genes rather than the metabolic ones. The graph family's best genome-wide input, ProtT5 on the node (Section 5.4), reads the same way (job 2697, six seeds): on the released split the single-experiment affine takes it from 0.859 to 0.891, higher on six of six seeds, and on the 4,674 genome-wide genes from 0.751 to 0.766, six of six, with the YPD ratio alone worth +0.014 and +0.004.

**Table 87:** The deletion's pFBA flux vector as $\log(1 + |v|)$ (the round-33 encoding, 4,131 columns, zero for a gene the store does not cover) as a node input: alone, beside the GO multi-hot, and beside GO with the four FBA ratios, on the MLP (200 epochs) and on graph convolution over the union of the nine graphs (400 epochs, as Table 9). The protocol of Table 5 otherwise; seeds 2700 to 2702. Bold marks the highest value in a column.

| model | graph | inputs | dim | params | mean | sd | seeds | selected epochs |
|---|---|---|---|---|---|---|---|---|
| MLP | none | solve log1p | 4,131 | 268,673 | 0.709 | 0.012 | 0.716, 0.692, 0.717 | 86, 17, 93 |
| GCN | union of nine | solve log1p (400 epochs) | 4,131 | 268,673 | 0.773 | 0.002 | 0.771, 0.772, 0.775 | 393, 393, 388 |
| MLP | none | GO + solve log1p | 6,493 | 419,841 | 0.851 | 0.000 | 0.851, 0.852, 0.851 | 23, 25, 26 |
| GCN | union of nine | GO + solve log1p (400 epochs) | 6,493 | 419,841 | 0.909 | 0.006 | 0.900, 0.912, 0.914 | 376, 397, 397 |
| MLP | none | GO + FBA ratios + solve log1p | 6,497 | 420,097 | 0.851 | 0.001 | 0.851, 0.852, 0.850 | 25, 27, 26 |
| GCN | union of nine | GO + FBA ratios + solve log1p (400 epochs) | 6,497 | 420,097 | **0.910** | 0.001 | 0.911, 0.912, 0.909 | 396, 397, 397 |

The transformer side of the same question is round 47 (jobs 2434 and 2438; Table 88, Figure 89): the GO multi-hot on the node beside the pathway block and ProtT5 with the $\log(1 + |v|)$ vector in the affine, six seeds, and the same with GO in place of the pathway block, beside a same-pack control with GO on the node and the four ratios only. GO costs the transformer what it cost in round 26, with the solve in the affine as with the ratios: 0.866 (sd 0.009; pooled 0.872 [0.81, 0.93]) against the round-33 cell's 0.908 and 0.912, with GO in place of the pathway block 0.846 (pooled 0.855), and the control 0.852 (pooled 0.854), the round-26 cells' 0.849; the GO cells select at epochs 42 to 194 where the round-33 cell selects at 33 to 58. So neither family takes the other's input as the input enters it now, the convolution nothing from the solve columns on the node and the transformer a cost from GO on the node; whether GO enters the transformer the way the solve does, on the logit through the affine rather than on the node, is round 50's read.

**Table 88:** Round 47, the GO multi-hot beside the layer-free solve in one transformer. The round-33 cell and the round-26 GO cells (GO beside ProtT5, and GO beside the pathway block and ProtT5, the four ratios only) as the comparators, against GO beside the pathway block and ProtT5 with the $\log(1 + |v|)$ vector in the affine, GO in place of the pathway block with the vector, and GO beside the pathway block and ProtT5 without the vector run in the same pack. Batch 32, class weight, 200 epochs; seeds 2700 to 2705 for the round-33 cell, 2700 to 2702 for the rest. Bold marks the highest value in a column. ↑ (↓) marks an arm whose held-out AUROC is above (below) the first arm's on every one of the same three seeds.

| arm | mean | sd | seeds | pooled [95% CI] | selected epochs | val step |
|---|---|---|---|---|---|---|
| the vector as log1p, default initialization | **0.908** | 0.008 | 0.896 to 0.919 (6) | **0.912** [0.87, 0.95] | 33 to 58 | 0.0043 |
| GO multi-hot (propagated, 2,362) + ProtT5 + FBA ratios | 0.833 | 0.008 | 0.830, 0.827, 0.842 | 0.839 [0.76, 0.91] | 123, 48, 62 | 0.0044 |
| GO + pathway multi-hots + ProtT5 + FBA ratios | 0.849 | 0.005 | 0.852, 0.852, 0.844 | 0.851 [0.78, 0.92] | 131, 48, 126 | 0.0019 |
| GO + pathway + ProtT5, the four ratios only (round 47 control) | 0.852 | 0.011 | 0.838 to 0.871 (6) | 0.854 [0.78, 0.92] | 42 to 179 | 0.0028 |
| GO multi-hot + ProtT5 (no pathway block), the log1p vector in the affine | 0.846 | 0.009 | 0.834 to 0.857 (6) | 0.855 [0.78, 0.92] | 47 to 194 | 0.0034 |
| GO + pathway + ProtT5, the log1p vector in the affine, no layer | 0.866 | 0.009 | 0.857 to 0.880 (6) | 0.872 [0.81, 0.93] | 68 to 185 | 0.0026 |

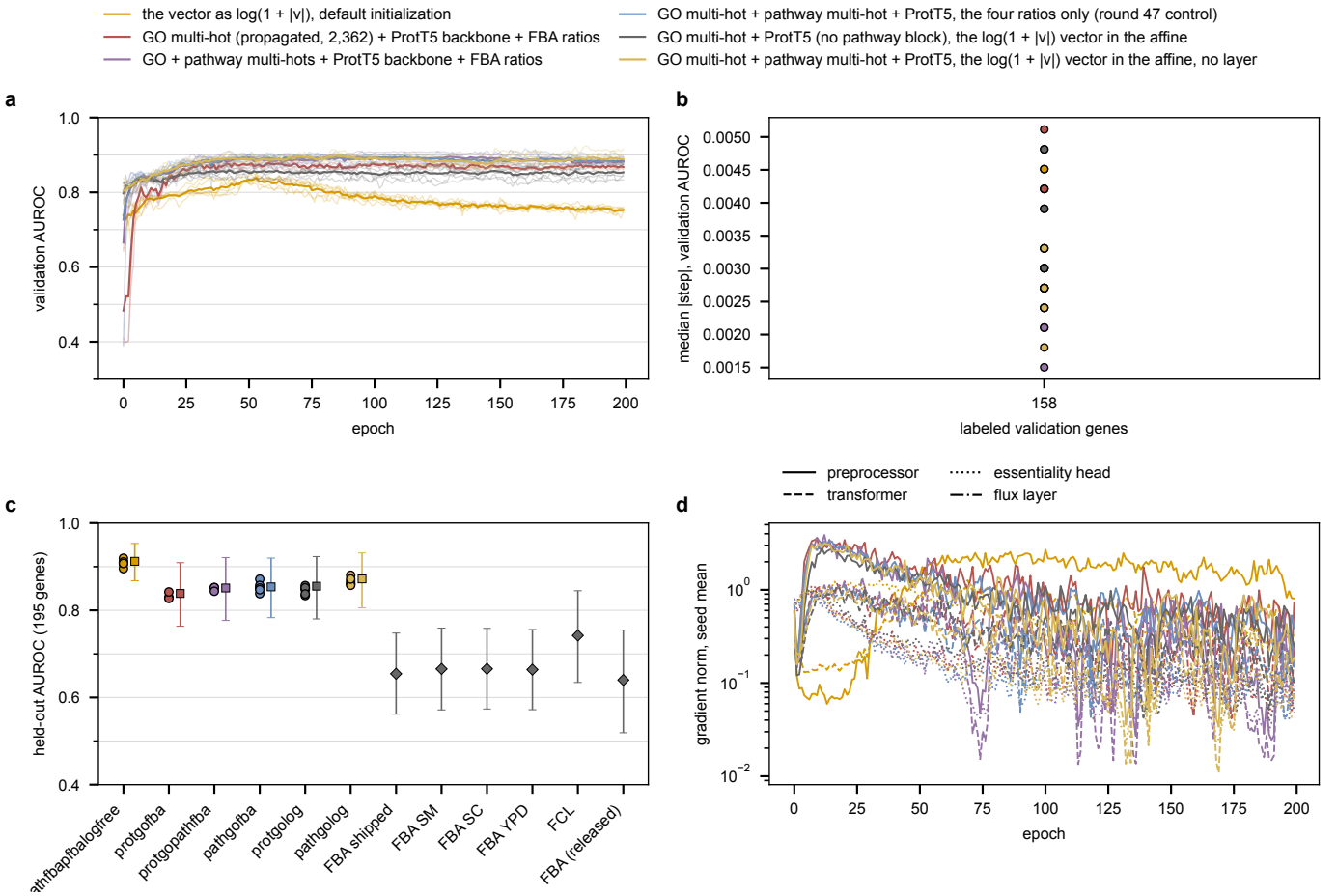



**Figure 89: Round 47, the GO multi-hot beside the layer-free solve in one transformer.** **a**, Validation AUROC per epoch, one thin line per seed and the seed mean per arm. **b**, The median absolute epoch-to-epoch step per cell against the validation size. **c**, Held-out AUROC per cell (points), the seed mean and the gene bootstrap of the pooled per-gene logits (square with interval), beside FBA on four media, Flux Cone Learning and the released FBA predictions on the same 195 genes. **d**, Gradient norm per module per epoch, seed mean. Written by `experiments/028-gene-essentiality/scripts/wave12_readout.py -round w63`.

GO enters the transformer on the logit at no cost and adds nothing there either (round 50, jobs 2440, 2444 and 2446; Table 89, Figure 90). The affine reads any store written in the solve store's shape, so the propagated GO multi-hot was written as one (`go_affine_store.py`: 2,362 columns over the 1,121 labeled genes, 1,120 with an annotation; the trainer's $\log(1 + |v|)$ maps a term to 0 or 0.693), alone and appended to the pFBA vector. GO alone as affine columns beside pathway and ProtT5 on the node reads 0.886 on six seeds (sd 0.012; pooled 0.892 [0.83, 0.94]) against 0.852 with GO on the node; GO with the $\log(1 + |v|)$ vector as affine columns reads 0.895, 0.901, 0.906, 0.900, 0.893 and 0.911 (mean 0.901, sd 0.007; pooled 0.903 [0.85, 0.95]) against 0.866 with GO on the node beside the same vector and 0.908 and 0.912 with the vector alone; the same on ProtT5 alone 0.876 (pooled 0.880); and the GO-on-node cell run to 400 epochs 0.860 (pooled 0.864), no better than at 200. So the logit path carries the annotation where the node path pays 0.04 for it, and the one model on the logit reads the solve-alone number, not the two-family average (0.948). What the GO columns do in the affine is visible in the training loss: with 2,362 binary columns over 629 training genes the cell's training loss crosses 0.6 at epoch 14 to 21 and reaches 0.01 to 0.04 by epoch 199, where the vector-alone cell crosses at

epochs 33 to 58 and ends near 0.1, and the selected epochs run 79 to 197 (251 to 379 at 400 epochs, which reads 0.895); the affine memorizes the training genes through the GO columns and the held-out ranking drifts on a fit plateau. Under the fold protocol (round 58, job 2453) the form reads 0.911, 0.907, 0.896, 0.886 and 0.884 on the folds and 0.918 [0.86, 0.97] as the ensemble on one seed, the solve-alone cell's 0.919 and 0.920 (Table 48), so the one model on the logit is one number with the solve alone on both protocols and 0.035 under the two-family average. A regularized read of the GO columns is what remains of the one-model question on this backbone.

**Table 89:** Round 50, GO on the logit instead of the node. The round-33 cell and the round-47 GO-on-node cells as the comparators, against the GO multi-hot (2,362 propagated terms) entering as affine columns through the solve store path, encoded as $\log(1 + |v|)$ (0 or 0.693 per term): GO alone beside the four ratios with pathway and ProtT5 on the node, GO appended to the pFBA vector (the one-model GO + layer-free form on the logit), the same on ProtT5 alone, and the round-47 GO-on-node cell run to 400 epochs. Batch 32, class weight; seeds 2700 to 2705 for the round-33 cell, 2700 to 2702 for the rest. Bold marks the highest value in a column. ↑ (↓) marks an arm whose held-out AUROC is above (below) the first arm's on every one of the same three seeds.

| arm | mean | sd | seeds | pooled [95% CI] | selected epochs | val step |
|---|---|---|---|---|---|---|
| the vector as log1p, default initialization | **0.908** | 0.008 | 0.896 to 0.919 (6) | **0.912** [0.87, 0.95] | 33 to 58 | 0.0043 |
| GO + pathway + ProtT5, the four ratios only (round 47 control) | 0.852 | 0.011 | 0.838 to 0.871 (6) | 0.854 [0.78, 0.92] | 42 to 179 | 0.0028 |
| GO + pathway + ProtT5, the log1p vector in the affine, no layer | 0.866 | 0.009 | 0.857 to 0.880 (6) | 0.872 [0.81, 0.93] | 68 to 185 | 0.0026 |
| round-47 GO + pathway + ProtT5 layer-free cell at 400 epochs | 0.860 | 0.004 | 0.864, 0.862, 0.855 | 0.864 [0.80, 0.93] | 111, 255, 265 | 0.0015 |
| pathway + ProtT5 on the node; the GO multi-hot as affine columns (no solve) | 0.886 | 0.012 | 0.870 to 0.897 (6) | 0.892 [0.83, 0.94] | 83 to 194 | 0.0017 |
| pathway + ProtT5 on the node; the log1p vector and the GO multi-hot as affine columns | 0.901 | 0.007 | 0.893 to 0.911 (6) | 0.903 [0.85, 0.95] | 79 to 197 | 0.0021 |
| the same cell at 400 epochs | 0.895 | 0.006 | 0.892, 0.902, 0.890 | 0.895 [0.84, 0.94] | 251, 286, 379 | 0.0024 |
| ProtT5 alone on the node; the log1p vector and the GO multi-hot as affine columns | 0.876 | 0.003 | 0.874, 0.880, 0.875 | 0.880 [0.81, 0.94] | 164, 165, 176 | 0.0015 |

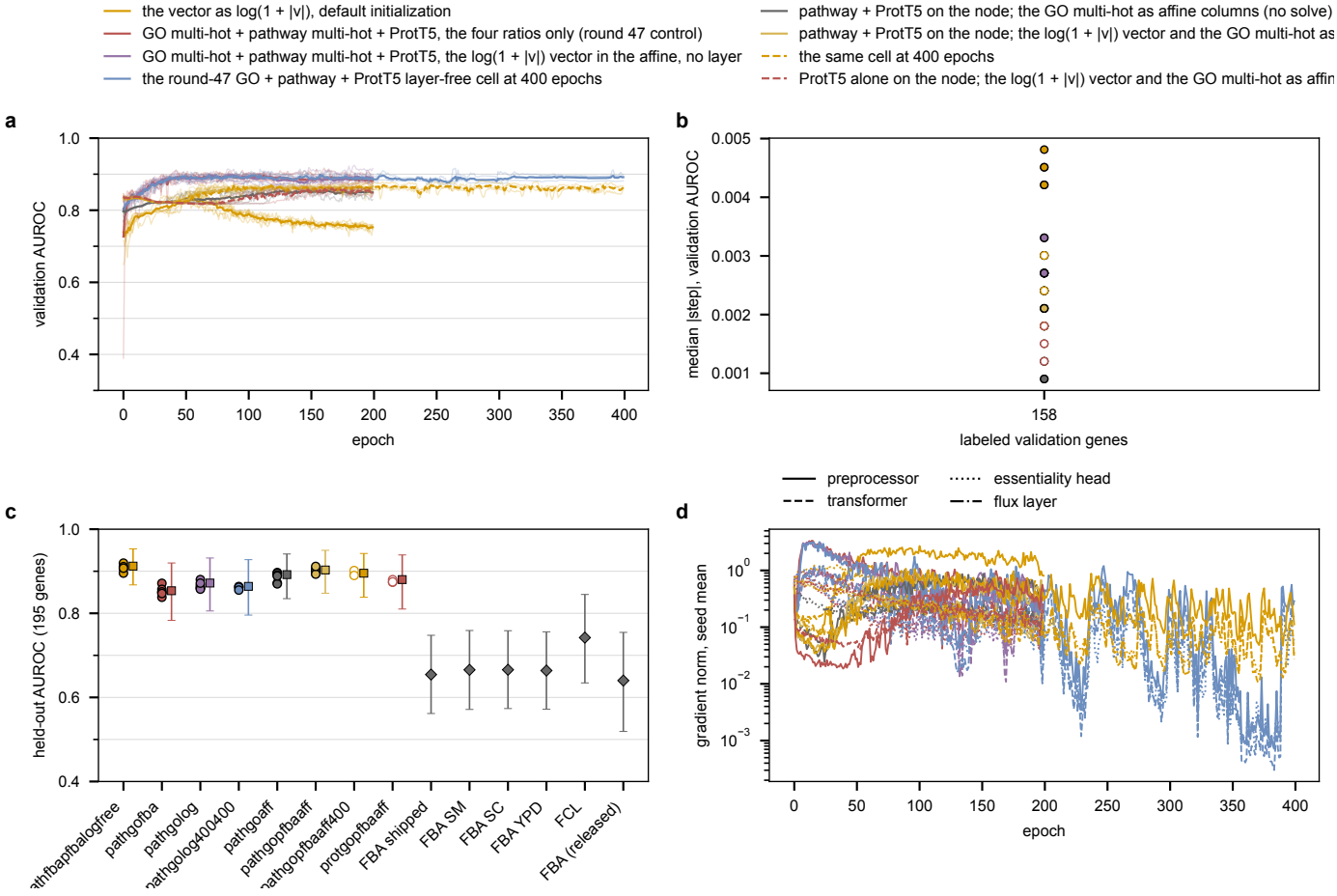



**Figure 90: Round 50, GO on the logit instead of the node.** **a**, Validation AUROC per epoch, one thin line per seed and the seed mean per arm. **b**, The median absolute epoch-to-epoch step per cell against the validation size. **c**, Held-out AUROC per cell (points), the seed mean and the gene bootstrap of the pooled per-gene logits (square with interval), beside FBA on four media, Flux Cone Learning and the released FBA predictions on the same 195 genes. **d**, Gradient norm per module per epoch, seed mean. Written by `experiments/028-gene-essentiality/scripts/wave12_readout.py -round w66`.

Weight decay is not the regularizer (round 59, job 2454; Table 90, Figure 91). The GO-and-vector affine cell with AdamW decay 0.1, a thousand times the base $10^{-4}$, still crosses training loss 0.6 at epochs 15 to 16, ends at 0.03 to 0.05 and selects at epochs 173 to 182, reading 0.890, 0.898 and 0.891 (mean 0.893; pooled 0.892 [0.83, 0.94]) against

0.901 at the base; at decay 1.0 the cell selects at epochs 8 to 11 and reads 0.849, 0.849 and 0.845 (mean 0.848), the decay pulling the affine back faster than the training genes can be fit. What the columns need is fewer of them, or a read that cannot memorize 629 genes through 2,362 indicators; the reduced form is round 60.

**Table 90:** Round 59, weight decay on the GO affine columns. The round-33 cell (the vector alone in the affine) and the round-50 cell (GO and the vector as affine columns, AdamW weight decay $10^{-4}$) as the comparators, against the round-50 cell with weight decay 0.1 and 1.0. Batch 32, class weight, 200 epochs; seeds 2700 to 2705 for the comparators, 2700 to 2702 for the rest. Bold marks the highest value in a column. ↑ (↓) marks an arm whose held-out AUROC is above (below) the first arm's on every one of the same three seeds.

| arm | mean | sd | seeds | pooled [95% CI] | selected epochs | val step |
|---|---|---|---|---|---|---|
| the vector as log1p, default initialization | **0.908** | 0.008 | 0.896 to 0.919 (6) | **0.912** [0.87, 0.95] | 33 to 58 | 0.0043 |
| pathway + ProtT5 on the node; the log1p vector and the GO multi-hot as affine columns | 0.901 | 0.007 | 0.893 to 0.911 (6) | 0.903 [0.85, 0.95] | 79 to 197 | 0.0021 |
| the same cell with AdamW weight decay 0.1 | 0.893 | 0.004 | 0.890, 0.898, 0.891 | 0.892 [0.83, 0.94] | 182, 173, 173 | 0.0021 |
| the same cell with AdamW weight decay 1.0 | 0.848 | 0.003 | 0.849, 0.849, 0.845 | 0.848 [0.75, 0.93] | 10, 11, 8 | 0.0006 |

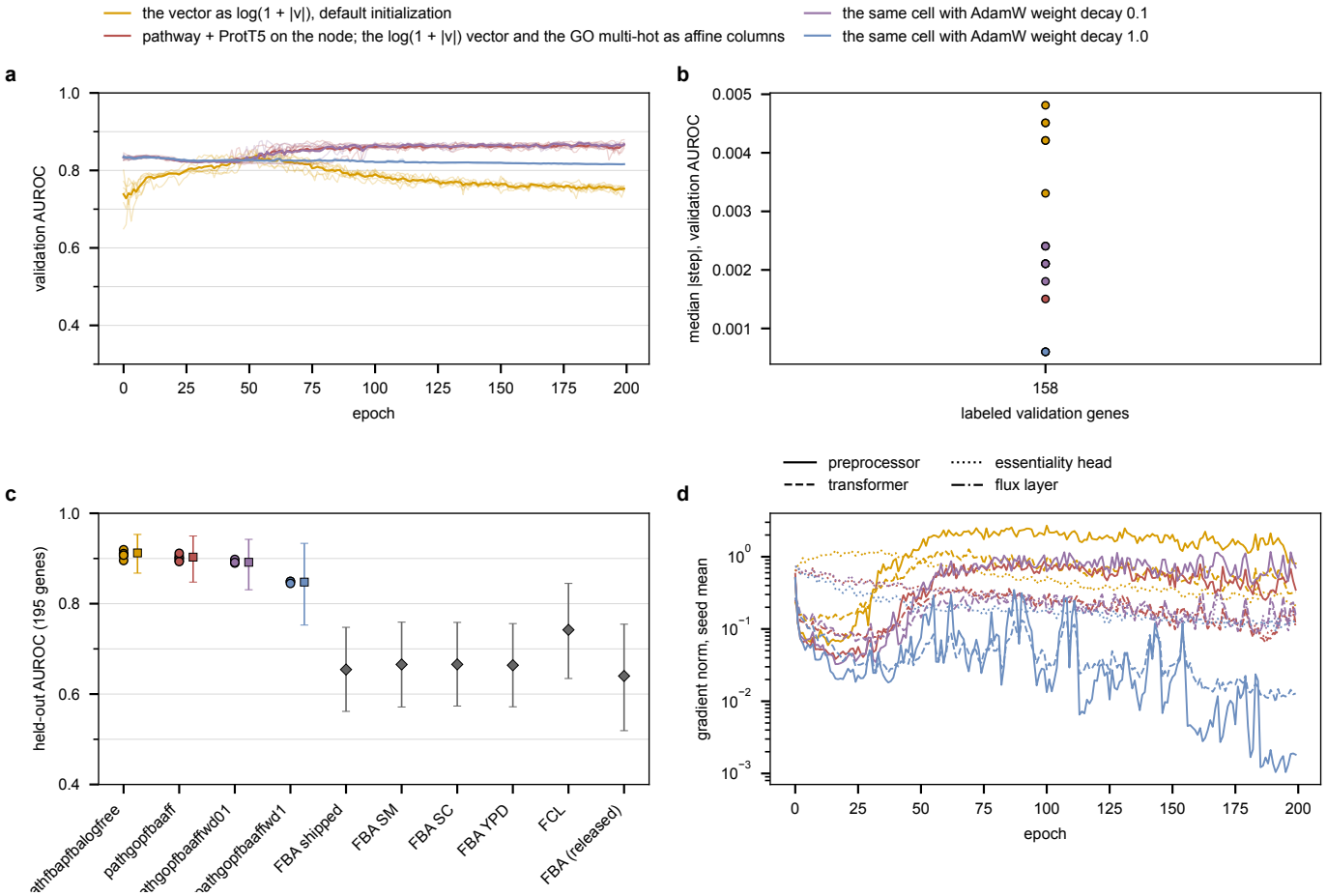



**Figure 91: Round 59, weight decay on the GO affine columns.** **a**, Validation AUROC per epoch, one thin line per seed and the seed mean per arm. **b**, The median absolute epoch-to-epoch step per cell against the validation size. **c**, Held-out AUROC per cell (points), the seed mean and the gene bootstrap of the pooled per-gene logits (square with interval), beside FBA on four media, Flux Cone Learning and the released FBA predictions on the same 195 genes. **d**, Gradient norm per module per epoch, seed mean. Written by `experiments/028-gene-essentiality/scripts/wave12_readout.py -round w75`.

Fewer columns stop the memorizing and read the solve-alone number (round 60, jobs 2455 and 2458; Table 91, Figure 92). The GO multi-hot projected on its first 64 principal components fitted over the 1,121 labeled genes (80 percent of the variance), split into positive and negative parts so that the $\log(1 + |v|)$ encoding keeps the sign (128 columns), reads 0.886 on six seeds alone (sd 0.022; pooled 0.902 [0.85, 0.95]) and 0.905, 0.911, 0.919, 0.900, 0.900 and 0.902 with the vector (mean 0.906, sd 0.008; pooled 0.913 [0.86, 0.96]), the vector-alone cell's 0.908 and 0.912, the cells crossing training loss 0.6 at epochs 40 to 66 and selecting at 33 to 74 as the vector-alone cell does; under the fold protocol (round 62, jobs 2459 to 2461) the same cell reads 0.914 [0.86, 0.96] on one seed and 0.917 [0.87, 0.96] on three (folds 0.917, 0.878, 0.843, 0.901 and 0.912 as seed-mean logits), the solve-alone 0.919 and 0.920 (Table 48). Sixteen components (round 63, jobs 2462 and 2467; 32 columns, 52 percent of the variance) read the same: 0.874 alone (sd 0.029; pooled 0.893) and 0.908 with the vector (sd 0.016; pooled 0.915 [0.86, 0.96]) on six seeds. The one model on this backbone is closed on the split file and the folds: GO on the node costs 0.04, GO on the logit costs nothing and adds nothing, in full it

memorizes and reduced to 64 or 16 components it reads the solve alone, so the annotation's 0.04 over the two families' average is not the transformer's to take through either path, and the two families stay two models.

**Table 91:** Rounds 60 and 63, the GO multi-hot reduced before the affine. The round-33 cell and the round-50 cells (the full GO multi-hot as affine columns, alone and with the vector) as the comparators, against the GO multi-hot projected on its first 64 and its first 16 principal components fitted over the 1,121 labeled genes and split into positive and negative parts (128 and 32 columns, so the $\log(1 + |v|)$ encoding keeps the sign), alone and with the vector. Batch 32, class weight, 200 epochs; seeds 2700 to 2705 (2700 to 2702 at 16 components). Bold marks the highest value in a column. ↑ (↓) marks an arm whose held-out AUROC is above (below) the first arm's on every one of the same three seeds.

| arm | mean | sd | seeds | pooled [95% CI] | selected epochs | val step |
|---|---|---|---|---|---|---|
| the vector as log1p, default initialization | **0.908** | 0.008 | 0.896 to 0.919 (6) | 0.912 [0.87, 0.95] | 33 to 58 | 0.0043 |
| pathway + ProtT5 on the node; the GO multi-hot as affine columns (no solve) | 0.886 | 0.012 | 0.870 to 0.897 (6) | 0.892 [0.83, 0.94] | 83 to 194 | 0.0017 |
| pathway + ProtT5 on the node; the log1p vector and the GO multi-hot as affine columns | 0.901 | 0.007 | 0.893 to 0.911 (6) | 0.903 [0.85, 0.95] | 79 to 197 | 0.0021 |
| pathway + ProtT5 on the node; GO reduced to 64 principal components (positive and negative parts) as affine columns | 0.886 | 0.022 | 0.848 to 0.909 (6) | 0.902 [0.85, 0.95] | 33 to 74 | 0.0034 |
| the same with the log1p vector as affine columns too | 0.906 | 0.008 | 0.896 to 0.919 (6) | 0.913 [0.86, 0.96] | 52 to 69 | 0.0028 |
| GO reduced to 16 principal components (positive and negative parts) as affine columns | 0.873 | 0.028 | 0.824 to 0.900 (6) | 0.893 [0.83, 0.95] | 33 to 108 | 0.0039 |
| the same with the log1p vector as affine columns too | 0.908 | 0.016 | 0.877 to 0.919 (6) | **0.915** [0.86, 0.96] | 38 to 77 | 0.0026 |

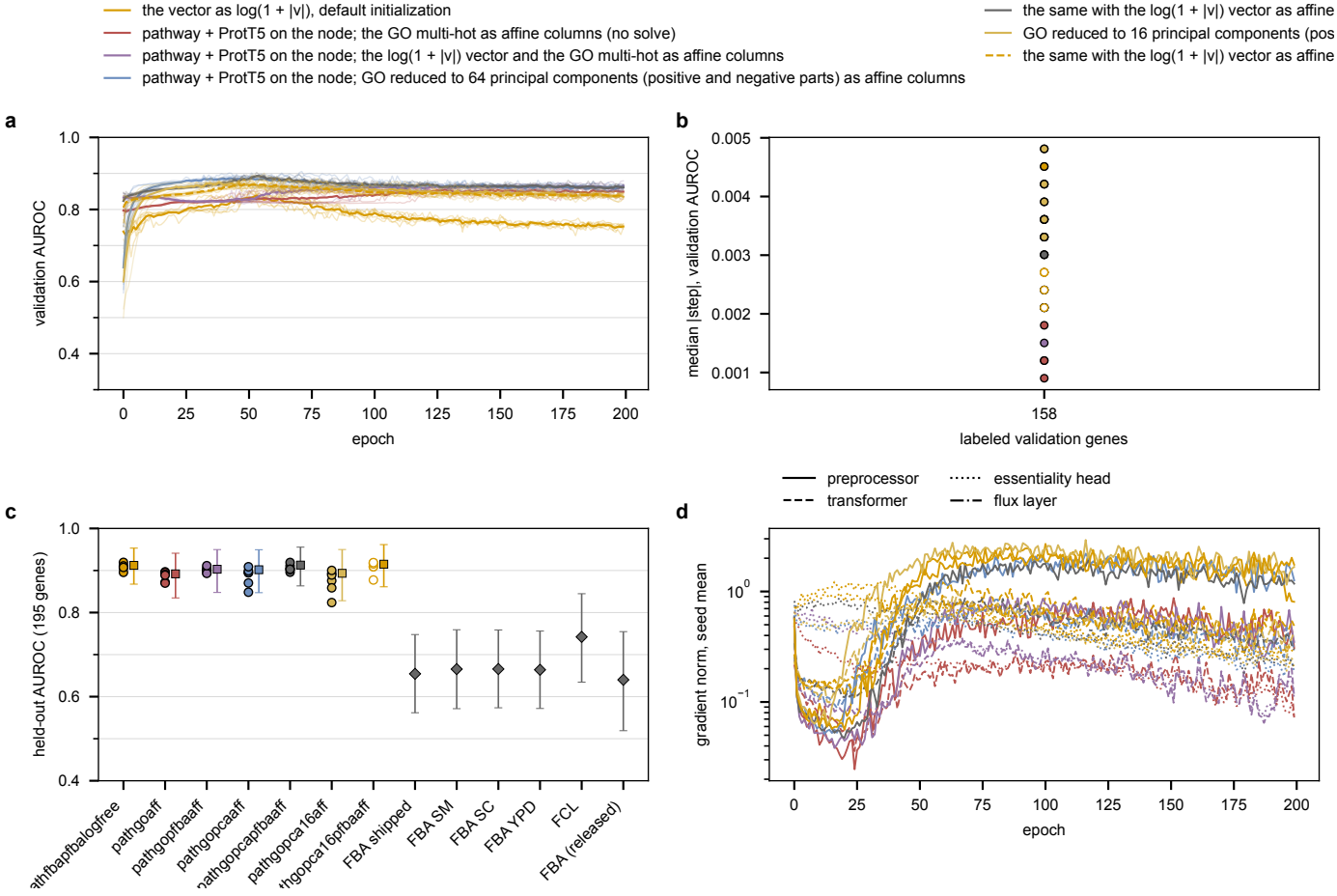



**Figure 92: Rounds 60 and 63, the GO multi-hot reduced before the affine.** **a**, Validation AUROC per epoch, one thin line per seed and the seed mean per arm. **b**, The median absolute epoch-to-epoch step per cell against the validation size. **c**, Held-out AUROC per cell (points), the seed mean and the gene bootstrap of the pooled per-gene logits (square with interval), beside FBA on four media, Flux Cone Learning and the released FBA predictions on the same 195 genes. **d**, Gradient norm per module per epoch, seed mean. Written by `experiments/028-gene-essentiality/scripts/wave12_readout.py -round w76`.

## 5.4 Round 30: the genome-wide labels as a larger held-out set ✗

Every number above sits on 195 genes, and the open list's first question was whether the graph convolution's margin survives a larger held-out set. The genome-wide labels of round 1 give one: 4,674 genes the released file never labeled, 974 essential by SGD annotation and 3,700 nonessential by a deletion-collection fitness record. Round 30 trains on the released labeled genes outside the released held-out set (the split `ess_genome_heldout_val0.2_s0`: 629 train, 158

validation, the 195 released held-out genes dropped so the read is not on them) and reads once on the 4,674 (Table 92). Two properties of the set bound what it can say. None of the 4,674 carries a yeast-GEM reaction, since the released file had labeled the GEM's genes, so the read is a transfer from metabolic to non-metabolic genes and FBA's knockout ratio is 1.0 on every one of them, an AUROC of exactly 0.5. And label origin coincides with class: every essential gene is an SGD annotation and every nonessential one a fitness record, so a model that can tell the two sources apart scores on this set without reading essentiality.

**Table 92:** Round 30, the genome-wide labels as a larger held-out set. Every model is trained on the released labeled genes outside the released held-out set (629 train, 158 val, the split `ess_genome_heldout_val0.2_s0`) and read once on the 4,674 genes the released file never labeled, 974 of them essential by SGD annotation and the rest nonessential by a fitness record, so label origin coincides with class on this set. None of these genes carries a yeast-GEM reaction, so FBA's knockout ratio is 1.0 on every one and the solve has no call. Cells are seeds 2700 to 2702; the pooled column scores the seed-mean logit with a 2,000-draw gene bootstrap. Bold marks the highest value in a column.

| arm | cells | per cell | mean | sd | pooled [95% CI] | selected epochs |
|---|---|---|---|---|---|---|
| pathway multi-hot + ProtT5 + FBA affine (round 22 recipe) | 6 | 0.571, 0.571, 0.579, 0.569, 0.571, 0.573 | 0.572 | 0.003 | 0.572 [0.55, 0.59] | – |
| cached solve + flux head, pathway + ProtT5 (round 27 form) | 6 | 0.543, 0.480, 0.531, 0.512, 0.579, 0.543 | 0.532 | 0.033 | 0.551 [0.53, 0.57] | – |
| layer-free log1p vector + affine, pathway + ProtT5 (round 33 form) | 6 | 0.542, 0.561, 0.569, 0.557, 0.568, 0.599 | 0.566 | 0.019 | 0.570 [0.55, 0.59] | – |
| layer-free, store over every gene (wild-type rows; round 76 control) | 6 | 0.542, 0.561, 0.569, 0.557, 0.568, 0.599 | 0.566 | 0.019 | 0.570 [0.55, 0.59] | – |
| layer-free + physical degree and clustering in the affine (round 76) | 6 | 0.564, 0.594, 0.588, 0.603, 0.656, 0.607 | 0.602 | 0.030 | 0.593 [0.57, 0.61] | – |
| MLP on ProtT5 alone | 3 | 0.539, 0.531, 0.532 | 0.534 | 0.003 | 0.534 [0.51, 0.55] | 62, 58, 56 |
| MLP on ProtT5 + FBA ratios | 3 | 0.494, 0.476, 0.488 | 0.486 | 0.008 | 0.481 [0.46, 0.50] | 71, 197, 76 |
| MLP on the three inputs | 3 | 0.511, 0.508, 0.507 | 0.509 | 0.002 | 0.509 [0.49, 0.53] | 55, 58, 55 |
| graph convolution, physical graph, all-ones input | 3 | 0.684, 0.683, 0.692 | 0.686 | 0.004 | 0.686 [0.67, 0.70] | 91, 100, 77 |
| graph convolution, physical graph, degree and clustering | 3 | 0.757, 0.752, 0.749 | 0.753 | 0.003 | 0.759 [0.74, 0.78] | 28, 31, 3 |
| graph convolution, physical graph, ProtT5 | 3 | 0.545, 0.542, 0.544 | 0.544 | 0.001 | 0.544 [0.52, 0.57] | 192, 197, 188 |
| graph convolution, union of nine, all-ones input | 3 | 0.643, 0.643, 0.643 | 0.643 | 0.000 | 0.643 [0.62, 0.66] | 20, 5, 3 |
| graph convolution, union of nine, degree and clustering | 3 | 0.660, 0.667, 0.660 | 0.663 | 0.003 | 0.663 [0.64, 0.68] | 197, 196, 197 |
| graph convolution, union of nine, ProtT5 | 3 | 0.755, 0.753, 0.756 | 0.755 | 0.001 | 0.755 [0.74, 0.77] | 197, 197, 197 |
| graph attention, physical graph, degree and clustering | 3 | 0.709, 0.692, 0.720 | 0.707 | 0.011 | 0.708 [0.69, 0.73] | 179, 190, 119 |
| graph attention, union of nine, degree and clustering | 3 | 0.715, 0.704, 0.719 | 0.713 | 0.006 | 0.717 [0.70, 0.74] | 197, 197, 197 |
| graph convolution on ProtT5 alone, 400 epochs | 3 | 0.746, 0.753, 0.749 | 0.749 | 0.003 | 0.749 [0.73, 0.77] | 373, 380, 393 |
| graph convolution on ProtT5 + FBA ratios | 3 | 0.759, 0.758, 0.758 | **0.759** | 0.000 | 0.759 [0.74, 0.78] | 196, 197, 197 |
| graph convolution on the three inputs, 400 epochs | 3 | 0.742, 0.744, 0.744 | 0.744 | 0.001 | 0.744 [0.72, 0.76] | 380, 396, 380 |
| graph convolution on GO alone, 400 epochs | 3 | 0.716, 0.697, 0.694 | 0.702 | 0.010 | 0.703 [0.68, 0.72] | 397, 395, 379 |
| graph convolution on GO + FBA ratios, 400 epochs | 3 | 0.699, 0.693, 0.707 | 0.700 | 0.006 | 0.700 [0.68, 0.72] | 397, 392, 397 |
| graph convolution on ProtT5 + GO + FBA ratios, 400 epochs | 3 | 0.700, 0.697, 0.707 | 0.701 | 0.004 | 0.702 [0.68, 0.72] | 397, 368, 391 |
| FBA knockout ratio on YPD (0 of 4674 genes carry a reaction) | 1 | 0.500 | 0.500 | 0.000 | 0.500 [0.50, 0.50] | – |

**Table 93:** The gene's degree in each graph as the essentiality score, nothing trained, on the 4,674 genome-wide held-out genes of round 30, the 195 released held-out genes, and the 787 released genes the models train and validate on. AUROC of degree with a 2,000-draw gene bootstrap, and the median degree of the essential against the nonessential genes in that set. Bold marks the highest AUROC in a column. Written by `experiments/028-gene-essentiality/scripts/degree_by_class.py`.

| | genome held-out (4,674) | | released held-out (195) | | released train + val (787) | |
|---|---|---|---|---|---|---|
| graph | AUROC | median ess / non | AUROC | median ess / non | AUROC | median ess / non |
| physical | **0.740** [0.72, 0.76] | 50 / 16 | 0.594 [0.47, 0.71] | 27 / 19 | 0.591 [0.53, 0.65] | 21 / 17 |
| regulatory | 0.478 [0.46, 0.50] | 4 / 5 | 0.513 [0.41, 0.62] | 6 / 7 | 0.474 [0.42, 0.53] | 6 / 6 |
| tflink | 0.562 [0.55, 0.58] | 30 / 27 | 0.424 [0.31, 0.53] | 37 / 46 | 0.424 [0.37, 0.48] | 37 / 42 |
| STRING neighborhood | 0.514 [0.50, 0.53] | 0 / 0 | **0.618** [0.50, 0.73] | 175 / 67 | **0.629** [0.57, 0.68] | 142 / 68 |
| STRING fusion | 0.625 [0.61, 0.64] | 1 / 0 | 0.605 [0.49, 0.71] | 7 / 4 | 0.555 [0.50, 0.61] | 6 / 5 |
| STRING cooccurence | 0.497 [0.48, 0.51] | 0 / 0 | 0.356 [0.27, 0.45] | 0 / 1 | 0.368 [0.32, 0.42] | 0 / 1 |
| STRING coexpression | 0.681 [0.66, 0.70] | 394 / 182 | 0.617 [0.49, 0.73] | 477 / 292 | 0.585 [0.52, 0.64] | 385 / 300 |
| STRING experimental | 0.638 [0.62, 0.66] | 263 / 121 | 0.594 [0.47, 0.70] | 203 / 158 | 0.517 [0.46, 0.57] | 139 / 149 |
| STRING database | 0.714 [0.70, 0.73] | 22 / 1 | 0.577 [0.48, 0.68] | 41 / 29 | 0.485 [0.43, 0.54] | 24 / 29 |
| union of nine | 0.655 [0.64, 0.67] | 534 / 297 | 0.611 [0.49, 0.73] | 660 / 412 | 0.572 [0.51, 0.63] | 540 / 449 |

The inputs do not transfer and the graph does. The MLP on the transformer's three inputs reads 0.509 on the 4,674 genes (0.486 on ProtT5 with the ratios), chance, from 0.870 on the released 195; the same inputs through two layers of graph convolution read 0.744, ProtT5 alone through the graph 0.749 and with the ratios 0.759, and GO alone through the graph 0.702, under the three-input convolution here where on the 195 it was 0.02 above it. The graph convolution on GO does not keep its margin off the metabolic genes; the graph itself carries what transfers, and the ratios add 0.01 to a convolution on ProtT5 when every ratio in the test set is 1.0, which measures the training set and not the test set. What the graph carries is the gene's neighborhood in the physical graph, measured directly with the ladder's structure-only inputs (job 2401): degree and clustering alone, through two layers of convolution over the physical graph, read 0.753 [0.74, 0.78], level with ProtT5 through the union graph (0.755) and with the best number on the set, while the all-ones input reads 0.686 on the physical graph and 0.643 on the union, ProtT5 without a graph 0.534, and ProtT5 through the physical graph at 200 epochs 0.544. On these genes the number is the degree of the gene in the protein interaction graph, which a convolution reads from any input that lets it count neighbors and a model without the graph

cannot; whether that is essentiality or the deletion collection's own coverage the label origin does not let this set decide. Degree needs no model to say so (Table 93): the gene's degree in the physical graph, ranked as it is, reads 0.740 [0.72, 0.76] on the 4,674 genes, a median of 50 interactions for an essential gene against 16 for a nonessential one, and 0.594 [0.47, 0.71] on the released 195 and 0.591 on the 787 released training genes, where the medians are 27 against 19 and 21 against 17. The STRING database channel's degree reads 0.714 on the genome-wide genes and 0.577 on the released ones, the union's 0.655 and 0.611, and no other graph's degree passes 0.69 anywhere. So the physical graph's degree is a genome-wide essentiality signal that the metabolic genes barely vary in, which is why no model in this document read it on the 195 and every graph model read it on the 4,674. The transformer does not transfer either: the round-22 recipe, the pathway multi-hot with ProtT5 and the FBA affine, trained on the same 629 genes reads 0.572 [0.55, 0.59] on the 4,674 (0.569 to 0.579 over six seeds, jobs 2396 and 2544), 0.07 above chance and 0.18 under the graph convolution on ProtT5 alone, from 0.890 on the released 195. Its inputs are the MLP's, and in this recipe the gene graphs enter training only through the graph regularizer, not as a message-passing input, so the three models that read 0.87 to 0.89 on the metabolic genes without the graph as an input all fall to chance or near it off them, and the one that convolves over the graph does not. The scalable module form on the same split (jobs 2397, 2545 and 2546, six seeds) reads 0.551 [0.53, 0.57] pooled, and is the one arm here whose seeds disagree: 0.480 to 0.579, a spread of 0.033 against the affine recipe's 0.003, with one seed below chance. It sits under the affine recipe: its solved flux is the wild-type vector on every one of these genes, a constant the head cannot rank by, so what it adds over the backbone is nothing by construction, and what it reads is the backbone with a head that fit the metabolic genes' flux. The layer-free form (round 65, jobs 2464 and 2466, six seeds) reads 0.566 (sd 0.019; pooled 0.570 [0.55, 0.59]) on the same split, between the two, for the same reason: the solve store covers the 1,121 labeled genes only, so the affine's solve columns read one all-zero row on every held-out gene (`n_genes_with_solve` 1,121 in the result files), a constant that shifts every logit alike.

The other direction says the larger label set is not the missing training data for the metabolic genes (Tables 94 and 95). Trained on the 4,368 genome-labeled genes and read on the released 195, the graph convolution on GO reads 0.913 on every seed (0.912, 0.913, 0.913), the number it reads from 629 genes with the seed spread closed, and the cheap models on the transformer's three inputs lose to their 629-gene runs: the MLP 0.837 from 0.870, the 400-epoch convolution 0.843 from 0.888. Seven times the labels, of a different origin for six sevenths of them, buy the GO convolution nothing and cost the three-input models 0.03 to 0.05, which is what round 1 measured on the transformer.

**Table 94:** The three rows of Table 9 trained on the genome-wide labels (split `ess_genome_val0.2_s0`: 4,368 train and 1,093 validation genes, the released labels where the file has them and SGD or a fitness record elsewhere) and read on the same 195 released held-out genes, seeds 2700 to 2702. Bold marks the highest value in a column.

| model | graph | inputs | dim | params | mean | sd | seeds | selected epochs |
|---|---|---|---|---|---|---|---|---|
| GCN | union of nine | GO (400 epochs) | 2,362 | 155,457 | **0.913** | 0.000 | 0.912, 0.913, 0.913 | 397, 397, 397 |
| GCN | union of nine | GO + FBA ratios (400 epochs) | 2,366 | 155,713 | 0.911 | 0.002 | 0.912, 0.914, 0.908 | 397, 397, 395 |
| GCN | union of nine | ProtT5 + GO + FBA ratios (400 epochs) | 3,390 | 221,249 | 0.907 | 0.002 | 0.906, 0.910, 0.906 | 396, 397, 397 |

**Table 95:** The MLP and graph convolution on the transformer's inputs (Table 5) trained on the genome-wide labels (split `ess_genome_val0.2_s0`) and read on the same 195 released held-out genes, seeds 2700 to 2702. Bold marks the highest value in a column.

| model | graph | inputs | dim | params | mean | sd | seeds | selected epochs |
|---|---|---|---|---|---|---|---|---|
| MLP | none | ProtT5 + chrom-pathway | 1,221 | 82,433 | 0.808 | 0.004 | 0.812, 0.803, 0.810 | 101, 71, 71 |
| GCN | union of nine | ProtT5 + chrom-pathway | 1,221 | 82,433 | 0.824 | 0.001 | 0.825, 0.825, 0.823 | 197, 197, 197 |
| MLP | none | ProtT5 + FBA ratios | 1,028 | 70,081 | 0.770 | 0.000 | 0.769, 0.770, 0.770 | 79, 80, 79 |
| GCN | union of nine | ProtT5 + FBA ratios | 1,028 | 70,081 | 0.818 | 0.002 | 0.819, 0.820, 0.815 | 197, 196, 197 |
| MLP | none | ProtT5 + chrom-pathway + FBA ratios | 1,225 | 82,689 | 0.837 | 0.002 | 0.837, 0.840, 0.834 | 68, 80, 71 |
| GCN | union of nine | ProtT5 + chrom-pathway + FBA ratios | 1,225 | 82,689 | 0.824 | 0.003 | 0.822, 0.822, 0.827 | 197, 197, 197 |
| GCN | union of nine | ProtT5 (400 epochs) | 1,024 | 69,825 | 0.831 | 0.002 | 0.828, 0.831, 0.833 | 397, 397, 396 |
| GCN | union of nine | ProtT5 + chrom-pathway + FBA ratios (400 epochs) | 1,225 | 82,689 | **0.843** | 0.002 | 0.841, 0.843, 0.846 | 395, 395, 395 |

# 6 Findings ✗

The two questions the document opened with have the following answers on the released Merzbacher 2025 split, every number at a validation-selected epoch on the 195 held-out genes with a gene bootstrap, and every gain attributed by a paired, pre-registered read.

**The held-out number is 0.893 [0.84, 0.94].** A width-12 cell graph transformer reading the chromosome-pathway indicator and ProtT5, with FBA's four growth ratios on its logit, under Flux Cone Learning's own five-fold ensemble protocol; 0.15 above Flux Cone Learning's 0.742 on the same genes and 0.06 above the classical ladder's best rung. Its per-cell mean over seven seeds is 0.869.

**The inputs are the number.** The gain over the backbone alone (0.774) is the four solver ratios (to 0.821) and the pathway multi-hot beside the protein embedding (to 0.871); the five gene properties in the same encoding,

expression included, add nothing, and neither do width, depth, the preprocessor, ESM2, codon frequency or the species-LM composite. A two-layer MLP on the same three inputs reads 0.870 per cell in six seconds and 0.886 as a fold ensemble; graph convolution over the nine gene graphs reads 0.888. Three architectures are one number at this resolution.

**The pathway multi-hot is the indicator.** Split into its blocks, the 197-column encoding's gain is the 173 pathway columns: with ProtT5 and the ratios they read 0.880 on the MLP and 0.890 on the transformer, above the full encoding (0.870 and 0.871), while the chromosome one-hot (0.809, 0.822) and the seven properties (0.798, 0.824) sit at the level of ProtT5 with the ratios. On the transformer the pathway columns need the sequence embedding beside them (0.766 alone); on the MLP they do not (0.828). The pooled transformer on ProtT5 with the pathway multi-hot and no readout reads 0.860. The solve does not substitute for the annotation: the deletion's flux vector on ProtT5 alone reads 0.868 pooled, the same vector with the pathway block 0.912, and the block with the ratios alone 0.890 (Section 4.28).

**Keeping the edge type is the one architectural choice that registers.** Of the message-passing forms of MATH-490 and Gene_Graph on the transformer's three inputs, relational attention over the nine graphs as nine edge types reads 0.889, GraphSAGE 0.887, relational convolution 0.860, graph convolution over the untyped union 0.847 and the graph isomorphism network 0.705, against the MLP's 0.870. The 1-WL-expressive aggregator is the weakest, and the typed-edge attention is level with the transformer's best single cell.

**GO through the graph is the highest number in the document.** Two layers of graph convolution over the union of the nine gene graphs on the propagated GO multi-hot alone, no sequence and no solve, read 0.912, 0.910 and 0.904 at 400 epochs, 43 seconds per seed, and under the reported five-fold protocol 0.917 [0.87, 0.96] against the transformer's 0.893 [0.84, 0.94], the folds above on four of five and the same 0.917 with three seeds per fold; the curve is flat past 400 epochs, the sequence embedding and the ratios add nothing to it, and on six seeds it reads 0.898 to 0.912 (mean 0.905, sd 0.005) with no run that misses. On the ladder GO is the strongest single encoding (0.808 linear, 0.844 with a random forest and the ratios). The transformer is the one model that does not take it: 0.833 beside ProtT5 and a cost of 0.04 beside the pathway block.

**The genes every model misses are the ones the solve misses.** Thirteen of the fifteen hardest essential held-out genes have a knockout growth ratio of 1.0 (CDC19 covered by PYK2, PET9 by AAC1 and AAC3), and the hardest nonessential genes are metabolic paralogs (TDH1, TDH2, TDH3, PFK1, PFK2) the annotation vectors make look essential; the seven transformer arms agree on their failures at a Spearman of 0.4 to 0.9, and the GO convolution fails elsewhere (0.15 to 0.4 with them), halving the misrank on FRQ1, AUR1, PGA3, LCB1 and LCB2 and repairing twelve of the misranked paralogs. Of the 19 essential held-out genes the solve calls viable, four are covered by an alternative gene set on every reaction and fifteen have a reaction of their own that the growth program does not need, so what the solve misses is mostly a requirement outside the biomass objective and only partly isozyme cover, and the two model families miss different parts of it (Section 5.1). Averaging their existing predictions with equal weight and nothing fitted scores 0.948 [0.91, 0.98] on the split-file cells (the six-seed GO convolution with the six-seed layer-free form, from 0.905 and 0.910 alone; 0.946 with the scalable form) and 0.954 [0.92, 0.98] under the fold protocol, from families at 0.917 and 0.922 alone (the GO convolution's fold models with the scalable form's): the maximum of the fifteen and eleven post hoc averages tried, reported as such; one model over both inputs, the solve's columns beside GO under the convolution, reads the GO-alone 0.909 and recovers none of it, GO on the node beside the solve in the transformer's affine reads 0.866, the round-26 cost, and GO as affine columns beside the solve reads 0.901, the solve-alone number, memorizing the training genes by epoch 20, and 0.906 reduced to 64 components, fitting on time; the two families stay two models (Section 5.3).

**Off the metabolic genes, only the graph transfers.** On the 4,674 genome-wide genes the released split never labeled (974 essential by SGD, 3,700 nonessential by a fitness record, none with a GEM reaction, so FBA reads exactly 0.5 and label origin coincides with class), models trained on the 629 released genes read: the MLP on the transformer's inputs 0.509, the transformer's reported recipe 0.572, the scalable module form 0.551 and the layer-free form 0.570 (the solve is one constant row on every one of them, the wild-type flux for the scalable form and an all-zero row for the layer-free form, whose store covers the labeled genes only, and the scalable form's six seeds span 0.480 to 0.579), graph convolution on ProtT5 alone 0.749, on the three inputs 0.744, on GO alone 0.702, degree and clustering alone through the physical graph 0.753, and the raw degree in the physical graph, nothing trained, 0.740 (0.594 on the released 195). The GO convolution's margin is a property of the metabolic genes; off them the number is the gene's degree in the protein interaction graph, a genome-wide signal the metabolic genes barely vary in, which any input through the graph reaches and no input without it does. The other way, the genome-wide labels as training data leave the GO convolution at 0.913 on the 195 and cost the three-input models 0.03 to 0.05 (Section 5.4).

**Structure is not the signal.** The 1-WL test separates 1,118 of 1,121 single-deletion graphs on the physical graph and all of them on the union, so expressivity does not bound a message-passing model here, and the WL kernel, all-ones and degree inputs all rank at chance. What is written on the nodes carries the number.

**Accuracy is the wrong statistic.** yeast-GEM's own benchmark reports 0.902 on a gene set whose majority class alone gives 0.856, with recall 0.41; FBA on 9.0.2 reproduces it at 0.907, the released labels agree with the Stanford list on 1,102 of 1,104 shared genes, and the transformer's balanced accuracy at the loss's decision boundary is 0.78 against 0.63 to 0.72 for the solvers.

**The solve carries the module's value, and the alternating layer cannot compute it.** Routed through the module's growth column the prediction never beats the backbone alone. The layer's alternating projections never reach the polytope the solve optimizes over: with the balance exact the box is violated on 58 percent of reactions, the second law on half, the protein pool by up to seven times, the dissipation limit by three; the growth column is a free signed coordinate the readout learns to use with either sign; the $k_{cat}$ content is unread (the shuffled table matches); and handed the solve's own point, the anchored constraint set discards it.

**The enzyme-capacity box as parameterized contains no growing flux.** A linear program under the layer's bounds is infeasible for every genotype, because the predicted $k_{cat}$ tier times the nominal abundances caps the median reaction at 0.084 mmol/gDW/h. Every anchored arm in this document read a flux off the polytope for that reason. Scaled by 30 or more the box admits growth; the ceilings still bind up to a scale near 3,000.

**On the polytope, the module helps.** On the GEM's own box the solve survives the projections and the program that solves is feasible and ranks the held-out genes at 0.73 before training, above FBA's 0.66; the gated module cell reaches the level of the solve appended as an input, 0.816 by warm start and 0.824 by the layer that solves, three of three seeds above the round-4 gated cell, with the growth column entering through a positive gate. On six seeds the two gated forms read 0.814 (GEM box) and 0.804 (elastic program, feasible by construction), twelve seeds of twelve above the gated cell without the solve and 0.01 to 0.02 under the affine.

**The ceilings cannot carry the ranking, so the forced growth column cannot be trained.** With the program elastic, feasible for every genotype and its gradient arriving on nine genotypes in ten, the forced column still reads 0.70 to 0.74 and its training loss never leaves 1.25, under ten times the learning rate, a deeper head, a wider reach and two ceiling scales. The dual-overlap probe measures why: the same six gene-owned ceilings bind whichever gene is deleted (median Jaccard 1.0), and the deleted gene's own reactions never bind because the deletion has closed them, so the head can move one growth level for every genotype and cannot move one genotype against another.

**The solve's flux vector, read by a linear head, is the module's number.** The gated elastic cell with a zero-initialized linear head over $\log(1 + |v^*|)$ of the solved flux reads 0.883, 0.853 and 0.868 on the ProtT5 backbone, the first module cell to pass a pre-registered read, 0.047 above the FBA affine on that backbone and with the growth column's gate ending near zero. Over the 274 exchange fluxes alone it reads 0.835. On ProtT5 with the pathway multi-hot the same cell reads 0.897 and 0.903 on two seeds, the two highest single cells in the document and above the affine on that input (0.890), and 0.716 on a third seed whose training loss never fell. The vector carries it: with nothing learned in the layer the same head reads 0.852, the ceilings add 0.01 to 0.02, and the gene-owned columns hold it (0.861 against 0.835 for the exchanges). Under the fold-ensemble protocol on the pathway input it reads 0.897 [0.84, 0.95], one number with the reported 0.893, from three fold models at 0.90 and two that never fit; three of this form's seventeen cells miss the fitting transition within 200 epochs, and the miss is a chance event of the run and not of the seed: the unfit seed rerun under the same program fits at epoch 70 and reads 0.889, the two runs diverging from their first epoch (Sections 4.20, 4.21 and 4.23). The chance is the card's kernels: with cuDNN's autotuner off, deterministic kernels on and cuBLAS's workspace fixed, two runs of that seed are identical to four decimals at every epoch, both fit at epoch 86 and read 0.894, where two free runs of the same seed read 0.920 and 0.782, at a five percent cost in wall time; pinned, six seeds fit six times and read 0.894 to 0.913 (Section 4.24). The form that scales, the program solved once per genotype with nothing learned in the layer and its flux read through a zero-initialized head, reads 0.897 to 0.917 on eight of nine runs over six seeds (pooled 0.917 [0.87, 0.96] on the six-seed rerun, above the elastic form's 0.910), and misses the transition on the ninth, a seed it had fit in the previous attempt, so the chance is not in the ceilings; under the fold protocol it reads 0.912 [0.86, 0.95] with all five folds fit, one number with the reported 0.893 (Section 4.22). Metabolism enters the cell as a perturbation-conditional transform that holds mass balance and the bounds exactly, costs nothing on the card, and cannot hurt at initialization. With no layer in the graph at all, the deletion's pFBA vector precomputed and appended to the affine's input (round 31), the cell reads 0.871 pooled alone and 0.901 with the solve statistics beside it, at 31 minutes per cell against 83, and four of the six vector cells select an epoch before the backbone's fitting transition over a default-initialized 4,135-column affine, where the two that select after it read 0.923 and 0.911 (Section 4.26). The encoding is the whole of that gap: the same vector as $\log(1 + |v|)$ reads 0.896 to 0.919 on twelve of twelve runs (0.912 pooled; zero initialization changes nothing, standardized columns zero-initialized still peak early), the cached-solve head's number with no layer, no solver in training, no missed transition and the lowest spread in the document, at 11 minutes alone on a card, and 0.920 [0.87, 0.96] as a fold ensemble, the document's highest transformer ensemble, one number with the GO convolution's 0.917; the eight solve statistics beside it cost 0.02 on every seed, and a hidden layer over it reads no more than the linear affine (0.915 and 0.909 pooled at widths 64 and 256), nor does the cell's capacity from 25 to 125 thousand parameters

(width 48 reads 0.914 on six seeds), while at width 96 the cell misses the fitting transition on two seeds of three (Section 4.27); with the card's kernels pinned five more seeds read 0.899 to 0.909, though two pinned runs of this cell were not one trajectory as the LP cell's were, through the pathway multi-hot itself (every pinned pair with the pathway block on the node diverged from epoch 0, every pair without it, ProtT5 alone or GO and ProtT5, was one trajectory), an input rebuilt from a Python set at every launch and served in the column order the previous process wrote, not a kernel; with the dataset fixed both cells pinned are one trajectory and the free run of this cell is the pinned run on all six seeds, so the switch is needed only on the LP form (Section 4.25). On the node instead of the logit, as a gene embedding through the preprocessor, the same vector reads 0.875 (pooled 0.894) alone and 0.889 with the affine columns too, under the affine without any vector, and fits late; reduced to 64 or 16 principal components it fits on time and reads 0.894 and 0.901 pooled, the affine-alone level, and 0.909 with the full vector in the affine as well, the logit form's number and no more (Section 4.29).

**The solve's argmax is not unique, and the head reads the solver's point.** The round-27 program solved for every labeled gene by five solvers gives one growth value (to $10^{-5}$, FBA's 0.60 to 0.68 on the 195 whichever solver) and five flux vectors: another exact vertex moves 4 percent of the reactions in the head's encoding, a point inside the optimal face 13 to 16 percent. A probe fitted on the reference vertices reads 0.756 on them and 0.50 to 0.69 on the other solvers' vectors, at chance on the other exact vertex; fitted within each solver it reads 0.65 to 0.78, the interior point's 0.776 level with the vertex's. The store and the head must come from one solver, and a point inside the face, which is what a first-order solve on the card returns, ranks as well as a vertex; across tolerances of one such solver the point is the same point (PDLP at $10^{-4}$ and $10^{-6}$ move 0.6 percent of the reactions against each other and a probe fitted on either scores the other at its own number), so the solver and not the tolerance is what the store and the head must share (Section 4.31). Trained over such a store the layer-free cell reads 0.911 on six of six seeds, the pFBA store's number, and 0.912 as a three-seed fold ensemble against the pFBA store's 0.919; over the PDLP store, the card's algorithm on the CPU, 0.903 on six of six (pooled 0.907) and 0.911 as a three-seed fold ensemble, and 0.905 over the same solver's store at $10^{-6}$, where the $10^{-8}$ store with 320 of 1,121 programs uncertified and kept as zero rows reads 0.887 and 0.906 once those rows are filled from the looser solves; over the pFBA store solved on synthetic complete medium instead of YPD, 0.912, the same number; where the growth program's own dual-simplex vertex reads 0.840 with two early peaks (Section 4.30): the store can be built by the batched solver on the card, and not every point of the face is a good store.

**Where the gradient goes is not the problem.** On the alternating layer the gradient reaches the flux layer on every arm that lets it; the backbone's preprocessor, three quarters of the parameters, is where fitting continues after the validation curve peaks.

What carries forward to a model trained over every phenotype in the database is the part of this that did not depend on the 195 genes. Metabolism enters as a perturbation-conditional data transform: the genome-scale program solved once per genotype and medium on the GEM's own box, cached, and read through a zero-initialized head over $\log(1 + |v^*|)$, which cost nothing on the card, held mass balance and the bounds exactly, read 0.90 to 0.92 on eight runs of nine where the affine read 0.89, and cannot hurt at initialization; over the 13.5 million records of the solid-growth build that transform is 469,474 programs, an hour on 20 cores, read as a gather at the control's step rate (Section 4.36); the learned enzyme-constrained layer, whose ceilings added 0.01 to 0.02 at five times the wall time and whose forced growth column could not be trained, is a fine-tuning module and not a backbone component. Annotation vectors enter through message passing over the gene graphs and not through the transformer's token: the GO multi-hot read 0.917 through two layers of convolution and 0.833 through the transformer, and the pathway block's gain on the transformer was the one annotation it took. The gene graphs enter as a message-passing input and not as a regularizer, since off the metabolic genes the models without the graph as an input fell to chance and the ones with it held 0.75. What the 195 genes could not decide, which of three architectures on the same inputs is better, the genome-wide set decided for the graph; what neither set can decide is whether that transfer is essentiality or the deletion collection's coverage.

# 7 Open ✗

The held-out number is close to saturated: the round-12 ensemble's interval, 0.893 [0.84, 0.94], spans what 195 genes with 31 essential can resolve, and every lever on the transformer side has been moved (width, depth, the preprocessor, the budget, the class weight, six encodings). What remains is attribution and the module.

- The pathway multi-hot carries the round-9 gain (rounds 14 and 22, and the split on the cheap models); the deletion's own flux vector does not substitute for it (round 35: 0.868 on ProtT5 alone against 0.912 with the block, and the block without the vector 0.890), so the two carry different information. Which pathways, and what in the annotation the solve cannot see, is not measured.

- The round-16 cell (the round-11 affine on the round-9 backbone) and the round-9 cell are one cell to the resolution of 195 genes (rounds 16 and 17); a larger held-out set, such as the genome-wide labels scored on genes the released split never touched, is what would separate them.

- The MLP, the graph convolution and the transformer on the same three inputs are one number as fold ensembles (0.886, 0.888, 0.893), and graph convolution on GO alone is 0.917 above them with overlapping intervals. On the 4,674 genome-wide genes the released split never touched (round 30) the architectures separate: the graph convolution transfers at 0.75 and the GO margin does not survive (0.70), while the MLP and the transformer's recipe read 0.51 and 0.57. Open on that set: whether a model trained on the metabolic genes transfers because it reads essentiality or because it reads what the graph encodes about SGD-annotated against deletion-collection genes (label origin coincides with class there, and for a viable deletion the two are close to one definition). Degree and clustering alone reach the 0.75 on that set, so the question is what degree in the physical graph measures there; a genome-wide set whose nonessential labels come from the same source as the essential ones would separate origin from class, and the physical graph's own coverage of the deletion collection against the SGD inviable set is the first thing to count.

- The GO multi-hot is the ladder's strongest encoding and the cheap models take it (graph convolution 0.887), and the transformer does not (0.833 beside ProtT5, a cost of 0.04 beside the pathway block, round 26), and a cell 350 times wider reads it the same (round 28), so capacity is not the reason. What separates 173 sparse pathway columns from 2,362 dense GO columns in this backbone is not measured; a per-block projection, the GO vector entering as its own token, and the pooled readout itself are the untested constructions, and the answer decides whether annotation vectors scale as inputs to this backbone.

- The forced growth column is closed by the dual-overlap probe (Section 4.16): the ceilings that bind are the same for every genotype, so no step size, head or reach trains it (rounds 18 to 20). A construction that changes which bounds bind per genotype, such as ceilings the deleted gene's neighbors own, or an objective other than biomass, since the layer that solves accepts any objective column, is untested.

- The head over the solved flux is the module's number, and the form that scales (round 27: the program solved once per genotype, nothing learned in the layer, a zero-initialized head) reads 0.90 to 0.92 on eight runs of nine and misses the fitting transition on one, where the learned-ceiling form misses it on three cells in seventeen, and reads 0.912 [0.86, 0.95] under the fold protocol with all five folds fit. Open on it: what in the forward makes the transition a chance event of the run rather than of the seed (round 29: the same seed and program fit at epoch 70 in one run and never in another, the two diverging from the first epoch, and the scalable form repeats it with the solve cached and nothing learned in the layer); round 32 closed it: pinning the card's kernels makes two runs of the seed one trajectory, so the chance was the kernels, and pinned, six seeds fit six times, so the miss is a property of the free run and not of any seed; the pooled layer-free cell's two pinned runs are not one trajectory (round 45), with no operation warning under the deterministic mode, and round 46 places the divergence with the pathway multi-hot in the pooled arm (the ProtT5-alone layer-free pair is one trajectory, the pathway-and-ProtT5 affine pair with no solve columns is not, and the LP cell carries both embeddings under another arm and was one trajectory), and round 49 with the pathway embedding itself (the pathway multi-hot alone with the affine and pathway-and-ProtT5 with the plain pooled head both diverge pinned, GO-and-ProtT5 with the affine is one trajectory); the pathway dataset was rebuilt at every launch and enumerated its 173 columns from a Python set, so each process served a column permutation the previous one wrote, and with the enumeration sorted and the rebuild removed both cells pinned are one trajectory (rounds 51 and 52) and the free run is the pinned run on every seed (round 64), so the chance in the pooled cell is closed, it was an input and not a kernel, and the pinned-kernel switch is needed only on the LP form; the cache's hit rate and where the wall time goes when the solve is cached, since six cached cells on one card ran no faster per cell than three uncached; and the multi-phenotype form, the flux in the cell representation rather than on the logit: a hidden layer over the vector reads no more than the linear affine (round 41), the raw vector as a node embedding through the preprocessor reads 0.03 under the logit form and under no vector at all (round 43), and reduced to 64 or 16 principal components it fits on time and reads the affine-alone number on the node and the logit form's number only when the vector is in the affine as well (round 44). So the representation path carries the vector without cost once it is compact and reads nothing from it that the logit does not; what is open is a head that reads a representation-borne vector, the deleted gene's own token rather than the pooled context, and the vector as its own token.

- The two model families, annotation through the graph and the solve through the transformer, miss different genes, and their post hoc average reads 0.954 under the fold protocol (the GO convolution with the scalable form). The one model tried, the solve's columns on the node beside GO under the convolution, reads the GO-alone number (0.909), so the average's gain is not the graph model's to take from those columns, and the transformer with GO on the node beside the solve in the affine reads 0.866, the round-26 cost with the solve as with the ratios (round 47); GO on the logit through the affine, the path the solve takes, costs nothing and adds nothing (0.901 with the vector, round 50), the affine memorizing the training genes through 2,362 binary columns by epoch 20, and neither weight decay (round 59) nor reducing GO to 64 components before the affine (round 60: 0.906, the

solve-alone number, fitting on time) recovers any of the average; so on this backbone the two families stay two models, and what is untested is a learned weighting of their logits and the transformer's affine read of the solve beside a graph-convolved GO token, while the average's number carries the selection of the best of eleven.

- The layer that solves has the wrong shape for a training loop: the program runs on the CPU inside the forward pass, the GPU waits on it (0 to 13 percent utilization sampled during round 19, against a 1.6 hour cell where the alternating layer takes 19 minutes), and the bytes moved are not the cost, the synchronization is. With nothing learned in the solve (round 27) the cost leaves the loop entirely: the program is solved once per distinct genotype and medium, offline, and stored as a feature, and no solver runs in training. Round 31 measured that form: the deletion's pFBA vector precomputed and appended to the affine reads 0.871 pooled alone and 0.901 with the solve statistics at 31 minutes per cell against 83 for the solve inside the graph, under the cached-solve head's 0.917, with four of six vector cells selecting an epoch before the fitting transition; round 33 closed the 0.05 with the encoding alone: as $\log(1 + |v|)$ the vector reads 0.896 to 0.919 on twelve of twelve runs, 0.912 pooled, with nothing in the graph and 11 minutes a cell, and 0.920 as a fold ensemble. What remains open on the form is why standardized columns make the backbone peak before it fits (and why one $\log(1+|v|)$ fold cell in seventeen still did). The count that sizes the store is measured (Section 4.36): over the 13.5 million records of the solid-growth build, 469,474 distinct (GEM subset, medium) keys, solved in 71 minutes on 20 cores, and read at the control's step rate by the transformer trained over that build, whose number is not yet read. On this endpoint the store solved on synthetic complete medium reads the YPD store's number (round 42), and the build's selection media have no FBA recipe, so whether the medium axis of the store carries anything is a question for a phenotype scored across media with recipes. Where a solve must stay in the loop, the fine-tuning form with learned ceilings, three constructions would remove the synchronization, in order of how much they change: a batched first-order solve on the card, which NVIDIA's cuOpt offers as PDLP with a batch mode for many small programs at a benchmark tolerance of $10^{-4}$ relative; a per-genotype basis warm start of the simplex through HiGHS's own interface, which keeps the exact solve and the exact duals; a smooth interior-point solve at a fixed barrier, which changes what the layer computes and belongs to its own rounds. The read that had to come first is in (Section 4.31): the program's argmax is degenerate, another exact vertex already breaks a head trained on the reference vertex (the probe at chance), and a point inside the optimal face, PDLP's or the interior point's, is as rankable as a vertex when the head is trained on it, and rounds 36 and 38 trained the cell over an interior-point store to 0.911 and over the PDLP store, the card's algorithm, to 0.907, the pFBA store's number, where the growth program's own vertex trained to 0.840. So the solver is part of the module's definition, the store and the head must be built from one solver, the card's first-order point is a store, and across tolerances of one first-order solver the point is the same point: PDLP at $10^{-4}$ and $10^{-6}$ move 0.6 percent of the reactions against each other and a probe fitted on either scores the other at its own within-solver number (0.709 and 0.717 against 0.716 and 0.702), where across solvers the same probe loses 0.05 to 0.35 (Section 4.31); the cell trains to the same number over the $10^{-6}$ store (round 53), and the $10^{-8}$ store with 320 programs uncertified reads 0.02 under it until those rows are filled from the looser solves (rounds 55 and 57), so the store's cost is its uncertified fraction and a second pass recovers it. Why the dual simplex's vertex is the worst store of the four is answered by an intervention rather than by an association (Section 4.34): zeroing every column that reaches the clamp, where a loop can spin, lifts that store from 0.840 to 0.888 on six of six seeds, while the same rule on the pFBA store, where it cuts one column, moves it by $-0.003$. What stays open is the last 0.020 to the pFBA store, and whether a store solved so that no column reaches the clamp beats one cut after the fact.

- Of the module's own parameter sources, the $k_{cat}$ tier is measured at initialization and the thermodynamic route is measured to be inert: with the constraint scale at zero and the sign constraint off, swapping the route leaves the program identical at every capacity scale, so it is provenance on these rounds and not a hyperparameter. The tier sets how tight the capacity box is before any scaling: the measured tier leaves 2,849 of 3,728 catalytic units on the organism default, whose turnover is above the predictors' median, so its median ceiling is twice the predicted tier's and its program admits growth at a capacity scale of 3 where the predicted tier admits none; at the scale round 15c ran, the measured tier's ceilings are within 0.003 mmol/gDW/h of not binding (Section 4.35). Trained at the scale where both tiers bind on six seeds, the looser box trades a failure for a ceiling: the gated cell gains +0.012, about its seed spread, and the forced growth column stops collapsing into the absorbing infeasible state, none of six against two of six, while no longer reaching above 0.76 where the tighter box reaches 0.84 and 0.80. Round 72 closes that: the bimodality cannot be removed without the high end going with it, by either construction that removes it. On twelve seeds the hard program collapses on two and clears 0.80 on four (mean 0.699, spread 0.115), while the looser tier and the elastic program each put six of six between 0.63 and 0.76 (means 0.707 and 0.715, spreads near 0.045). Three constructions share one mean and differ only in spread, so the collapse buys a tail rather than accuracy. What is open is a construction that changes which bounds bind per genotype, since every arm here binds the same ceilings for every genotype (Section 4.16).

## 7.1 What the closing rounds settled ✗

Four of the questions above were answered after this section was written, and the answers are in Sections 8 and 9.

A rule-aware input exists and helps, but not where it was expected: six counts read off the gene-protein-reaction rules raise the released split from 0.908 to 0.918 and improve the fifteen essential genes that hold a reaction of their own, while making the four an alternative gene set covers worse (Section 8.3); the six columns are redundant among themselves (Section 8.10).

The deleted gene's own token was tried. Routing the class token through the perturbation operator gives the readout a strain-specific token beside the pooled one at no extra parameters, and on single deletions it is a small consistent cost, not a gain (Section 8.4).

The medium axis of the store does carry something on this endpoint, and only where the flux vector is absent. With the vector present one medium scores what four do (+0.002 over ten repeated splits); with the vector gone the growth ratios of four media are worth +0.012 over one, and two media recover about a third of that (Section 8.10).

The averaging of the two families was read off the released 195 genes, where 0.955 is the maximum of thirteen post hoc averages. Fixed in advance on ten other held-out fifths, the equal-weight average still beats the better single model on nine of ten (Section 8.1), so the margin is not the selection.

What remains open is unchanged: a learned weighting of the two families' logits, a construction whose binding ceilings differ per genotype, and the store's value on a second phenotype, which belongs to that phenotype's experiment rather than to this one.

# 8 The closing rounds: repeated splits, the rules, and what each simplification costs ✗

The released held-out set has 31 essential genes, so every conclusion drawn on it alone is a conclusion about 31 genes. The rounds in this section reread the finalists and the module's controls on ten other held-out fifths of the labeled genes, add the gene-protein-reaction rules beside the solve, and price each way of making the module cheaper or of fitting it inside one experiment per record.

## 8.1 Round 75: the finalists on ten repeated outer splits ✗

The released held-out set has 31 essential genes, the finalists' intervals on it overlap, and the average's 0.954 is the maximum of eleven averages chosen on those genes. Round 75 (jobs 2547 to 2567; Table 96) reads the same models on ten other held-out sets, each a class-stratified fifth of the released labeled genes with validation a stratified fifth of the rest (`build_essentiality_splits.py -resplit`), three seeds per model per split. The models, the equal-weight average and the statistics were committed before any cell finished (`resplit_readout.py`).

**Table 96:** Round 75, the finalists on ten repeated outer splits. Each test block is a class-stratified fifth of the released labeled genes (198 genes, 32 essential), validation a stratified fifth of the rest. Left: the AUROC of the three-seed family score (standardized logits averaged over seeds) of the transformer with the layer-free solve, of graph convolution on GO at 400 epochs, and of their equal-weight average, fixed before any cell finished; best of the three in bold. Right: the seed-mean AUROC per cell of the layer-free cell and its two controls on the same backbone. Below: the paired differences across the ten splits. The test blocks overlap, so the differences are not independent and no $p$-value is given. Written by `resplit_readout.py`.

| split | layer-free | GO conv. | average | layer-free, cells | FBA ratios, cells | no metabolism, cells |
|---|---|---|---|---|---|---|
| 0 | 0.864 | 0.884 | **0.901** | 0.863 | 0.864 | 0.840 |
| 1 | 0.844 | 0.884 | **0.899** | 0.843 | 0.803 | 0.794 |
| 2 | 0.857 | 0.872 | **0.893** | 0.854 | 0.844 | 0.854 |
| 3 | 0.850 | 0.857 | **0.895** | 0.849 | 0.835 | 0.774 |
| 4 | 0.826 | **0.868** | 0.868 | 0.824 | 0.843 | 0.773 |
| 5 | 0.857 | 0.868 | **0.892** | 0.849 | 0.830 | 0.741 |
| 6 | 0.857 | 0.889 | **0.911** | 0.842 | 0.835 | 0.814 |
| 7 | 0.786 | 0.838 | **0.844** | 0.783 | 0.737 | 0.729 |
| 8 | 0.904 | 0.872 | **0.936** | 0.899 | 0.878 | 0.826 |
| 9 | 0.831 | 0.860 | **0.918** | 0.824 | 0.853 | 0.821 |
| mean | 0.848 | 0.869 | 0.896 | 0.843 | 0.832 | 0.796 |

| | |
|---|---|
| average minus layer-free | +0.048 (sd 0.017), positive on 10 of 10 |
| average minus GO convolution | +0.026 (sd 0.021), positive on 9 of 10 |
| layer-free minus GO convolution | −0.022 (sd 0.024), positive on 1 of 10 |
| average minus the better single model | +0.023 (sd 0.017), positive on 9 of 10 |
| layer-free minus FBA ratio affine (seed means) | +0.011 (sd 0.023), positive on 7 of 10 |
| layer-free minus no metabolism (seed means) | +0.047 (sd 0.034), positive on 10 of 10 |
| layer-free minus FBA ratio affine (six-seed means) | +0.010 (sd 0.024), positive on 8 of 10 |

The average is the best method and the margin is its own, not a selection. It reads 0.896 as the mean of ten splits against 0.869 for graph convolution on GO and 0.848 for the transformer with the layer-free solve; it is above the transformer on ten of ten (+0.048, sd 0.017), above the graph model on nine (+0.026, sd 0.021), and above the better of the two on nine (+0.023, sd 0.017). The two single models are not the tie the released genes showed: graph convolution on GO is above the transformer on nine of ten (−0.022, sd 0.024 for the transformer minus the graph model).

With the single-experiment form of Section 8.10 as the transformer family (the YPD ratio, the flux vector and the rule counts, six seeds per split, against graph convolution on GO at six), the same rule reads 0.907 for the average against 0.871 for the graph model and 0.869 for the transformer. The average is above the better of the two on ten of ten splits (+0.029, sd 0.013), and the two single models tie (−0.003, the transformer higher on four of ten). On the released genes its five-fold ensemble reads 0.929 alone and 0.953 [0.92, 0.98] averaged with the graph family's, against 0.955 for the four-media recipe; that is one more post hoc average on those genes and it is not the maximum. The form that goal 2 keeps is therefore also the better partner for goal 1.

Every model reads lower on these fifths than on the released held-out genes, the transformer's cells 0.843 against 0.908 and the graph model's 0.868 against 0.905 as seed means, so the released held-out set ranks more easily than a random fifth of the labeled genes. What makes it so is not measured.

The module's contribution splits in two on the same backbone. The solve as an input is worth +0.047 over no metabolism, on ten of ten splits (sd 0.034). Within that, the deletion's flux vector over FBA's four growth ratios alone is +0.011 (sd 0.023, seven of ten), where the released split read +0.018 (0.908 against 0.890): the sign agrees and the difference does not separate from zero. Doubling the seeds does not separate it either (round 78, jobs 2586 to 2595): at six seeds per arm per split the difference is +0.010 (sd 0.025, eight of ten), the same mean with the same spread, so the power was not what the three-seed read lacked. So what the ten splits support is that the solve helps and that the four ratios carry most of it, while the 4,131 flux columns add little beyond them; six counts of gene-protein-reaction redundancy, on the same splits, add about twice as much (+0.019, Section 8.3).

## 8.2 Round 76: the physical graph's degree as an input to the transformer ✗

Off the metabolic genes the raw degree in the physical graph ranks the 4,674 genome-wide genes at 0.740 with nothing trained, and the transformer with the layer-free solve read 0.570. Round 76 (jobs 2568 to 2573, six seeds per arm; Table 92) gives the transformer that degree the cheapest way, as two more affine columns beside the flux vector, the gene's degree and clustering coefficient in the physical graph, over a store that covers all 6,607 genes (`degree_affine_store.py`; a gene with no deletion vector reads the wild-type vector instead of the all-zero row the earlier store gave it).

The control changes nothing, as a constant row should: with wild-type rows and no degree the six cells read the earlier arm's numbers to four decimals (mean 0.566, pooled 0.570). With the two columns the cells read 0.564 to 0.656 (mean 0.602, sd 0.030; pooled 0.593 [0.57, 0.61]), above the control on six of six seeds (+0.036), and 0.16 under the graph convolution's 0.755. On the released split the same store reads 0.906 (sd 0.009) against 0.908, so the columns cost

nothing there. The degree enters, and an affine fitted on 629 metabolic genes takes a fraction of it: among the 787 released training and validation genes degree ranks essentiality at 0.59, so the slope the affine can learn from them is small. Hypothesis, untested: the gap to the convolution is the training set and not the path, and a convolution trained on the same metabolic genes reaches 0.755 because its two layers read the neighbors' degrees as well as the gene's own.

## 8.3 Round 77: the gene-protein-reaction rules as affine columns ✗

Section 5.1 named what the solve cannot mark: the 19 essential held-out genes whose knockout growth ratio is 1.0, four of them covered by an alternative gene set on every reaction and fifteen holding a reaction of their own that the biomass objective does not need. The flux vector is close to the wild type's on all nineteen either way, so round 77 (jobs 2574 to 2585; Table 97) appends what the vector does not carry: six counts read off yeast-GEM's gene-protein-reaction rules in disjunctive normal form (`isozyme_affine_store.py`), the gene's reactions, how many its deletion alone disables, how many stay enabled, the minimum and mean number of alternative gene sets, and its largest complex. All are non-negative, so the affine's $\log(1 + |v|)$ keeps their order and the columns cost six weights.

They help, and by the smaller of the two mechanisms the section named. On the released split six seeds read 0.918 against 0.908 without them, above on six of six; on the ten repeated outer splits the seed-mean per split is 0.862 against 0.843, above on eight of ten (+0.019, sd 0.020). Over the 19 genes the mean misrank falls from 0.134 to 0.120.

The split inside those 19 is where the columns say what they are doing. The fifteen genes holding a reaction of their own improve on six seeds of six, from 0.124 to 0.089, while the four an alternative gene set covers get worse on six of six, from 0.171 to 0.234. The columns state what the deletion's flux vector cannot, that a gene holds a reaction nothing else catalyzes, because the program routes around those reactions without needing them; and where the rules say a gene is covered the columns push the cell further toward calling it dispensable, which is the wrong call on exactly those four. That pattern suggested the count of sole reactions was the one useful column; round 84 tested it and the six columns are redundant among themselves (Section 8.10). The hypothesis Section 5.1 raised, that a paralog-aware input would move the covered genes, is refuted on this construction: it moves the others.

**Table 97:** Round 77, six columns of gene-protein-reaction redundancy appended to the deletion's flux vector in the affine (`isozyme_affine_store.py`: reactions, sole reactions, covered reactions, minimum and mean alternative gene sets, co-subunits), against the same cell without them. Rows one and two are held-out AUROC, six seeds on the released split and ten split means of three seeds each; rows three to five are the mean misrank of the essential held-out genes whose FBA knockout growth ratio is 1.0, for which a LOWER value is better and the sign count therefore counts the seeds made worse. Better of the two in bold. Written by `gpr_readout.py`.

| quantity | with | without | difference | sd | positive |
|---|---|---|---|---|---|
| released split, held-out AUROC | **0.918** | 0.908 | +0.0102 | 0.0053 | 6 of 6 |
| ten repeated outer splits, AUROC | **0.862** | 0.843 | +0.0192 | 0.0198 | 8 of 10 |
| misrank, the 19 essential genes FBA calls viable | **0.120** | 0.134 | -0.0139 | 0.0111 | 1 of 6 |
| of them, the 15 holding a reaction of their own | **0.089** | 0.124 | -0.0343 | 0.0133 | 0 of 6 |
| of them, the 4 an alternative gene set covers | 0.234 | **0.171** | +0.0628 | 0.0288 | 6 of 6 |

## 8.4 Round 79: the class token through the perturbation operator ✗

Every pooled cell in this document reads the genotype through a mean over the perturbed gene tokens, because the class token was sliced off BEFORE the perturbation operator and is therefore the encoding of the unperturbed graph, the same vector for every strain. A flag that sends it through the operator as one more query over the same deleted-gene keys existed on another branch, unlanded and absent from this worktree until it was ported on 2026-09-20; it adds no parameters and leaves the gene rows bit-identical. Round 79 (jobs 2597 to 2600; Table 98) is the paired read on the released split.

The token becomes strain-specific and the ranking gets slightly worse. Its across-strain spread ends at 0.52 against exactly 0 without the flag, so the mechanism is doing what it claims. On the seeds of round 79 alone neither arm separated from zero; round 81 (jobs 2603 to 2605) completed the pooled arm's pairing and took the layer-free arm to nine pairs, and at fifteen pairs the cost is established: the layer-free cell reads 0.905 against 0.911, a paired difference of −0.0054 (95 percent interval −0.0100 to −0.0009, two of nine above), the pooled cell 0.855 against 0.858 (−0.0028, one of six above), and the two together −0.0044 (interval −0.0075 to −0.0012, twelve of fifteen below). It is a small cost, about a third of this cell's seed spread, and it is consistent.

The construction explains it. A strain here deletes ONE gene, so the class token's query attends over exactly one key and the perturbed token is a function of the wild-type token and that gene's token, which the perturbed-gene pool already carries. The flag adds no information on a single deletion and widens the head's input, which is the shape of a small consistent cost rather than a gain. It has something to add only where a genotype perturbs several genes and

the pool's mean over them discards their arrangement, which is the setting it was written for and not this one. Here it stays off.

**Table 98:** Round 79, the class token sent through the perturbation operator as query row 0 over the same deleted-gene keys (`perturb_cls`, ported from the 025 branch), against the identical cell without it, paired by seed on the released split. No parameters are added and the gene rows are unchanged. The last column is the across-strain spread of the class token at the final epoch, which is exactly 0 without the flag because the token is then the encoding of the unperturbed graph. Written by `cls_readout.py`.

| arm | seeds | with | without | difference | sd | positive | token sd |
|---|---|---|---|---|---|---|---|
| layer-free solve vector in the affine | 9 | 0.905 | **0.911** | -0.0054 | 0.0059 | 2 of 9 | 0.52 |
| pooled cell, no solve and no affine | 6 | 0.855 | **0.858** | -0.0028 | 0.0054 | 1 of 6 | 0.60 |

## 8.5 Round 80: the rule counts under the reported protocol ✗

Round 77 read the gene-rule columns on single cells. Round 80 (jobs 2601 and 2602) runs the same cell on the five folds, so the document's best module form has a number on the protocol the reported 0.893 was computed under, and so the two-family average can be formed from it.

It is the highest single model in the document on both counts. The five fold models read 0.9335 [0.890, 0.968] as an ensemble (fold mean 0.9133), against the layer-free form's 0.920 and graph convolution on GO's 0.917, and 0.930 as a standardized family. Averaged with the GO convolution's fold models it reads 0.9550 [0.922, 0.982] (Table 86), a little above the 0.9538 of the cached-solve average and the 0.9532 of the layer-free one. That number is now the maximum of thirteen post hoc averages rather than eleven, and carries the selection that implies; the three best are within 0.002 of one another and their intervals are nearly identical, so what the folds establish is that the two families average above either alone, not an ordering among the three module forms.

## 8.6 Round 82: what each half of the module is worth ✗

Round 77 added six gene-rule counts beside the 4,131-column flux vector and round 78 found the vector itself worth little over the four growth ratios, so the two halves have never been measured against the same baseline. Round 82 (jobs 2606 to 2609) runs the counts with NO flux vector, six seeds and the five folds, completing the factorial (Table 99).

Both halves help and the solve is the larger of the two. Against the four growth ratios alone the counts are worth +0.012 and the vector +0.024, paired on the three seeds all four arms share, and together they reach 0.918. The fold ensembles order the same way: 0.909 for the counts alone, 0.920 for the vector alone, 0.934 for both. The counts are worth more once the vector is already there than the vector is worth once the counts are (+0.020 against +0.008), which is what two partly overlapping signals look like.

This corrects a comparison made in passing after round 78. The counts' +0.019 there was measured ON TOP of the vector while the vector's +0.010 was measured on top of the ratios, so the two are different conditionals and cannot be read as a ranking. Against one baseline the solve is worth about twice the counts on the released split, and the ten repeated splits put them within each other's spread. What survives both readings is that each half is worth about a hundredth to two hundredths of AUROC, that they are complementary, and that the pair is the best form of the module.

Neither reading changes the practical conclusion. The counts cost no solver, no medium and no per-genotype computation, and alone they carry half of what the module contributes here; the solve carries the other half and needs one linear program per genotype and medium, run once offline.

**Table 99:** Round 82, what each half of the module is worth on the same backbone, the pathway multi-hot with ProtT5 and the four FBA growth ratios on the logit. The solved flux vector is 4,131 columns from one linear program per genotype; the gene-rule counts are six columns read off yeast-GEM's rules with no solver. Columns: the number of seeds, their mean held-out AUROC and its spread, and the five-fold ensemble where one was run. Best of each column in bold. Written by `notes_tex_tables.py` from the cells' own result files.

| affine columns | seeds | cell mean | sd | fold ensemble |
|---|---|---|---|---|
| four growth ratios only | 3 | 0.890 | 0.005 | – |
| + gene-rule counts, no solve | 6 | 0.902 | 0.005 | 0.909 |
| + solved flux vector, no counts | 9 | 0.911 | 0.007 | 0.920 |
| + both | 9 | **0.920** | 0.006 | **0.934** |

## 8.7 Round 83: every column inside one experiment ✗

A record in the database is one genotype in one environment, and the four FBA growth ratios are four different media, so a cell carrying all four reads four experiments at once and cannot be written as one record. The other metabolic columns are already admissible: the solved flux vector is a function of genotype and medium and this store is YPD, and the gene-rule counts carry no environment at all. Round 83 (jobs 2610 to 2615, six seeds per arm) keeps the YPD ratio, drops the other three, and reads the same three recipes (Table 100).

The restriction is free wherever the solve is present. The full recipe reads 0.916 against 0.918 at the round's six seeds (the table's row carries the nine seeds of round 107, 0.912 against 0.920, Section 8.10), the vector without the counts 0.912 against 0.911, and only the arm with nothing else loses anything: the ratio alone reads 0.877 against 0.890. So about a hundredth of what the four ratios contributed was the model reading four environments, and the deletion's own flux vector supplies it from within one.

That settles the question the endpoint could not answer on its own. Nothing in the module as it now stands needs more than one experiment per record, so the form carries to a database where the environment is part of the key rather than part of the feature vector.

**Table 100:** Round 83, every affine column restricted to what ONE experiment defines. The four FBA growth ratios are four different media, so a cell carrying all four reads four experiments at once, which a genotype-by-environment record cannot; the solved flux vector is already one medium (YPD) and the gene-rule counts carry no environment at all. The right block keeps the YPD ratio and drops the other three. Seed means of the held-out AUROC, better of each row in bold. Written by `notes_tex_tables.py` from the cells' own result files.

| affine columns | seeds | four media | seeds | YPD only | change |
|---|---|---|---|---|---|
| growth ratio alone | 3 | **0.890** | 6 | 0.877 | -0.012 |
| + solved flux vector | 9 | 0.911 | 6 | **0.912** | +0.001 |
| + vector and rule counts | 9 | **0.920** | 9 | 0.912 | -0.009 |

## 8.8 Round 108: DCell's ontology hierarchy over these splits ✗

Graph convolution on the Gene Ontology multi-hot is the strongest single encoding here, which raises the question of whether the ontology's own structure reads it better than message passing does. DCell answers that structure directly: one small subsystem per GO term, wired child to parent along the ontology, each term's state a function of its children's states and of its own gene input, with a linear head on every term at weight 0.3 beside the root head.

As published it cannot be run on this endpoint. DCell gives every gene its own weight into every term it belongs to, and a held-out gene's deletion never appears in training, so those weights keep their initialization; measured, that arm reads 0.578 over six seeds and 0.651 as a fold ensemble, and it loses to the graph on ten of ten repeated splits (Table 101). Replacing the per-gene weights with one shared weight on whether the deleted gene carries the term makes the same hierarchy able to rank unseen genes, and it learns: 0.825 over six seeds, 0.848 as a fold ensemble. The auxiliary per-term losses are worth +0.026 on the released split.

It still loses to message passing on the same annotations, by 0.067 over the ten repeated splits, below on ten of ten. The hierarchy was sized on validation over ten configurations, and the width the data supports is 8, well under the published rule of $\max(20, 0.3n)$ genes per term, whose uncapped form would give a widest subsystem of 1,651 units and 74.9 million effective parameters against 629 training genes; capped at 32 it is the worst configuration tried. So the ontology's grouping is not what carries the GO signal on this endpoint, and the same annotations read through the gene graphs carry it further.

**The ontology of the trigenic DCell runs.** The DCell runs of experiments 005 and 006 read a different Gene Ontology from the 2,362-term multi-hot above: SGD's annotations with those inferred from genetic interaction removed, redundant terms collapsed, and terms kept at four or more genes, which leaves 2,655 subsystems. Built here as a second store (`go006_store.py`, jobs 2688 and 2689), it keeps 2,651 terms as columns, of which 1,421 are among the current 2,362; a labeled gene lights 34.5 of its columns on average against 46.5 in the current store, and the hierarchy is flatter, 14 strata with a widest fan-in of 106 children against 15 and 24. Graph convolution does not care which: on the ten repeated splits at three seeds it reads −0.002 on the trigenic runs' ontology (higher on four of ten) and 0.905 against 0.909 on the released split. The hierarchy does: the shared-input DCell reads 0.714 against 0.841 on the released split over the same three seeds and −0.020 over the ten splits (higher on two of ten, sd 0.041). Those reads used the configuration tuned on the current store; tuned on the trigenic store over the same twelve configurations (job 2694), validation prefers width 8 with normalization before the nonlinearity (0.876 against 0.873 for the current store's best), and at that configuration the released split reads 0.709 over the same three seeds (job 2700), no better than the first read; the ten-split read at it is job 2701. The current store is therefore the better of the two for the hierarchy, and neither ontology brings it to the graph's level.

**Table 101:** Round 108, DCell's ontology hierarchy on this endpoint. Every term of the Gene Ontology gets a small subsystem whose state is a function of its children's states and its own gene input, wired along the same 2,362-term multi-hot the graph baselines read, with DCell's auxiliary per-term heads at weight 0.3. The published input gives each gene its own weight into each term, which a held-out gene's deletion never trains, so the first two rows replace it with one shared weight on whether the deleted gene carries the term. Columns: seeds on the released split, their mean held-out AUROC and its spread, the five-fold ensemble on the same 195 genes, and the paired difference against graph convolution on the same annotations over ten repeated outer splits with the count of splits the row wins. Test blocks overlap across splits, so no p-value. Width and normalization were chosen on validation over ten configurations. Written by `notes_tex_tables.py` from `dcell_style_baseline.py`.

| model | seeds | cell mean | sd | fold ensemble | against the graph, ten splits |
|---|---|---|---|---|---|
| DCell hierarchy, GO membership on each term | 6 | **0.825** | 0.026 | **0.848** | −0.067 (0.037), 0/10 |
| the same, no auxiliary term losses | 6 | 0.799 | 0.037 | 0.844 | – |
| DCell's published per-gene input | 6 | 0.578 | 0.057 | 0.651 | −0.294 (0.034), 0/10 |
| graph convolution on the same GO | 6 | 0.905 | 0.005 | 0.918 | 0.868 over ten splits |

## 8.9 Rounds 109 and 110: the module on a backbone with no pathway annotation ✗

Every transformer cell from round 75 onward reads the curated pathway multi-hot beside ProtT5, and a dataset outside yeast may carry no such annotation. Rounds 109 and 110 (jobs 2685, 2686 and 2692) drop it: ProtT5 alone with the single-experiment module, the YPD growth ratio, the YPD flux vector and the six gene-rule counts as the affine, against ProtT5 alone with nothing on the logit. On the released split at six seeds the module takes the cell from 0.771 to 0.883, +0.112 paired by seed and higher on six of six, the largest gain of any pair in this document; the same pair on the pathway backbone reads +0.058 (0.858 to 0.916). With the module the annotation-free backbone sits 0.033 under the pathway backbone, without it 0.087 under, so the module recovers about two thirds of what the pathway annotation supplies. The fold protocol (round 111, job 2699, one seed per fold) reads 0.892 [0.83, 0.94] for the annotation-free backbone with the module against 0.795 [0.71, 0.88] without, beside 0.932 and 0.878 for the pathway backbone. The ten repeated splits (round 109, three seeds per split) agree: the module reads +0.102 over no metabolism, higher on ten of ten splits and on 30 of 30 split-and-seed pairs, against +0.061 for the same pair on the pathway backbone; with the module the annotation-free backbone sits −0.027 under the pathway backbone (higher on one of ten), without it −0.063 (one of ten).

## 8.10 Rounds 84 to 98: what each simplification costs ✗

Three simplifications make the module cheaper or easier to carry to another dataset: reading one medium instead of four, so that a record stays one genotype in one environment; dropping the stored flux vector, so that a record carries one growth ratio per medium and no 4,131-column row; and dropping rule-count columns. A growth ratio is itself one FBA program per genotype and medium, so dropping the vector saves the second program of the parsimonious solve and the storage, not the solver. Rounds 84 to 98 (jobs 2616 to 2667, job 2622 aside) measure each against the same baselines on the released split (Table 102) and, paired by split, on the ten repeated outer splits of Section 8.1 (Table 103). The repeated-split arms that carry the solve were extended to nine seeds per split (rounds 89 to 93 and 102 to 103), and the two recipes with neither the vector nor four media to six (rounds 100 and 105). Round 101 swapped the second medium and round 106 added the fold ensemble with no metabolism.

**Every form of the module beats no metabolism.** Against the same cell with no metabolic column, on the ten repeated splits at nine seeds each, the one-medium recipe with the vector and the counts reads +0.061 and the four-media recipe +0.059, each higher on ten of ten splits and on 89 and 88 of 90 split-and-seed pairs. The vector without the counts reads +0.040, the counts without the vector +0.041 and the four ratios alone +0.028, each on ten of ten. At six seeds the two-media counts recipe reads +0.036 and the one-medium counts recipe +0.032, each on nine of ten. Round 106 gave the no-metabolism cell its fold ensemble at three seeds per fold, 0.878, against 0.932 for the one-medium recipe under the same protocol.

**One medium costs nothing on two protocols and about 0.01 on the third.** With the flux vector and the counts kept, the YPD ratio alone against all four ratios reads +0.002 over the ten splits at nine seeds each, higher on six of ten. The fold ensembles agree, 0.932 against 0.934 at one seed per fold and 0.932 against 0.933 at three. The cells of the released split do not: at round 83's six seeds they read 0.916 against 0.918, and round 107 (job 2683) took both recipes to nine seeds, where they read 0.912 against 0.920, −0.009 paired by seed with the one-medium cell higher on one of nine. Two of the three added one-medium seeds read under 0.90. So round 83's conclusion holds on ten other held-out fifths and as an ensemble, and on the released 195 genes the single-seed cost is at most a hundredth.

**The solve and the counts are complementary and about equal.** Over the four ratios alone the counts are worth +0.013 (six of ten) and the vector +0.012 (eight of ten) on the repeated splits, nine seeds each. Added to the other, the counts are worth +0.019 (nine of ten) and the vector +0.018 (nine of ten). The released split orders them the other way, 0.911 for the vector over nine seeds against 0.902 for the counts over six, and so do the fold ensembles, 0.920 against 0.909. Across the three protocols neither half is reliably the larger, and the pair is reliably above either.

**The simplifications do not stack.** Once the vector is gone the other media carry signal the counts do not replace. With the counts and no flux vector, four ratios read 0.902 on the released split, the YPD and SC ratios 0.896, and the YPD ratio alone 0.888 (fold ensembles at three seeds per fold 0.909, 0.909 and 0.898), and on the repeated splits at six seeds one medium for four costs −0.011 (higher on two of ten) while two media for four cost −0.006 (two of ten), so the second medium buys back about a third of the gap. Round 101 asked whether the second medium matters: the YPD and SM ratios read 0.892 per cell and 0.907 as a one-seed fold ensemble, −0.004 against YPD and SC paired by seed (higher on two of six), so synthetic minimal medium is no better a partner than synthetic complete. The recipe that respects one experiment and stores no flux vector, one ratio with the six counts, reads 0.888 per cell and 0.902 as a fold ensemble against 0.920 and 0.934 for the full recipe, and −0.029 on the repeated splits, lower on ten of ten. The cheapest admissible recipe that keeps the full score is therefore the one-medium recipe with the vector. Its growth ratio already requires one linear program per genotype and medium, run once offline, and the parsimonious solve that yields the vector adds a second program on the same model, so the vector is nearly free once the ratio is computed.

**The six counts are redundant among themselves.** Round 77's misrank pattern pointed at the sole-reaction count as the one useful column. Round 84 tested that: beside the vector, that count alone reads 0.914, the other five without it 0.918, and all six 0.918, over six seeds each. The five without it recover the whole gain (−0.001 against all six) and the one alone recovers most of it (+0.006 over none, higher on five of six seeds), so no single column carries it.

**The training budget is not a limit.** Round 97 reran the full recipe and the counts-alone recipe at 400 epochs. Validation selects the scored epoch, which falls between 30 and 76 in all twelve cells, and each 400-epoch cell reproduced its 200-epoch partner's trajectory, selected the same epoch and returned the same held-out AUROC to the last digit. Nothing after epoch 76 is scored under this selection rule. That reproducibility held within one code version on one card; ten fold cells of the layer-free form rerun in round 92 under a later code version differ from their first runs by up to 0.066 on a single cell, while the three-seed fold ensemble reads 0.921 against 0.919 on the first runs (round 34).

**Three seeds per fold do not move the fold ensembles.** Rounds 92 and 94 took each fold model from one seed to the mean logit of three. The ensembles read 0.933 for the full recipe, 0.932 for one medium, 0.921 for the vector without counts and 0.909 for the counts without the vector, each within 0.001 of its one-seed value.

**Table 102:** What the affine columns cost and what they score, on the same backbone (pathway multi-hot with ProtT5). Media is the number of FBA growth ratios on the logit, each a different environment; vector says whether the recipe also reads the stored flux vector of the parsimonious solve. Every ratio is itself one FBA program per genotype and medium, run once offline. Columns: seeds, their mean held-out AUROC and its spread, and the five-fold ensemble at one seed per fold and at three. Best of each column in bold. Written by `notes_tex_tables.py` from `results/simplification_readout.json`.

| affine columns | media | vector | seeds | cell mean | sd | folds, 1 seed | folds, 3 seeds |
|---|---|---|---|---|---|---|---|
| four ratios, flux vector, rule counts | 4 | yes | 9 | **0.920** | 0.006 | **0.934** | **0.933** |
| YPD ratio, flux vector, rule counts | 1 | yes | 9 | 0.912 | 0.009 | 0.932 | 0.932 |
| four ratios, flux vector | 4 | yes | 9 | 0.911 | 0.007 | 0.920 | 0.921 |
| four ratios, rule counts | 4 | no | 6 | 0.902 | 0.005 | 0.909 | 0.909 |
| YPD and SC ratios, rule counts | 2 | no | 6 | 0.896 | 0.013 | 0.905 | 0.909 |
| YPD and SM ratios, rule counts | 2 | no | 6 | 0.892 | 0.008 | 0.907 | – |
| YPD ratio, rule counts | 1 | no | 6 | 0.888 | 0.013 | 0.902 | 0.898 |
| four ratios | 4 | no | 3 | 0.890 | 0.005 | – | – |
| YPD ratio | 1 | no | 6 | 0.877 | 0.011 | – | – |
| no metabolism | 0 | no | 6 | 0.858 | 0.004 | – | 0.877 |
| ProtT5 only: YPD ratio, flux vector, rule counts | 1 | yes | 6 | 0.883 | 0.003 | 0.892 | – |
| ProtT5 only: no metabolism | 0 | no | 6 | 0.771 | 0.011 | 0.795 | – |

**Table 103:** Every paired difference the simplification rounds fixed in advance, first recipe minus second. On the repeated splits the unit is a split, scored as the mean over the seeds both arms ran; the test blocks overlap, so no p-value is given. Columns: units, seeds per unit, the two means, the mean difference with its standard deviation over units, and the units on which the first recipe scored higher. Written by `notes_tex_tables.py` from `results/simplification_readout.json`.

| first minus second | units | seeds | first | second | difference (sd) | higher |
|---|---|---|---|---|---|---|
| *ten repeated outer splits, unit = split* | | | | | | |
| ProtT5 only: one medium, flux vector and counts, over no metabolism | 10 | 3 | 0.835 | 0.734 | +0.1016 (0.0395) | 10/10 |
| ProtT5 only against the pathway backbone, both with the module | 10 | 3 | 0.835 | 0.862 | −0.0266 (0.0206) | 1/10 |
| ProtT5 only against the pathway backbone, neither with the module | 10 | 3 | 0.734 | 0.796 | −0.0626 (0.0370) | 1/10 |
| one medium for four, flux vector and counts kept | 10 | 9 | 0.863 | 0.861 | +0.0021 (0.0071) | 6/10 |
| counts for the flux vector, four media | 10 | 9 | 0.843 | 0.842 | +0.0010 (0.0238) | 5/10 |
| counts over the four ratios alone | 10 | 9 | 0.843 | 0.830 | +0.0133 (0.0208) | 6/10 |
| flux vector over the four ratios alone | 10 | 9 | 0.842 | 0.830 | +0.0123 (0.0211) | 8/10 |
| counts added to the flux vector, four media | 10 | 9 | 0.861 | 0.842 | +0.0188 (0.0165) | 9/10 |
| no flux vector, counts kept, four media | 10 | 9 | 0.843 | 0.861 | −0.0178 (0.0146) | 1/10 |
| one medium for four, no flux vector | 10 | 6 | 0.833 | 0.843 | −0.0105 (0.0093) | 2/10 |
| two media for four, no flux vector | 10 | 6 | 0.837 | 0.843 | −0.0059 (0.0047) | 2/10 |
| two media for one, no flux vector | 10 | 6 | 0.837 | 0.833 | +0.0046 (0.0089) | 6/10 |
| no flux vector and one medium, against the full recipe | 10 | 6 | 0.833 | 0.862 | −0.0289 (0.0146) | 0/10 |
| one medium, flux vector and counts, over no metabolism | 10 | 9 | 0.863 | 0.802 | +0.0607 (0.0275) | 10/10 |
| four media, flux vector and counts, over no metabolism | 10 | 9 | 0.861 | 0.802 | +0.0587 (0.0254) | 10/10 |
| four media and the flux vector, over no metabolism | 10 | 9 | 0.842 | 0.802 | +0.0398 (0.0267) | 10/10 |
| four media and the counts, over no metabolism | 10 | 9 | 0.843 | 0.802 | +0.0409 (0.0247) | 10/10 |
| two media and the counts, over no metabolism | 10 | 6 | 0.837 | 0.801 | +0.0365 (0.0267) | 9/10 |
| one medium and the counts, over no metabolism | 10 | 6 | 0.833 | 0.801 | +0.0319 (0.0234) | 9/10 |
| four ratios alone, over no metabolism | 10 | 9 | 0.830 | 0.802 | +0.0276 (0.0141) | 10/10 |
| *rule-count subsets on the released split, unit = seed* | | | | | | |
| sole-reaction count alone, against all six | 6 | 1 | 0.914 | 0.918 | −0.0046 (0.0088) | 2/6 |
| five without it, against all six | 6 | 1 | 0.917 | 0.918 | −0.0010 (0.0076) | 3/6 |
| sole-reaction count alone, against none | 6 | 1 | 0.914 | 0.908 | +0.0056 (0.0114) | 5/6 |
| five without it, against none | 6 | 1 | 0.917 | 0.908 | +0.0092 (0.0068) | 5/6 |
| *400 epochs against 200 on the released split, unit = seed* | | | | | | |
| flux vector and counts | 6 | 1 | 0.918 | 0.918 | +0.0000 (0.0000) | 0/6 |
| counts alone | 6 | 1 | 0.902 | 0.902 | +0.0000 (0.0000) | 0/6 |

# 9 The results on one page ✗

The study had two goals, and this section answers them in that order.

**Goal 1: the best held-out score, against Flux Cone Learning and FBA.** On the 195 released held-out genes the best model scores 0.955 and the best single model 0.934, against 0.742 for Flux Cone Learning and 0.640 for FBA on the same genes (Figure 94**a**). The margin of the averaged model over the better single model was fixed in advance and holds on nine of ten other held-out fifths (Section 8.1).

**Goal 2: a metabolic module that carries to other datasets.** A record in the database is one genotype in one environment, so a portable module may read one medium only, and it may not solve a program inside the training loop at database scale. The form that meets both is an input and no layer: one parsimonious FBA solve per genotype and medium, run once offline and cached, read as $\log(1 + |v|)$ by a zero-initialized affine on the logit, beside that medium's growth ratio and six counts read off the gene-protein-reaction rules. On the ten repeated splits, nine seeds each, that single-medium form scores +0.061 over the same cell with no metabolism, higher on ten of ten splits and on 89 of 90 split-and-seed pairs (Figure 95**e** and **f**), and it matches the four-media form there (+0.002) and as a five-fold ensemble (0.932 against 0.933), while single cells on the released split read 0.912 against 0.920 at nine seeds (Section 8.10). Every form that kept a linear program or a projection inside the training loop scored lower and less reliably than this one (Figure 95**a**), so the layer is retired and the solve is kept as data. The store for the 13.5 million records of the solid-growth build is 469,474 programs and 71 minutes on 20 cores (Figure 95**c**).

**The best method is two models averaged, and the best single model is a tie.** Under the fold protocol on the released 195 genes the equal-weight average of graph convolution on GO with the transformer reading the solve scores 0.955 with the solve vector and the gene-rule counts beside it (Section 8.5), 0.954 with the cached solve head and 0.953 with the layer-free solve, the maximum of thirteen post hoc averages. Alone, the rule-count cell scores 0.934, the layer-free form 0.920 and graph convolution on GO 0.917, with overlapping intervals, against Flux Cone Learning's 0.742 and FBA's 0.640 on the same genes (Figure 94**a** and **b**). On ten other held-out fifths

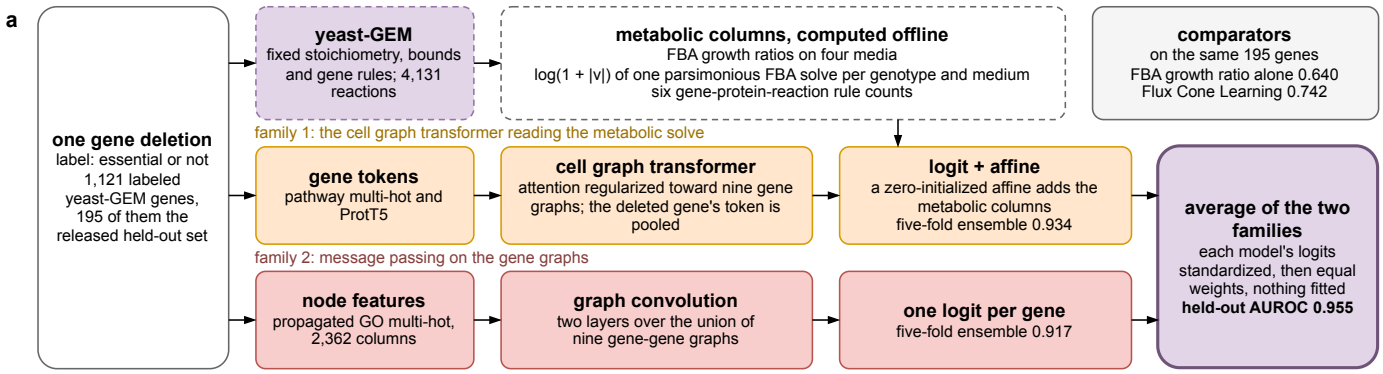



**Figure 93: The two model families and their average.** Family 1 is the cell graph transformer: gene tokens from the pathway multi-hot and ProtT5, attention regularized toward nine gene graphs, the deleted gene's token pooled to a logit, and a zero-initialized affine that adds the metabolic columns computed offline from yeast-GEM. Family 2 is a two-layer graph convolution over the union of nine gene-gene graphs on the propagated Gene Ontology multi-hot. Each family is a five-fold ensemble, and the average standardizes each model's logits and weights them equally with nothing fitted. The two comparators score the same 195 genes. Drawn by `notes/assets/drawio/028-families.gen.py`.

of the labeled genes, with the average fixed in advance, the average beats the better single model on nine (mean +0.023) and graph convolution on GO beats the transformer on nine (Section 8.1).

**The module form that helps is the one with no layer, and the rules help beside it.** On one backbone the cell with no metabolism reads 0.858 on six seeds, FBA's growth ratios on the logit 0.890, and the deletion's solved flux vector as $\log(1 + |v|)$ on the logit 0.911 on nine (sd 0.007, minimum 0.896; the three extra seeds of each are round 81's controls, which is why the paired reads below quote the six they share with their partner). Every form that keeps a linear program in the training loop reads lower and less reliably: the cached solve with a flux head 0.882 with one seed at 0.772, learned ceilings with a flux head 0.845 with two seeds unfit, the gated growth column 0.867, and the forced growth column 0.699 on twelve seeds. The layer-free form costs 30 minutes per cell with three cells on a card against 83 to 100 for the forms that solve in the loop (Figure 95**a** and **b**). On the ten other held-out fifths the vector's gain over no metabolism holds on ten of ten (+0.047) and its gain over the four growth ratios alone does not separate from zero (+0.011, sd 0.023, seven of ten; Section 8.1). Six counts of gene-protein-reaction redundancy beside the vector read 0.918 on the released split against 0.908, and 0.862 against 0.843 on the ten fifths, above on six of six seeds and eight of ten splits (Section 8.3). Against the four ratios alone the counts are worth +0.012 and the solve +0.024 on the released split (Section 8.6) and +0.010 and +0.010 on the ten fifths, so the two halves are complementary and neither is reliably the larger (Section 8.10).

**Every metabolic column fits inside one experiment.** The four growth ratios are four media, so they read four experiments at once; restricted to the one medium the solve store is built on, the full recipe reads 0.916 against 0.918 at six seeds (0.912 against 0.920 at nine) and the solve-only recipe 0.912 against 0.911, while the ratio alone falls from 0.890 to 0.877 (Section 8.7). The solve supplies from within one environment what the four media were carrying, and the ten fifths agree (+0.002 for one medium, above on six of ten). Without the flux vector the restriction is not free: one ratio with the rule counts reads 0.888 against 0.902 for four, and 0.030 under the full recipe on the ten fifths (Section 8.10).

**The interface is an affine on the logit, and it carries across model families.** The graph family reads the same columns. As node features the flux vector does not help graph convolution on GO (−0.009 over the ten fifths, higher on one); as a zero-initialized affine on the node's logit, the transformer's form, the YPD ratio alone reads +0.011 (ten of ten), the single-experiment recipe +0.018 (nine of ten) and the four-media recipe +0.021 (nine of ten) (Figure 95**g**, Section 5.3). The module is therefore a set of offline columns and one linear map added to whatever logit a model produces. Handing both families the columns does not improve their average (0.897 against 0.907 on the ten fifths), so the best method keeps the module on one family and averages it with the plain graph convolution. Where every held-out record reads the constant row, the genome-wide 4,674 genes, the affine still does not hurt: graph convolution on GO reads 0.767 with it against 0.702 without.

**The layer-free form scales, and it covers only the genes with a reaction.** Over the 13.5 million records of the solid-growth build the transform is 469,474 programs, solved once in 71 minutes on 20 cores and read as a gather in training (Figure 95**c**). Off the metabolic genes every deletion reads one constant row, and the transformer with the layer-free solve scores 0.570 on the 4,674 genome-wide genes where graph convolution on ProtT5 scores 0.755 and the raw degree in the physical graph 0.740 (Figure 94**c**). A model over the whole database therefore needs both inputs: the solve store for the genes with a GEM reaction, and the gene graphs as a message-passing input for the rest.

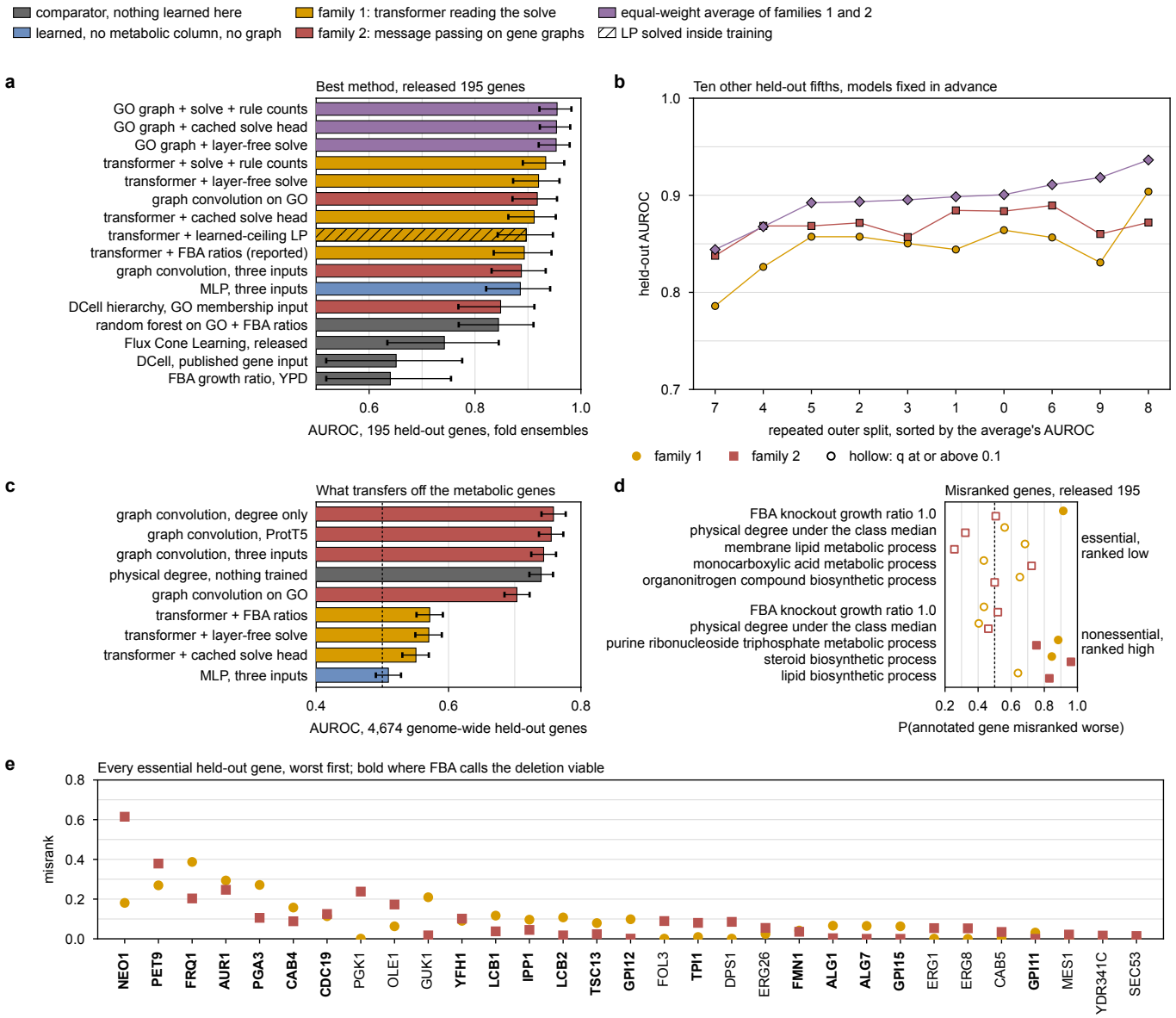



**Figure 94: Goal 1: the best method on gene essentiality, against Flux Cone Learning and FBA.** The families are drawn in Figure 93. One color means one thing in every panel: gray a comparator with nothing learned here, blue a learned model reading no metabolic column and no graph, orange family 1, red family 2, purple their average; a hatch marks a linear program solved inside the training loop. **a**, Every method on the 195 released held-out genes, best first; the first three rows are averages of family 2 with a form of family 1, and the two DCell rows are round 108. Learned models are five-fold ensembles, the random forest is one fit on the split file, and intervals are gene bootstraps. **b**, The layer-free form of family 1 (circles), graph convolution on the Gene Ontology multi-hot (squares) and their average (diamonds), fixed in advance, on ten other held-out fifths of the labeled genes, three seeds per split, sorted by the average's AUROC. **c**, The same models trained on the 629 released genes and read on the 4,674 genome-wide genes that carry no GEM reaction, pooled over seeds with a gene bootstrap; the dashed line is chance. **d**, What the misranked genes of the released 195 share, per class: the chance that a gene carrying the annotation is misranked worse than one without it, 0.5 (dashed) being no association, for family 1 with the layer-free solve (circles) and family 2 (squares), filled where the Benjamini-Hochberg $q$ within the family is under 0.1; the genome-wide set is in Table 104. **e**, Every one of the 31 essential genes in the released held-out set, worst first by the two families' mean misrank, with the fraction of nonessential genes each family ranks above it; a gene's name is bold where FBA calls its deletion viable on YPD, the feature panel **d** tests. Written by `experiments/028-gene-essentiality/scripts/summary_figure.py`.

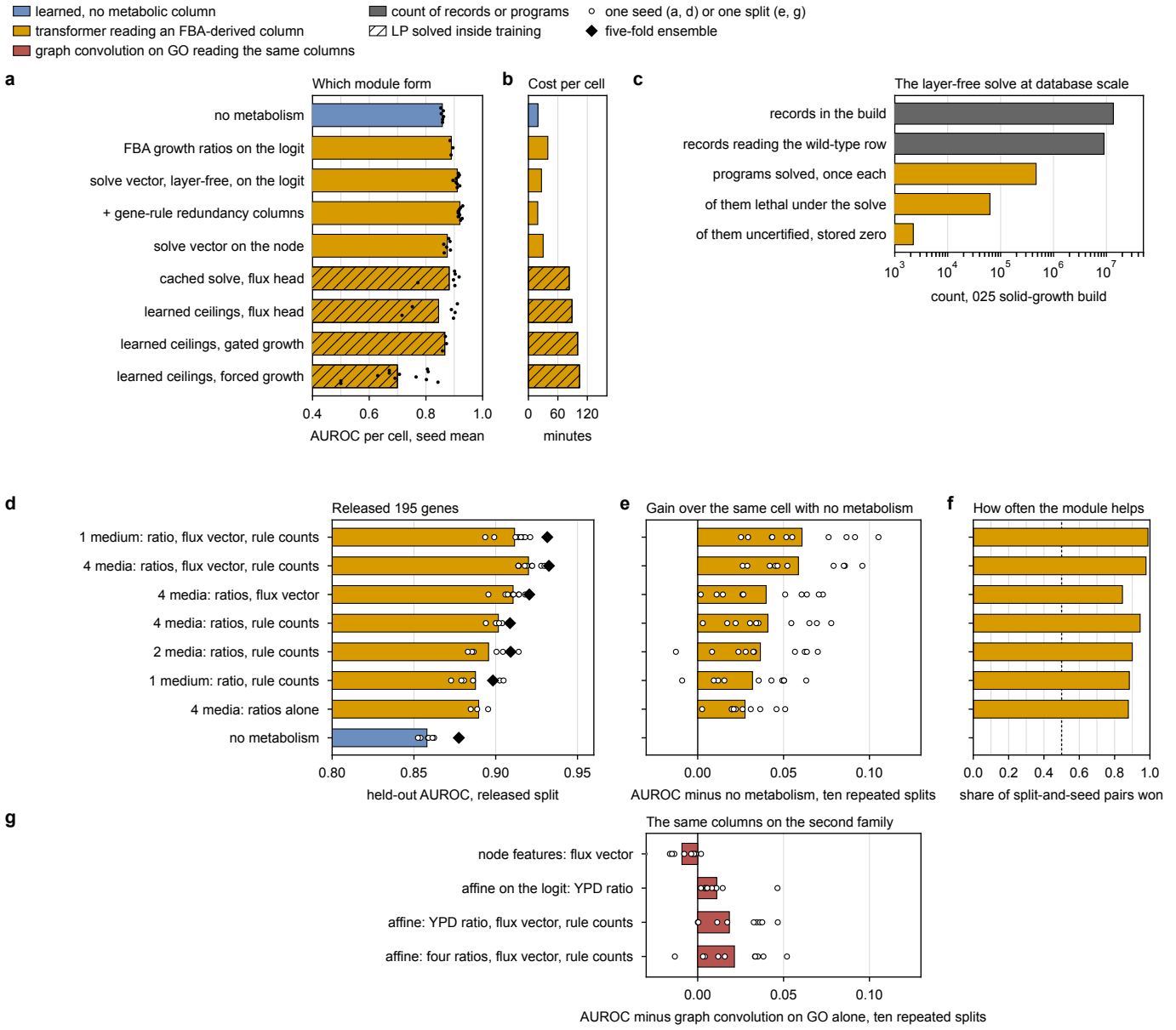



**Figure 95: Goal 2: the form of the metabolic module that helps, what it costs, and what a record of one genotype in one medium can carry.** Orange is the transformer reading any FBA-derived column, blue the same cell with no metabolic column, gray a count; a hatch marks a linear program solved inside the training loop. **a**, The forms of the module on one backbone, the pathway multi-hot with ProtT5: the bar is the seed mean of the held-out AUROC and each point one seed; the forced growth column ran on ProtT5 alone, twelve seeds. **b**, Wall minutes per cell for the forms of **a**, as run, with up to three cells sharing a card. **c**, What the layer-free solve costs over the 025 solid-growth build, logarithmic: the records, those whose deletion touches no GEM gene and read the wild-type row, the distinct programs solved once each, and the lethal and uncertified ones among them. **d**, The layer-free forms by the number of media whose growth ratio they read and by whether they also read the stored flux vector and the six gene-rule counts, on the released split: the bar is the seed mean, each point one seed, the diamond the five-fold ensemble. **e**, Each form minus the same cell with no metabolism on the ten repeated outer splits: the bar is the mean over splits and each point one split. **f**, The share of split-and-seed pairs on which the form scores above the cell with no metabolism; the dashed line is a module that helps on half the runs. **g**, The same columns handed to the second family, graph convolution on GO, on the ten repeated splits at three seeds: as node features (the flux vector, job 2684) and as a zero-initialized affine on the node's logit (job 2691), each minus the same convolution on GO alone. Written by `experiments/028-gene-essentiality/scripts/summary_figure.py` from `results/simplification_readout.json` and `results/resplit_readout.json`.

**The two families miss different genes, and the annotations say which.** On the released 195 the essential genes the transformer ranks low are the ones FBA calls viable: the 19 of 31 with a knockout growth ratio of 1.0 have a mean misrank of 0.134 against 0.025 for the rest (effect 0.91, $q$ 0.003), and under graph convolution on GO the same feature has no effect (0.51). No biological-process term reaches $q < 0.1$ among the 31 essential genes under either model. The nonessential genes both models rank high sit in steroid biosynthesis (effect 0.85 for the transformer and 0.96 for the graph model) and, for the transformer, in nucleoside triphosphate metabolism (0.88) and monosaccharide biosynthesis (0.83). On the genome-wide set the graph model ranks essential genes low when their degree in the physical graph is under the class median (effect 0.70) and when they act in cytoskeleton organization (0.79) or a microtubule-based process (0.81), and ranks nonessential genes high in ncRNA metabolism (0.90), rRNA processing (0.92), RNA modification (0.92) and cytoplasmic translation (0.85) (Figure 94**d** and **e**, Table 104). Hypothesis, untested: the graph model scores a gene by its neighborhood, and the nonessential members of the ribosome biogenesis and translation machinery sit in neighborhoods made of essential genes. No essential gene is missed by every predictor: over the ten fifths, where 140 essential genes are held out, none sits below the median nonessential gene under all four learned predictors, and the residual they share is not thin annotation (its correlation with the count of GO terms is $-0.002$) but transporters and cofactor supply that FBA calls viable, which modeling the deletion collection's strain background does not change (Section 5.2).

**Table 104:** What the misranked genes of each held-out set share, per model and class. A gene's misrank is the fraction of the other class on the wrong side of it, averaged over the model's cells; a term is tested within one class by a one-sided Mann-Whitney $U$ of the annotated genes' misrank against the rest, and the effect is $U$ as a probability, the chance an annotated gene is misranked worse than an unannotated one (0.5 is no association), with the Benjamini-Hochberg $q$ within the family in parentheses. Rows per family: the two named features where both sides hold five genes, then the four best specific biological-process terms (at most 5 percent of the class or 12 genes, near-duplicates removed). A row is bold with ↑ where the association is both established and substantive, $q < 0.1$ and an effect of 0.75 or more; on the genome-wide set almost every term clears $q$ alone, so the effect is what separates them. Written by `misclass_enrichment.py`.

| term | genes | misrank, annotated | misrank, rest | effect ($q$) |
|---|---|---|---|---|
| *released 195 (metabolic genes); transformer, layer-free solve; essential, ranked low (31 genes, 44 terms, 1 at $q < 0.1$)* | | | | |
| **FBA knockout growth ratio 1.0** | 19 | **0.134** ↑ | 0.025 | **0.91 (0.003)** |
| physical degree under the class median | 15 | 0.103 | 0.082 | 0.56 (0.944) |
| membrane lipid metabolic process | 6 | 0.119 | 0.085 | 0.68 (0.944) |
| organonitrogen compound biosynthetic process | 8 | 0.116 | 0.083 | 0.65 (0.944) |
| carbohydrate derivative biosynthetic process | 8 | 0.084 | 0.095 | 0.56 (0.944) |
| generation of precursor metabolites and energy | 5 | 0.092 | 0.092 | 0.52 (0.944) |
| *released 195 (metabolic genes); transformer, layer-free solve; nonessential, ranked high (164 genes, 211 terms, 70 at $q < 0.1$)* | | | | |
| FBA knockout growth ratio 1.0 | 152 | 0.082 | 0.214 | 0.44 (0.892) |
| physical degree under the class median | 81 | 0.058 | 0.124 | 0.40 (0.988) |
| **purine ribonucleoside triphosphate metabolic process** | 7 | **0.528** ↑ | 0.072 | **0.88 (0.006)** |
| **steroid biosynthetic process** | 5 | **0.324** ↑ | 0.084 | **0.85 (0.025)** |
| **monosaccharide biosynthetic process** | 5 | **0.472** ↑ | 0.080 | **0.83 (0.026)** |
| macromolecule modification | 9 | 0.140 | 0.089 | 0.72 (0.043) |
| *released 195 (metabolic genes); graph convolution on GO; essential, ranked low (31 genes, 44 terms, 0 at $q < 0.1$)* | | | | |
| FBA knockout growth ratio 1.0 | 19 | 0.111 | 0.071 | 0.51 (0.990) |
| physical degree under the class median | 15 | 0.052 | 0.136 | 0.32 (1.000) |
| monocarboxylic acid metabolic process | 5 | 0.128 | 0.089 | 0.72 (0.758) |
| small molecule biosynthetic process | 5 | 0.114 | 0.092 | 0.69 (0.758) |
| generation of precursor metabolites and energy | 5 | 0.164 | 0.082 | 0.68 (0.758) |
| purine ribonucleotide metabolic process | 7 | 0.091 | 0.097 | 0.60 (0.773) |
| *released 195 (metabolic genes); graph convolution on GO; nonessential, ranked high (164 genes, 211 terms, 57 at $q < 0.1$)* | | | | |
| FBA knockout growth ratio 1.0 | 152 | 0.095 | 0.099 | 0.52 (0.598) |
| physical degree under the class median | 81 | 0.070 | 0.120 | 0.46 (0.940) |
| **steroid biosynthetic process** | 5 | **0.554** ↑ | 0.081 | **0.96 (0.004)** |
| **lipid biosynthetic process** | 12 | **0.329** ↑ | 0.077 | **0.83 (0.004)** |
| macromolecule modification | 9 | 0.191 | 0.090 | 0.75 (0.052) |
| **pyruvate metabolic process** | 7 | **0.220** ↑ | 0.090 | **0.77 (0.059)** |
| *genome-wide 4,674 (no GEM reaction); transformer, layer-free solve; essential, ranked low (974 genes, 731 terms, 61 at $q < 0.1$)* | | | | |
| physical degree under the class median | 476 | 0.404 | 0.462 | 0.43 (1.000) |
| establishment of protein localization to mitochondrion | 25 | 0.651 | 0.428 | 0.74 (0.004) |
| **inner mitochondrial membrane organization** | 5 | **0.869** ↑ | 0.431 | **0.95 (0.012)** |
| nucleosome organization | 17 | 0.640 | 0.430 | 0.74 (0.016) |
| localization within membrane | 42 | 0.562 | 0.428 | 0.65 (0.024) |
| *genome-wide 4,674 (no GEM reaction); transformer, layer-free solve; nonessential, ranked high (3,700 genes, 1,427 terms, 59 at $q < 0.1$)* | | | | |
| physical degree under the class median | 1,810 | 0.495 | 0.375 | 0.61 (<0.001) |
| DNA recombination | 113 | 0.592 | 0.429 | 0.65 (<0.001) |
| homologous recombination | 47 | 0.675 | 0.431 | 0.73 (<0.001) |
| RNA modification | 65 | 0.630 | 0.430 | 0.69 (<0.001) |
| DNA repair | 135 | 0.563 | 0.429 | 0.63 (<0.001) |
| *genome-wide 4,674 (no GEM reaction); graph convolution on ProtT5; essential, ranked low (974 genes, 731 terms, 252 at $q < 0.1$)* | | | | |
| physical degree under the class median | 476 | 0.337 | 0.158 | 0.70 (<0.001) |
| **cytoskeleton organization** | 43 | **0.438** ↑ | 0.236 | **0.79 (<0.001)** |
| **microtubule-based process** | 36 | **0.459** ↑ | 0.237 | **0.81 (<0.001)** |
| mitochondrion organization | 37 | 0.379 | 0.240 | 0.74 (<0.001) |
| **attachment of spindle microtubules to kinetochore** | 20 | **0.488** ↑ | 0.240 | **0.81 (<0.001)** |
| *genome-wide 4,674 (no GEM reaction); graph convolution on ProtT5; nonessential, ranked high (3,700 genes, 1,427 terms, 599 at $q < 0.1$)* | | | | |
| physical degree under the class median | 1,810 | 0.166 | 0.321 | 0.26 (1.000) |
| **ncRNA metabolic process** | 173 | **0.635** ↑ | 0.226 | **0.90 (<0.001)** |
| **rRNA processing** | 85 | **0.648** ↑ | 0.236 | **0.92 (<0.001)** |
| **cytoplasmic translation** | 98 | **0.551** ↑ | 0.237 | **0.85 (<0.001)** |
| **RNA modification** | 65 | **0.717** ↑ | 0.237 | **0.92 (<0.001)** |

## 9.1 What was run from the closing list, and what is left out ✗

Six experiments were listed as able to change a conclusion above. Four are run, one is left as a design note, and one is out of scope.

**Repeated outer splits for the finalists: run.** Ten stratified re-splits of the 1,121 labeled genes at the released proportions, with the finalists and the average fixed in advance (Section 8.1), later extended to the module's forms and to nine seeds per split (Section 8.10).

**A rule-aware input for the essential genes the solve calls viable: run.** Six counts read off the gene-protein-reaction rules, beside the vector (Section 8.3), under the fold protocol (Section 8.5), alone (Section 8.6) and column by column (Section 8.10). They help on the fifteen genes holding a reaction of their own and hurt on the four an alternative gene set covers.

**The gene graphs as an input to the transformer that reads the solve: not run.** Round 76 tried the cheapest form, degree and clustering in the physical graph as affine columns (Section 8.2). A faithful version would have to take all nine gene graphs, and the transformer already receives those graphs through the regularization of its attention toward them (these cells run with no hard mask), so graph-derived node inputs would duplicate them; testing the inputs properly means relaxing that regularizer at the same time, which is a change to the backbone and not to the module. The averaged model already recovers what the graph family knows.

**The module on a second model family: run.** Graph convolution on GO reads the columns as an affine on its logit and gains on nine of ten fifths (Section 5.3); as node features it does not.

**DCell's hierarchy and the ontology of the trigenic runs: run.** The ontology's own structure reads the GO multi-hot worse than message passing on either of the two ontologies tried (Section 8.8).

**The store on a second phenotype: out of scope.** The endpoint here is essentiality. Panel **c** of Figure 95 shows the store is affordable over the solid-growth build; whether it helps there belongs to that experiment.

The construction in which the binding ceilings differ per genotype (Section 7) stays a design question and is not on this list.